

Static Risk Sensitive Reinforcement Learning

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To my friends and family

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ABSTRACT

Static Risk Sensitive Reinforcement Learning

by

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Reinforcement Learning (RL) is a core area of artificial intelligence (AI) that enables systems to make decisions in complex environments. RL algorithms have shown promise in various domains, including resource management, robotics, and games. However, most existing RL approaches fail to account for the risks associated with decision-making. This oversight becomes particularly critical in high-stakes settings, such as healthcare, finance, criminal justice, and autonomous driving, where poor decisions can have significant consequences.

To address risk in decision-making, another line of research explores the properties (axioms) of “desirable” risk measures. This desirability is defined as the ability to quantify the capital requirement for random losses. Monetary risk measures are widely accepted to serve as the basic requirement for the study of risk measures. In this work, we focus on four prominent monetary risk measures: Value at Risk (VaR), Conditional Value at Risk (CVaR), Entropic Value at Risk (EVaR), and the Entropic Risk Measure (ERM).

Despite various attempts to incorporate risk into RL, many existing methods focus on dynamic risks and distributional Markov risks, leading to algorithms that fail to optimize meaningful risk metrics, limiting their effectiveness in real-world scenarios. This thesis introduces novel algorithms

that accurately optimize risks within the general RL framework of discounted Markov Decision Processes (MDPs). In particular, we prove specific dynamic programming principles for each risk measure and develop reliable risk-sensitive RL algorithms.

This thesis makes three key contributions:

1. We disprove the efficacy of a widely adopted approach that directly applies coherent risk decomposition to MDP policy optimization.
2. We propose two novel fully polynomial-time approximation scheme (FPTAS) algorithms to optimize entropic-based risk MDPs.
3. We propose practical VaR RL algorithms with performance bounds and convergence analysis.

These contributions are based on three peer-reviewed conference publications. By addressing both theoretical and practical challenges in risk-sensitive reinforcement learning, this work aims to develop fundamentally sound models and algorithms to better measure and handle uncertainty in high-risk domains.

The methods presented in this thesis have potential applications in AI, healthcare, finance, and autonomous vehicles. Experimental results demonstrate that our algorithms optimized the desired risk measures accurately and consistently outperform existing methods in optimizing risk in standard RL benchmarks, including inventory management, pest population control, gambler’s ruin, cliff walking, machine replacement, and river swim environments.

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CHAPTER 1

Introduction

What should I do to become financially independent and retire at the earliest possible age? Should I buy a lottery ticket? Should I get a college degree? Should I choose a major that I enjoy, such as education, or should I choose a more lucrative major, such as accounting? Learning, planning, and making decisions to complete tasks are indispensable processes in our daily lives. Ever wish you had a genius assistant who could tell you the best decision that leads you closer to your desired goal? Artificial intelligence (AI) is not just about robots and sci-fi movies, it's the reason you get the perfect Spotify playlist, Amazon two-day delivery, and desirable results when searching on Google. This branch of computer science focuses on creating systems capable of performing tasks that typically require human intelligence. By letting AI handle repetitive learning and planning tasks, humans are freed up to focus on researching more efficient methods, integrating different tools, and creating new innovative applications to make life easier.

1.1 Reinforcement Learning

Reinforcement Learning (RL) is an important subfield of AI that encompasses both autonomous learning and autonomous planning. RL agents interact autonomously with the environment, learn and then plan to optimize a specific objective. This work focuses on a specific type of RL in which the policy makers are interested in maximizing a risk sensitive performance measure of the total reward.

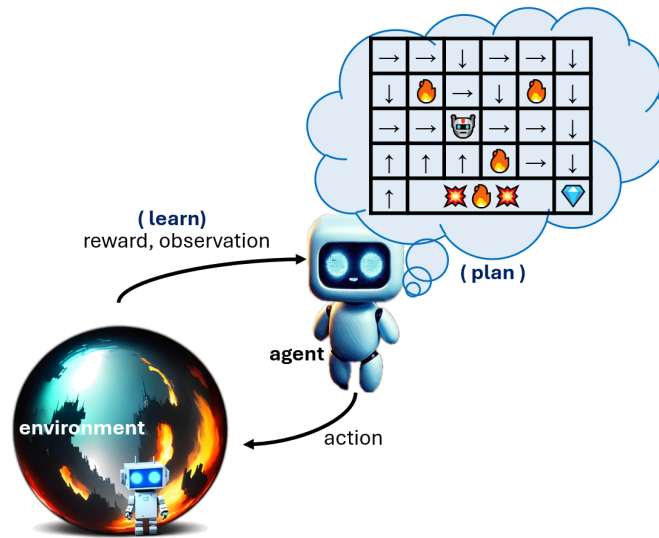


Figure 1.1: RL process

1.1.1 Autonomous Learning

Instead of being explicitly programmed, the RL agent learns autonomously by trial and error. In the beginning, the agent chooses a random available action in each step, observes the outcome and rewards associated with its action. After many iterations, the agent learns to recognize the preferable strategy and act more intelligently to maximize the performance measure of the total reward. Since most environments are stochastic, the same action or decision may not always end with the same results, and there may be trade-offs between different possible outcomes. Stochastic approximation or stochastic gradient descent is used to approximate the performance measure from a stream of samples.

1.1.2 Autonomous Planning

Life is full of decisions that unfold one after another. Imagine playing a video game where each move you make (choosing a path, picking up a power-up, or fighting a boss) affects what happens next. That's sequential decision-making, every choice sets the stage for future outcomes. Typical planning problems are characterized by a structured state space, a set of possible actions, possible outcomes, and a performance objective function. This thesis treats planning as an optimization problem: once the problem is formalized, the computer can evaluate the performance efficiently

to find the best plan. Specifically, we focus on general planning methods that can apply to all environments and domains under the Markov Decision Process (MDP) framework.

MDPs are a general mathematical framework, which has been studied since the late 1950s [16], for modeling sequential decision-making with stochastic outcomes. MDPs describe the environment using key parameters like transition probabilities and rewards aligned for each decision. Most works on MDPs have focused on optimizing the expected total discounted rewards attained in the interaction with the environment. This goal of maximizing the expected total discounted rewards is known as the *risk-neutral* objective. Risk-neutral RL algorithms have shown great performance in a wide range of domains including digital advertisement, computer games, and robotics.

1.1.3 Challenges

In high-stakes environments, agents must account for the risk of catastrophic failures in decision-making, it is becoming apparent that risk-neutral objectives are often insufficient. In these domains, policymakers generally prefer less risky alternatives, even if they may sacrifice some mean reward. The one-size-fits-all risk-neutral model is insufficient in many use cases, especially when dealing with the risk preferences of diverse problems and decision-makers.

Risk-sensitive RL has become an active research topic in recent years driven by the desire to bring automated, data-driven decision-making to high-stakes domains. One stream of research has focused on specific risk measures or methods for evaluating the quality of an uncertain decision. Over the past 20 years, *monetary risk measures* have become the most popular approach for accounting for the risk associated with uncertain outcomes. Some of the most popular monetary risk measures in RL include value at risk (VaR) [79, 82], conditional value at risk (CVaR) [13, 33], entropic risk measure (ERM) [24], and entropic value at risk (EVaR) [93]. The risk-sensitive objective utilizes risk measures that map a distribution of possible outcomes to a real number to quantify the risk associated with each action given the current state.

Another stream of research has focused on developing algorithms for specific application domains, such as motion planning [3, 27, 61], autonomous systems [71, 128], healthcare [77, 117, 135],

investment [89] and others.

We elaborate on two applications to illustrate the importance of risk-sensitive decision-making and the challenges involved.



Figure 1.2: Autonomous vehicle



Figure 1.3: Pest management

Autonomous vehicles Navigating real-world traffic poses a formidable challenge for autonomous vehicles due to their inherently chaotic nature. Factors like unpredictable human behavior, construction zones, and adverse weather conditions create a highly stochastic environment where even a minor misjudgment can result in traffic gridlocks or catastrophic collisions. Autonomous vehicles must navigate a delicate balance between conflicting objectives: (1) ensuring safety, obeying traffic laws, and providing a comfortable ride often require slower, more cautious driving, while (2) minimizing travel time, optimizing traffic flow, and reducing congestion favor a faster pace. In such an environment, risk-averse decision-making becomes paramount, as the cost of failure can be devastating—leading to loss of life, regulatory setbacks, diminished public trust, and reduced investment. Therefore, the challenge lies in developing strategies that can effectively mitigate risks while still delivering passengers to their destinations safely, comfortably, and in a timely manner. This balance is essential for the successful integration of autonomous vehicles into society.

Pest Population Control Management Deploying decision support systems in agriculture is

a complex challenge due to the highly stochastic nature of ecological systems and the variability of weather across seasons. Farmers are inherently risk-averse, as failures in agriculture can have severe consequences, including food shortages, famine, soil erosion, disruption of local ecosystems, and public health crises. Developing risk-averse management strategies is therefore critical. In pest management, the population dynamics of pests can be modeled as an evolving state. Land managers must decide whether to apply pesticides based on the current pest population. Failing to act when necessary could lead to exponential pest population growth, causing irreversible harm to the ecosystems. On the other hand, excessive pesticide use incurs immediate costs and risks long-term consequences such as soil degradation and loss of biodiversity. Adding to this complexity, the effectiveness of any treatment action is highly uncertain, influenced by numerous environmental factors. The central challenge is to develop risk-averse strategies that manage pests effectively while preserving soil health, enabling sustainable and resilient agricultural systems.

Domain specific methods are great for their intended tasks but often perform poorly outside their narrow categories. This thesis takes a different approach, focusing on general MDP framework that work across a variety of settings. Our algorithms are build to tackle a wide range of problems without reinventing the wheel for each new domain. We design scalable algorithms that can handle risk-sensitive objectives while being both accurate and computationally efficient.

1.2 Contributions and Outline

Risk-averse reinforcement learning (RL) research has bifurcated into two distinct streams. One stream focuses on the theoretical aspects of risk-averse decision-making but remains limited to small domains without developing scalable algorithms [13, 79]. The other stream emphasizes large-scale implementations but often neglects the theoretical rigor of the proposed methods, which frequently lack formal guarantees and fail to optimize meaningful risk metrics [39, 81]. Little effort has been made to bridge the gap between these two extremes. This dissertation builds on three conference papers [62–64], aiming to address the missing theoretical components in risk-averse RL and develop algorithms that are both theoretically sound and scalable.

This dissertation aims to develop risk-averse reinforcement learning (RL) algorithms for general discounted Markov Decision Processes (MDPs) that can be applied to complex, high-stakes real-world domains. These algorithms must be fundamentally sound, particularly in applications where failures can result in significant financial losses, injuries, or even loss of life. My research aims to bridge the gap between theoretically robust and practically scalable algorithms in risk-averse RL. In Chapter 2, we will introduce the preliminaries for the thesis, Chapter 6 concludes the thesis, while Chapters 3 to 5 will detail each of our contributions, which are summarized in the following paragraphs.

Chapter 3 proves that the popular risk-averse decompositions, particularly for coherent risk measures such as CVaR and EVaR, fail when applied to policy optimization tasks. Prior works [29, 30, 33, 46, 47, 93, 105, 118] have proposed these algorithms, which were thought to be optimal when the risk level is sufficiently discretized. However, we show that these popular decompositions for CVaR and EVaR are inherently suboptimal regardless of the discretization level. These methods tend to overestimate their effectiveness, as they cannot provide reliable performance guarantees in risk-sensitive decision-making. However, a policy optimization decomposition does hold for VaR, and our proof demonstrates how this risk measure differs from CVaR and EVaR. Our findings are significant because risk-averse algorithms are used in high-stakes environments, making their correctness much more critical. These results have been published in [62].

Building on the solid fundamentals of the VaR dynamic program (DP) introduced in Chapter 3, we propose practical algorithms for VaR MDPs in Chapter 4. Previous works on model-free risk-averse RL typically fall into one of two categories [81]: distributional risk-averse RL [39, 72] and dynamic risk-averse RL [35, 69, 116]. Both methods often lack theoretical rigor and produce solutions that fail to optimize the intended objectives reliably, sometimes leading to unpredictable failures even in small domains, limiting their practical use. Compared with prior works, our new VaR DP decomposition requires neither known transition probabilities nor solving complex saddle point equations and serves as a strong foundation for model-free RL algorithms like VaR-Q-learning. Our VaR-Q-learning algorithm has strong convergence and performance guarantees, making it

suitable for real-world applications. Although this algorithm can be seen as an enhancement to the risk-sensitive Q-learning methods developed in distributional RL, we show that our modification makes a meaningful difference in the quality of the computed policy. These results have been published in [64].

The limitation of works explained in Chapter 3 also motivates us to derive a practical and accurate algorithm for entropic-based risk presented in Chapter 5. We propose new novel polynomial-time delta-optimal algorithms to solve MDPs for ERM and EVaR objectives. Since EVaR provides the tightest bound to CVaR, our EVaR MDP algorithm also serves as a close approximation for CVaR optimization. This study has been published in [63].

CHAPTER 2

Preliminary

This chapter establishes the foundational notations and introduces key risk measures that will be utilized throughout the thesis. Subsequently, we formalize the Markov Decision Process (MDP) framework with a risk-sensitive objective, which we refer to as the risk-sensitive MDP. This framework represents a simplification of the general reinforcement learning paradigm, specifically focusing on autonomous planning, as the model parameters are assumed to be given as part of the problem definition.

In this chapter, we deliberately set aside the autonomous learning aspect of reinforcement learning, where models are unknown and must be inferred through iterative interaction with the environment. Neglecting autonomous learning simplifies the analysis significantly, as we can simply treat the risk-sensitive objective as an optimization problem without needing to consider the exploration and exploitation tradeoff. This more intricate facet of reinforcement learning will be thoroughly explored in Section 4.4, where we address the challenges and methodologies involved in learning from the environment.

2.1 Notation

The equality operator ($=$) signifies that the left-hand side and right-hand side are identical in value. For instance, $a = b$ implies that a and b are equivalent. The definition operator ($:=$) is used to denote the assignment by definition, indicating that the left-hand side is defined as the right-hand side. For example, $f(x) := x^2$ defines the function $f(x)$ as the square of x . Let

$a = (a_1, \dots, a_n)$ be a vector, the p -norm of a for $p \geq 1$ is $\|a\|_p := (\sum_{i=1}^n |a_i|^p)^{1/p}$ and the ∞ -norm $\|a\|_\infty := \max_{i \in \{1, \dots, n\}} |a_i|$.

Set The augmented reals are $\bar{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$ and we denote by $\mathcal{I}$ the class of closed intervals in $\mathbb{R}$. Given sets $\mathcal{M}$ and $\mathbb{S}$, we write $\mathbb{S}^{\mathcal{M}}$ for the set of all functions mapping $\mathcal{M}$ to $\mathbb{S}$. Given a finite set $\mathcal{Z} := \{1, \dots, Z\}$, the probability simplex is $\Delta_{\mathcal{Z}} := \{y \in \mathbb{R}_+^Z \mid \mathbf{1}^\top y = 1\}$. Given a probability space $(\Omega, \mathcal{F}, P)$ where Ω defines the set of all possible outcomes, the σ -algebra $\mathcal{F}$ defines the collection of all events of interest (a subset of sample space) and the probability measure $P: \mathcal{F} \rightarrow [0, 1]$ is a set function that returning an event's probability. We denote $\mathbb{X}: \Omega \rightarrow \mathbb{R}$ as the space of $\mathcal{F}$ -measurable real-valued random variables. Random variables are marked with a tilde, e.g., $\tilde{x} \in \mathbb{X}$.

Sequence A sequence of integers from $m \in \mathbb{Z}$ to $n \in \mathbb{Z}$ is denoted by $m:n = (m, m+1, \dots, n) = (i)_{i=m}^n$. For conciseness, we denote by $[n]$ the sequence of integers from 0 to n .

2.2 Risk Measures

Decisions often face uncertain outcomes encoded as probability distributions over returns. Although these distributions capture the full stochastic nature of rewards, directly comparing two distinct distributions presents a significant challenge. Without a structured framework, determining which option is “better” becomes ambiguous, risk measures address this by transforming a distribution to a scalar value which is pivotal for comparison. Risk measures enable agents to systematically evaluate policies in alignment with desired risk preferences, providing the mathematical foundation to formalize and automate risk-sensitive choices in uncertain environments.

For example, to answer the question “Which distribution on the left of Fig. 2.1 is preferable?” We must first specify the risk measure or objective of interest. From the right panel of Fig. 2.1, the distribution of $\tilde{x}_1$ is preferable under the risk measures $\text{CVaR}_{50\%}$ and $\text{EVaR}_{25\%}$ while the distribution of $\tilde{x}_2$ is preferable under $\mathbb{E}$ and $\text{VaR}_{75\%}$. We provide a detailed explanation of each of

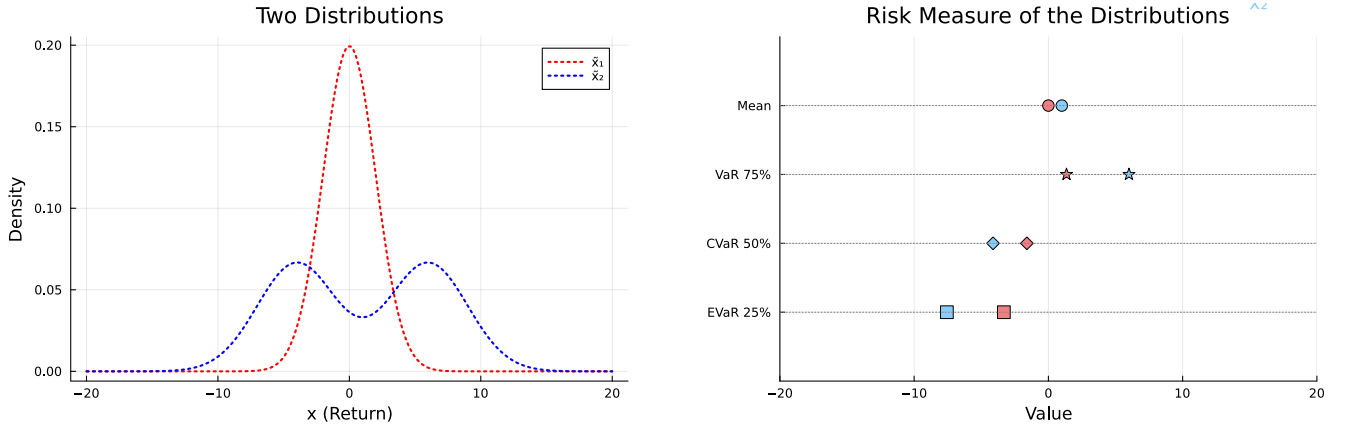


Figure 2.1: Comparing distributions

these risk measures in Sections 2.2.1 to 2.2.5 after introducing the fundamental desirable properties of risk measures. Throughout this thesis, the random variable $\tilde{x} \in \mathbb{X}$ is defined as the reward or return, higher values are preferred. Other formulations of risk measures exist in the literature for random variable $\tilde{x}$ representing losses but induce identical preferences for appropriately chosen rewards and risk level α .

To address risk in decision-making, a distinct line of research focuses on axiomatic properties that define desirable risk measures. Such measures are deemed desirable when they can reliably quantify the capital required to cover uncertain losses. We summarize the **important axioms** of risk measures (denoted by ψ) that generalize the expectation operator to account for the variability of the random variable, as follows:

$$\text{A-1 Monotonicity:} \quad \tilde{x}_1 \leq \tilde{x}_2 \implies \psi[\tilde{x}_1] \leq \psi[\tilde{x}_2], \quad \forall \tilde{x}_1, \tilde{x}_2 \in \mathbb{X}$$

$$\text{A-2 Translation invariance:} \quad \psi[c + \tilde{x}] = c + \psi[\tilde{x}], \quad \forall c \in \mathbb{R}, \forall \tilde{x} \in \mathbb{X}$$

$$\text{A-3 Super-additivity:} \quad \psi[\tilde{x}_1 + \tilde{x}_2] \geq \psi[\tilde{x}_1] + \psi[\tilde{x}_2], \quad \forall \tilde{x}_1, \tilde{x}_2 \in \mathbb{X}$$

$$\text{A-4 Positive homogeneity:} \quad \psi[c \cdot \tilde{x}] = c \cdot \psi[\tilde{x}], \quad \forall c \in \mathbb{R}_+, \forall \tilde{x} \in \mathbb{X}$$

$$\text{A-5 Law invariant:} \quad \tilde{x}_1 \sim \tilde{x}_2 \implies \psi[\tilde{x}_1] = \psi[\tilde{x}_2], \quad \forall \tilde{x}_1, \tilde{x}_2 \in \mathbb{X}$$

$$\text{A-6 Time consistency:} \quad \psi[\tilde{x}_1] = \psi[\psi[\tilde{x}_1 \mid \tilde{x}_2]], \quad \forall \tilde{x}_1, \tilde{x}_2 \in \mathbb{X}$$

Axiom / Property	$\mathbb{E}$, Min, Max	ERM	VaR	CVaR, EVaR
A-1 Monotonicity	✓	✓	✓	✓
A-2 Translation Invariant	✓	✓	✓	✓
A-3 Super-Additive	✓	✓	✗	✓
A-4 Positive Homogeneity	✓	✗	✓	✓
A-5 Law Invariant	✓	✓	✓	✓
A-6 Time Consistency	✓	✓	✗	✗

Table 2.1: Properties of representative monetary risk measures in discounted setting.

Definition 2.2.1 (Monetary Risk Measure). A monetary risk measure is a function $\psi: \mathbb{X} \rightarrow \mathbb{R}$ that maps a random variable $\tilde{x} \in \mathbb{X}$ to a real number and satisfies axioms A-1 and A-2.

Definition 2.2.2 (Coherent risk measure). A monetary risk measure $\psi: \mathbb{X} \rightarrow \mathbb{R}$ is *coherent* if it satisfies the axioms A-1 to A-4.

Well-known monetary risk measures, like Conditional Value at Risk (CVaR) and Entropic Value at Risk (EVaR), are coherent. Every coherent risk measure is a concave risk measure but a concave risk measure may not be coherent. For instance, the Entropic Risk Measure (ERM), is concave but not coherent.

Definition 2.2.3 (Concave risk measure). A monetary risk measure $\psi: \mathbb{X} \rightarrow \mathbb{R}$ is *concave* if it satisfies the following property:

$$\text{A-7 Concavity: } \psi[c \cdot \tilde{x}_1 + (1 - c)\tilde{x}_2] \geq c \cdot \psi[\tilde{x}_1] + (1 - c) \cdot \psi[\tilde{x}_2], \quad \forall c \in [0, 1], \forall \tilde{x}_1, \tilde{x}_2 \in \mathbb{X}.$$

Next, we summarize some other important properties of monetary risk measures that are relevant to our work. A risk measure satisfies A-5, *law invariance* if its value depends only on the probability distribution of the random variable as opposed also on the values the random variable assigns to particular elements of the probability space [113]; we provide an example later in Fig. 2.12 to demonstrate that nested VaR (nVaR) is not law invariant. A risk measure is A-6, *time consistent* also known as dynamically consistent if it satisfies the tower property [113] which can be optimized using a dynamic program [10, 38, 45, 49, 58, 101, 104].

Table 2.1 summarizes the properties satisfied by each risk measure. Unfortunately, expectation ($\mathbb{E}$), minimum ($\min$) and maximum ($\max$) are the only measures that satisfy all axioms A-1 to A-6, which allow them to be solved accurately with computationally convenient methods. Monetary risk measures generalize the average criterion to account for uncertain outcomes. All the risk measures (VaR, CVaR, EVaR, and ERM) we investigate in this thesis are monetary risk measures, which satisfy axioms A-1 and A-2. We first introduce the risk-neutral mean operator, followed by a brief introduction on VaR, CVaR, ERM, and EVaR.

2.2.1 Expected Value ($\mathbb{E}$)

The mean (or expected value) is one of the most popular measures of a random variable $\tilde{x} \in \mathbb{X}$ due to its simplicity, intuitive interpretation, and desirable properties in both theoretical and practical applications in probability theory, statistics, and measure theory. The expected value of a discrete random variable $\tilde{x}$ with a countably many possible outcomes $x_1, x_2, \dots, x_j$ is defined as

$$\mathbb{E}[\tilde{x}] := \sum_{i=1}^j x_i \cdot \mathbb{P}(\tilde{x} = x_i) . \quad (2.1)$$

More generally for a real-valued random variable $\tilde{x}$ defined on a probability space $(\Omega, \mathcal{F}, P)$, the expected value may be expressed as $\mathbb{E}[\tilde{x}] = \int_{\Omega} \tilde{x} dP = \int_{\Omega} \tilde{x}(\omega) P(d\omega)$ [98].

One key reason why the expected value is so widely used and valued is that it represents the first moment of a distribution, providing a central measure that characterizes a distribution that satisfies all desirable axioms A-1 to A-6. Moreover, several important convergence analyses and theorems, including the Law of Iterated Expectations (which captures time-consistency) and the Law of Large Numbers highlight the computational advantages and general applicability of the expected value.

The expected value is known to be risk-neutral, as it does not account for the risk inherent in a distribution. In other words, high risk can be canceled out by an equally high reward. Fig. 2.2 illustrates two different distributions with identical means and variance. A risk-neutral expected value or even mean-variance method would show no preference between them. When we consider

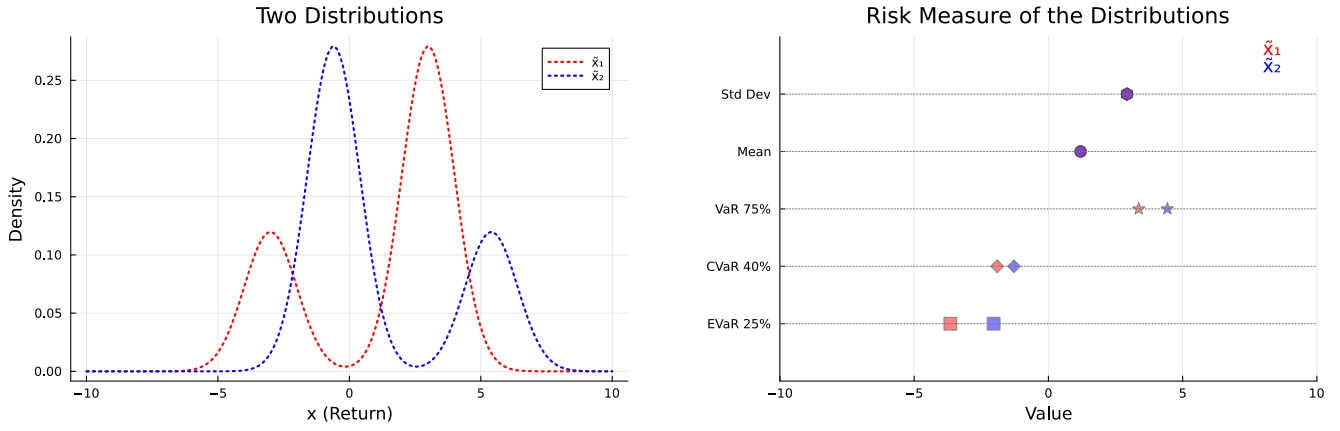


Figure 2.2: Same mean and variance distributions

25%-EVaR, 40%-CVaR and 75%-VaR, the blue distribution $\tilde{x}_2$ has higher values compared to the red distribution $\tilde{x}_1$, makes distribution $\tilde{x}_2$ preferable for these risk measures.

2.2.2 Quantiles and Value-at-Risk (VaR)

The quantile of a random variable $\tilde{x} \in \mathbb{X}$ at level $\alpha \in [0, 1]$ is any $\tau \in \bar{\mathbb{R}}$ such that $\mathbb{P}[\tilde{x} \leq \tau] \geq \alpha$ and $\mathbb{P}[\tilde{x} \geq \tau] \geq 1 - \alpha$. For non absolutely continuous random variables, quantile may not be unique but lies in the interval $[\mathbf{q}_\alpha^-(\tilde{x}), \mathbf{q}_\alpha^+(\tilde{x})]$, where

$$\begin{aligned} \mathbf{q}_\alpha^-(\tilde{x}) &:= \min \{ \tau \in \bar{\mathbb{R}} \mid \mathbb{P}[\tilde{x} \leq \tau] \geq \alpha \}, \\ \text{VaR}_\alpha[\tilde{x}] &:= \mathbf{q}_\alpha^+(\tilde{x}) := \max \{ \tau \in \bar{\mathbb{R}} \mid \mathbb{P}[\tilde{x} < \tau] \leq \alpha \}. \end{aligned} \quad (2.2)$$

Among them, $\mathbf{q}^+$ also known as VaR is the most common [57, 113]. The maximum in Eq. (2.2) exists because the mapping $\tau \mapsto \mathbb{P}[\tilde{x} < \tau]$ is lower semi-continuous. So does the minimum, since $\tau \mapsto \mathbb{P}[\tilde{x} \leq \tau]$ is upper semi-continuous. Also, $\mathbf{q}_0^-(\tilde{x}) = -\infty$ and $\mathbf{q}_1^+(\tilde{x}) = \infty$, while $\mathbf{q}_\alpha^-(\tilde{x}) \in \mathbb{R}$ for all $\alpha \in (0, 1]$ and $\mathbf{q}_\alpha^+(\tilde{x}) \in \mathbb{R}$ for all $\alpha \in [0, 1)$. Visualization of quantile for absolutely continuous and discrete distributions are presented in Fig. 2.3 respectively. For the discrete distribution, when the quantile level splits neatly between two distinct observed values, the quantile is a range (i.e., 50% quantile = $[2, 3]$ rather than a single scalar value, VaR refers to the maximum value of the range $\text{VaR}_{50\%}[\tilde{x}] = 3$).

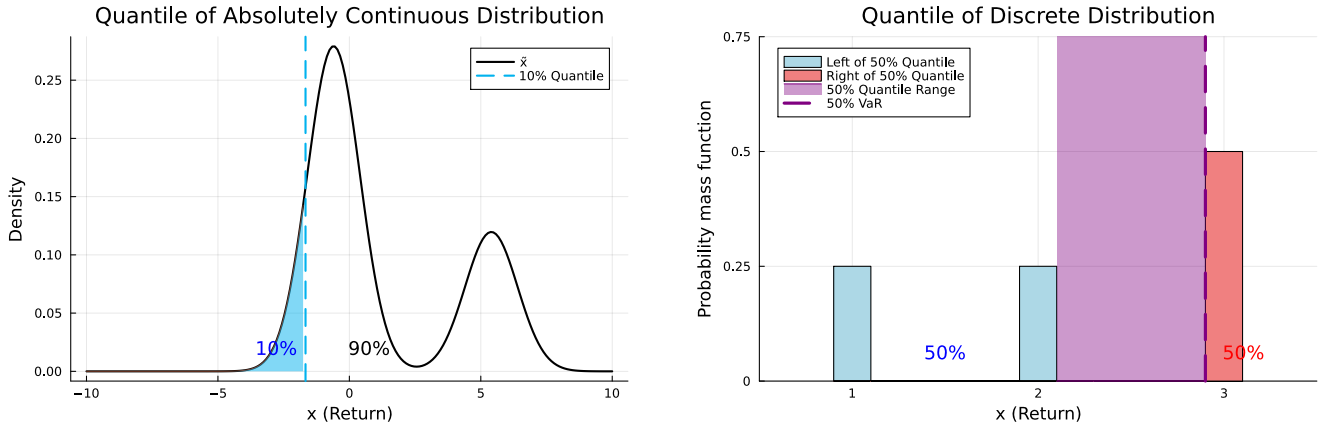


Figure 2.3: Distributions to illustrate quantile values

Suppose the 5% VaR of a portfolio is \$100, meaning there is at least 95% chance the portfolio's reward will be at least \$100, and less than 5% chance that the reward will fall below \$100. VaR is used in data centers to ensure the reliability of cloud computing services [41], in epidemiology to understand how exposure to disease differs across continuous health outcomes distributions [130], or more generally in financial markets to assess counterparty risk [7, 51]. The Basel Committee introduced 1% VaR for market risk capital requirements in Basel II [36] and requires banks to evaluate their potential losses due to market risk.

Although VaR is widely used for its simplicity and interpretability, it has critical limitations as it provides only a point estimate at the α -quantile level, it neglects the magnitude and severity of losses in the tail of the distribution, failing to account for extreme, catastrophic outcomes below the α -quantile. The global financial crisis of 2007–2008 highlighted these shortcomings, as institutions relying on VaR underestimated the true risk exposure and the devastating impact of tail events. In response to these weaknesses, the Basel III framework introduced 2.5% Conditional Value at Risk (CVaR) as the new more robust standard for market risk measurement [50, 95].

Fig. 2.4 illustrate two distributions with similar upside potential, but $\tilde{x}_1$ exhibits significantly higher tail risk. However, the 5% VaR of these two distributions are identical, because VaR focuses solely on value at a specific quantile level, it disregards the severity of risks below that threshold. This limitation means VaR may fail to adequately mitigate catastrophic outcomes occurring below the quantile level.

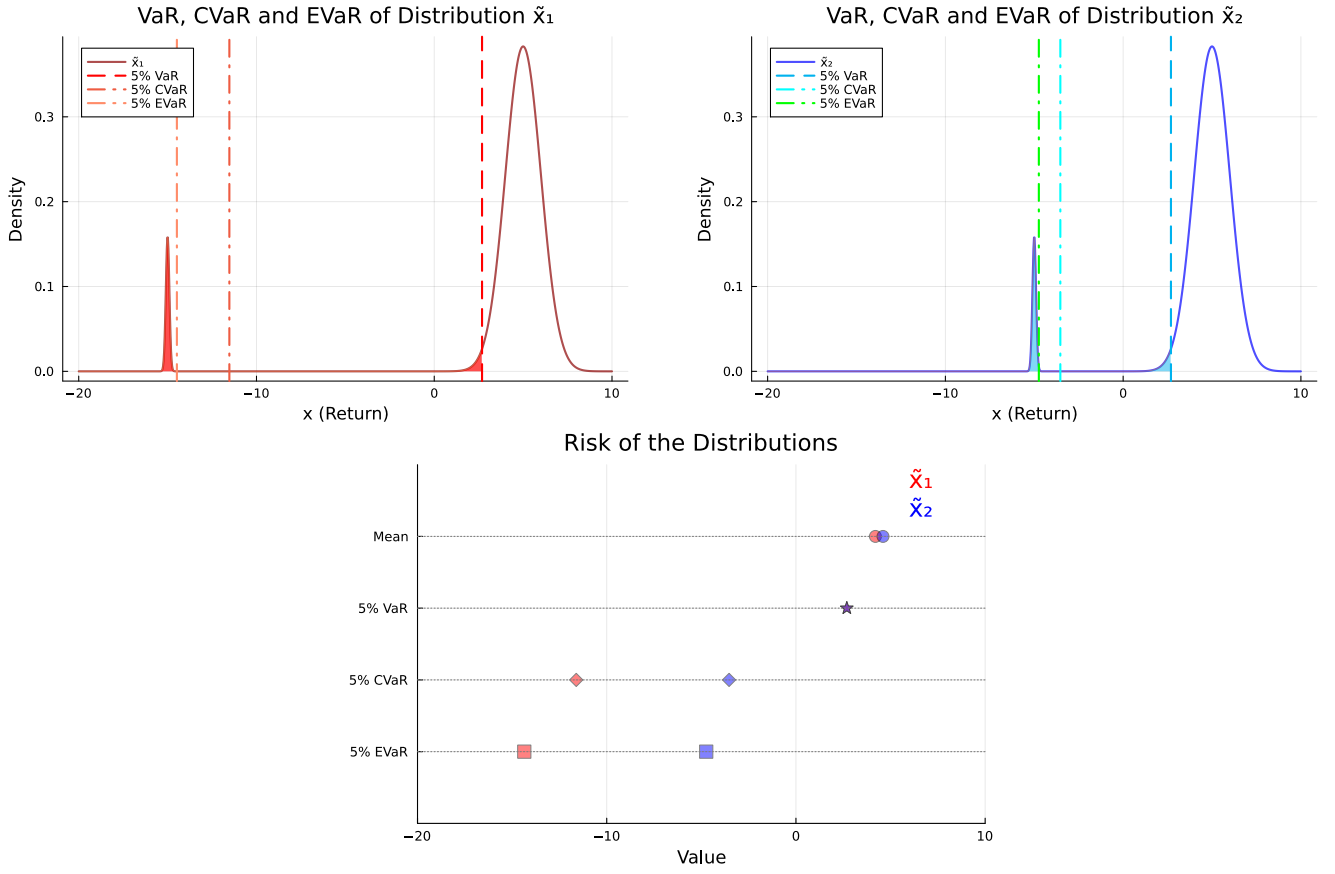


Figure 2.4: Distributions to illustrate VaR limitation

2.2.3 Conditional Value at Risk (CVaR)

CVaR, also known as expected shortfall (ES) or average value at risk (AVaR), considers the average of the α left tail of the distribution, capturing the full extent of tail risk and providing a more comprehensive and robust assessment of potential financial losses. For a risk level $\alpha \in (0, 1]$ and a random variable $\tilde{x} \in \mathbb{X}$ distributed as $\tilde{x} \sim \mathbf{q}$ then

$$\text{CVaR}_\alpha[\tilde{x}] \stackrel{(a)}{:=} \sup_{z \in \mathbb{R}} (z - \alpha^{-1} \mathbb{E}[z - \tilde{x}]_+) \stackrel{(b)}{=} \inf \left\{ \boldsymbol{\xi}^\top \mathbf{x} \mid \boldsymbol{\xi} \in \Delta_m, \alpha \cdot \boldsymbol{\xi} \leq \mathbf{q} \right\} . \quad (2.3)$$

It is also convenient to extend $\text{CVaR}_0[\tilde{x}] := \lim_{\alpha \rightarrow 0^+} \text{CVaR}_\alpha[\tilde{x}] = \text{ess inf}[\tilde{x}]$, and useful to note that $\text{CVaR}_1[\tilde{x}] = \mathbb{E}[\tilde{x}]$. Fig. 2.5 illustrates the CVaR measure, visually contrasting its two definitions. Definition (a) above is known as the primal representation of CVaR which computes the mean of the α tail distribution. Notably, the supremum is achieved at the VaR value, $\text{VaR}_\alpha(\tilde{x}) =$

$\arg \sup_{z \in \mathbb{R}} (z - \alpha^{-1} \mathbb{E}[z - \tilde{x}]_+)$. The dual definition (b) follows from standard Fenchel conjugate for finite probability spaces [57, 113].

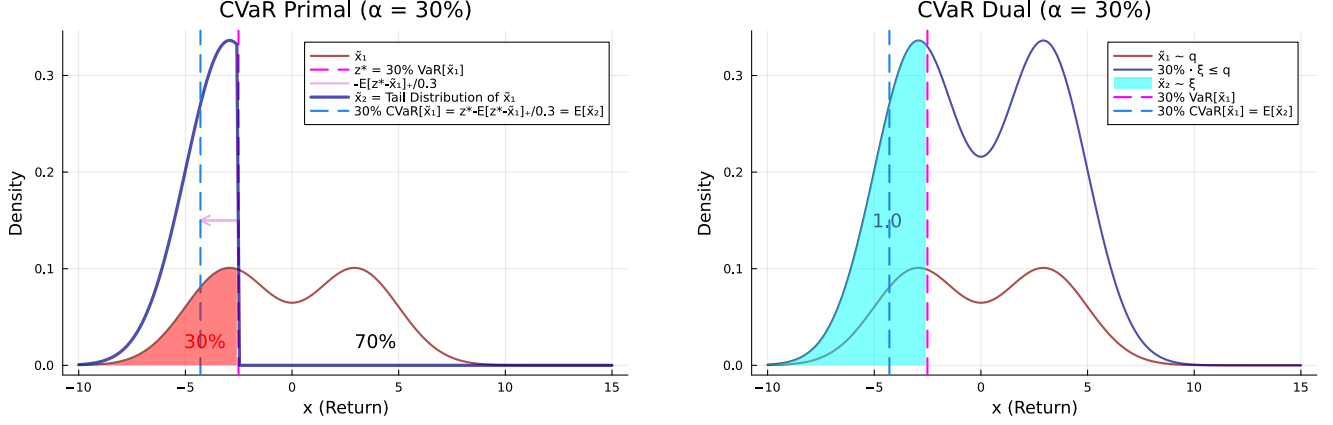


Figure 2.5: CVaR illustration

CVaR is also a coherent risk measure that exhibits concavity, an appealing mathematical property for computation advantages and encourages diversification in portfolio management. Its robustness in risk assessment and computational efficiency for optimization has contributed to its widespread adoption across various fields, including finance, inventory management, supply chain management, network design, transportation, and traffic control [55].

2.2.4 Entropic Risk Measure (ERM)

ERM is another popular risk measure based on expected utility under the exponential utility function $U_\beta(x) = \frac{1-e^{-\beta x}}{\beta}$ for $\beta > 0$, where $\lim_{\beta \rightarrow 0^+} U_\beta(x) = x$ [17, 57]. ERM is the only law-invariant risk measure that satisfies the time consistency property [78]. This property enables ERM to be computed recursively, a concept that will be further explored in Chapter 5. Applications of ERM and exponential utility are widespread includes finance [57], revenue management [80], robotics [90] and insurance [18, 111].

ERM is defined as $\text{ERM}_\beta: \mathbb{X} \rightarrow \mathbb{R}$ for a risk aversion parameter $\beta \geq 0$ and a random variable

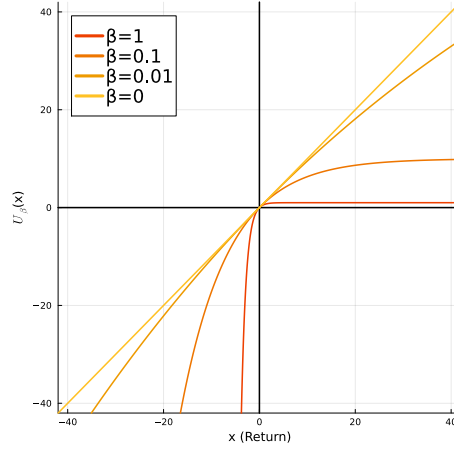


Figure 2.6: ERM illustration

$\tilde{x} \in \mathbb{X}$ as

$$\text{ERM}_\beta[\tilde{x}] := \begin{cases} \mathbb{E}[\tilde{x}] & \text{if } \beta = 0, \\ -\beta^{-1} \cdot \log\left(\mathbb{E}[e^{-\beta \cdot \tilde{x}}]\right) & \text{if } \beta \in (0, \infty), \\ \text{ess inf}[\tilde{x}] & \text{if } \beta = \infty. \end{cases} \quad (2.4)$$

We note that ERM can be derived as the certainty equivalent $U_\beta^{-1}(\mathbb{E}[U_\beta(\tilde{x})])$ when $0 < \beta < \infty$. The ERM with β parameter could also approximate mean-variance $(\mathbb{E}, \sigma^2)$ criteria since $\text{ERM}_\beta[\tilde{x}] = \mathbb{E}(\tilde{x}) - \frac{\beta}{2}\sigma^2(\tilde{x}) + o(\beta^2)$, and for Gaussian distribution $o(\beta^2) = 0$ [53, 126]. Although the parameter β in ERM is less intuitive, as illustrated in Fig. 2.7 the β parameter can be linked to Entropic Value at Risk (EVaR), which uses the α -quantile as a parameter, similar to VaR and CVaR.

2.2.5 Entropic Value at Risk (EVaR)

Introduced in 2012, EVaR is a relatively new risk measure that is more sensitive to extreme risks while also accounting for rewards above the quantile level of interest [4, 5]. EVaR for a quantile parameter $\alpha \in (0, 1]$ is defined as

$$\begin{aligned} \text{EVaR}_\alpha[\tilde{x}] &\stackrel{(a)}{:=} \sup_{\beta > 0} -\frac{1}{\beta} \log(\alpha^{-1} \mathbb{E}[e^{-\beta \tilde{x}}]) = \sup_{\beta > 0} \left(\text{ERM}_\beta[\tilde{x}] + \frac{\log(\alpha)}{\beta} \right) \\ &\stackrel{(b)}{:=} \inf \{ \boldsymbol{\xi}^T \mathbf{x} \mid \boldsymbol{\xi} \in \Delta_m, \boldsymbol{\xi} \ll \mathbf{q}, \text{KL}(\boldsymbol{\xi} \parallel \mathbf{q}) \leq -\log \alpha \}, \end{aligned} \quad (2.5)$$

where KL is the standard KL-divergence defined for each $\boldsymbol{\xi}, \mathbf{q} \in \Delta_m$ as $\text{KL}(\boldsymbol{\xi} \parallel \mathbf{q}) = \sum_{\omega \in \Omega} \xi_\omega \log \left(\frac{\xi_\omega}{q_\omega} \right)$. To ensure the KL divergence is well-defined $\boldsymbol{\xi}$ is absolutely continuous with respect to $\mathbf{q}$, which is denoted as $\boldsymbol{\xi} \ll \mathbf{q}$ and corresponds to $q_\omega = 0 \Rightarrow \xi_\omega = 0$ for each $\omega \in \Omega$. We extend the definition above for convenience with $\text{EVaR}_0[\tilde{x}] := \lim_{\alpha \rightarrow 0^+} \text{EVaR}_\alpha[\tilde{x}] = \text{ess inf}[\tilde{x}]$ and note that $\text{EVaR}_1[\tilde{x}] = \mathbb{E}[\tilde{x}]$. Similar to CVaR, EVaR can be expressed through two equivalent formulations: (a) the primal representation and (b) the dual representation. The primal representation interprets risk through a penalize exponential utility functions, capturing the trade-off between risk level and reward. In contrast, the dual representation has a compelling interpretation in terms of a worst-case expectation of a distribution from a KL-divergence ball [4].

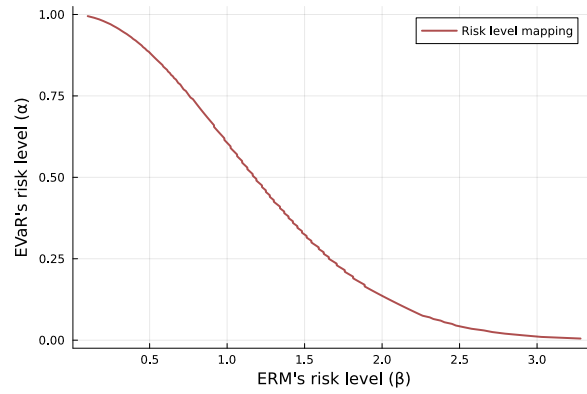


Figure 2.7: Risk Parameter mapping for standard Normal distribution: ERM to EVaR ($\beta \rightarrow \alpha$).

EVaR provides the tightest Chernoff bound on VaR, making it a more robust measure for assessing tail risks in both financial and operational contexts. Compared to VaR and CVaR, EVaR offers several key advantages:

1. **Greater Sensitivity to Catastrophic Outcomes:** EVaR is always more conservative than CVaR and VaR, meaning it better accounts for extreme negative events. Specifically, for any reward distribution $\tilde{x} \in \mathbb{X}$ and quantile risk level $\alpha \in (0, 1]$, it holds that $\text{EVaR}_\alpha[\tilde{x}] \leq \text{CVaR}_\alpha[\tilde{x}] \leq \text{VaR}_\alpha[\tilde{x}]$ (as shown in Fig. 2.4).
2. **Consideration of the Entire Distribution:** Unlike VaR and CVaR, which focus only on the tail of the distribution, EVaR evaluates the entire distribution. This allows it to differentiate

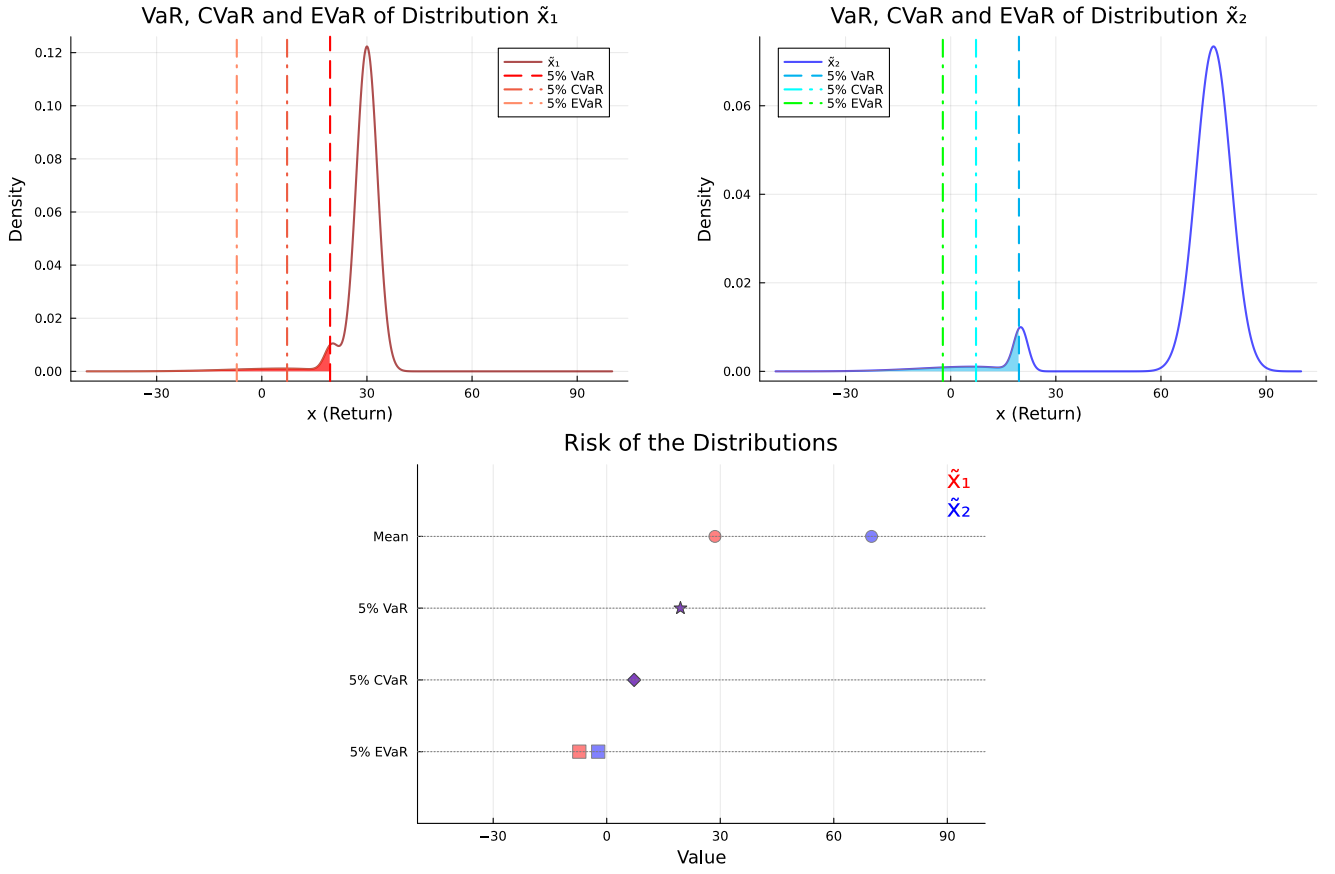


Figure 2.8: Distributions to illustrate VaR and CVaR limitation

between distributions with similar tail behavior but different upside potential, providing a more comprehensive risk assessment (as shown in Fig. 2.8).

3. **Alignment with Time-Consistent Decision Frameworks:** While EVaR itself is not inherently time-consistent, it can be formulated as an optimization over the ERM objective, which possesses this property. This connection to ERM allows EVaR to be incorporated into recursive decision-making frameworks, an aspect further explored in Chapter 5.

These advantages make EVaR a powerful alternative to VaR and CVaR, particularly in applications requiring a more nuanced and robust risk measure.

2.3 Markov Decision Process (MDP)

We formulate the decision process as an MDP $(\mathcal{S}, \mathcal{A}, r, p, \gamma, T)$, which is composed of a finite nonempty set states $\mathcal{S} = \{1, \dots, S\}$, a finite nonempty set of actions $\mathcal{A} = \{1, \dots, A\}$, a bounded reward function $r: \mathcal{S} \times \mathcal{A} \rightarrow [\underline{R}, \bar{R}]$, and a transition probability $p: \mathcal{S} \times \mathcal{A} \rightarrow \Delta_{\mathcal{S}}$, where $p(s, a, s') = p_{sas'}$ denotes the probability to transit from $s \in \mathcal{S}$ to $s' \in \mathcal{S}$ after taking action $a \in \mathcal{A}$. We consider the discount factor $\gamma \in [0, 1]$ for $T < \infty$ and restrict the discount factor to $0 \leq \gamma < 1$ when $T = \infty$ to guarantee the total discounted reward is finite. While a discounted ($\gamma < 1$) finite-horizon objective is seldom used in practice, we use it in the dissertation as an intermediate step for solving the infinite-horizon objective. We use

$$\Delta_{\mathfrak{R}} = \begin{cases} (\bar{R} - \underline{R})/(1 - \gamma) & \text{if } T = \infty \\ T \cdot (\bar{R} - \underline{R}) & \text{if } \gamma = 1 \\ (\bar{R} - \underline{R}) \cdot (1 - \gamma^T)/(1 - \gamma) & \text{otherwise} \end{cases}$$

to denote the maximum range (span semi-norm) of the MDP return.

We denote the t -horizon random total discounted reward also known as *return* of a policy π as

$$\tilde{\mathfrak{R}}_t^\pi(s) := \sum_{\tau'=0}^{t-1} \gamma^{\tau'} r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \quad (2.6)$$

where $\tilde{s}_0 = s$, and the random variables $\tilde{s}_{\tau'} \sim p(\tilde{s}_{\tau'-1}, \tilde{a}_{\tau'-1})$, $\tilde{a}_{\tau'} \sim \pi_{\tau'}(\tilde{s}_{\tau'})$, and $r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'})$ are state visited, action taken, and reward received at a time-step $\tau' \in 0:t-1$ respectively. More generally, we use $\tilde{\mathfrak{R}}_{\tau:T}^\pi(s)$ to denote the random *return* of a policy π from time-step τ to T starting at state $s \in \mathcal{S}$, and define it as $\tilde{\mathfrak{R}}_{\tau:T}^\pi(s) := \sum_{\tau'=\tau}^{T-1} \gamma^{\tau'-\tau} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'})$.

The goal of an MDP is to determine a strategy, known as the policy $\pi \in \Pi_{\text{HR}}$ in the set of history-dependent randomized policies, that specifies the action to take maximizing the *static* objective

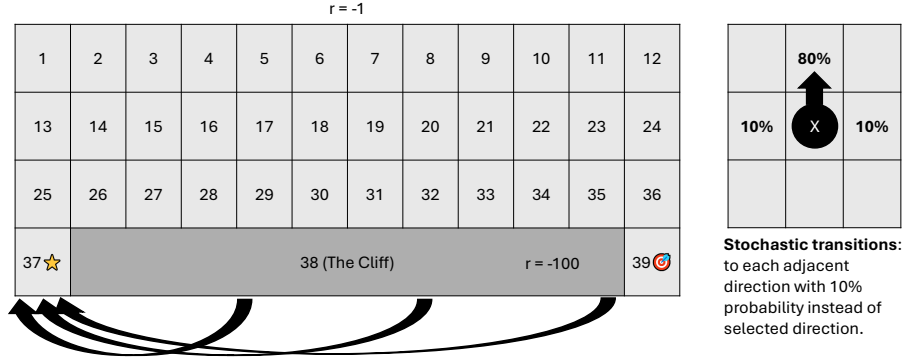


Figure 2.9: The cliffwalk domain

measure (ψ) that evaluates the total discounted sum of rewards as

$$\max_{\pi \in \Pi_{\text{HR}}} \psi^{\pi, s_0} \left[\sum_{k=0}^{T-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] = \max_{\pi \in \Pi_{\text{HR}}} \psi \left[\tilde{\mathfrak{R}}_T^{\pi}(s_0) \right], \quad (2.7)$$

for some initial state $s_0 \in \mathcal{S}$. In Eq. (2.7), the distribution of $\tilde{s}_k$ for $k \geq 1$ is implicitly governed by the transition model p while the superscript s_0 fixes the initial state.

2.4 MDP Example

In Fig. 2.9, we use the Cliffwalk (CW) domain similar to the one described in [122, Ex. 6.6], to illustrate the components of an MDP. The state space ($\mathcal{S}$) consists of 39 grid positions, with four possible actions ($\mathcal{A}$): left, down, right, up. The agent receives a reward of -1 in states 1 to 37, -100 if it falls off the cliff (state 38), and 0 upon reaching the goal (state 39). In states 1 to 37, the agent transitions to each adjacent direction with a 10% probability and follows the chosen action 80% of the time. If the agent transitions outside the grid, it remains in its current state. Falling into the cliff (state 38) forces the agent to restart at state 37 with 100% probability, regardless of the action taken. Once the agent reaches the goal (state 39), it remains there permanently. Let $T = 100$ define the time horizon of interest and the discount factor $\gamma = 0.9$ discount future rewards exponentially, which means that earlier rewards are valued more than later ones. Fig. 2.10

→	→	→	→	→	→	→	→	→	→	→	→
→	→	→	→	→	→	→	→	→	→	→	→
↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↓
↑	→										→

Figure 2.10: A stationary policy of cliffwalk domain

illustrates a deterministic stationary policy, which maps each state to the action the agent would take when in that state. Next, we will introduce a broader class of policy sets and the concept of the value function, which will help us learn the optimal policy to maximize our objective.

2.4.1 Policies and Value Functions

As mentioned above, the solution of a MDP is a policy $\pi \in \Pi_{\text{HR}}$ governs the selection of actions $\tilde{a}_k$ at each time step. Let $T \in \mathbb{N}$ be the decision horizon, the most general form of policy is the *history-dependent policy* $\pi_{[T-1]} := (\pi_k)_{k=0}^{T-1}$, which represents a sequence of decision rules $\pi_k: \mathcal{H}_k \rightarrow \mathcal{A}$ from histories to actions. The history at time $k \in [T-1]$ is defined as $h_k := (s_0, a_0, s_1, a_1, \dots, s_k) \in \mathcal{H}_k := (\mathcal{S} \times \mathcal{A})^k \times \mathcal{S}$, appending history h_k to $a \in \mathcal{A}$ and $s' \in \mathcal{S}$ is denoted by $\langle h_k, a, s' \rangle \in \mathcal{H}_{k+1}$.

Definition 2.4.1. A deterministic policy $\pi: \mathcal{H}_k \rightarrow \mathcal{A}$ selects a specific action for each possible history at time step k . In contrast, a stochastic policy $\pi: \mathcal{H}_k \rightarrow \Delta_{\mathcal{A}}$ assigns a probability distribution over the set of possible actions for each history. Stochastic policy introduces randomness enabling actions to be chosen probabilistically.

We use Π_{HR} , Π_{MR} , and Π_{SR} to denote history-dependent, Markovian, and stationary randomized policy and Π_{HD} , Π_{MD} , and Π_{SD} to determine their deterministic variant respectively. We note that Delage et al. [44] established the existence of optimal deterministic policies for all the risk measures we are interested in this thesis, so we can ignore stochastic policies without impairing optimality.

Π_{HD}^T denotes the set of all history-dependent deterministic policies for horizon T , Π_{MD}^T denotes the set of all T -horizon Markov deterministic policies where $\pi: [T-1] \times \mathcal{S} \rightarrow \mathcal{A}$, $\forall \pi \in \Pi_{\text{MD}}^T$, and Π_{SD}

denotes the set of all stationary deterministic policies where $\pi: \mathcal{S} \rightarrow \mathcal{A}$, $\forall \pi \in \Pi_{\text{SD}}$. The sets of policies satisfy $\Pi_{\text{SD}} \subseteq \Pi_{\text{MD}} \subseteq \Pi_{\text{HD}}$. Standard risk-neutral MDPs ($\psi = \mathbb{E}$) primarily focus on the class of Markov and stationary policies, as these are proven to be optimal for finite-horizon and discounted infinite-horizon problems respectively [103]. However, for risk-averse objectives they are generally not, so we must consider the history-dependent policies.

Rather than directly solving for the optimal policy over a fixed horizon T , it is often more practical to consider a simpler sub-problem. We introduce the concept of a value function, denoted as $v_t^\pi(s)$ or $v^\pi(t, s)$ interchangeably, where $0 \leq t \leq T$ represents the remaining horizon. $v_t^\pi(s)$ quantifies the objective measure of the total discounted reward when following a policy $\pi \in \Pi_{\text{HD}}$ starting from state $s \in \mathcal{S}$, formally defined as

$$v_t^\pi(s) := \psi^{\pi, s} \left[\sum_{k=0}^{t-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right]. \quad (2.8)$$

A policy $\pi^* \in \Pi_{\text{HD}}^t$ such that $v_t^{\pi^*}(s) \geq v_t^\pi(s) \forall \pi \in \Pi_{\text{HD}}^t$ is referred to as the optimal policy for t remaining horizon. The corresponding value function $v_t^{\pi^*}(s)$ is known as the optimal value function often denoted interchangeably $v_t^*(s) := v_t^{\pi^*}(s) = \max_{\pi \in \Pi_{\text{HD}}} v_t^\pi(s)$.

2.4.2 Risk Neutral MDP

Risk-neutral MDPs, also known as the standard MDPs, are the most commonly studied and widely used class of Markov Decision Processes, where the objective is to maximize the expected total discounted reward without considering risk preferences. In risk-neutral MDPs, Markov policies are proven to be optimal for the finite horizon, and stationary policies are proven to be optimal for the infinite horizon [103]. The objective of a T -horizon risk-neutral MDP is defined as

$$\max_{\pi \in \Pi_{\text{MD}}^T} \mathbb{E}^{\pi, s_0} \left[\sum_{k=0}^{T-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] = \max_{\pi \in \Pi_{\text{MD}}} \mathbb{E} \left[\tilde{\mathfrak{R}}_T^\pi(s_0) \right], \quad (2.9)$$

where s_0 refers to the initial states. One of the most widely used methods for solving risk-neutral MDPs is dynamic programming (DP), specifically the value iteration algorithm, which iteratively

computes the value function of a simpler subproblem, gradually building up the solution to the full horizon T . For T -horizon MDP, we define the value function $v^\pi: [T] \times \mathcal{S} \rightarrow \mathbb{R}$ for all Markovian deterministic policies $\pi \in \Pi_{\text{MD}}$ and the optimal value function $v^*: [T] \times \mathcal{S} \rightarrow \mathbb{R}$ for a state $s \in \mathcal{S}$ as follows:

$$v^\pi(t, s) = \mathbb{E}^{\pi, s} \left[\sum_{k=0}^{t-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right], \quad (2.10)$$

$$v^*(t, s) = \max_{\pi \in \Pi_{\text{MD}}^t} \mathbb{E}^{\pi, s} \left[\sum_{k=0}^{t-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right]. \quad (2.11)$$

The following propositions show how the value function defined in Eqs. (2.10) and (2.11) can be efficiently computed by a DP. We will first consider the finite horizon MDP ($T < \infty$) and then discuss the extension to infinite horizon discounted MDPs.

Proposition 2.4.2 (Finite Horizon Risk Neutral Value Iteration). *For any policy $\pi \in \Pi_{\text{MD}}$, discount factor $\gamma \in [0, 1]$ and time horizon $T < \infty$, the optimal value function $v^*: [T] \times \mathcal{S} \rightarrow \mathbb{R}$ defined in Eq. (2.11) is the unique solution to the following system of equations for all $s \in \mathcal{S}$,*

$$v^*(t+1, s) = \max_{a \in \mathcal{A}} \mathbb{E} [r(s, a) + \gamma v^*(t, \tilde{s}')] , \forall t \in [T-1] \quad (2.12)$$

where $\tilde{s}' \sim p(s, a)$ and note that $v^*(0, s_T) := 0, \forall s_T \in \mathcal{S}$. Moreover, the optimal policy is the optimizer of the system of equations as $\pi^*(t, s) = \arg \max_{a \in \mathcal{A}} \mathbb{E} [r(s, a) + \gamma v^*(t, \tilde{s}')] , \forall t \in [T-1]$.

Proof. We will prove the following with mathematical induction. Let's assume

$$v^*(t, s) := \max_{\pi \in \Pi_{\text{MD}}^t} \mathbb{E}^{\pi, s} \left[\sum_{k'=0}^{t-1} \gamma^{k'} r(\tilde{s}_{k'}, \tilde{a}_{k'}) \right] \quad (2.13)$$

we will show that $v^*(t+1, s) := \max_{\pi \in \Pi_{\text{MD}}^{t+1}} \mathbb{E}^{\pi, s} \left[\sum_{k=0}^t \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] = \max_{a \in \mathcal{A}} \mathbb{E} [r(s, a) + \gamma v^*(t, \tilde{s}')] :$

$$\begin{aligned} & \max_{a \in \mathcal{A}} \mathbb{E} [r(s, a) + \gamma v^*(t, \tilde{s}')] \\ & \stackrel{(2.13)}{=} \max_{a \in \mathcal{A}} \mathbb{E} \left[r(s, a) + \gamma \max_{\pi \in \Pi_{\text{MD}}^t} \mathbb{E}^{\pi, \tilde{s}'} \left[\sum_{k'=0}^{t-1} \gamma^{k'} r(\tilde{s}_{k'}, \tilde{a}_{k'}) \right] \right] \end{aligned}$$

$$\begin{aligned}
& \stackrel{(A-2, A-4)}{=} \max_{a \in \mathcal{A}} \mathbb{E} \left[\max_{\pi \in \Pi_{\text{MD}}^t} \mathbb{E}^{\pi, \tilde{s}'} \left[r(s, a) + \gamma \sum_{k'=0}^{t-1} \gamma^{k'} r(\tilde{s}_{k'}, \tilde{a}_{k'}) \right] \right] \\
& \stackrel{k=k'+1}{=} \max_{a \in \mathcal{A}} \mathbb{E} \left[\max_{\pi \in \Pi_{\text{MD}}^t} \mathbb{E}^{\pi, \tilde{s}'} \left[\sum_{k=0}^t \gamma^k r(\tilde{s}_k, \tilde{a}_k) \mid \tilde{s}_0 = s, \tilde{a}_0 = a \right] \right] \\
& \stackrel{A-1, \text{Thm 14.60 of [106]}}{=} \max_{a \in \mathcal{A}} \max_{\pi \in \Pi_{\text{MD}}^t} \mathbb{E} \left[\mathbb{E}^{\pi, \tilde{s}'} \left[\sum_{k=0}^t \gamma^k r(\tilde{s}_k, \tilde{a}_k) \mid \tilde{s}_0 = s, \tilde{a}_0 = a \right] \right] \\
& \stackrel{(A-6)}{=} \max_{\pi \in \Pi_{\text{MD}}^{t+1}} \mathbb{E}^{\pi, s} \left[\sum_{k=0}^t \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] := v^*(t+1, s)
\end{aligned}$$

□

Formulating the DP equations in (2.4.2) is only possible because of four important properties of the expectation operator [103, 113]. In particular, the expectation operator $\mathbb{E}[\cdot]$ is monotone A-1, translation invariant A-2, positive homogeneous A-4, and satisfies the tower property A-6.

Proposition 2.4.3 (Finite Horizon Risk Neutral Policy Evaluation DP). *For any policy $\pi \in \Pi_{\text{MD}}$, discount factor $\gamma \in [0, 1]$ and time horizon $T < \infty$, its value function $v^\pi: [T] \times \mathcal{S} \rightarrow \mathbb{R}$ defined in Eq. (2.10) is the unique solution to the following system of equations for all $s \in \mathcal{S}$,*

$$v^\pi(t+1, s) = \mathbb{E}[r(s, \pi(t, s)) + \gamma v^\pi(t, \tilde{s}')] , \forall t \in [T-1] \quad (2.14)$$

where $\tilde{s}' \sim p(s, a)$ and note that $v^\pi(0, s_T) = 0$, $\forall s_T \in \mathcal{S}$.

Proof. The proof of Proposition 2.4.3 is identical to the proof of Proposition 2.4.2 without the maximum operator. □

Based on Propositions 2.4.2 and 2.4.3, we can efficiently evaluate a given policy and compute both the optimal value function and the optimal policy through dynamic programming. Finally, we present Algorithm 1 for solving standard finite-horizon MDPs.

For infinite horizon $T = \infty$ and a discount factor $\gamma \in [0, 1)$, the value function of any stationary policy $\pi \in \Pi_{\text{SD}}$, where $\pi: \mathcal{S} \rightarrow \mathcal{A}$ can be easily computed. Fixing the policy π reduces the MDP to a Markov chain, with transition probabilities defined as $P^\pi(s, s') := p(s, \pi(s), s')$ representing the

Algorithm 1: Finite Horizon Risk Neutral Value Iteration

Input: Planning horizon $T < \infty$

Output: Optimal value $v^* : [T] \times \mathcal{S} \rightarrow \mathbb{R}$ and policy $\pi^* : [T - 1] \times \mathcal{S} \rightarrow \mathcal{A}$

1 Initialize $v^*(0, s) \leftarrow 0, \forall s \in \mathcal{S}$

2 **for** $t \in 0 : (T - 1)$ **do**

3 $\pi^*(t, s) \leftarrow \arg \max_{a \in \mathcal{A}} \mathbb{E}^s [r(s, a) + \gamma \cdot v^*(t, \tilde{s}')] , \forall s \in \mathcal{S}$

4 $v^*(t + 1, s) \leftarrow \max_{a \in \mathcal{A}} \mathbb{E}^s [r(s, a) + \gamma \cdot v^*(t, \tilde{s}')] , \forall s \in \mathcal{S}$

5 **return** v^*, π^* ;

probability of moving from state s to next state s' under π . Likewise, the reward for each state, since the policy is stationary, is given by $r^\pi(s) := r(s, \pi(s))$. Considering v_∞^π and r^π as vectors in $\mathbb{R}^{\mathcal{S}}$, and P^π as a matrix in $\mathbb{R}^{\mathcal{S} \times \mathcal{S}}$ induced by their respective functions. Since the spectral radius $\mathfrak{s}(\gamma P^\pi) \leq \gamma < 1$, the value function of a policy can be computed via Neumann series [67] as

$$v_\infty^\pi := \left(\mathbb{E}^{\pi, s} \left[\sum_{k=0}^{\infty} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] \right)_{s \in \mathcal{S}} = \sum_{k=0}^{\infty} (\gamma P^\pi)^k r^\pi = (I - \gamma P^\pi)^{-1} r^\pi ,$$

where the spectral radius of a $n \times n$ matrix A is defined as the maximum of absolute eigenvalues $\mathfrak{s}(A) := \max\{|\lambda_1|, \dots, |\lambda_n|\}$.

Proposition 2.4.4 (Contraction mapping theorem). *For any $v_1, v_2 \in \mathbb{R}^{\mathcal{S}}$, the standard Bellman operator $\mathcal{B}_{\mathbb{E}} : \mathbb{R}^{\mathcal{S}} \rightarrow \mathbb{R}^{\mathcal{S}}$ defined as*

$$(\mathcal{B}_{\mathbb{E}}v)(s) := \max_{a \in \mathcal{A}} \mathbb{E}[r(s, a) + \gamma v(\tilde{s}')] = \max_{a \in \mathcal{A}} \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s, a, s') \cdot v(s') \right] \quad (2.15)$$

with discount factor $\gamma \in [0, 1)$ satisfies contraction mapping

$$\|\mathcal{B}_{\mathbb{E}}v_1 - \mathcal{B}_{\mathbb{E}}v_2\|_\infty \leq \gamma \|v_1 - v_2\|_\infty \quad (2.16)$$

and has a unique fixed point v_∞^ satisfying $\mathcal{B}_{\mathbb{E}}v_\infty^* = v_\infty^*$.*

Proof. We first show that the standard Bellman Operator $\mathcal{B}_{\mathbb{E}}$ is a contraction mapping, then we

will show that there is only a unique fixed point v_∞^* .

$$\begin{aligned}
& \|\mathcal{B}_\mathbb{E} v_1 - \mathcal{B}_\mathbb{E} v_2\|_\infty \\
& \max_{s \in \mathcal{S}} |\mathcal{B}_\mathbb{E} v_1(s) - \mathcal{B}_\mathbb{E} v_2(s)| \\
& = \max_{s \in \mathcal{S}} \left| \max_{a \in \mathcal{A}} \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s, a, s') \cdot v_1(s') \right] - \max_{a \in \mathcal{A}} \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s, a, s') \cdot v_2(s') \right] \right| \\
& \stackrel{(a)}{\leq} \max_{s \in \mathcal{S}} \max_{a \in \mathcal{A}} \left[\left| \left(r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s, a, s') \cdot v_1(s') \right) - \left(r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s, a, s') \cdot v_2(s') \right) \right| \right] \\
& \stackrel{(b)}{=} \gamma \cdot \max_{s, a \in \mathcal{S} \times \mathcal{A}} \left[\sum_{s' \in \mathcal{S}} p(s, a, s') \cdot |v_1(s') - v_2(s')| \right] \\
& \stackrel{(c)}{\leq} \gamma \cdot \|v_1 - v_2\|_\infty
\end{aligned}$$

The inequality (a) comes from reverse triangle inequality of ∞ -norm. Equality (b) follows the positive homogeneity of absolute value and maximum operator, and inequality (c) upper bound mean with max. We now show the fixed point v_∞^* is unique. Suppose $v_1^* = \mathcal{B}_\mathbb{E} v_1^*$ and $v_2^* = \mathcal{B}_\mathbb{E} v_2^*$ are two fixed points, then

$$\|v_1^* - v_2^*\|_\infty = \|\mathcal{B}_\mathbb{E} v_1^* - \mathcal{B}_\mathbb{E} v_2^*\|_\infty \leq \gamma \|v_1^* - v_2^*\|_\infty ,$$

$0 \leq \gamma < 1$ implies $\|v_1^* - v_2^*\|_\infty = 0$ and so $v_1^* = v_2^*$. □

The contraction mapping theorem (Proposition 2.4.4) ensures that repeatedly applying the Bellman operator T in Eq. (2.15) guarantees convergence to the optimal value function v_∞^* . Furthermore, the optimizer of the Bellman operator $\mathcal{B}_\mathbb{E} v_\infty^*$ is an optimal policy.

The algorithms of infinite horizon risk neutral MDP are presented in Algorithm 2. To address the non-associative nature of floating-point arithmetic, a small tolerance threshold $\mathcal{E} > 0$ is introduced as a stopping criterion. The value function is considered to have reached a fixed point when $\|v - \mathcal{B}_\mathbb{E} v\|_\infty \leq \mathcal{E}$.

Algorithm 2: Infinite horizon risk neutral MDP

Input: Infinite horizon stopping criterion $\mathcal{E}$

Output: Optimal value $v^* : \mathcal{S} \rightarrow \mathbb{R}$ and policy $\pi^* : \mathcal{S} \rightarrow \mathcal{A}$

```
1 Initialize  $v^*(s) \leftarrow 0$  and  $\hat{v}(s) \leftarrow \infty, \forall s \in \mathcal{S}$ 
2 while  $\|v^* - \hat{v}\|_\infty > \mathcal{E}$  do
3    $\hat{v} \leftarrow v^*$ 
4    $v^*(s) \leftarrow \max_{a \in \mathcal{A}} \mathbb{E}^s [r(s, a) + \gamma \cdot \hat{v}(\tilde{s}')] , \forall s \in \mathcal{S}$ 
5    $\pi^*(s) \leftarrow \arg \max_{a \in \mathcal{A}} \mathbb{E}^s [r(s, a) + \gamma \cdot v^*(\tilde{s}')] , \forall s \in \mathcal{S}$ 
6 return  $v^*, \pi^*$  ;
```

2.4.2.1 The Q-value Function

In RL, when the environment model is known, we can compute the optimal action at each state using the Bellman update Eq. (2.15). However, in many real-world scenarios, the model is not explicitly given nor learned in advance, making it necessary for the RL agent to interact with the environment and learn through trial and error. This scenario falls under model-free reinforcement learning, where the agent must balance the trade-off between exploration (gathering new information) and exploitation (using current knowledge to maximize rewards).

To make optimal decisions in this setting, we need a way to evaluate how good each action is in a given state. The Q-value function (state-action value function) is a fundamental concept in RL that extends beyond state-value functions by explicitly associating values with state-action pairs. Formally, let the infinite horizon optimal Q-value function $q_\infty^* : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ be defined as:

$$q_\infty^*(s, a) := \max_{\pi \in \Pi_{\text{SD}} : \tilde{a}_0 = a} \mathbb{E}^{\pi, s} \left[\sum_{k'=0}^{\infty} \gamma^{k'} r(\tilde{s}_{k'}, \tilde{a}_{k'}) \right] . \quad (2.17)$$

This function quantifies the expected cumulative reward if action a is taken at state s , followed by the optimal policy for the remaining time steps. From the optimal Q-value function, we can derive the optimal state-value function

$$v_\infty^*(s) = \max_a q_\infty^*(s, a)$$

and the optimal policy

$$\pi(s) = \arg \max_a q_\infty^*(s, a)$$

for all $s \in \mathcal{S}$.

When the environment model is unavailable, the initial estimates of the Q-values $\hat{q}_0^*(s, a)$ may be inaccurate, the subscript in the estimates represents the number of times $\hat{q}^*$ was updated. Q-learning is a model-free RL algorithm that allows an agent to iteratively update Q-values based on observed experience. By maintaining and updating Q-values for state-action pairs, the agent incrementally improves its value function estimates based on observed transitions and rewards. A common exploration strategy is ϵ -greedy action selection, with an exploration rate $\epsilon_i \in (0, 1)$ scheduled to decay over time, gradually shifting the agent's behavior from exploration toward exploitation as learning progresses. At iteration i , the agent selects a random action with probability ϵ_i and selects the action with the highest estimated Q-value with probability $1 - \epsilon_i$.

The Q-value update rule follows the stochastic approximation method, also known as the Robbins-Monro algorithm. At each step, the agent observes the next state s' , receives an immediate reward r , and updates its Q-value estimate via

$$\hat{q}_{i+1}^*(s, a) \leftarrow (1 - \eta) \hat{q}_i^*(s, a) + \eta_i (r + \gamma \max_{a'} \hat{q}_i^*(s', a')) , \quad (2.18)$$

where η_i is the learning rate, which controls the weight given to new experiences versus prior estimates. This iterative update enables the agent to refine its Q-value estimates over time and converges toward the optimal q function under certain conditions that can be satisfied by choosing $\eta_i = \epsilon_i = \frac{1}{i}$ where $i \in \mathbb{N}$ refers to the number of times the Q-value was updated.

2.5 Risk Sensitive RL

The interest in risk-sensitive reinforcement learning (RL) has surged in recent years, driven by the imperative to automate high-stakes decision-making processes. Here, we present two notable quotes from decision-makers: the U.S. Army Corps of Engineers' perspective on mitigating natural

disasters and General Motors' (GM) approach to optimizing production capacity.

“Much of the work of the U.S. Army Corps of Engineers (Corps) is involved with managing the risk associated with natural processes, such as flooding and drought. Safety and quality of human life, large amounts of money and environmental issues are all at stake in the decisions made by the Corps relating to the management of natural risks.” - [86]

“In any decision under risk, expected profit is not the only objective. Management is also concerned about the risk involved. This is especially true for a set of decisions like the one faced by GM, where large amounts of money and the careers of many individuals are involved.” - [52]

Most real-world decision-making extends beyond traditional expected value objectives by incorporating alternative risk-sensitive criteria.

2.5.1 Static Risk MDP

In this thesis, we are interested in static risk-sensitive MDPs objective which is similar to the one in (2.9) for the risk-neutral case with the expectation operator $\mathbb{E}$ replaced with the desired risk measure ψ includes VaR, CVaR, ERM, EVaR as

$$\max_{\pi \in \Pi_{\text{HR}}^T} \psi^{\pi, s_0} \left[\sum_{k=0}^{T-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] = \max_{\pi \in \Pi_{\text{HR}}} \psi \left[\tilde{\mathfrak{R}}_T^\pi(s_0) \right]. \quad (2.19)$$

Although this may appear as a small change compared to (2.9), it has profound implications on the complexity of the solution. Recall that the DP equations for MDPs exist because $\mathbb{E}[\cdot]$ satisfies all axioms A-1 to A-6. Alas, most interesting general monetary risk measures do not satisfy all these properties simultaneously, as shown in Table 2.1, which makes it difficult to solve the optimization problem in (2.19). As a result, standard risk-neutral MDP methods cannot be directly applied to static risk objectives. Furthermore, developing a single generalized algorithm that works for all these risk measures is challenging, as each measure fails to satisfy different axioms.

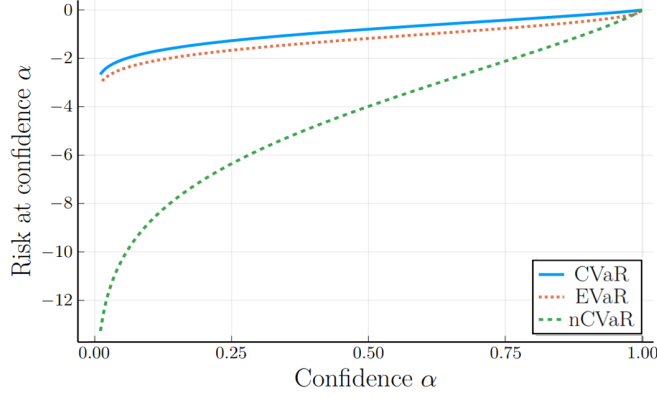


Figure 2.11: Risk of the sum of standard normal distributions $\sum_{t=0}^4 \tilde{r}_t$.

Some research in risk-sensitive RL overlooks these theoretical inconsistencies and instead focuses on naively modifying the standard Bellman operator, leading to what is known as dynamic risk or distributional Markovian risk. Despite the simplicity and computation benefit of these algorithms, the value functions and policies derived from such approaches lack formal guarantees and fail to optimize any meaningful risk objective in terms of the total discounted reward.

2.5.2 Nested / Dynamic Risk

A common way to formulate DP equations for risk-averse MDPs is to use *nested* risk measures [113], also known as Markov [110], iterated [96], or recursive [12] risk measures. For instance, the nested CVaR is informally defined as

$$\text{nCVaR}_\alpha[\tilde{\mathcal{R}}_T^\pi] = \text{CVaR}_\alpha^\pi[r(\tilde{s}_0, \tilde{a}_0) + \gamma \text{CVaR}_\alpha^\pi[r(\tilde{s}_1, \tilde{a}_1) + \dots]].$$

When the nested risk measure is properly formalized, one can compute the optimal value function using the DP equation

$$v_t^*(s) = \max_{a \in \mathcal{A}} \text{CVaR}_\alpha [r(s, a) + \gamma \cdot v_{t-1}^*(\tilde{s}')],$$

which is similar to that in (2.12) for risk-neutral MDPs.

Risk measures like VaR, CVaR, and EVaR are not time consistent, indicating that naively nesting

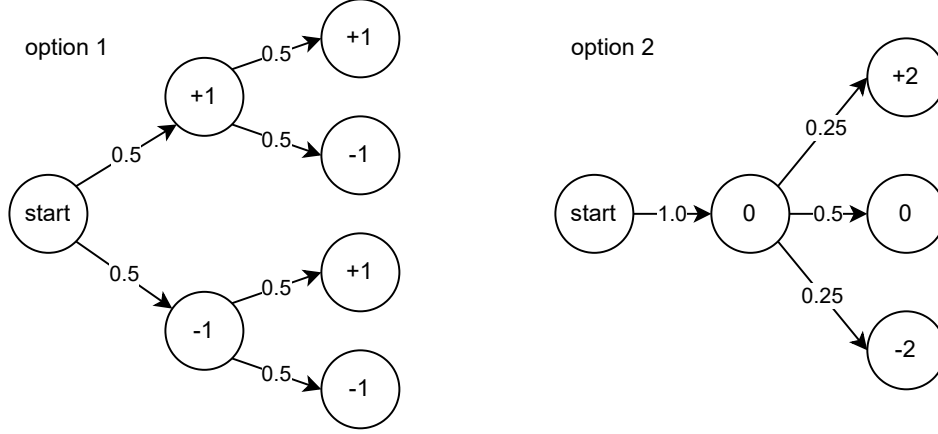


Figure 2.12: Example used to illustrate the law invariance of risk measures

the risk measure is inaccurate. For example, let us consider the risk measure (CVaR) we have that

$$v_{t+1}^\pi(s) := \text{CVaR}_\alpha^{\pi,s} \left[\sum_{k=0}^t \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] \quad (2.20)$$

$$\neq \underbrace{\text{CVaR}_\alpha^{\pi,s} [r(s, a) + \gamma \hat{v}_t^\pi(\tilde{s}')]]}_{\text{nCVaR}} := \hat{v}_{t+1}^\pi(s), \forall t \in [T-1], \quad (2.21)$$

note that the value function for zero-horizon MDP is merely zero, $v_0(s) = \hat{v}_0 = 0$. The DP induced by Eq. (2.21) is known as nested/dynamic risk. This nested CVaR (nCVaR) is not equivalent to the static CVaR. In general, nested risk measures are poor approximations of static risk measures. To illustrate this fact, Fig. 2.11 depicts the values of CVaR, EVaR, and nCVaR for evaluating a simple distribution $\sum_{t=0}^4 \tilde{r}_t$ such that all immediate rewards ($\tilde{r}_t$ for all $t \in [4]$) follows standard normal distribution $N(0, 1)$, demonstrates that nCVaR is a poor approximation of the static CVaR.

Despite their favorable computational properties, nested (VaR, CVaR, EVaR, ERM) suffer from an important drawback in that they are not *law-invariant*. Therefore, the risk value is not solely a function of the distribution of the return $\tilde{\mathfrak{R}}_T^\pi$ and is affected by the MDP's dynamics in a non-trivial way that is difficult to anticipate and interpret. The impact of law invariance on risk preferences is well-documented in the risk literature and can actually cause an agent to prefer returns with higher variability [70].

Non-law-invariant risk measures may evaluate identical cumulative discounted reward distributions differently, leading to inconsistent decision-making. For example, consider a discount factor of $\gamma = 1$. Fig. 2.12 presents two options with the same joint distribution. In both cases, the outcomes are -2, 0, or +2, occurring with probabilities of 25%, 50%, and 25%, respectively. However, the evaluation of options 1 and 2 of nested VaR (nVaR) at $\alpha = 0.6$ are two (2) and zero (0) respectively. Despite having identical joint distributions, the differing nVaR assessments highlight inconsistencies in risk evaluation, raising concerns about the validity and meaningfulness of the nested risk decision-making framework.

2.5.3 Distributional Markovian risk

As explained in Section 2.4.2.1, standard model-free RL learns the optimal Q-value function $\hat{q}^*(s, a)$ for each state-action pair through Q-learning. In contrast, distributional RL [14], aims to capture the entire optimal Q-value distribution denoted as $\tilde{Q}^*: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{X}$ which represents the distribution of cumulative discounted rewards of the optimal policy, whereas the standard Q-value function $q^*(s, a) := \mathbb{E}[\tilde{Q}^*(s, a)]$ corresponds to the expected value of this distribution, providing a mean estimate of the cumulative discounted rewards.

The distributional Bellman operator $\tilde{\mathcal{B}}_{\mathbb{E}}^*: \mathbb{X}^{\mathcal{S} \times \mathcal{A}} \rightarrow \mathbb{X}^{\mathcal{S} \times \mathcal{A}}$ on the Q-value distributions $\tilde{Q}^*$ is

$$(\tilde{\mathcal{B}}_{\mathbb{E}}^* \tilde{Q}^*)(s, a) := r(s, a) + \gamma \tilde{Q}^*(\tilde{s}', a^*(\tilde{s}')) \quad (2.22)$$

where the target optimal action

$$a^*(s') := \arg \max_{a'} \mathbb{E}[\tilde{Q}^*(s', a')] , \forall s' \in \mathcal{S} \quad (2.23)$$

and the random variable $\tilde{s}'$ is induced by probability simplex $p(s, a)$. This distributional Bellman operator is shown to be a γ -contraction in $\mathbb{E}[\tilde{Q}^*]$ as

$$\left\| \mathbb{E}[\tilde{\mathcal{B}}_{\mathbb{E}}^* \tilde{Q}_1] - \mathbb{E}[\tilde{\mathcal{B}}_{\mathbb{E}}^* \tilde{Q}_2] \right\|_{\infty} \leq \gamma \left\| \mathbb{E}[\tilde{Q}_1] - \mathbb{E}[\tilde{Q}_2] \right\|_{\infty} . \quad (2.24)$$

However, the distributional Bellman operator is not a contraction in $\tilde{Q}^*$, meaning that multiple optimal policies may exist with distinct distributions of cumulative discounted rewards while sharing the same expected value.

Given access to the full distribution rather than just the expectation, a natural approach to risk-sensitive decision-making is to replace the expectation with alternative measures during the construction of $\tilde{Q}^*$ [39, 72]. This leads to a distributional Markov action-selection defined as

$$(\tilde{\mathcal{B}}_\psi^* \tilde{Q}^*)(s, a) := r(s, a) + \gamma \tilde{Q}^*(\tilde{s}', a^*(\tilde{s}')), \quad a^*(s') := \arg \max_{a'} \psi[\tilde{Q}^*(s', a')] , \forall s' \in \mathcal{S} . \quad (2.25)$$

For the expected value measure, this distributional Markov action-selection strategy converges to both static and nested risk. However, for general risk measures, it does not converge to either static or nested (dynamic) risk [81]. There are two primary reasons this strategy fails to converge to static risk: (1) The optimal policy under general risk measures is not necessarily Markovian, and (2) the risk level in static risk for sequential decision-making depends on time or history. Consequently, restricting policies to the Markov framework and assuming a fixed risk level impairs both the optimality and correctness of the optimization process.

CHAPTER 3

On Dynamic Programming Decompositions of Static Risk Measures in MDPs

3.1 Introduction

Risk-averse reinforcement learning (RL) seeks to provide a risk-averse policy for high-stakes real-world decision problems. These high-stake domains include autonomous driving [71, 115], robot collision avoidance [3, 61], liver transplant timing [77], HIV treatment optimization [72, 135], unmanned aerial vehicles (UAV) control [31], and investment liquidation [89], to name a few. Because these domains call for reliable solutions, risk-averse algorithms must be based on solid theoretical foundations. This is one reason why monetary risk measures, such as Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR), have become pervasive in risk-averse RL [102]. Indeed, risk measures such as CVaR are known to be coherent [9] with respect to a set of fundamental axioms that define how risk should be quantified and have been adopted as gold standards in banking regulations [95].

Introducing risk-averse objectives in Markov decision processes (MDPs)—the primary model used in RL—is challenging. Dynamic programming, the linchpin of most RL algorithms, cannot be used directly to optimize a risk measure like VaR or CVaR in MDPs. One line of work tackles this challenge by exploiting the primal representation of risk measures and augmenting the state space of their dynamic programs (DPs) with an additional parameter that typically represents the total cumulative reward up to the current point [13, 21, 32, 54, 63, 82, 131, 132]. Even when the

original MDP is finite, this DP requires computing the value function for a continuous state space, and thus, has been considered inefficient in practice [30, 33, 79].

Another line of recent work leverages the dual representation to produce a *risk-level decomposition* of risk measures [99]. Using this decomposition, numerous authors have derived DPs for common risk measures and integrated them within various RL algorithms [29, 30, 33, 46, 47, 93, 105, 118]. Although this risk-level decomposition requires augmenting the state space with a continuous parameter, this parameter is naturally bounded between 0 and 1. It has been generally accepted, given several proofs supporting this claim [33, 79], that these DPs recover the optimal policy if we can discretize the augmented state space sufficiently finely. Moreover, it is believed that one can use the optimal value function from this DP to recover the policies that are optimal for the full range of risk levels.

In this chapter, we make a surprising discovery that numerous claims of optimality of risk-level decompositions published in the past several years are incorrect. Even when one discretizes the augmented state space arbitrarily finely, most risk-level DPs are not guaranteed to recover the optimal value function and policy. There are several reasons why existing arguments fail. As the most common reason, several papers assume that a certain saddle point property holds, either explicitly [33] or implicitly [46, 47]. We show that this property does not generally hold, invalidating the optimality of DPs, as hinted at in Chapman et al. [29, 30]. This finding directly refutes the claimed or hypothesized optimality of algorithms proposed in many recent research papers and pre-prints, such as Chapman et al. [29], Chow et al. [33], Ding and Feinberg [46, 47], Ni and Lai [93], Rigter et al. [105], Stanko and Macek [118]. Our results also affect applications of these algorithms, such as automated vehicle motion planning [71]. We also identify gaps in related decompositions [79, 93] and propose how to fix them.

3.1.1 Contributions

We make the following contributions in this chapter. First, we show in Section 3.3 that the popular DP for optimizing CVaR in MDPs may not recover the optimal value function and policy

regardless of how finely one discretizes the risk level in the augmented states. This method was first proposed in Chow et al. [33] but adopted widely afterwards [29, 46, 47, 105, 118]. The simple counterexample in this section contradicts the optimality claims in Chow et al. [33], Ding and Feinberg [46]. We hypothesize that prior work missed this issue because the CVaR DP works for policy evaluation and only fails when one uses it to optimize policies based on the “risk-to-go value function”. Therefore, our results do not contradict the original decomposition in Pflug and Pichler [99] that only applies to policy evaluation. We give a new independent and simple proof that the CVaR decomposition indeed works when evaluating a fixed policy.

Second, we show in Section 3.4 that the DP for optimizing the Entropic-Value-at-Risk in MDPs, proposed by Ni and Lai [93], does not compute the correct value function even when the policy is fixed. Although EVaR has not been as popular as CVaR, it has been gaining attention in recent years [63]. We give an example that contradicts the correctness claims of the risk-level decomposition for EVaR in Ni and Lai [93]. The gap that we identify with this objective applies to both policy evaluation and policy optimization. Furthermore, we prove a new, correct EVaR decomposition for policy evaluation. Unfortunately, the EVaR decomposition fails and is sub-optimal when applied to policy optimization, similar to CVaR.

Third, we propose an *optimal dynamic program* for policy optimization of VaR and q^- in Section 3.5. Our DP is based on a risk-level decomposition that closely resembles the quantile MDP decomposition in Li et al. [79] but corrects for several technical inaccuracies. The derivation shows why VaR stands apart from coherent risk measures like CVaR and EVaR. VaR is unique in that the decomposition can be constructed directly from the primal formulation of the risk measure, which avoids the complications that arise in the robust formulations used in CVaR and EVaR decompositions.

It is important to note that the correctness of DPs that augment the state space with the accumulated rewards is unaffected by our results [13, 32, 34, 63]. These DPs use the *primal* risk measure representation and do not suffer from the same saddle point issue as the augmentation methods that use the *dual* representation of the risk measures, such as the one in Chow et al. [33].

3.2 Preliminaries and Problem Formulation

For a formal definition and notation, refer to the general framework of Markov Decision Processes (MDPs) in Section 2.3. This chapter focuses on the *finite-horizon objective* in which the agent aims to compute policies that optimize the sum of rewards over a known horizon T . We demonstrate with reward function of the form $r: \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [\underline{R}, \bar{R}]$, a single time horizon $T = 1$ is sufficient to construct counterexamples that challenge existing dynamic programming methods. Notably, our counterexample also applies to reward functions $r: \mathcal{S} \times \mathcal{A} \rightarrow [\underline{R}, \bar{R}]$ but an extra horizon (i.e., $T = 2$) is required to demonstrate the failure. Moreover, deriving the decompositions with $T = 1$ makes it possible to avoid technicalities caused by history-dependent policies, which could distract us from the main ideas presented in this work. Extension of our results to general horizons $T > 1$ can be found in Chapter 4 and extension to discounted infinite-horizon objectives can be done using standard techniques [33].

With horizon $T = 1$, the set of randomized history-dependent policies is $\Pi = \{\boldsymbol{\pi}: \mathcal{S} \rightarrow \Delta_A\}$. The symbol $\pi(s, a)$ denotes the probability of action a in a state s , and $\boldsymbol{\pi}(s) = \pi(s, \cdot) \in \Delta_A$ denotes the A -dimensional vector of action probabilities in a state s . Given a risk measure $\psi \in \text{VaR}, \text{CVaR}, \text{EVaR}$ with a risk level $\alpha \in [0, 1]$, the finite-horizon risk-averse value of a policy $\pi \in \Pi$ is computed as

$$v_1^\pi(\alpha) = \psi_\alpha^{\tilde{a} \sim \boldsymbol{\pi}(\tilde{s})} [r(\tilde{s}, \tilde{a}, \tilde{s}')], \quad (3.1)$$

Throughout the chapter, we generally use $\tilde{s}$ to denote the random initial state at start and $\tilde{s}'$ to denote the random state at termination. In risk-neutral objectives, when $\psi = \mathbb{E}$, one can use the tower property of the expectation operator and define a value function v_t for each time step t [103], but this property does not hold in most static risk measures [13]. The term *policy evaluation* in the remainder of the chapter refers to computing the value in (3.1).

The goal in an MDP is to compute an *optimal* value function and a policy that attains it. In

risk-averse MDPs, this goal is formalized as the following risk-averse optimization

$$v_1^*(\alpha) = \max_{\pi \in \Pi} v_0^\pi(\alpha) = \max_{\pi \in \Pi} \psi_\alpha^{\tilde{a} \sim \pi(\tilde{s})} [r(\tilde{s}, \tilde{a}, \tilde{s}')], \quad (3.2)$$

with the *optimal policy* π^* being any policy that attains the maximum in (3.2). As with policy evaluation, when $\psi = \mathbb{E}$, the optimal value function v_t^* can be defined for each time-step t [103], but this is impossible in general for common risk measures, like VaR and CVaR. The term *policy optimization* in the remainder of the chapter refers to computing the value and the maximizer in (3.2).

In the remainder of the chapter, we study dynamic programming algorithms proposed to solve the policy evaluation problem in (3.1) and policy optimization problem in (3.2). In general, these algorithms build on risk-level decomposition [99] of coherent risk measures to define a value function $v_t^\pi(s, \alpha)$ for each time step $t \in [T]$, state $s \in \mathcal{S}$, and risk-level $\alpha \in [0, 1]$. The value function represents the risk-adjusted sum of rewards that can be obtained if starting in a state $s \in \mathcal{S}$ with remaining horizon t and a risk level α . For example, one would define the value function as $v_0^\pi(s, \alpha) = \psi_\alpha^{\tilde{a} \sim \pi(s)} [r(s, \tilde{a}, \tilde{s}')]$ and compute v_1^π using a Bellman operator T_α^π as $v_1^\pi(\alpha) = (T^\pi v_0^\pi)(\alpha)$. In risk-neutral objectives, the Bellman operator is defined as $(T^\pi(v_0^\pi))(\alpha) = \mathbb{E}[v_0^\pi(\tilde{s}, \alpha)]$, with $\alpha \in \{1\}$, but in risk-averse formulations the operator definition is more complex. The remainder of the chapter discusses the decompositions and the operator for CVaR, EVaR, and VaR risk measures respectively.

3.3 CVaR: Decomposition Fails in Policy Optimization

In this section, we show that a common CVaR decomposition proposed in Chow et al. [33] and used to optimize risk-averse policies is inherently sub-optimal regardless of how closely one discretizes the state space. The following proposition represents one of the key results used to decompose the risk measure in multi-stage decision-making.

Proposition 3.3.1 (lemma 22 in Pflug and Pichler 99). *Suppose that $\pi \in \Pi$ and $\tilde{s} \sim \hat{\mathbf{p}}$, $\tilde{a} \sim \pi(\tilde{s})$,*

$\tilde{s}' \sim \mathbf{p}_{s,a}$. Then,

$$\text{CVaR}_\alpha[r(\tilde{s}, \tilde{a}, \tilde{s}')] = \min_{\boldsymbol{\zeta} \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \zeta_s \text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s], \quad (3.3)$$

where the state s on the right-hand side is not random and

$$\mathcal{Z}_C = \{\boldsymbol{\zeta} \in \Delta_S \mid \alpha \cdot \boldsymbol{\zeta} \leq \hat{\mathbf{p}}\}. \quad (3.4)$$

The notation in Proposition 3.3.1 differs superficially from lemma 22 in Pflug and Pichler [99]. Specifically, our CVaR is defined for rewards rather than costs, the meaning of our α corresponds to $1 - \alpha$ in Pflug and Pichler [99], and we use $\zeta_s = z_s \hat{p}_s$ as the optimization variable. We include a simple proof of Proposition 3.3.1 for completeness in Appendix A.1.

The decomposition in Proposition 3.3.1 is important because it shows that the CVaR evaluation can be formulated as a dynamic program. The theorem shows that CVaR for horizon $T = 1$ decomposes into a convex combination of CVaR values for horizon $T = 0$. Recursively repeating this process, one can formulate a dynamic program for any finite time horizon T . Because the risk level at horizon $T = 0$ differs from the level at $T = 1$ and depends on the optimal $\boldsymbol{\zeta}$, one must compute CVaR values for all (or many) risk-levels $\alpha \in [0, 1]$ at time $T = 0$. As a result, the dynamic program includes an additional state variable that represents the current risk level.

Using this decomposition, numerous authors [29, 30, 33, 46, 47, 93, 105, 118] integrated the decomposition in Proposition 3.3.1 to policy optimization as

$$\begin{aligned} \max_{\pi \in \Pi} \text{CVaR}_\alpha^{\tilde{a} \sim \pi(\tilde{s})}[r(\tilde{s}, \tilde{a}, \tilde{s}')] &= \max_{\pi \in \Pi} \min_{\boldsymbol{\zeta} \in \mathcal{Z}_C} \underbrace{\sum_{s \in \mathcal{S}} \zeta_s \left(\text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim \pi(s)}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \right)}_{\theta_\pi(\boldsymbol{\zeta})} \\ &\stackrel{??}{=} \min_{\boldsymbol{\zeta} \in \mathcal{Z}_C} \max_{\pi \in \Pi} \underbrace{\sum_{s \in \mathcal{S}} \zeta_s \left(\text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim d}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \right)}_{\theta_\pi(\boldsymbol{\zeta})} \\ &= \min_{\boldsymbol{\zeta} \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \zeta_s \left(\max_{\pi \in \Pi} \text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim d}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \right). \end{aligned} \quad (3.5)$$

They used the decomposition in (3.5) to formulate a dynamic program with the current risk level as an additional state variable. We prove in the following theorem that the second equality in (3.5) marked with question marks is false in general.

Theorem 3.3.2. *There exists an MDP and a risk level $\alpha \in [0, 1]$ such that*

$$\max_{\pi \in \Pi} \text{CVaR}_{\alpha}^{\tilde{a} \sim \pi(\tilde{s})}[r(\tilde{s}, \tilde{a}, \tilde{s}')] < \min_{\zeta \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \zeta_s \left(\max_{\mathbf{d} \in \Delta_{\mathcal{A}}} \text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim \mathbf{d}}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \right). \quad (3.6)$$

Before proving Theorem 3.3.2, we discuss its implications. First, Theorem 3.3.2 contradicts theorems 5 and 7 in Chow et al. [33] and shows that their algorithm is inherently sub-optimal regardless of the resolution of the discretization. Theorem 3.3.2 also contradicts the optimality of the accelerated dynamic program proposed in Stanko and Macek [118]. The result of Chow et al. [33] was exploited as is in Chapman et al. [29], Ding and Feinberg [46], and Jin et al. [71] to propose DP reductions, and extended, without proof, in [105] to the context of a Bayesian MDP.

Finally, it is important to emphasize that Theorem 3.3.2 only applies to the policy optimization setting and does not contradict Proposition 3.3.1, which holds for the evaluation of policies that assign the same action distribution to each history of states and actions (i.e., policies that are independent of the hypothesized values of ζ).

Proof. Let's consider the CVaR objective with risk level $\alpha = 0.5$ and the MDP in Fig. 3.1 with $M = 600$. In state s_1 , three actions a_1, a_2 and a_3 are available, and in state s_2 , only action a_1 is available. To simplify the notation, we define $\theta_{\pi}: \mathcal{Z}_C \rightarrow \mathbb{R}$ for each $\pi \in \Pi$ and $\zeta \in \mathcal{Z}_C$ as

$$\theta_{\pi}(\zeta) = \sum_{s \in \mathcal{S}} \zeta_s \text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim \pi(s)}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s].$$

Because CVaR is convex in the distribution [44] and any distribution for $r(\tilde{s}, \tilde{a}, \tilde{s}')$ obtained from a policy $\pi \in \Pi$ is a mixture of the distributions of $r(\tilde{s}, a_1, \tilde{s}')$ and $r(\tilde{s}, a_2, \tilde{s}')$, it is sufficient to consider only *deterministic* policies (there exists an optimal deterministic policy). Thus, we can

reformulate the *left-hand side* of (3.6) in terms of $\theta_\pi(\zeta)$ as

$$\begin{aligned}
\max_{\pi \in \Pi} \text{CVaR}_\alpha^{\tilde{a} \sim \pi(\tilde{s})}[r(\tilde{s}, \tilde{a}, \tilde{s}')] &= \max_{\pi \in \{\pi_1, \pi_2, \pi_3\}} \text{CVaR}_\alpha^{\tilde{a} \sim \pi(\tilde{s})}[r(\tilde{s}, \tilde{a}, \tilde{s}')] \\
&= \max_{\pi \in \{\pi_1, \pi_2, \pi_3\}} \min_{\zeta \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \zeta_s \cdot \text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim \pi(s)}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \\
&= \max_{\pi \in \{\pi_1, \pi_2, \pi_3\}} \min_{\zeta \in \mathcal{Z}_C} \theta_\pi(\zeta),
\end{aligned}$$

with $\pi_1(s_1, a_1) = 1$, $\pi_2(s_1, a_2) = 1$ and $\pi_3(s_1, a_3) = 1$. The functions $\theta_{\pi_1}(\cdot)$, $\theta_{\pi_2}(\cdot)$ and $\theta_{\pi_3}(\cdot)$ are depicted in Fig. 3.2. Similarly, the *right-hand side* of (3.6) can be expressed using the convexity of CVaR in the distribution by algebraic manipulation as

$$\min_{\zeta \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \zeta_s \max_{\mathbf{d} \in \Delta_A} \text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim \mathbf{d}}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] = \min_{\zeta \in \mathcal{Z}_C} \max_{\pi \in \{\pi_1, \pi_2, \pi_3\}} \theta_\pi(\zeta).$$

Using the notation introduced above and the sufficiency of optimizing over deterministic policies only, the inequality in (3.6) becomes

$$\max_{\pi \in \{\pi_1, \pi_2, \pi_3\}} \min_{\zeta \in \mathcal{Z}_C} \theta_\pi(\zeta) < \min_{\zeta \in \mathcal{Z}_C} \max_{\pi \in \{\pi_1, \pi_2, \pi_3\}} \theta_\pi(\zeta). \quad (3.7)$$

Figure 3.2 demonstrates the inequality in (3.7) numerically, with the green dot $\bullet$ representing the left-hand side maximum and the violet dot $\bullet$ representing the right-hand side minimum. The dashed line represents the function $\zeta \mapsto \max_{\pi \in \{\pi_1, \pi_2, \pi_3\}} \theta_\pi(\zeta)$.

To show the strict inequality in (3.7) formally, we evaluate the functions $\theta_{\pi_1}(\cdot)$, $\theta_{\pi_2}(\cdot)$ and $\theta_{\pi_3}(\cdot)$ for MDP M_C . The function $\theta_{\pi_2}(\cdot)$ is linear because the CVaR applies to a constant, and CVaR is translation invariant. The function $\theta_{\pi_1}(\cdot)$ is piecewise-linear and convex, and its slope can be computed using the subgradient that for each $s \in \mathcal{S}$ and $\hat{\zeta} \in \mathcal{Z}_C$ satisfies [33]

$$\partial_{\zeta_s} \hat{\zeta}_s \text{CVaR}_{\alpha \hat{p}_s^{-1} \hat{\zeta}_s}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \ni \text{VaR}_{\alpha \hat{p}_s^{-1} \hat{\zeta}_s}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s].$$

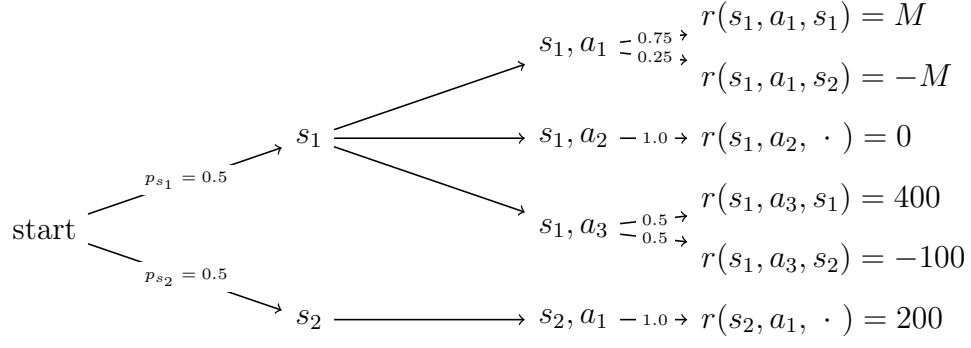


Figure 3.1: An example used to show the sub-optimality of a policy in Eq. (3.5).

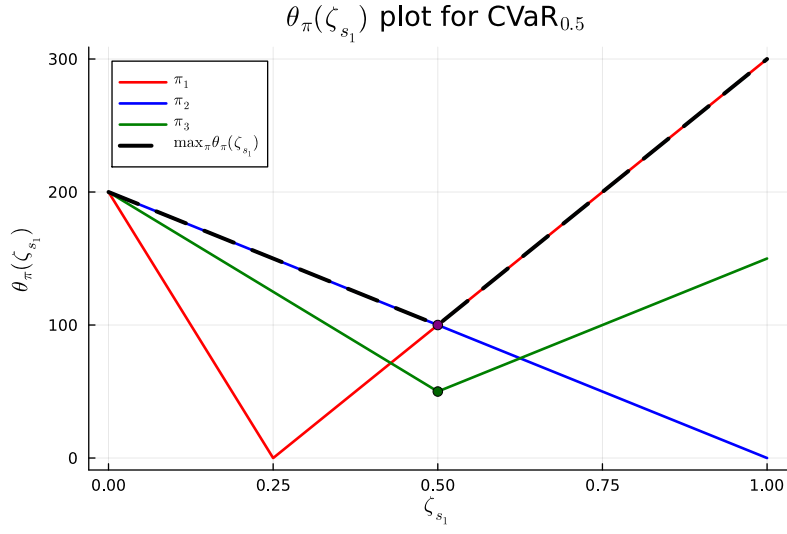


Figure 3.2: The functions $\theta_{\pi}(\cdot)$ and ζ_{s_1} in CVaR decomposition for the counterexample Fig. 3.1 with $M = 600$ and $\alpha = 0.5$.

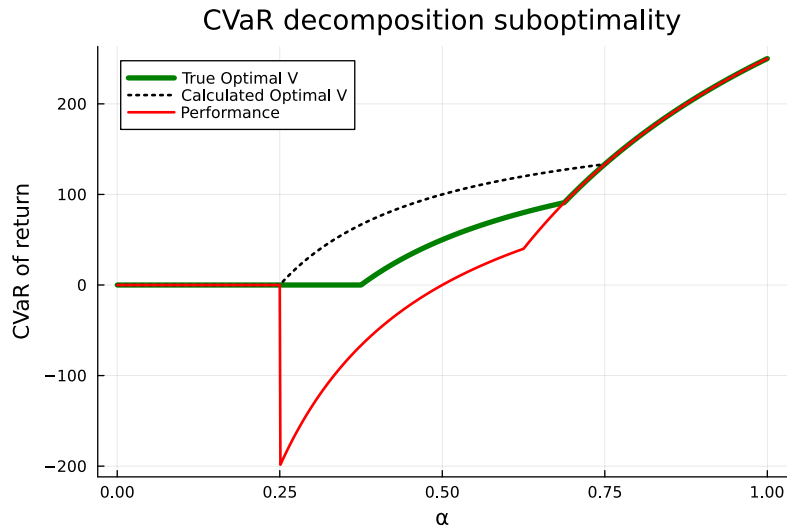


Figure 3.3: Return computed in Chow et al. [33] vs optimal CVaR policy for the example in Fig. 3.1 with $M = 600$.

Simple algebraic manipulation then shows that

$$\begin{aligned}\theta_{\pi_1}(\zeta) &= \max \{200 - 800 \zeta_{s_1}, -100 + 400 \zeta_{s_1}\} \\ \theta_{\pi_2}(\zeta) &= 200 - 200 \zeta_{s_1} \\ \theta_{\pi_3}(\zeta) &= \max \{200 - 300 \zeta_{s_1}, -50 + 200 \zeta_{s_1}\}\end{aligned},$$

and $\mathcal{Z}_C = \Delta_{\mathcal{S}}$, which implies that $\zeta_{s_1} \in (0, 1)$. Therefore, by algebraic manipulation, we get the desired strict inequality

$$50 = \max_{\pi \in \{\pi_1, \pi_2, \pi_3\}} \min_{\zeta \in \mathcal{Z}_C} \theta_{\pi}(\zeta) < \min_{\zeta \in \mathcal{Z}_C} \max_{\pi \in \{\pi_1, \pi_2, \pi_3\}} \theta_{\pi}(\zeta) = 100,$$

where 50 and 100 are represented by the green dot $\bullet$ and violet dot $\bullet$ in Fig. 3.2, respectively. $\square$

In summary, the decomposition in Proposition 3.3.1 cannot be exploited in policy optimization because the inequality in the derivation above may not be tight:

$$\begin{aligned}\max_{\pi \in \Pi} \text{CVaR}_{\alpha}^{\tilde{a} \sim \pi(\tilde{s})}[r(\tilde{s}, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] &= \max_{\pi \in \Pi} \min_{\zeta \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \zeta_s \text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim \pi(\tilde{s})}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \\ &\leq \min_{\zeta \in \mathcal{Z}_C} \max_{\pi \in \Pi} \sum_{s \in \mathcal{S}} \zeta_s \text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim \pi(\tilde{s})}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \\ &= \min_{\zeta \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \zeta_s \max_{d \in \Delta_A} \text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim d}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s],\end{aligned}$$

where the last equality follows from the interchangeability property of optimization and expected value [113, theorem 7.92].

Figure 3.3 depicts the sub-optimality demonstrated by the CVaR decomposition. CVaR decomposition for policy optimization would over-approximate (black dash) the true value function (green solid) and commit to a sub-optimal solution (red solid). This simple example answers the following important concerns on the suboptimality.

First, CVaR policy optimization decomposition in Chow et al. [33] can lead one to choose a suboptimal policy. In particular, Fig. 3.3 shows that the CVaR policy optimization decomposition

CVaR decomposition suboptimality

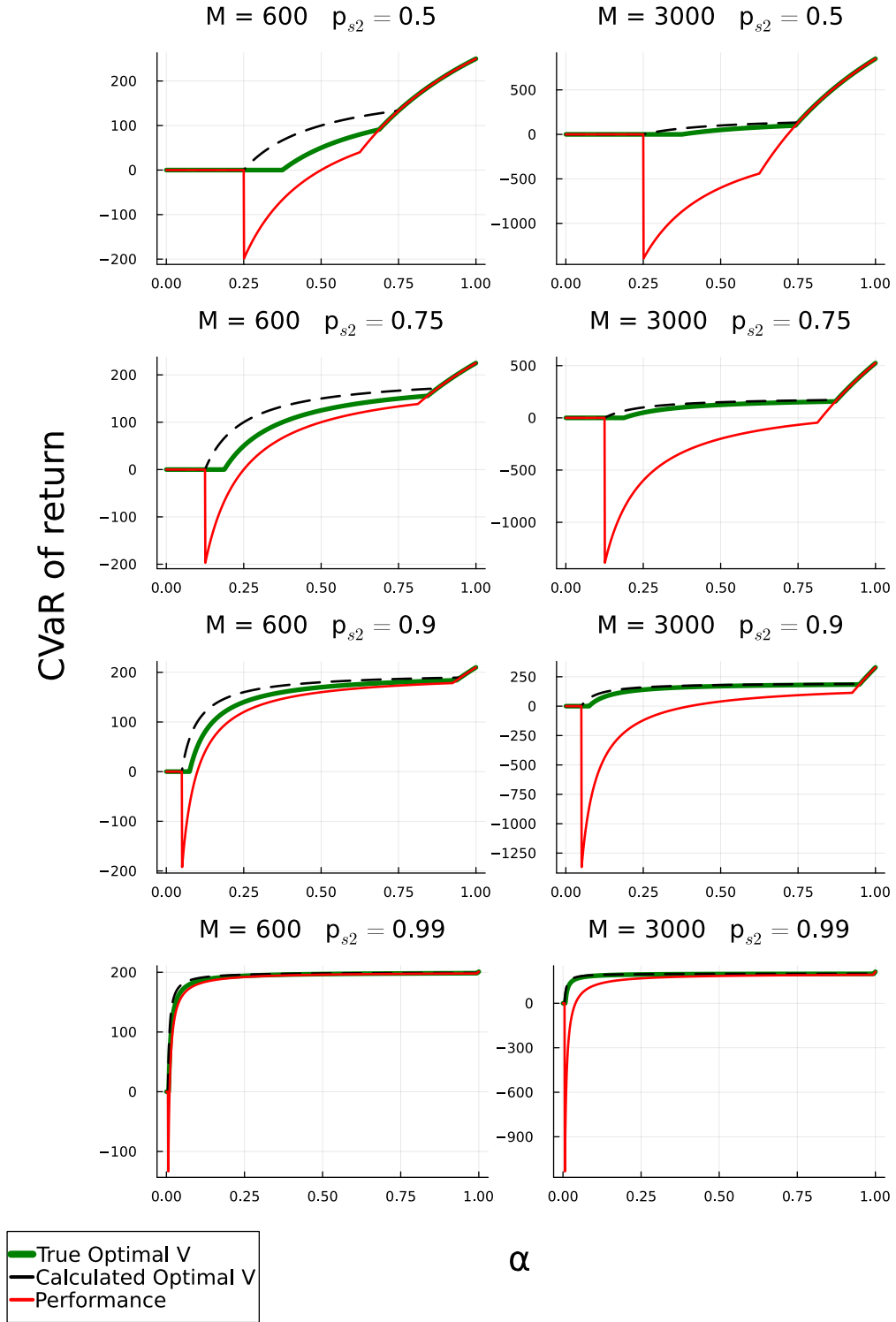


Figure 3.4: CVaR decomposition suboptimality for different M and p_{s2} for MDP in Fig. 3.1

is an overestimate of the return and the decision maker commits to the worst action for CVaR objective $\alpha \in (0.25, 0.625)$. Furthermore, action a_3 is never chosen for [33] even though it is the only optimal action for $\alpha \in (0.375, 0.6875)$. We can see from Fig. 3.2 when we inspect the ζ function for $\alpha = 0.5$ that a_3 is optimal with CVaR return of 50. However, the CVaR decomposition would select a_1 or a_2 and over-estimate the CVaR return to 100, but the sub-optimal actions a_1 or a_2 would have the true CVaR return of 0 instead.

Second, there is no upper bound on the sub-optimality of the action chosen with respect CVaR return of the optimal action. The regret of the sub-optimal return is dependent on the distribution of the sub-optimal action and unbounded with respect to the true optimal return, as the performance of the sub-optimal return for $\alpha \in (0.25, 0.5)$ is worse than the worst case scenario of the best worst-case policy. Increasing the magnitude ($M > 400$) for the reward instead of the sub-optimal action a_1 in s_1 in Fig. 3.1 increases the sub-optimality and can be arbitrarily large.

Third, the sub-optimality can occur for any $\alpha \in (0, 1)$. As above, we use the MDP in Fig. 3.1 and perturb two values to demonstrate the suboptimality by (i) increasing the magnitude ($M > 400$) for the reward of the sub-optimal action s_1, a_1 , or (ii) increasing the initial state probability (p_{s_2}). That leads to increasing the range where the CVaR decomposition for policy optimization yields a sub-optimal policy as shown in Fig. 3.4.

This finding directly challenges the claimed or hypothesized optimality of several recently proposed algorithms, as presented in various research papers and preprints, including [29, 30, 33, 46, 47, 93, 105, 118]. We omit to comment on the validity of the CVaR decomposition in Li et al. [79] given that it considers a different measure than the CVaR defined in Eq. (2.3). Namely, their measure takes the form: $\widetilde{\text{CVaR}}_\alpha[\tilde{x}] = \inf_{z \in \mathbb{R}} (z + (1 - \alpha)^{-1} \mathbb{E}[\tilde{x} - z]_+) = -\text{CVaR}_{1-\alpha}[-\tilde{x}]$, which is not a coherent risk measure.

3.4 EVaR: Decomposition for Policy Evaluation

In this section, we show that a decomposition for EVaR proposed in Ni and Lai [93] is inexact even when considering the policy evaluation setting. Ni and Lai [93] recently proposed a decomposition

of EVaR for a fixed $\pi \in \Pi$ with $\tilde{a} \sim \boldsymbol{\pi}(\tilde{s})$ and a risk level $\alpha \in (0, 1]$ as

$$\text{EVaR}_\alpha[r(\tilde{s}, \tilde{a}, \tilde{s}')] \stackrel{??}{=} \min_{\boldsymbol{\xi} \in \mathcal{Z}_E} \sum_{s \in \mathcal{S}} \xi_s \text{EVaR}_{\alpha \xi_s \hat{p}_s^{-1}}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s], \quad (3.8)$$

where $\tilde{s} \sim \hat{\boldsymbol{p}}$, $\tilde{s}' \sim p(\tilde{s}, \tilde{a}, \cdot)$, and

$$\mathcal{Z}_E = \left\{ \boldsymbol{\xi} \in \Delta_{\mathcal{S}} \mid \sum_{s \in \mathcal{S}} \xi_s \log(\xi_s / \hat{p}_s) \leq -\log \alpha, \quad \overbrace{\alpha \cdot \boldsymbol{\xi} \leq \hat{\boldsymbol{p}}}^{\text{implicit in Ni and Lai [93]}} \right\}. \quad (3.9)$$

Note that we use variables $\xi_s = z_s \hat{p}_s$ in comparison with z_s in Ni and Lai [93].

The constraint $\alpha \cdot \boldsymbol{\xi} \leq \hat{\boldsymbol{p}}$ in (3.9) was not stated explicitly in Ni and Lai [93] but is necessary because $\text{EVaR}_{\alpha'}[\cdot]$ is defined only for $\alpha' \in [0, 1]$. When $\alpha' = \alpha \xi_s \hat{p}_s^{-1}$ in (3.8) it must also satisfy for each $s \in \mathcal{S}$ that

$$\alpha' \leq 1 \quad \Leftrightarrow \quad \alpha \xi_s \hat{p}_s^{-1} \leq 1 \quad \Leftrightarrow \quad \alpha \cdot \xi_s \leq \hat{p}_s.$$

This additional constraint on $\boldsymbol{\xi}$ implies that $\mathcal{Z}_E \subseteq \mathcal{Z}_C$, for the $\mathcal{Z}_C$ defined in (3.4).

We claim in the following theorem (see Appendix A.2 for the proof) that the equality in (3.8) does not hold even in the policy evaluation setting.

Theorem 3.4.1. *There exists an MDP with a single action and $\alpha \in (0, 1]$ such that*

$$\text{EVaR}_\alpha[r(\tilde{s}, a_1, \tilde{s}')] < \min_{\boldsymbol{\xi} \in \mathcal{Z}_E} \sum_{s \in \mathcal{S}} \xi_s \text{EVaR}_{\alpha \xi_s \hat{p}_s^{-1}}[r(s, a_1, \tilde{s}') \mid \tilde{s} = s], \quad (3.10)$$

the set $\mathcal{Z}_E$ defined by (3.9).

Theorem 3.4.1 demonstrates a stronger failure mode than the one in Theorem 3.3.2 (for CVaR policy optimization) since it applies to both policy evaluation and policy optimization settings.

We propose a correct decomposition of EVaR for policy evaluation in the following theorem and employ it to establish that the decomposition in (3.8) overestimates the actual value of EVaR (see Appendix A.3 for a proof).

Theorem 3.4.2. *Given any finite MDP with horizon $T = 1$ and $\alpha \in (0, 1]$, we have that*

$$\text{EVaR}_\alpha [r(\tilde{s}, \tilde{a}, \tilde{s}')] = \inf_{\zeta \in (0, 1]^S, \xi \in \mathcal{Z}'_E(\zeta)} \sum_{s \in \mathcal{S}} \xi_s \text{EVaR}_{\zeta_s} [r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] ,$$

where

$$\mathcal{Z}'_E(\zeta) = \left\{ \xi \in \Delta_S \mid \xi \ll \hat{p}, \sum_{s \in \mathcal{S}} \xi_s (\log(\xi_s / \hat{p}_s) - \log(\zeta_s)) \leq -\log \alpha \right\} .$$

Moreover, EVaR can be upper-bounded as

$$\text{EVaR}_\alpha [r(\tilde{s}, \tilde{a}, \tilde{s}')] \leq \min_{\xi \in \mathcal{Z}_E} \sum_{s \in \mathcal{S}} \xi_s \text{EVaR}_{\alpha \xi_s \hat{p}_s^{-1}} [r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] . \quad (3.11)$$

Despite correcting the decomposition of EVaR for policy evaluation, approaches similar to those used in CVaR optimization [33, 79, 93] may still over-approximate the EVaR objective

$$\max_{\pi \in \Pi} \text{EVaR}_\alpha [r(\tilde{s}, \tilde{a}, \tilde{s}')] \leq \inf_{\zeta \in (0, 1]^S, \xi \in \mathcal{Z}'_E(\zeta)} \sum_{s \in \mathcal{S}} \xi_s \left(\max_{\pi \in \Pi} \text{EVaR}_{\zeta_s} [r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \right) , \quad (3.12)$$

due to the same underlying duality gap identified in Theorem 3.3.2.

3.5 VaR: Decomposition Holds for Policy Evaluation and Optimization

In this section, we discuss a dynamic program decomposition for VaR whose decomposition resembles those for CVaR and EVaR described in Sections 3.3 and 3.4. However, the VaR decomposition can be used for policy evaluation Theorem 3.5.2 as well as policy optimization Theorem 3.5.3. We provide a new proof of the VaR decomposition to elucidate the differences that make it optimal in contrast to CVaR and EVaR decompositions. Our VaR decomposition closely resembles the quantile MDP approach in Li et al. [79] with a few technical modifications that can significantly impact the computed value.

To contrast the typical definition of VaR with the quantile definition in Li et al. [79], it is helpful

to summarize how VaR is related to the quantile of a random variable. Let $\mathbf{q} \in \mathbb{R}$ define as the α -quantile of $\tilde{x} \in \mathbb{X}$ when

$$\mathbb{P}[\tilde{x} \leq \mathbf{q}] \geq \alpha \quad \text{and} \quad \mathbb{P}[\tilde{x} < \mathbf{q}] \leq \alpha . \quad (3.13)$$

In general, the set of quantiles is an interval $[\mathbf{q}_{\tilde{x}}^-(\alpha), \mathbf{q}_{\tilde{x}}^+(\alpha)]$ with the bounds computed as [57, appendix A.3]

$$\begin{aligned} \mathbf{q}_{\tilde{x}}^-(\alpha) &= \sup \{z \mid \mathbb{P}[\tilde{x} < z] < \alpha\} = \inf \{z \mid \mathbb{P}[\tilde{x} \leq z] \geq \alpha\} , \\ \mathbf{q}_{\tilde{x}}^+(\alpha) &= \inf \{z \mid \mathbb{P}[\tilde{x} \leq z] > \alpha\} = \sup \{z \mid \mathbb{P}[\tilde{x} < z] \leq \alpha\} . \end{aligned}$$

Note that when the distribution of $\tilde{x}$ is absolutely continuous (atomless), then $\mathbf{q}_{\tilde{x}}^+(\alpha) = \mathbf{q}_{\tilde{x}}^-(\alpha)$ and the quantile is unique. The following example illustrates a simple setting in which the quantile is not unique.

Example 3.5.1 (Bernoulli random variable). Consider a Bernoulli random variable $\tilde{e}$ such that $\tilde{e} = 1$ and $\tilde{e} = 0$ with equal (50%) probabilities. Then, any value $q \in [0, 1]$ is a valid 0.5-quantile because

$$\begin{aligned} \mathbf{q}_{\tilde{e}}^-(0.5) &= \inf_{z \in \mathbb{R}} \{z \mid \mathbb{P}[\tilde{e} \leq z] \geq 0.5\} = \inf_{z \in \mathbb{R}} \{z \mid z \geq 0\} = 0 \\ \mathbf{q}_{\tilde{e}}^+(0.5) &= \sup_{z \in \mathbb{R}} \{z \mid \mathbb{P}[\tilde{e} \geq z] \geq 0.5\} = \sup_{z \in \mathbb{R}} \{z \mid z \leq 1\} = 1. \end{aligned}$$

The objective in Li et al. [79] is to maximize the quantile operator $Q_\alpha: \mathbb{X} \rightarrow \mathbb{R}$ defined for a reward random variable $\tilde{x} \in \mathbb{X}$ and a risk level $\alpha \in [0, 1]$ as

$$Q_\alpha(\tilde{x}) = \inf_{z \in \mathbb{R}} \{z \mid \mathbb{P}[\tilde{x} \leq z] \geq \alpha\} . \quad (3.14)$$

The quantile operator Q_α and VaR differ in which quantile of the random variable they consider:

$$Q_\alpha(\tilde{x}) = \mathbf{q}_{\tilde{x}}^-(\alpha), \quad \text{but} \quad \text{VaR}_\alpha[\tilde{x}] = \mathbf{q}_{\tilde{x}}^+(\alpha) . \quad (3.15)$$

As a result, the quantile MDP objective in (3.14) coincides with the VaR value only when the quantile is unique, which is not always the case, as shown in Example 3.5.1.

Theorem 3.5.2. *Let $\tilde{y}: \Omega \rightarrow [N]$ be a random variable distributed as $\hat{\mathbf{p}} = (\hat{p}_i)_{i=1}^N$ with $\hat{p}_i > 0$. Then for any random variable $\tilde{x} \in \mathbb{X}$, we have*

$$\text{VaR}_\alpha[\tilde{x}] = \sup_{\boldsymbol{\zeta} \in \Delta_N} \left\{ \min_i \text{VaR}_{\alpha \zeta_i \hat{p}_i^{-1}}[\tilde{x} \mid \tilde{y} = i] \mid \alpha \cdot \boldsymbol{\zeta} \leq \hat{\mathbf{p}} \right\}, \quad (3.16)$$

where we interpret the minimum to evaluate to ∞ if all terms are infinite, which only occurs if $\alpha = 1$.

Proof. We decompose VaR using the definition in (2.2) as

$$\begin{aligned} \text{VaR}_\alpha[\tilde{x}] &= \sup \{z \in \mathbb{R} \mid \mathbb{P}[\tilde{x} < z] \leq \alpha\} \stackrel{(a)}{=} \sup \left\{ z \in \mathbb{R} \mid \sum_{i=1}^N \mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] \cdot \hat{p}_i \leq \alpha \right\} \\ &\stackrel{(b)}{=} \sup \left\{ z \in \mathbb{R} \mid \boldsymbol{\zeta} \in [0, 1]^N, \mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] \leq \zeta_i, \forall i \in [N], \sum_{i=1}^N \zeta_i \hat{p}_i \leq \alpha \right\} \\ &\stackrel{(c)}{=} \sup \left\{ \sup \{z \in \mathbb{R} \mid \mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] \leq \zeta_i, \forall i \in [N]\} \mid \boldsymbol{\zeta} \in [0, 1]^N, \sum_{i=1}^N \zeta_i \hat{p}_i \leq \alpha \right\} \\ &= \sup \left\{ \sup_{i \in [N]} \left\{ z \in \mathbb{R} \mid \mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] \leq \zeta_i \right\} \mid \boldsymbol{\zeta} \in [0, 1]^N, \sum_{i=1}^N \zeta_i \hat{p}_i \leq \alpha \right\} \\ &\stackrel{(d)}{=} \sup \left\{ \min_{i \in [N]} \sup \{z \in \mathbb{R} \mid \mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] \leq \zeta_i\} \mid \boldsymbol{\zeta} \in [0, 1]^N, \sum_{i=1}^N \zeta_i \hat{p}_i \leq \alpha \right\} \\ &\stackrel{(e)}{=} \sup \left\{ \min_{i \in [N]} \text{VaR}_{\zeta_i}[\tilde{x} \mid \tilde{y} = i] \mid \boldsymbol{\zeta} \in [0, 1]^N, \sum_{i=1}^N \zeta_i \hat{p}_i \leq \alpha \right\}. \end{aligned}$$

We decompose the probability $\mathbb{P}[\tilde{x} < z]$ as a marginal of the conditional probabilities $\mathbb{P}[\tilde{x} < z \mid \tilde{y} = i]$ and $\hat{p}_i = \mathbb{P}[\tilde{y} = i]$ in step (a) and then lower-bound them by an auxiliary variable ζ_i in step (b). In step (c) we replace the joint supremum over z and $\boldsymbol{\zeta}$ by sequential suprema, and then we replace the supremum of an intersection by the minimum of the suprema of sets in (d). The equality in (d) holds because $\mathbb{P}[\tilde{x} < z]$ is monotone and, therefore, the sets $\{z \in \mathbb{R} \mid \mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] \leq \zeta_i\}$ are nested. Finally, step (e) follows from the definition of VaR in (2.2). $\square$

Focusing on the finite MDP with horizon $T = 1$, we can show that the decomposition proposed in Theorem 3.5.2 is amenable to policy optimization. The main difference between the VaR decomposition and CVaR is that the former VaR was expressed as a supremum instead of an infimum over quantile levels ζ . For VaR, changing the order of maximum (π) and supremum (ζ) does not suffer from a potential gap, but changing the order of maximum (π) and infimum/minimum (ζ) in CVaR does suffer from such a gap as shown in Theorem 3.3.2.

The following theorem (proved in Appendix A.4) summarizes the decomposition for VaR.

Theorem 3.5.3. *Given any finite MDP with horizon $T = 1$ and $\alpha \in [0, 1]$, we have*

$$\max_{\pi \in \Pi} \text{VaR}_{\alpha}^{\tilde{a} \sim \pi(\tilde{s})} [r(\tilde{s}, \tilde{a}, \tilde{s}')] = \sup_{\zeta \in \Delta_S} \left\{ \min_{s \in S} \max_{d \in \Delta_A} \text{VaR}_{\alpha \zeta s \hat{p}_s^{-1}}^{\tilde{a} \sim d} [r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \mid \alpha \cdot \zeta \leq \hat{p} \right\}.$$

Section 4.3 extends Theorem 3.5.3 to horizon $T > 1$, providing dynamic programming equations and a definition of the optimal policy. In summary, the VaR MDP can be solved using the DP formulation described as follows: Let initialize the terminal value function as $v_0^*(s_T, \alpha_T) := \text{VaR}_{\alpha_T}[0] \forall s_T \in \mathcal{S}, \alpha_T \in [0, 1]$. The optimal value function can be computed iteratively for all $t \in [T - 1]$ and $\alpha \in [0, 1]$ with

$$v_{t+1}^*(s, \alpha) := \max_{\pi \in \Pi_{\text{HD}}} \text{VaR}_{\alpha}^{\pi, s} \left[\sum_{k=0}^t \gamma^k r(\tilde{s}_k, \tilde{a}_k, \tilde{s}_{k+1}) \right] = \max_{a \in \mathcal{A}} \sup_{\zeta \in \Delta_S} \left\{ \min_{s' \in S} r(s, a, s') + \gamma v_t^*(s', \frac{\alpha \zeta_{s'}}{p_{sas'}}) \right\} \quad (3.17)$$

and the optimal policy $\pi_t^*(s)$ is the optimizer of the equation. Despite having a DP decomposition for VaR MDP, one key issue remains unresolved for developing a practical algorithm. Since the risk level $\alpha \in [0, 1]$ is continuous, the state space becomes effectively infinite. This makes it impractical to compute Eq. (3.17) for all $t \in [T - 1]$ and $\alpha \in [0, 1]$. We will further discuss and address this issue in Chapter 4.

We finally present below a valid decomposition for the lower quantile MDP, used officially as the objective in [79] (see Appendix A.5 for a proof).

Proposition 3.5.4. *Given any finite MDP with horizon $T = 1$ and some $\alpha \in [0, 1]$, we have that:*

$$\max_{\pi \in \Pi} Q_{\alpha}^{\tilde{a} \sim \pi(\tilde{s})}(r(\tilde{s}, \tilde{a}, \tilde{s}')) = \sup_{\zeta \in [0,1]^S} \left\{ \min_{s \in \mathcal{S}: \zeta_s < 1} \max_{d \in \Delta_A} Q_{\zeta_s}^{\tilde{a} \sim d}(r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s) \mid \sum_{s=1}^S \zeta_s \hat{p}_s < \alpha \right\}.$$

We note that the difference with the result presented in [79] resides in the constraint imposed on ζ that replaces the weak inequality with a strict one. In fact, this strict versus weak inequality is the main distinguishing factor between the decompositions for the lower and upper quantiles.

3.6 Chapter Summary

This chapter shows that a popular decomposition approach to solving MDPs with CVaR and EVaR objectives is suboptimal despite the claims to the contrary. This suboptimality arises from a saddle-point gap when *optimizing policy*. We also prove that a similar decomposition approach is optimal for policy optimization and evaluation when solving MDPs with the VaR objective. The decomposition is optimal because VaR does not involve the same saddle point problem as CVaR and EVaR.

Our findings are significant because practitioners who make risk-averse decisions in high-stakes scenarios need to have confidence in the correctness of the algorithms they use. Our work raises awareness that popular static CVaR and EVaR MDP algorithms are suboptimal, and their analyses are inaccurate. We hope the results we present in this chapter will increase the scrutiny of dynamic programming methods for risk-averse MDPs and motivate research into alternative approaches, such as the parametric dynamic programs.

CHAPTER 4

Q-learning for Quantile MDPs: A Decomposition, Performance, and Convergence Analysis

4.1 Introduction

The practicality of reinforcement learning (RL) models has led to their widespread integration in autonomous decision-making [74, 75]. Traditional metrics focusing on expected value fail to address the uncertainties of random returns, highlighting the limitations of one-size-fits-all models [68, 123]. This observation is apparent in various applications such as motion control [3, 27, 61], autonomous systems [71, 128], healthcare [77, 117], or capital investment [89], where each necessitates a specific risk-sensitive objective. Risk-averse RL (RARL) endeavors to align with the decision-maker’s preferences by tailoring the objective function according to their risk interests [133]. Common measures in RARL include exponential utility [24, 63], target value [82, 131], value-at-risk (VaR) [34, 62, 79], conditional VaR (CVaR) [13, 32, 81], or mean-variance objectives [84, 123], among others.

Quantile measures such as VaR are used in data centers to ensure the reliability of cloud computing services [41], in epidemiology to understand how exposure to disease differs across continuous health outcomes distributions, e.g., BMI [130], or more standardly in financial markets to assess counterparty risk [7, 51]. Although VaR does not enjoy the mathematical properties of CVaR and thus is not a coherent risk, its numerical advantages for backtesting have led the Basel Committee to retain it for credit value adjustment [50, 95]. Unlike expected value, quantile risk measures account for the return distribution. For example, the expectation may favor low probability

outcomes yielding a high return, whereas the α -level-VaR guarantees the gain is greater or equal to that value with high probability $1 - \alpha$. In RARL, recent years have witnessed an increasing interest in quantile measures, especially since the emergence of distributional RL [14, 40] and its successful application in robotics [85].

A major challenge in deriving practical RL solutions is that a full knowledge of the environment is often inaccessible. This can be even more problematic in RARL where risk-averse strategies must involve the full return distribution rather than just its expectation. Previous works have presented model-free methods to find optimal CVaR policies [39, 72, 81], but all assumed the existence of an optimal Markov policy. Such assumption is valid for mean and entropic risk objectives but generally not for quantile-based objectives such as CVaR or VaR [17, 62]. Indeed, optimal-VaR policies can potentially all be history-dependent, so restricting the search to Markov policies can produce suboptimal return [62, 79].

4.1.1 Contributions

In this chapter, we propose a new dynamic programming (DP) formulation for quantile MDPs, which we refer to as VaR-MDP since it seeks a policy that is optimal with respect to the VaR of the total discounted reward. Compared with Eq. (3.17), this new DP decomposition requires neither known transition probabilities nor solving complex saddle point equations. As a second contribution, we introduce VaR-Q-learning, the first model-free method that provably optimizes the return distribution’s VaR. We develop a rigorous proof of convergence to an optimal policy, thus ensuring the validity of our approach. Although our VaR-Q-learning can be seen as a simple variant of the risk-sensitive Q-learning methods developed in distributional RL [14, 39, 40], we show that our slight modification makes a meaningful difference in the quality of the computed policy.

We first delineate our research setting and preliminary concepts in Section 4.2. Then, we introduce new DP equations in Section 4.3 to solve VaR-MDPs while being simpler than existing methods. This enables us to propose a novel quantile Q-learning algorithm in Section 4.4, which optimizes the

VaR-MDP from sampled trajectories, and establishes its convergence property. Finally, numerical experiments presented in Section 4.5 illustrate the effectiveness of our algorithm.

4.2 Preliminaries and Problem Formulation

We first introduce the relevant properties for quantile and VaR risk measures, then formalize the MDP framework with the VaR objective in Section 4.2.1, which we shorten as VaR-MDP. We prove the claims of this section in Appendix B.1.

Quantiles and Value-at-Risk As presented in Section 2.2.2, the quantile of a random variable $\tilde{x} \in \mathbb{X}$ at level $\alpha \in [0, 1]$ is any $\tau \in \bar{\mathbb{R}}$ such that $\mathbb{P}[\tilde{x} \leq \tau] \geq \alpha$ and $\mathbb{P}[\tilde{x} \geq \tau] \geq 1 - \alpha$. It may not be unique but lies in the interval $[\mathbf{q}_\alpha^-(\tilde{x}), \mathbf{q}_\alpha^+(\tilde{x})]$, where

$$\begin{aligned} \mathbf{q}_\alpha^-(\tilde{x}) &:= \min \left\{ \tau \in \bar{\mathbb{R}} \mid \mathbb{P}[\tilde{x} \leq \tau] \geq \alpha \right\}, \\ \text{VaR}_\alpha[\tilde{x}] &:= \mathbf{q}_\alpha^+(\tilde{x}) := \max \left\{ \tau \in \bar{\mathbb{R}} \mid \mathbb{P}[\tilde{x} < \tau] \leq \alpha \right\}. \end{aligned} \quad (4.1)$$

Monetary risk measures generalize the average criterion to account for uncertain outcomes. Among them, quantile-based measures like VaR are the most common [57, 113]. Given a risk level $\alpha \in [0, 1]$, the VaR function $\text{VaR}_\alpha: \mathbb{X} \rightarrow \bar{\mathbb{R}}$ is defined as the largest $1 - \alpha$ confidence lower bound on the value of $\tilde{x}$. By convention, $\text{VaR}_\alpha[0] = 0$ if $\alpha \in [0, 1)$ and ∞ otherwise. Also, $\mathbf{q}_0^-(\tilde{x}) = -\infty$ and $\mathbf{q}_1^+(\tilde{x}) = \infty$, while $\mathbf{q}_\alpha^-(\tilde{x}) \in \mathbb{R}$ for all $\alpha \in (0, 1]$ and $\mathbf{q}_\alpha^+(\tilde{x}) \in \mathbb{R}$ for all $\alpha \in [0, 1)$.

Elicitability. Based on the works of [60] and [15], a risk measure is *elicitable* if it is the solution of an empirical risk minimization problem. In particular, a quantile can be estimated via quantile regression with the loss [76]:

$$\ell_\alpha(\delta) := \max \left\{ \alpha\delta, -(1 - \alpha)\delta \right\}, \quad (4.2)$$

where δ represents the residual error between the prediction and the noisy observation. Thus, quantile measures are elicitable as stated below.

Lemma 4.2.1. *For any $\tilde{x} \in \mathbb{X}$ and $\alpha \in [0, 1]$, it holds that*

$$\arg \min_{y \in \mathbb{R}} \mathbb{E}[\ell_\alpha(\tilde{x} - y)] = [\mathbf{q}_\alpha^-(\tilde{x}), \mathbf{q}_\alpha^+(\tilde{x})] \cap \mathbb{R}.$$

As explained in Bellini and Bignozzi [15, Ex. 3.8], VaR is not elicitable unless $\tilde{x}$ is continuous, i.e. $\mathbf{q}_\alpha^-(\tilde{x}) = \mathbf{q}_\alpha^+(\tilde{x})$. Hence, methods that rely on statistical estimation of conditional VaR using Lemma 4.2.1 must contend with potential underestimation.

4.2.1 MDPs with Value-at-Risk Objective

For a formal definition and notation, refer to the general framework of Markov Decision Processes (MDPs) in Section 2.3. In VaR-MDP, the agent aims to find a policy π that optimizes the *static* VaR of the discounted sum of returns,

$$\rho(\pi) := \text{VaR}_{\alpha_0}^{\pi, s_0} \left[\sum_{k=0}^{T-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] = \text{VaR}_{\alpha_0} \left[\tilde{\mathfrak{R}}_T^\pi(s_0) \right], \quad (4.3)$$

for some initial state $s_0 \in \mathcal{S}$ and reference risk level $\alpha_0 \in (0, 1)$. In Eq. (4.3), the distribution of $\tilde{s}_k$ for $k \geq 1$ is implicitly governed by the transition model p while the superscript s_0 fixes the initial state. The policy π governs the realization of actions $\tilde{a}_k$ at all steps k , which we formalize next. Let Π_{HD}^t be the set of all history-dependent deterministic policies over horizon t . All in all, given a quantile level α_0 and an initial state s_0 , we aim to find $\max_{\pi \in \Pi_{\text{HD}}^T} \rho(\pi)$.

Although the optimal policy for Eq. (4.3) is history-dependent, it can still be computed using DP and value iteration [62, 79]. As we will see, the difference with standard DP lies in the fact that the optimal state-action value function must also adapt the quantile level $\alpha \in [0, 1]$ at each state $s \in \mathcal{S}$ and $t \in [T]$. Let thus $q_t^*: \mathcal{S} \times [0, 1] \times \mathcal{A} \rightarrow \bar{\mathbb{R}}$ be the optimal state-action value function for $t \in [T]$:

$$q_t^*(s, \alpha, a) := \max_{\substack{\pi \in \Pi_{\text{HD}}^t: \\ \pi_0(s) = a}} \text{VaR}_\alpha^{\pi, s} \left[\sum_{k=0}^{t-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right]. \quad (4.4)$$

It is also convenient to define the optimal state value function $v_t^*: \mathcal{S} \times [0, 1] \rightarrow \bar{\mathbb{R}}$ for horizon $t \in [T]$ as

$$v_t^*(s, \alpha) := \max_{\pi \in \Pi_{\text{HD}}^t} \text{VaR}_\alpha^{\pi, s} \left[\sum_{k=0}^{t-1} \gamma^k \cdot r(\tilde{s}_k, \tilde{a}_k) \right].$$

Similar to risk-neutral MDPs, the state value function is related to the state-action value function through $v_t^*(s, \alpha) = \max_{a \in \mathcal{A}} q_t^*(s, \alpha, a), \forall t \in [T]$ (see Appendix B.1.2).

4.3 VaR Dynamic Programming

In this section, we devise a DP method to compute an optimal policy for the static VaR objective in (4.3). This section assumes that the model is known and builds on the analysis in [62, 79]. Section 4.4 extends the approach to the model-free setting. Regardless of the methodology, the key idea of VaR-DP is to augment the state space with a risk-level input and to perform Bellman recursions on the augmented state-action value function. This augmentation should not be arbitrary, as the ‘risk-level state’ must evolve in a specific way to yield an optimal policy. We report the proofs of this section in Appendix B.2.

4.3.1 Model-based VaR-DP

We present the VaR-DP introduced by [79] and revised in [62]. Let $B_{\max}: \bar{\mathbb{R}}^{\mathcal{S} \times [0, 1] \times \mathcal{A}} \rightarrow \bar{\mathbb{R}}^{\mathcal{S} \times [0, 1] \times \mathcal{A}}$ be the following Bellman operator. For all $s \in \mathcal{S}$, $\alpha \in [0, 1]$, and $a \in \mathcal{A}$:

$$(B_{\max} q)(s, \alpha, a) := r(s, a) + \gamma \cdot \max_{o \in \mathcal{O}_{sa}(\alpha)} \min_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} q(s', o_{s'}, a'), \quad (4.5)$$

$$\mathcal{O}_{sa}(\alpha) := \left\{ o \in [0, 1]^{\mathcal{S}} \mid \sum_{s' \in \mathcal{S}} o_{s'} \cdot p(s, a, s') \leq \alpha \right\}.$$

Further consider the sequence $q_0(s, \alpha, a) := \text{VaR}_\alpha[0]$, and $q_{t+1} := B_{\max} q_t$ for all $t \in [T - 1]$. The constraint set above depends on the transition model p so Bellman recursions are model-based. Correspondingly, a proper recursion on risk levels leads to an optimal policy in terms of VaR return. Namely, for $k \in [T - 1]$, let $\alpha_k: \mathcal{H}_k \rightarrow [0, 1]$ such that $\alpha_0(s) = \alpha_0, \forall s \in \mathcal{S}$, while

$\alpha_{k+1}(h_{k+1})$ satisfies both $\alpha_{k+1}(\langle h_k, a, \cdot \rangle) \in \mathcal{O}_{sa}(\alpha_k(h_k))$ and

$$q_{T-k}(s, \alpha_k(h_k), a) = r(s, a) + \gamma \min_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} q_{T-k-1}(s', \alpha_{k+1}(\langle h_k, a, s' \rangle), a').$$

Then, one can derive a greedy policy $\pi := (\pi_k)_{k=0}^{T-1}$ from the constructed risk level mappings $(\alpha_k)_{k=0}^{T-1}$, i.e., at each step $k \in [T-1]$ as

$$\pi_k(h_k) \in \arg \max_{a \in \mathcal{A}} q_{T-k}(s, \alpha_k(h_k), a), \quad \forall h_k \in \mathcal{H}_k. \quad (4.6)$$

The following theorem shows that the above Bellman operator is optimal, in the sense that the sequence $q := (q_t)_{t=0}^{T-1}$ resulting from these recursions yields Eq. (4.4). Accordingly, a policy π that is greedy for that sequence q of state-action value functions at the constructed risk-level sequence is optimal for the VaR objective (4.3).

Theorem 4.3.1. *Let a sequence $q = (q_t)_{t=0}^T$ be such that $q_0(s, \alpha, a) = \text{VaR}_\alpha[0]$ and $q_{t+1} := B_{\max} q_t$ for $t \in [T-1]$. Then, $q_t = q_t^*$ for all $t \in [T]$, where q_t^* is defined in (4.4). Moreover, if a policy $\pi = (\pi_k)_{k=0}^{T-1}$ is greedy w.r.t. q as in (4.6), then it maximizes the VaR objective (4.3).*

Theorem 4.3.1 enables solving MDPs with static VaR objectives through DP equations. An important limitation of this representation is that it requires access to the underlying transition model. This implicitly appears in the constraint set $\mathcal{O}_{sa}(\alpha)$ from Eq. (4.5), which is required to find an optimal greedy policy according to the risk level mappings $(\alpha_k)_{k=0}^{T-1}$. Additionally, each Bellman update requires solving a constrained optimization problem, which slows down the learning process.

The next section proposes a model-free learning algorithm with theoretical convergence guarantees. To assess the optimality of our value function updates, we reformulate VaR-DP equations in terms of Bellman operators in Section 4.3.2. Our VaR-DP formulation can be seamlessly used for known or unknown transition probability models, as it allows for statistical estimation from sampled trajectories.

4.3.2 Nested VaR-DP

Let $B_u: \bar{\mathbb{R}}^{\mathcal{S} \times [0,1] \times \mathcal{A}} \rightarrow \bar{\mathbb{R}}^{\mathcal{S} \times [0,1] \times \mathcal{A}}$ be the following VaR Bellman operator:

$$(B_u q)(s, \alpha, a) := \text{VaR}_\alpha^{a,s}[r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q(\tilde{s}_1, \tilde{u}, a')],$$

where the VaR is based on the joint distribution of $(\tilde{s}_1, \tilde{u})$ with $\tilde{s}_1 \sim p(s, a, \cdot)$ and an independent $\tilde{u}$ uniformly distributed on $[0, 1]$ ($\tilde{u} \sim U([0, 1])$). Correspondingly, consider the following DP equations:

$$\begin{aligned} q_0^u(s, \alpha, a) &:= \text{VaR}_\alpha[0], \quad \forall s \in \mathcal{S}, \alpha \in [0, 1], a \in \mathcal{A}, \\ q_{t+1}^u &:= B_u q_t^u, \quad \forall t \in [T-1]. \end{aligned} \tag{4.7}$$

We must also adapt the risk level mappings to this new Bellman recursion. Let $\hat{\alpha}_0^u(s_0) := \alpha_0$ and for $k \in [T-1]$,

$$\hat{\alpha}_{k+1}^u(h_{k+1}) := \min \left\{ o \in [0, 1] \mid \max_{a \in \mathcal{A}} q_{T-k-1}^u(s_{k+1}, o, a) \geq \frac{q_{T-k}^u(s_k, \hat{\alpha}_k(h_k), a_k) - r(s_k, a_k)}{\gamma} \right\}. \tag{4.8}$$

The greediness criterion remains unchanged: at each step $k \in [T-1]$, construct

$$\pi_k^u(h_k) \in \arg \max_{a \in \mathcal{A}} q_{T-k}^u(s, \hat{\alpha}_k^u(h_k), a), \quad h_k \in \mathcal{H}_k. \tag{4.9}$$

The following theorem states that this alternative method is still valid for finding an optimal solution.

Theorem 4.3.2. *Let a sequence $q^u = (q_t^u)_{t=0}^T$ be such that $q_0^u(s, \alpha, a) := \text{VaR}_\alpha[0]$ and $q_{t+1}^u := B_u q_t^u$ for $t \in [T-1]$. Then, $q_t^u = q_t^*$ for all $t \in [T]$, where q_t^* is defined in (4.4). Moreover, if a policy $\pi^u = (\pi_k^u)_{k=0}^{T-1}$ is greedy w.r.t. q^u as in (4.9), then it maximizes the VaR objective in (4.3).*

In contrast to [62, 79], Theorem 4.3.2 does not require knowing the transition model. More

importantly, it reduces VaR-MDPs to a nested VaR conditional mapping:

$$v_T^*(s_0, \alpha_0) = \max_{a_0 \in \mathcal{A}} \text{VaR}_{\alpha_0}^{a_0, s_0} \left[r(s_0, a_0) + \gamma \cdot \max_{a_1 \in \mathcal{A}} \text{VaR}_{\tilde{u}_1} [r(\tilde{s}_1, a_1) + \cdots + \gamma \cdot \right. \\ \left. \max_{a_{T-2} \in \mathcal{A}} \text{VaR}_{\tilde{u}_{T-2}} [r(\tilde{s}_{T-2}, a_{T-2}) + \gamma \cdot \max_{a_{T-1} \in \mathcal{A}} r(\tilde{s}_{T-1}, a_{T-1}) | \tilde{s}_{1:T-2}, \tilde{u}_{1:T-2}] \dots | \tilde{s}_1, \tilde{u}_1] \right].$$

The value is still over an augmented state space, but this time, the risk tolerance is independently and uniformly drawn from $[0, 1]$ at each step. This is particularly suitable for sample-based RL and more amenable to deep settings such as [39, 40, 81]. Yet, two issues remain to produce a Q-learning procedure. First, the state space is infinite because the risk level α is continuous. Second, the elicitation procedure described in Lemma 4.2.1 underestimates the VaR when the risk-to-go variable is discrete, so we cannot directly employ quantile regression. To address these issues, in the next section, we propose an approximation scheme that replaces the Bellman operator of Eq. (4.7) with either lower or upper bounds of the appropriate quantiles.

4.4 Q-learning Algorithm and Analysis

This section builds on the DP of Section 4.3 to derive a new Q-learning algorithm for the static VaR objective. Section 4.4.1 introduces an approximate Bellman operator that can be used to compute Eq. (4.7) in a tractable way. Then, Section 4.4.2 proposes the VaR-Q-learning algorithm and shows its convergence guarantees.

4.4.1 Discretized Quantile Q-functions

The challenge in computing the value function in Eq. (4.7) stems from the fact that it is defined over a continuous $\alpha \in [0, 1]$. To make the computation tractable, we propose to approximate the risk level α with properly defined functions that yield lower and upper bounds on the quantile value function.

Definition 4.4.1. Let $\bar{f}, \underline{f}: [0, 1] \rightarrow [0, 1]$ be two non-decreasing right-continuous functions such that $\bar{f}(\alpha) > \alpha \geq \underline{f}(\alpha)$ for all $\alpha \in [0, 1)$, while $\bar{f}(1) = 1 \geq \underline{f}(1)$.

The following result shows how the functions $\underline{f}, \bar{f}$ can yield upper and lower bounds on VaR, and thus, on the state-action value function q . This development exploits that $\alpha \mapsto q(s, \alpha, a)$ is non-decreasing for each $s \in \mathcal{S}$ and $a \in \mathcal{A}$.

Lemma 4.4.2. *For any $\tilde{x} \in \mathbb{X}$ and $\alpha \in (0, 1)$, we have*

$$\text{VaR}_\alpha(\tilde{x}) \geq \mathbf{q}_{\underline{f}(\alpha)}^+(\tilde{x}) = \max \arg \min_{q \in \mathbb{R}} \mathbb{E}[\ell_{\underline{f}(\alpha)}(\tilde{x} - q)],$$

$$\text{VaR}_\alpha(\tilde{x}) \leq \mathbf{q}_{\bar{f}(\alpha)}^-(\tilde{x}) = \min \arg \min_{q \in \mathbb{R}} \mathbb{E}[\ell_{\bar{f}(\alpha)}(\tilde{x} - q)],$$

where ℓ_α is defined as in Eq. (4.2) and $\underline{f}, \bar{f}$ as in Definition 4.4.1.

We now derive Bellman operators that facilitate the construction of the Q-learning algorithm. First, the set-valued operator $\mathcal{B}_u: \mathbb{R}^{\mathcal{S} \times [0,1] \times \mathcal{A}} \rightarrow \mathcal{I}^{\mathcal{S} \times [0,1] \times \mathcal{A}}$ generalizes B_u as empirical risk minimizer, i.e.

$$(\mathcal{B}_u q)(s, \alpha, a) := \arg \min_{x \in \mathbb{R}} \mathbb{E}^{a,s} \left[\ell_\alpha \left(r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q(\tilde{s}_1, \tilde{u}, a') - x \right) \right],$$

where $\tilde{u} \sim U([0, 1])$. Second, we define the upper and lower bounding Bellman operators $\mathcal{B}_u^{\bar{f}}$ and $\mathcal{B}_u^f$ for each $b = (s, \alpha, a)$, $s \in \mathcal{S}$, $\alpha \in [0, 1]$, and $a \in \mathcal{A}$ as

$$\begin{aligned} (\mathcal{B}_u^{\bar{f}} q)(b) &:= \begin{cases} (\mathcal{B}_u q)(s, \bar{f}(\alpha), a) & \text{if } \bar{f}(\alpha) < 1, \\ \bar{R} + \max_{s' \in \mathcal{S}, a' \in \mathcal{A}} q(s', 1, a') & \text{if } \bar{f}(\alpha) = 1, \end{cases} \\ (\mathcal{B}_u^f q)(b) &:= \begin{cases} (\mathcal{B}_u q)(s, \underline{f}(\alpha), a) & \text{if } \underline{f}(\alpha) > 0, \\ \underline{R} + \min_{s' \in \mathcal{S}, a' \in \mathcal{A}} q(s', 0, a') & \text{if } \underline{f}(\alpha) = 0. \end{cases} \end{aligned}$$

The following theorem shows how to use these operators to bound the value functions and the performance of the computed policy.

Theorem 4.4.3. *Suppose that $\bar{q}_0^u = \underline{q}_0^u = 0$, and that $\bar{q}_{t+1}^u$ and $\underline{q}_{t+1}^u$ are right-continuous non-decreasing functions in α satisfying $\bar{q}_{t+1}^u \in (\mathcal{B}_u^{\bar{f}} \bar{q}_t^u)$ and $\underline{q}_{t+1}^u \in (\mathcal{B}_u^f \underline{q}_t^u)$ for all $t \in [T - 1]$. Then,*

$\bar{q}_t^u \geq q_t^* \geq \underline{q}_t^u$ for all $t \in [T]$, where q_t^* is defined in (4.4). Moreover, if a policy $\underline{\pi} := (\underline{\pi}_k)_{k=0}^{T-1}$ is greedy for $\underline{q}^u$, in the sense that

$$\underline{\pi}_k(h_k) \in \arg \max_{a \in \mathcal{A}} \underline{q}_{T-k}^u(s, \underline{\alpha}_k^u(h_k), a),$$

with $\underline{\alpha}^u$ as in (4.8), then it satisfies $\max_{a \in \mathcal{A}} \underline{q}_T^u(s_0, \alpha_0, a) \leq \rho(\underline{\pi})$. Note that, Right-continuity and non-decreasingness can be obtained by ensuring $\bar{q}_t(s, \alpha, a) = \bar{q}_t(s, \alpha', a)$ if $\bar{f}(\alpha) = \bar{f}(\alpha')$.

To simplify the exposition, we focus on the simple approximation scheme that discretizes the risk-level α using a uniform grid as below.

Example 4.4.4 (J -uniform discretization). Define $\underline{f}, \bar{f}: [0, 1] \rightarrow [0, 1]$ as

$$\begin{aligned} \underline{f}(\alpha) &:= \max \{j/J \mid j/J \leq \alpha, j \in [J-1]\}, \\ \bar{f}(\alpha) &:= \max \{j^{+1}/J \mid j/J \leq \alpha, j \in [J-1]\}, \end{aligned}$$

for $J \geq 2$. These functions satisfy the conditions of Definition 4.4.1.

Under this uniform discretization scheme, $\mathcal{B}_u$ becomes $\mathcal{B}_u^d: \mathbb{R}^{\mathcal{S} \times [J-1] \times \mathcal{A}} \rightarrow \mathcal{I}^{\mathcal{S} \times [J-1] \times \mathcal{A}}$, defined for $s \in \mathcal{S}$, $j \in [J-1]$, and $a \in \mathcal{A}$ as

$$(\mathcal{B}_u^d q)(s, j, a) := \arg \min_{x \in \mathbb{R}} \frac{1}{J} \sum_{j'=0}^{J-1} \mathbb{E}^{a,s} \left[\ell_{\frac{j}{J}} \left(r(s, a) + \gamma \max_{a' \in \mathcal{A}} q(\tilde{s}_1, j', a') - x \right) \right].$$

Similarly, the Bellman operator $\mathcal{B}_u^f$ becomes $\underline{\mathcal{B}}_u^d: \mathbb{R}^{\mathcal{S} \times [J-1] \times \mathcal{A}} \rightarrow \mathcal{I}^{\mathcal{S} \times [J-1] \times \mathcal{A}}$, which is defined for $b = (s, j, a)$, $s \in \mathcal{S}$, $j \in [J-1]$, and $a \in \mathcal{A}$ as

$$(\underline{\mathcal{B}}_u^d q)(b) := \begin{cases} (\mathcal{B}_u^d q)(s, j, a) & \text{if } j \geq 1, \\ \underline{R} + \min_{s \in \mathcal{S}, a \in \mathcal{A}} q(s, 0, a) & \text{if } j = 0. \end{cases} \quad (4.10)$$

The following proposition states the correctness of the discretized Bellman operators.

Proposition 4.4.5. *Given the J -uniform discretization of Example 4.4.4, let $q_0^d := 0$ and $q_{t+1}^d \in \mathcal{B}_u^d q_t^d, t \in [T-1]$. Then, the sequence $\underline{q} := (q_t)_{t=0}^T$ defined for $t \in [T]$ as*

$$q_t(s, \alpha, a) := q_t^d(s, J \cdot \underline{f}(\alpha), a), \forall s \in \mathcal{S}, \alpha \in [0, 1], a \in \mathcal{A},$$

provides a lower-bound for the optimal value function q^ defined in Eq. (4.4). Moreover, it can be used to build a policy $\underline{\pi} := (\pi_t)_{t=0}^T$ that achieves this lower-bound.*

Algorithm 3 and Algorithm 4 present a procedure that constructs the policy $\underline{\pi}$ described in Proposition 4.4.5. Note that the VaR operator in Line 4 of Algorithm 3 is applied over the joint distribution of $\tilde{s}' \sim p(s, a)$ and $\tilde{j}'$, which follows a discrete uniform distribution over $[J-1]$.

Algorithm 3: Finite Horizon VaR MDP

Input: Risk level $\alpha \in [0, 1)$, discretization $J \in \mathbb{Z}_+$, planning horizon $T < \infty$
Output: Optimal value function $q^d := (q_t^d)_{t=0}^T$ where $q_t^d: \mathcal{S} \times [J-1] \times \mathcal{A} \rightarrow \mathbb{R}$

- 1 Initialize $\alpha_j \leftarrow j/J, \forall j \in [J-1]$
- 2 Initialize $q_0^d(s, j, a) \leftarrow 0, \forall s \in \mathcal{S}, j \in [J-1], a \in \mathcal{A}$
- 3 **for** $t \in [T-1]$ **do**
- 4 $q_{t+1}^d(s, j, a) \leftarrow \text{VaR}_{\alpha_j} \left[r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q_t^d(\tilde{s}', \tilde{j}', a') \right], \forall s \in \mathcal{S}, j \in [J-1], a \in \mathcal{A}$
- 5 **return** q^d ;

Algorithm 4: Static VaR Policy Execution

Input: $s_0 \in \mathcal{S}, \alpha_0 \in (0, 1), T, J \in \mathbb{N}, \underline{q}^d: [T] \times \mathcal{S} \times [J-1] \times \mathcal{A} \rightarrow \mathbb{R}$

- 1 $(s, j) \leftarrow (s_0, \lfloor J \cdot \alpha_0 \rfloor)$
- 2 **for** $t = T, \dots, 1$ **do**
- 3 $a^* \leftarrow \arg \max_{a \in \mathcal{A}} q_t^d(s, j, a)$
- 4 Execute a^* and observe r and s'
- 5 $\tau \leftarrow \gamma^{-1}(q_t^d(s, j, a^*) - r)$
- 6 $\mathcal{J} \leftarrow \left\{ j' \in [J-1] \mid \max_{a' \in \mathcal{A}} q_{t-1}^d(s', j', a') \geq \tau \right\}$
- 7 $j \leftarrow J-1$; // arbitrary initialization
- 8 **if** $\mathcal{J}$ is not empty **then**
- 9 $j \leftarrow \min \mathcal{J}$
- 10 $s \leftarrow s'$

Note that we limit our analysis to schemes that calculate a posteriori error bounds due to the

discretization of α . It is also important to study a priori structural error bounds, which can guide the choice of discretization. However, such analysis is beyond the scope of this work.

4.4.2 VaR-Q-learning Algorithm

We now use the Bellman operators from Section 4.4.1 to develop and analyze a new Q-learning algorithm that solves the VaR objective in a model-free way.

An important obstacle in developing the Q-learning algorithm is that the Bellman operators in Section 4.4.1 are set-valued and lack a unique solution. The operators are set-valued because the quantile is not unique. As a result, even a well-designed iterative algorithm may oscillate among multiple possible solutions. It is common in RARL to replace the quantile loss function with Huber's loss to guarantee differentiability [39]; however, as we show in Appendix B.3.4, Huber's loss is insufficient to guarantee the uniqueness of value function. Instead, we replace the loss ℓ_α of Eq. (4.2) with the *soft-quantile loss* $\ell_\alpha^\kappa: \mathbb{R} \rightarrow \mathbb{R}$ defined for $\kappa \in (0, 1]$ and $\alpha \in (0, 1)$ as

$$\ell_\alpha^\kappa(\delta) := \begin{cases} \frac{(1-\alpha)\kappa}{2} \left((\delta + \kappa)^2 - \frac{2\delta}{\kappa} - 1 \right) & \text{if } \delta < -\kappa, \\ (1-\alpha) \left(\frac{\delta^2}{2\kappa} \right) & \text{if } \delta \in [-\kappa, 0), \\ \alpha \left(\frac{\delta^2}{2\kappa} \right) & \text{if } \delta \in [0, \kappa), \\ \frac{\alpha\kappa}{2} \left((\delta - \kappa)^2 + \frac{2\delta}{\kappa} - 1 \right) & \text{if } \delta \geq \kappa. \end{cases} \quad (4.11)$$

We also need the derivative $\partial \ell_\alpha^\kappa$ of ℓ_α^κ , which is

$$\partial \ell_\alpha^\kappa(\delta) := \begin{cases} (1-\alpha)(\kappa\delta + \kappa^2 - 1) & \text{if } \delta < -\kappa, \\ \frac{1-\alpha}{\kappa} \delta & \text{if } \delta \in [-\kappa, 0), \\ \frac{\alpha}{\kappa} \delta & \text{if } \delta \in [0, \kappa), \\ \alpha(\kappa\delta - \kappa^2 + 1) & \text{if } \delta \geq \kappa. \end{cases} \quad (4.12)$$

As the following lemma states, the function ℓ_α^κ is strongly convex and has a Lipschitz-continuous

gradient. These properties are instrumental in showing the value function's uniqueness and analyzing our Q-learning algorithm.

Lemma 4.4.6. *The function $m \mapsto \mathbb{E}[\ell_\alpha^\kappa(\tilde{x} - m)]$ with $\tilde{x}$ discrete is μ -strongly convex and has an L -Lipschitz continuous derivative for each $\alpha \in (0, 1), \kappa \in (0, 1]$ with*

$$\mu = \min\{\alpha, 1 - \alpha\} \kappa, \quad L = \max\{\alpha, 1 - \alpha\} \kappa^{-1}.$$

Algorithm 5: VaR Q-learning Algorithm

Input: Step sizes β_i , stream of sampled transitions $(t_i, s_i, j_i, a_i, s'_i)$, for all $i \in \mathbb{N}$

Initialize: $\underline{q}_0^d(b) \leftarrow t\underline{R}$, $\forall b \in [T] \times \mathcal{S} \times [J - 1] \times \mathcal{A}$

```

1 for  $i = 0, 1, 2, \dots$  do
2    $b_i \leftarrow (t_i, s_i, j_i, a_i)$ 
3   if  $j_i > 0$  and  $t_i > 0$  then
4      $\underline{q}_{i+1}^d(b_i) \leftarrow \underline{q}_i^d(b_i) + \frac{\beta_i}{J} \sum_{j'=0}^{J-1} \partial \ell_{\frac{j_i}{J}}^\kappa(r(s_i, a_i) + \gamma \cdot \max_{a' \in \mathcal{A}} \underline{q}_i^d(t_i - 1, s'_i, j', a') - \underline{q}_i^d(b_i))$ 
5   else
6     // Effectively do nothing
      $\underline{q}_{i+1}^d(b_i) \leftarrow \underline{q}_i^d(b_i) + \beta_i(t_i \underline{R} - \underline{q}_i^d(b_i))$ 

```

Having introduced the soft-quantile function in (4.11), we are now ready to adapt the Bellman operators to ensure that the Q-learning algorithm converges to a unique solution. In particular, we replace $\mathcal{B}_u^d$ with B_κ^d which is defined for $b = (t, s, j, a)$, $t \in 1:T$, $s \in \mathcal{S}$, $j \in 1:J - 1$, and $a \in \mathcal{A}$ as

$$(B_\kappa^d q)(b) := \arg \min_{x \in \mathbb{R}} \mathbb{E}^{a,s} \left[\ell_{\frac{j}{J}}^\kappa \left(r(s, a) + \gamma \max_{a' \in \mathcal{A}} q(t - 1, \tilde{s}_1, \tilde{j}', a') - x \right) \right],$$

and we replace the lower-bound operator $\underline{\mathcal{B}}_u^d$ by $\underline{B}_\kappa^d$:

$$(\underline{B}_\kappa^d q)(b) := \begin{cases} \underline{R} \cdot t & \text{if } j = 0 \vee t = 0, \\ (B_\kappa^d q)(b) & \text{otherwise.} \end{cases}$$

Note that the operators B_κ^d and $\underline{B}_\kappa^d$ are not calligraphic because their objective functions possess unique minimizers. In addition, B_κ^d and $\underline{B}_\kappa^d$ are applied to value functions defined across all time

π	CW	INV1	INV2	MR	POP	RS	GR
$\bar{q}^d$	-9.11	237.19	970.08	-2.79	-14348.60	50.0	4.78
Algorithm 4	-9.11	237.02	968.01	-2.84	-14348.60	50.0	4.78
$\underline{q}^d$	-9.11	236.88	967.60	-2.85	-14348.60	50.0	4.78
nVaR	-87.20	202.46	950.60	-20.00	-14348.60	50.0	0.00
VaR-IQN	-9.20	234.61	950.60	-18.21	-14348.60	50.0	0.00
E	-9.72	234.43	953.00	-2.96	-15101.04	33.3	3.14
Chow	-9.72	232.18	952.72	-4.13	-14348.60	50.0	3.14
CVaR	-9.72	235.27	953.07	-3.00	-14348.60	50.0	3.14
EVaR	-9.72	234.90	953.00	-2.96	-14348.60	50.0	2.82

Table 4.1: 25%-quantile of policy return: $\rho(\pi)$ for $\text{VaR}_{0.25}$.

steps simultaneously. This representation is convenient because we focus on the finite-horizon objective and must separate the time step from the Q-learning iteration. That is, $\underline{q}_i^d(t, s, j, a)$ represents the value function in the i -th iteration evaluated at time t , state s , risk level j , and action a .

Equipped with the above definitions, we now introduce the *VaR-Q-learning* algorithm in Algorithm 5. The algorithm seeks to identify the fixed-point $\underline{q}^d = \underline{B}_\kappa^d \underline{q}^d$, which is unique, as we show in Appendix B.3. The algorithm adapts the standard Q-learning approach to the risk-averse setting. It is well known that the standard Q-learning algorithm can be seen as a stochastic gradient descent on the quadratic loss function (e.g., [11]). Algorithm 5 also follows a sequence of stochastic gradient steps, but it replaces the quadratic loss function with the soft-quantile loss function ℓ_κ^α .

Algorithm 5 takes a stream of samples as input, and thus, implies that it is an offline algorithm. However, the algorithm and its analysis also apply to the online setting in which the sample $(t_i, s_i, j_i, a_i, s'_i)$ and step size β_i can depend on the values $\underline{q}_0^d, \dots, \underline{q}_i^d$ and are generated during the execution of the algorithm.

In an actual implementation, Line 6 in Algorithm 5 may be omitted since it does not modify the value function. We include this step to simplify our convergence analysis, which considers each update as a stochastic gradient step toward a contractive minimizer.

We require the following standard assumption to prove the convergence of Algorithm 5.

Assumption 4.4.7. The input to Algorithm 5 satisfies $\forall i \in \mathbb{N}$:

$$\mathbb{P} \left[\tilde{s}'_i = s' \mid \mathcal{G}_{i-1}, \tilde{t}_i, \tilde{s}_i, \tilde{j}_i, \tilde{a}_i, \tilde{\beta}_i \right] = p(\tilde{s}_i, \tilde{a}_i, s'), \quad \forall s' \in \mathcal{S},$$

almost surely, where $\mathcal{G}_{i-1} := (\tilde{\beta}_l, (\tilde{t}_l, \tilde{s}_l, \tilde{j}_l, \tilde{a}_l, \tilde{s}'_l))_{l=0}^{i-1}$.

The following theorem shows that Algorithm 5 enjoys convergence guarantees that are comparable to those in standard Q-learning.

Theorem 4.4.8. *Let $\kappa \in (0, 1]$. Assume that the sequences $\{\tilde{\beta}_i\}_{i=0}^\infty$ and $\{(\tilde{t}_i, \tilde{s}_i, \tilde{j}_i, \tilde{a}_i, \tilde{s}'_i)\}_{i=0}^\infty$ used in Algorithm 5 satisfy Assumption 4.4.7 and the step-size conditions*

$$\sum_{i \in \tilde{\mathcal{J}}(t,s,j,a)}^\infty \tilde{\beta}_i = \infty, \quad \sum_{i \in \tilde{\mathcal{J}}(t,s,j,a)} \tilde{\beta}_i^2 < \infty, \quad a.s. ,$$

where $\tilde{\mathcal{J}}(t, s, j, a) := \{i \in \mathbb{N} \mid (\tilde{t}_i, \tilde{s}_i, \tilde{j}_i, \tilde{a}_i) = (t, s, j, a)\}$. Then, the sequence $(\tilde{q}_i^d)_{i=0}^\infty$ produced by Algorithm 5 converges almost surely to $\underline{q}_\infty^d$ such that $\underline{q}_\infty^d = \underline{B}_\kappa^d \underline{q}_\infty^d$.

The proof of Theorem 4.4.8 follows an approach similar to that in the proofs of standard Q-learning [19] with two main differences. First, the algorithm converges even when $\gamma = 1$ and $\underline{B}_\kappa^d$ is not an L_∞ contraction. Instead, we show that $\underline{B}_\kappa^d$ is a contraction w.r.t. a particular weighted norm. Second, the use of a non-quadratic function ℓ_α^κ requires a more careful choice of the step-sizes than the standard analysis. Moreover, our analysis of the non-quadratic function extends the Q-learning analysis for risk-sensitive RL with nested risk measures in [116].

4.5 Numerical Experiments

In this section, we empirically test our theoretical results and algorithms on 7 tabular domains: machine replacement (MR) [43], gamblers ruin (GR) [13, 79], two inventory management problems (INV1 and INV2) [66], riverswim (RS) [121], population (POP) [73, 127], and cliffwalk (CW) [122]. We set the horizon to $T = 100$ with $\gamma = 0.9$ to evaluate the risk of the random discounted return.

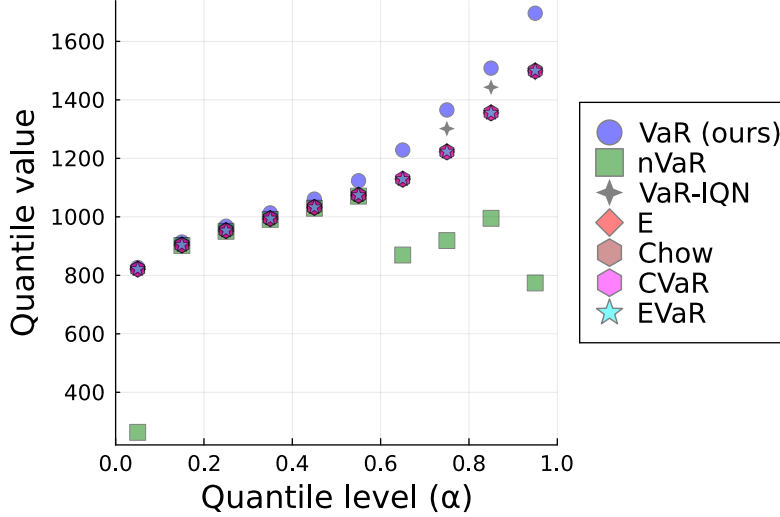


Figure 4.1: Policy performance $\text{VaR}_\alpha [\tilde{\mathcal{R}}_T^\pi]$ on INV2.

More details on each experiment can be found in Appendix B.4 and our code can be found at <https://github.com/MonkieDein/DRA-Q-LA>.

Policy execution. We first validate the discretization scheme presented in Algorithm 4 for model-free policy execution. To this aim, we compare the performance of Algorithm 4 VaR (ours) with other risk-averse algorithms. As a standard baseline, we include the risk-neutral objective (E). Other baselines are nested VaR (nVaR) [110], conditional VaR (CVaR) [13], Chow [33], distributional VaR (VaR-IQN) [39, 40], and entropic VaR (EVaR) [63] explained in Chapter 5. We take a quantile discretization level of $J = 4096$ to train Algorithm 4, Chow, and VaR-IQN. We discretize CVaR [13] that suffers from the challenge of a continuous, infinite state space using the same intuition presented in Section 4.4.1. Details of all algorithms can be found in Appendix B.4.2.

Table 4.1 shows the 25%-quantile value obtained after training, where each entry is the performance obtained from 100,000 episodes generated from the final policy. As we can see, our algorithm consistently outperforms all other algorithms across all tested domains. We also test our algorithm on a range of quantile levels $\alpha_0 \in \{0.05, 0.15, \dots, 0.85, 0.95\}$. Fig. 4.1 shows the quantile value obtained on INV2 after training each baseline. Our method shows an insensitive behavior to risk levels. All other domains exhibit a similar trend across quantile levels (see Appendix B.4), thus illustrating the robustness of Algorithm 4 to different environments and risk levels.

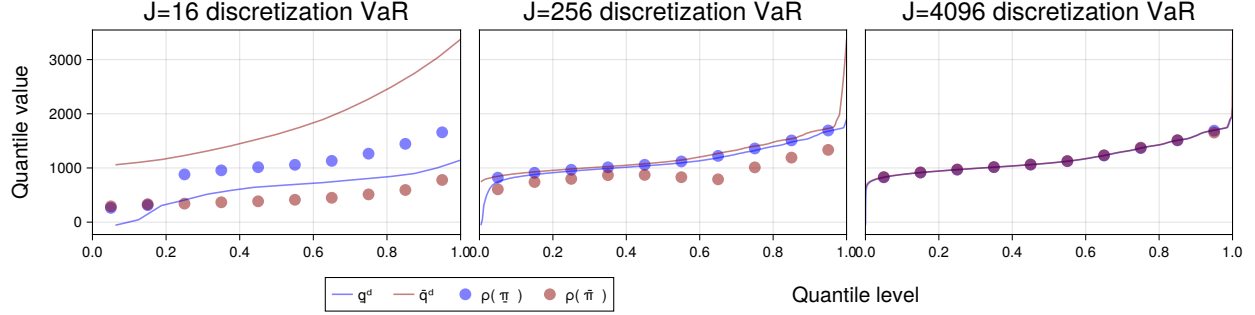


Figure 4.2: Impact of discretization level J on VaR-MDP performance and Q-functions.

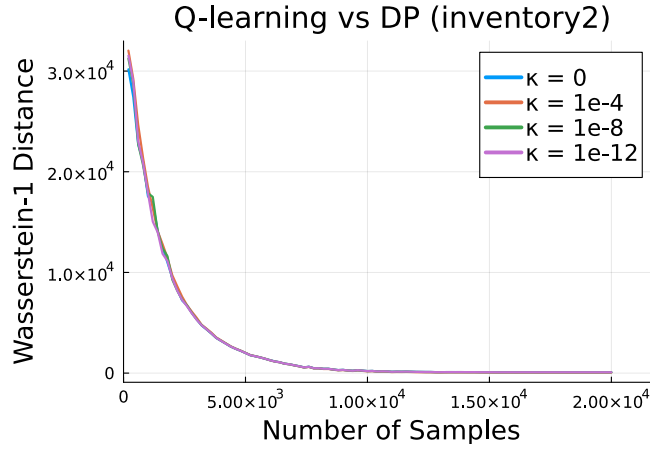


Figure 4.3: Distance between Q-values of Algorithm 5 and DP value function for varying κ .

We perform an ablative study to understand how the discretization and selection of $\underline{q}^d$ in Algorithm 4 contribute to the solution quality. We compare the performance of $\underline{\pi}$ with that of $\bar{\pi}$ (defined analogously to Eq. (4.9)) by confronting them to the bounds $\underline{q}^d$ and $\bar{q}^d$ from Example 4.4.4. We take $J \in \{16, 256, 4096\}$. Fig. 4.2 demonstrates that the performance of $\underline{\pi}$ lies within $[\underline{q}^d, \bar{q}^d]$, whereas $\bar{\pi}$ sometimes performs worse than $\underline{q}$. Furthermore, as the discretization level increases, the bounding gap $\bar{q}^d - \underline{q}^d$ shrinks, suggesting that $\underline{\pi}$ converges to π^* .

Quantile Q-learning. We now check the convergence and performance of Algorithm 5, which approximates the VaR computed from DP. In Section 4.4.2, we introduced a general VaR-Q-learning handling sampled time horizons. However, in practice, we remove the time index to reduce the computation overhead associated with updating time-indexed value functions. We take $\kappa \in \{10^{-4}, 10^{-8}, 10^{-12}, 0\}$ for the κ -soft quantile loss with a uniform discretization of $J = 256$.

For $\kappa = 0$, the loss is that of Eq. (4.2) while for positive values, it is that of Eq. (4.11). Fig. 4.3 displays the 1-Wasserstein distance between the quantile value estimated from VaR-Q-learning and the quantile value $\underline{q}^d$ computed via DP (Eq. (4.10)). For all κ 's, we see that the distance converges to zero as the number of samples increases. Furthermore, the VaR-Q-learning policy performs similarly to DP (see also Fig. B.4 in Appendix B.4).

To summarize, our experiments illustrate that the policy returned by the VaR-Q-learning algorithm: (1) Outperforms other baselines across both domains and quantile levels; (2) Lies in $[\underline{q}, \bar{q}]$; and (3) Performs similarly as the DP optimal policy $\underline{\pi}$.

4.6 Related Work and Discussion

Several works propose model-free methods for RARL. [88] introduces a temporal difference scheme for prediction and control with convergence guarantees, but focuses on a specific utility-based shortfall risk. Converging Q-learning algorithms are further extended in [25, 116] to a larger class of utility functions. However, all these works focus on a *nested* risk measure, which provides Bellman equations at the expense of interpretable policies. Differently, [119] study Q-learning for *static* CVaR, but their analysis relies on the DP equations of [33], which were shown to be incorrect by [62]. When considering static risk measures, one must use a proper state augmentation to guarantee that an optimal policy is identified [13, 62]. Otherwise, limiting assumptions such as the existence of an optimal Markov policy are required [81].

[39] present a similar goal as our study and propose to train risk-sensitive policies using a quantile representation of the return distribution. By leveraging Q-learning for distributional RL [14], they introduce IQN, an algorithm capable of achieving risk-sensitive behavior. Yet, as first pointed out in [81], the greedy step employed in IQN lacks a clear criterion of optimality for the trained policy. In this regard, our Q-learning algorithm modifies the optimal action selected at each state-risk level pair, thus fixing IQN's deficiency in policy optimization. For policy evaluation, our algorithm reduces to a variant of IQN ensuring that the resulting approximated distribution functions underestimate (in terms of stochastic ordering) the return distribution. A detailed discussion on the

differences between IQN and our VaR-Q-learning algorithm can be found in Section 4.6.1.

Most related to our work is the one by [59] in which the authors propose a Q-learning algorithm to identify a VaR-optimal policy on a special class of MDPs with end states. There, preferences are expressed using an ordering over end states. Following [23], their algorithm is based on stochastic approximation with two time-scales and its convergence is only empirically demonstrated. They leave open the question of how to generalize the approach to other forms of MDPs and raise the question of whether quantile regression methods could be used, which we address in this work.

Looking forward, the question of extending our results to an infinite horizon setting with continuous state and/or action spaces is definitely interesting. One might also be able to adapt the convergence analysis of policy evaluation for distributional RL in [108] to formally establish the convergence properties of Algorithm 5 under the non-strongly convex objective ℓ_α .

4.6.1 Comparison of VaR-Q-learning and IQN

Algorithm 6: Risk Sampled-based Quantile Q-learning Loss (adapted from Algorithm 5)

```

1 Require:  $K, K', \kappa$ , and functions  $\underline{q}^d$ 
   Input:  $t, s, a, s'$ 
2 # Sample current quantile thresholds
3  $\tau_k \sim U([0, 1])$ ,  $1 \leq k \leq K$ 
4 # Sample next quantile thresholds
5  $\tau'_{k'} \sim U([0, 1])$ ,  $1 \leq k' \leq K'$ 
6 # Compute greedy next quantile-based actions
7  $\mathbf{a}_{k'}^* \leftarrow \arg \max_{a' \in \mathcal{A}} \underline{q}^d(t-1, s', \lfloor \tau'_{k'} J \rfloor, a')$ ,  $1 \leq k' \leq K'$ 
8 # Compute distributional temporal differences
9  $\delta_{kk'} \leftarrow r(s, a) + \gamma \underline{q}^d(t-1, s', \lfloor \tau'_{k'} J \rfloor, \mathbf{a}_{k'}^*) - \underline{q}^d(t, s, \lfloor \tau_k J \rfloor, a)$ ,  $1 \leq k \leq K, 1 \leq k' \leq K'$ 
10 # Compute  $\kappa$ -soft quantile loss
11 Output:
    
$$\sum_{k=1}^K (1/K) \sum_{k'=1}^{K'} \ell_{\lfloor \tau_k J \rfloor / J}^\kappa(\delta_{kk'}) \cdot \mathbb{I}_{\tau_k \in [1/J, 1]} + (\underline{q}^d(t, s, 0, a) - \underline{R}t)^2 \cdot \mathbb{I}_{\tau_k \in [0, 1/J]}$$


```

We now compare the Implicit Quantile Network (IQN) Q-learning algorithm proposed in [39] to a variant of our Q-learning algorithm that stochastically approximates the expected value operation over the sampling of next risk $j' \sim U([J-1])$ using K' sampled risk level. Specifically, we focus on a version of IQN Q-learning that considers a finite horizon problem (using an additional t state

Algorithm 7: Implicit Quantile Network Loss (adapted from [39])

```

1 Require:  $K, K', h$ , and functions  $\Gamma, \hat{q}^d$ 
   Input:  $t, s, a, s'$ 
2 # Sample current quantile thresholds
3  $\tau_k \sim U([0, 1])$ ,  $1 \leq k \leq K$ 
4 # Sample next quantile thresholds
5  $\tau'_{k'} \sim U([0, 1])$ ,  $1 \leq k' \leq K'$ 
6 # Compute greedy next uniform action
7  $\mathbf{a}^* \leftarrow \arg \max_{\mathbf{a}' \in \mathcal{A}} (1/J) \sum_{j'=0}^{J-1} \Gamma(j') \hat{q}^d(t-1, s', j', \mathbf{a}')$ 
8 # Compute distributional temporal differences
9  $\delta_{kk'} \leftarrow r(s, a) + \gamma \hat{q}^d(t-1, s', \lfloor \tau'_{k'} J \rfloor, \mathbf{a}^*) - \hat{q}^d(t, s, \lfloor \tau_k J \rfloor, a)$ ,  $1 \leq k \leq K, 1 \leq k' \leq K'$ 
10 # Compute Huber quantile loss
11 Output:  $\sum_{k=1}^K (1/K') \sum_{k'=1}^{K'} \ell_{\tau_k}^h(\delta_{kk'})$ 

```

dimension), models the state-value function using a piecewise constant function of the risk level, i.e. $q(t, s, \alpha, a) := \hat{q}^d(t, s, \lfloor \alpha J \rfloor, a)$ with $\hat{q}^d \in \mathbb{R}^{[T] \times \mathcal{S} \times [J-1] \times \mathcal{A}}$, and models the risk aversion using a distorted risk measure parameterized by some non-decreasing $\beta_{IQN}: [0, 1] \rightarrow [0, 1]$ and implied $\Gamma(j) := \mathbb{E}[\beta_{IQN}(\tilde{u}) \mid j \leq \tilde{u} \leq (j+1)/J]$ for $\tilde{u} \sim U([0, 1])$. Algorithms 6 and 7 present in a comparable format how a quantile MDP and IQN approach compute their respective loss when updating their respective approximate state-value functions.

The biggest distinction between the two algorithms lies in the computation of the action or actions associated to state s' . On one hand, IQN seeks for each sampled (s, t, a, r, s') tuple a single action that captures a form of risk aversion portrayed by $\arg \max_{\mathbf{a}' \in \mathcal{A}} \mathbb{E}[\hat{q}^d(t-1, s', \beta_{IQN}(\tilde{u}), \mathbf{a}')] = \arg \max_{\mathbf{a}' \in \mathcal{A}} (1/J) \sum_{j'=0}^{J-1} \Gamma(j') \hat{q}^d(t-1, s', j', \mathbf{a}')$, where $\tilde{u} \sim U([0, 1])$. On the other hand, the variant of our Quantile Q-learning algorithm seeks an optimal action for each sampled next state quantile level $j'_{k'}$. The latter reflects our finding that the quantile MDP can be solved by solving a nested VaR DP where the risk level is independently sampled from a uniform distribution at each time step. In comparison, it is not clear what criterion of optimality is satisfied by the policy evaluated by IQN; see [81] for a discussion regarding the case where $\beta_{IQN}(\cdot)$ reflect a CVaR measure.

As part of the finer differences between the two algorithms, one can observe that our quantile Q-learning employs our κ -soft quantile loss, whereas IQN uses a Huber quantile loss, denoted by $\ell_{\tau_k}^h(\cdot)$. We also use the quantile loss function associated with the discretized level $\lfloor \tau_k J \rfloor / J$ instead

of using τ_k directly in order to guarantee a conservative approximation (instead of an estimation) of the value-at-risk of level τ_k . We finally handle samples with $\tau_k < 1/J$ differently than the rest given that our Algorithm 5 prescribes steering the value of $\underline{q}^d(t, s, 0, a)$ towards the value of $\underline{R}t$ in order to produce a natural lower bound on the value-at-risk for risk levels in that lower range.

CHAPTER 5

Entropic Risk Optimization in Discounted MDPs

5.1 Introduction

A major concern in high-stakes applications of reinforcement learning (RL), such as those in healthcare or finance, is to quantify the risk associated with the variability of returns. Since the standard expected objective does not capture the risk of random returns, *concave risk measures* have emerged as one of the most popular tools to quantify this risk in RL and beyond. They are sufficiently general to capture a wide range of stakeholder preferences and are more computationally convenient than many other alternatives [57]. Conditional value-at-risk (CVaR) is the best-known concave risk measure [57, 113] and the most commonly used to model risk aversion in MDPs [8, 12, 13, 20, 28, 32–34, 65, 83, 97, 112, 124, 125, 134].

The popularity of CVaR is mainly due to its intuitive interpretation as the expectation of the undesirable tail of the return random variable. Alas, solving risk-averse MDPs with the CVaR objective (CVaR-MDP) poses a difficult optimization problem. One can only formulate a dynamic program (DP) and a value function when the state space is augmented with an additional continuous variable [13, 79, 99, 100], which significantly complicates the computation of the value function and the implementation of the policy.

A popular remedy for the computational complexity of solving CVaR-MDPs is to use *nested*, also known as *iterated* or *Markov*, CVaR risk measure [12, 42, 96]. MDPs with a nested CVaR objective admit a value function that can be solved efficiently using DP. Unfortunately, nested

CVaR approximates CVaR poorly and has several properties that make it impractical, e.g., it is difficult to interpret and is not law-invariant. The latter property is because the risk value also depends on the model dynamics and not only on the probability distribution of the returns [57].

5.1.1 Contributions

In this chapter, we propose new algorithms for solving risk-averse discounted MDPs with two *entropic concave risk measures*: the entropic risk measure (ERM) [57] and the entropic value-at-risk (EVaR) [4, 57]. Entropic risk measures are important alternatives to CVaR but their behavior in dynamic decision domains, like MDPs, is not yet well understood. Prior work on entropic risk measures in dynamic decision-making has been limited to ERM in undiscounted and average-reward MDPs [22, 92], ERM for stochastic programs [49], and nested EVaR-constrained models [1, 2, 48]. We believe this is the first work that investigates non-nested entropic risk measures in the *discounted* setting, which is the typical objective in RL.

We make two main contributions in this section. As the first one, we show in Section 5.3 that in a discounted ERM-MDP, there exists an *optimal deterministic Markov policy* and an *optimal value function* that can be computed using *dynamic programming*. It is well-known that ERM is unique among law-invariant risk measures [78] in that it satisfies the tower property (see Theorem 5.2.2). However, the challenge with deriving DP equations with ERM in the discounted setting is that ERM is not positively homogeneous, which makes it impossible to account for the discount factor. We hypothesize that this is the reason most prior works on ERM-MDPs have focused on undiscounted objectives [22, 92], despite the popularity of discounting. Our main innovation in deriving the DP formulation is to compute a value function that uses *time-dependent risk levels* that decay exponentially over time to compensate for the discount factor. The DP is optimal for finite-horizon objectives and computes optimal infinite-horizon policies to a tolerance δ in $O(S^2 A \log(1/\delta))$ time, where S and A are the number of states and actions in the MDP.

As the second contribution, we show in Section 5.4 that in a discounted EVaR-MDP, there exists an *optimal deterministic Markov policy* and a policy that is arbitrarily close to optimal can be

computed using a sequence of *dynamic programs*. This is particularly surprising because EVaR does not satisfy the tower property (Theorem 5.2.2), which is required for the existence of DP optimality equations. To show this result, we reduce solving the EVaR-MDP to solving a specific sequence of ERM-MDPs. In particular, our EVaR algorithm runs in $O(S^2 A (\frac{\log(1/\delta)}{\delta})^2)$ time. To the best of our knowledge, this is the first polynomial-time approximate algorithm for computing history-independent policies for coherent law-invariant risk measures in discounted MDPs.

Concurrently with our work, a state-augmentation approach has been proposed for solving discounted EVaR-MDPs [93, 94]. This approach to EVaR-MDPs is inspired by a state augmentation method for solving CVaR-MDPs [33]. Ni and Lai [93] states that the augmented EVaR-MDP Bellman operator is optimal with a sufficiently fine discretization of the augmented state. However, we provide counterexamples that demonstrate the suboptimality of these approaches in Chapter 3.

Our empirical results in Section 5.5 confirm the efficacy of our algorithms. They outperform other techniques not only when evaluated in terms of ERM and EVaR metrics but also in terms of CVaR and VaR. This is not surprising because EVaR-MDP is easier to optimize than CVaR-MDP, and EVaR closely approximates CVaR and VaR [4].

5.2 Preliminaries and Problem Formulation

Overviews of MDP can be found in Section 2.3. This section introduces and derives key properties of entropy-based risk measures that will be used in the remainder of the chapter.

5.2.1 Entropic Based Risk Measures

Entropic-based risk measures studied in this chapter include Entropic Risk Measure (ERM) and Entropic Value at Risk (EVaR), which are concave risk measures. Concave risk measures are a generalization of the expectation operator $\mathbb{E}[\cdot]$ that can account for the variability of random variables. Formally, a concave risk measure $\psi[\cdot]$ is defined as a mapping $\psi: \mathbb{X} \rightarrow \mathbb{R}$ from the set of real-valued random variables $\mathbb{X}$ to real numbers that is *concave*, *monotone*, and *translation invariant* [57]. We summarize some of the most relevant risk measures and their properties in

Section 2.2.

Entropic Risk Measure (ERM) is the first risk measure we study in this chapter. As presented in the preliminary Section 2.2.4, entropic risk measure $\text{ERM}_\beta[\tilde{x}]$ for a random variable $\tilde{x} \in \mathbb{X}$ with the risk parameter $\beta \in [0, \infty]$ is defined as

$$\text{ERM}_\beta[\tilde{x}] := \begin{cases} \mathbb{E}[\tilde{x}] & \text{if } \beta = 0, \\ -\beta^{-1} \cdot \log\left(\mathbb{E}[e^{-\beta \cdot \tilde{x}}]\right) & \text{if } \beta \in (0, \infty), \\ \text{ess inf}[\tilde{x}] & \text{if } \beta = \infty. \end{cases} \quad (5.1)$$

For the risk level $\beta = 0$, ERM equals to the expectation: $\text{ERM}_0[\tilde{x}] = \lim_{\beta \rightarrow 0^+} \text{ERM}_\beta[\tilde{x}] = \mathbb{E}[\tilde{x}]$, while for $\beta \rightarrow \infty$, ERM equals to the minimum value of the random variable $\tilde{x}$: $\text{ERM}_\infty[\tilde{x}] = \text{ess inf}[\tilde{x}]$. ERM is *monotone* and satisfies the *tower property*. For the purpose of constructing the tower property of ERM, we define conditional ERM as follows:

Definition 5.2.1. The conditional ERM is defined for $\tilde{x}_1, \tilde{x}_2 \in \mathbb{X}$ as

$$\text{ERM}_\beta[\tilde{x}_1 \mid \tilde{x}_2] = -\frac{1}{\beta} \log\left(\mathbb{E}[e^{-\beta \tilde{x}_1} \mid \tilde{x}_2]\right).$$

In fact, ERM is the only law-invariant risk measure that satisfies the tower property [78]. Since we heavily use this property of ERM in our results, we state it in the following theorem and report its proof in Appendix C.1.1.

Theorem 5.2.2 (Tower Property). *For any two random variables $\tilde{x}_1, \tilde{x}_2 \in \mathbb{X}$, we have*

$$\text{ERM}_\beta[\tilde{x}_1] = \text{ERM}_\beta[\text{ERM}_\beta[\tilde{x}_1 \mid \tilde{x}_2]],$$

where the conditional ERM is defined analogously to a conditional expectation (see Definition 5.2.1).

Despite the properties listed above, ERM is rarely employed in practice because it violates A-4 the positive homogeneous property in coherent risk (see Definition 2.2.2), which gives rise to

undesirable risk preferences.

Proposition 5.2.3 (ERM is not positive homogeneity). *There exists a random variable $\tilde{x} \in \mathbb{X}$ such that $\text{ERM}_\beta[c \cdot \tilde{x}] \neq c \cdot \text{ERM}_\beta[\tilde{x}]$.*

For instance, a decision-maker guided by ERM may prefer an outcome $\tilde{x}$ over $\tilde{y}$ when the profit is measured in dollars: $\text{ERM}_\beta[\tilde{x}] > \text{ERM}_\beta[\tilde{y}]$, and yet the same decision-maker may prefer $\tilde{y}$ over $\tilde{x}$ when the profit is measured in cents: $\text{ERM}_\beta[100 \cdot \tilde{x}] < \text{ERM}_\beta[100 \cdot \tilde{y}]$. We analyze ERM primarily because it has favorable properties in dynamic decision-making, such as the tower property (mentioned above), and more importantly, we use it as a building block for our EVaR analysis.

We use the following lemma in the analysis of EVaR solution approximation by discretization in this work.

Lemma 5.2.4. *The function $\beta \mapsto \text{ERM}_\beta[\tilde{x}]$ for any random variable $\tilde{x} \in \mathbb{X}$ and $\beta > 0$ is continuous and non-increasing.*

The following lemma, which represents a new result to the best of our knowledge, plays an important role in bounding the difference between ERM and expectation. This result serves to bound the error of replacing the risk-averse value function with a risk-neutral value function in Algorithm 9.

Lemma 5.2.5. *Let $\tilde{x} \in \mathbb{X}$ be a bounded random variable such that $x_{\min} \leq \tilde{x} \leq x_{\max}$ a.s. Then, for any risk level $\beta > 0$, $\text{ERM}_\beta[\cdot]$ can be bounded as*

$$\mathbb{E}[\tilde{x}] - \frac{\beta(x_{\max} - x_{\min})^2}{8} \leq \text{ERM}_\beta[\tilde{x}] \leq \mathbb{E}[\tilde{x}].$$

Proof reported in Appendix C.1.2.

The following lemma helps to show that a deterministic policy can attain the same return as a randomized policy when the objective is an ERM. This result is not surprising and derives from the fact that the $\text{ERM}_\beta[\tilde{x}] \leq \max_{\omega \in \Omega} \tilde{x}(\omega)$ for any random variable $\tilde{x} \in \mathbb{X}$ defined over a finite probability space.

Lemma 5.2.6. *Let $\tilde{x}: \Omega \rightarrow \mathcal{A}$ be a random variable defined over a finite action set $\mathcal{A}$ and let $g: \mathcal{A} \rightarrow \mathbb{R}$ be a function defined for each action. Then, for any $\beta \geq 0$, we have*

$$\max_{a \in \mathcal{A}} g(a) = \max_{d \in \Delta^{\mathcal{A}}} \text{ERM}_{\beta} [g(\tilde{x}) \mid \tilde{x} \sim d] .$$

Proof reported in Appendix C.1.3

Entropic Value-at-Risk (EVaR) is the second risk measure we study. As presented in the preliminary Section 2.2.5, EVaR is *concave*, and unlike ERM, *positively homogeneous*, which makes it a *coherent risk measure*. EVaR with a confidence parameter $\alpha \in (0, 1]$ for a random variable $\tilde{x} \in \mathbb{X}$ is defined as [4, 57]

$$\text{EVaR}_{\alpha} [\tilde{x}] = \sup_{\beta > 0} \left(\text{ERM}_{\beta} [\tilde{x}] + \frac{\log(\alpha)}{\beta} \right) . \quad (5.2)$$

The meaning of EVaR's confidence level α is consistent with the level of value-at-risk (VaR) and conditional value-at-risk (CVaR). Computing the supremum in (5.2) is relatively easy because it involves maximizing a concave function over a single parameter (see [6, proposition 2.11]). It is easy to see that $\text{EVaR}_1 [\tilde{x}] = \mathbb{E}[\tilde{x}]$ and $\lim_{\alpha \rightarrow 0} \text{EVaR}_{\alpha} [\tilde{x}] = \text{ess inf}[\tilde{x}]$. In addition, EVaR is a non-decreasing function in α and is bounded as:

$$\text{ess inf}[\tilde{x}] \leq \text{EVaR}_{\alpha} [\tilde{x}] \leq \mathbb{E}[\tilde{x}] .$$

EVaR was proposed as the tightest Chernoff-style lower bound on the popular VaR and CVaR with the same confidence level α . It is also a lower bound CVaR as the following lemma shows.

Lemma 5.2.7 (proposition 3.2 in [4]). *The following inequalities hold for any $\alpha \in (0, 1)$ and a random variable $\tilde{x} \in \mathbb{X}$:*

$$\text{EVaR}_{\alpha} [\tilde{x}] \leq \text{CVaR}_{\alpha} [\tilde{x}] \leq \text{VaR}_{\alpha} [\tilde{x}] .$$

Although EVaR is not dynamically consistent, we show in Section 5.4 that it can be optimized

using a DP by representing it in terms of ERM. Unlike ERM, EVaR is positively-homogeneous, and thus, coherent, which makes its riskiness independent of the scale of the random variable. The following lemma plays a crucial role in efficiently computing EVaR when expressed in terms of ERM.

Lemma 5.2.8 ([4]). *The function $t \mapsto \text{ERM}_{t-1}[\tilde{x}]$ for any random variable $\tilde{x} \in \mathbb{X}$ and $t > 0$ is concave and non-decreasing.*

The following lemma, which shows how the optimal solution of (5.2) scales with the scale of the random variable is necessary when analyzing the properties of EVaR solutions.

Lemma 5.2.9. *Suppose that the supremum in (5.2) is attained by some $\beta^* > 0$ for a random variable $\tilde{x}$. Then, the supremum in (5.2) is attained at $c^{-1} \cdot \beta^*$ for any random variable $c \cdot \tilde{x}$ and a constant $c > 0$. Proof reported in Appendix C.1.4.*

5.3 Discounted MDPs with ERM Objective

In this section, we study the fundamental properties of discounted ERM-MDPs and describe a new DP formulation. In particular, we show that if one defines an optimal value function for ERM-MDP with a specific time-dependent risk, then it can be computed using DP. We report the proofs for this section in Appendix C.2.

The objective in this section is to compute a policy that maximizes the ERM of the return random variable $\tilde{\mathfrak{R}}_T^\pi$ at some given risk level $\beta \geq 0$. That is, the objective is the optimization in (2.19) for risk-averse MDPs with the risk measure $\psi[\cdot]$ set to $\text{ERM}_\beta[\cdot]$:

$$\max_{\pi \in \Pi_{\text{HR}}} \text{ERM}_\beta^{\pi, s_0} \left[\sum_{k=0}^{T-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] = \max_{\pi \in \Pi_{\text{HR}}} \text{ERM}_\beta \left[\tilde{\mathfrak{R}}_T^\pi(s_0) \right]. \quad (5.3)$$

Although we formulate the objective in (5.3) in terms of history-dependent randomized policies, we will prove later in this section that there always exists a Markov (history-independent) deterministic policy for (5.3). In the remainder of this section, we first treat the finite-horizon case

($T < \infty$) and then extend the obtained results to the discounted infinite-horizon case ($T = \infty$ and $\gamma < 1$).

The closest objective to (5.3) studied in prior work is the ERM-MDP with an *average-reward* objective [22]. Value iteration, policy iteration, and even q-learning [24] have been studied for this objective. However, the average reward criterion is not as popular in RL as the discounted infinite horizon objective. In addition, the example below illustrates why the existing formulations do not readily extend to discounted ERM-MDPs.

Before defining the value function and deriving the corresponding DP equations for ERM-MDP, we describe a simple example that illustrates how one may use the tower property to derive such equations. The example also illustrates the challenge that discounting poses in ERM-MDP.

Example 5.3.1. Consider an MDP with a single action a , and thus, a single policy π . Assume that the horizon is $T = 2$ and the initial state $\tilde{s}_0$ is random. Recall that the return is defined as $\tilde{\mathfrak{R}}_2^\pi = r(\tilde{s}_0, a) + \gamma \cdot r(\tilde{s}_1, a)$. When $\gamma = 1$, one can directly use the tower property to decompose the return into value functions as

$$\begin{aligned}
\text{ERM}_\beta \left[\tilde{\mathfrak{R}}_2^\pi \right] &= \text{ERM}_\beta [r(\tilde{s}_0, a) + \gamma \cdot r(\tilde{s}_1, a)] \\
&\stackrel{A-6}{=} \text{ERM}_\beta [\text{ERM}_\beta [r(\tilde{s}_0, a) + \gamma \cdot r(\tilde{s}_1, a) \mid \tilde{s}_0]] \\
&\stackrel{A-2}{=} \text{ERM}_\beta [r(\tilde{s}_0, a) + \text{ERM}_\beta [\gamma \cdot r(\tilde{s}_1, a) \mid \tilde{s}_0]] \\
&= \text{ERM}_\beta [r(\tilde{s}_0, a) + \gamma \cdot \text{ERM}_\beta [r(\tilde{s}_1, a) \mid \tilde{s}_0]] \\
&= \text{ERM}_\beta [r(\tilde{s}_0, a) + \gamma \cdot v_0(\tilde{s}_1)],
\end{aligned} \tag{5.4}$$

where $v_0(\tilde{s}_1) = \text{ERM}_\beta [r(\tilde{s}_1, a) \mid \tilde{s}_0]$. While the above derivation readily generalizes to the MDP with $\gamma = 1$, it is not valid when $\gamma < 1$, because the equality in (5.4) requires ERM to be positive homogeneous.

5.3.1 Finite Horizon ERM-MDP

Although ERM is not positively homogeneous, in the following new result we show that it has a similar property if we allow for a change in the risk level.

Theorem 5.3.2 (Positive Quasi-homogeneity). *For any random variable $\tilde{x} \in \mathbb{X}$ and any constant $c \geq 0$, we have*

$$\text{ERM}_\beta [c \cdot \tilde{x}] = c \cdot \text{ERM}_{\beta \cdot c} [\tilde{x}] . \quad (5.5)$$

Proof reported in Appendix C.2.1.

Theorem 5.3.2 indicates that we can propagate the discount factor γ out of ERM in (5.4), if we change the risk level of the inner ERM to $\beta\gamma$. In particular, if we define the value function as $v_0(\tilde{s}_1) = \text{ERM}_{\beta \cdot \gamma} [r(\tilde{s}_1, a) \mid \tilde{s}_0]$, then the derivation in Example 5.3.1 works for any $\gamma \in (0, 1)$. Generalizing this intuition to the full ERM-MDP, we define the value function $v^\pi = (v_t^\pi)_{t=0}^T$ for a policy $\pi = (\pi_t)_{t=1}^T \in \Pi_{\text{MR}}$ and the optimal value function $v^\star = (v_t^\star)_{t=0}^T$ in a state $s \in \mathcal{S}$ as follows:

$$v_t^\pi(s) = \text{ERM}_{\beta \cdot \gamma^{T-t}} [\tilde{\mathfrak{R}}_{0:t}^\pi(s)] = \text{ERM}_{\beta \cdot \gamma^{T-t}} [\tilde{\mathfrak{R}}_{(T-t):T}^\pi(s)] , \quad (5.6)$$

$$v_t^\star(s) = \max_{\pi \in \Pi_{\text{MR}}^t} \text{ERM}_{\beta \cdot \gamma^{T-t}} [\tilde{\mathfrak{R}}_t^\pi(s)] = \max_{\pi \in \Pi_{\text{MR}}^t} \text{ERM}_{\beta \cdot \gamma^{T-t}} [\tilde{\mathfrak{R}}_{(T-t):T}^\pi(s)] , \quad (5.7)$$

where $\tilde{\mathfrak{R}}_{(T-t):T}^\pi(s)$ is defined in (2.6) and Π_{MR}^t is the set of randomized Markov policies for t -horizon MDP, represent in terms of time-steps $\tau \in (T-t):T$ in the T -horizon MDP.

As discussed above, it is important that the risk level in the definition of value function in (5.6) depends on the time-step τ . As time progresses, the risk level $\beta\gamma^\tau$ decreases monotonically and the value function becomes less risk-averse. Recall that in the risk-neutral setting, the risk level is $\beta = 0$ and $\text{ERM}_0[\tilde{x}] = \mathbb{E}[\tilde{x}]$. When we set $\beta = 0$ in (5.6), the value function becomes independent of t and the value function reduces to that in risk-neutral MDPs.

The following theorem states the main result of this section. It shows how the value functions defined in (5.6) and (5.7) can be efficiently computed by a DP when $T < \infty$.

Theorem 5.3.3 (Bellman Equations in ERM-MDP). *For any policy $\pi \in \Pi_{\text{MR}}$, its value function $v^\pi = (v_t^\pi)_{t=0}^T$ defined in (5.6) is the unique solution to the following system of equations for all $s \in \mathcal{S}$,*

$$v_t^\pi(s) = \text{ERM}_{\beta, \gamma^{T-t}} [r(s, \tilde{a}) + \gamma \cdot v_{t-1}^\pi(\tilde{s}')], \quad (5.8)$$

where $\tilde{a} \sim \pi_t(\cdot|s)$, $\tilde{s}' \sim p(\cdot|s, \tilde{a})$, and $v_0(s) := 0$. Moreover, the optimal value function $v^* = (v_t^*)_{t=0}^T$ defined in (5.7) is the unique solution to the following system of equations for all $s \in \mathcal{S}$,

$$v_t^*(s) = \max_{a \in \mathcal{A}} \text{ERM}_{\beta, \gamma^{T-t}} [r(s, a) + \gamma \cdot v_{t-1}^*(\tilde{s}')]. \quad (5.9)$$

Proof reported in Appendix C.2.2.

Theorem 5.3.3 suggests several new important and surprising properties for ERM-MDP. First, it shows the existence of value functions, both for any Markov policy and also the optimal value function. Unlike other risk measures, these value functions do not require that the state space is augmented. Second, the theorem shows that the value function can be computed efficiently using a dynamic program. And finally, the next theorem built on Theorem 5.3.3 shows that there always exists an optimal Markov (as opposed to history-dependent) deterministic policy for the ERM-MDP, and this policy is greedy w.r.t. the optimal value function.

Theorem 5.3.4 (Optimal Policy in ERM-MDP). *There exists a Markov deterministic optimal policy $\pi^* = (\pi_t^*)_{t=1}^T \in \Pi_{\text{MD}}$ for the optimization problem (5.3), which is greedy w.r.t. the optimal value function v^* defined by (5.9), that is,*

$$\pi_t^*(s) \in \arg \max_{a \in \mathcal{A}} \text{ERM}_{\beta, \gamma^{T-t}} [r(s, a) + \gamma v_{t-1}^*(\tilde{s}')], \quad (5.10)$$

for all $s \in \mathcal{S}$. Moreover, the optimal value function satisfies that $v^{\pi^*} = v^*$. Proof reported in Appendix C.2.3.

The existence of a deterministic optimal policy in ERM-MDP is surprising since many risk-averse

formulations require randomization [44, 83, 120]. Also surprisingly, ERM-MDP does not admit a stationary optimal policy (π^* in (5.10) is time-dependent) even when the horizon T is large or infinite. This is in contrast to risk-neutral discounted infinite-horizon MDPs which admit stationary optimal policies.

Algorithm 8: Finite-horizon ERM-MDP

Input: Risk level $\beta > 0$, planning horizon $T < \infty$, initial value function v_0^* (default:0)
Output: Optimal value $(v_t^*)_{t=0}^T$ and policy $(\pi_t^*)_{t=1}^T$
1 for $t = 1 : T$ **do**
2 $\pi_t^*(s) \leftarrow \arg \max_{a \in \mathcal{A}} \text{ERM}_{\beta\gamma^{T-t}} [r(s, a, \tilde{s}') + \gamma \cdot v_{t-1}^*(\tilde{s}')] , \forall s \in \mathcal{S}$
3 $v_t^*(s) \leftarrow \max_{a \in \mathcal{A}} \text{ERM}_{\beta\gamma^{T-t}} [r(s, a, \tilde{s}') + \gamma \cdot v_{t-1}^*(\tilde{s}')] , \forall s \in \mathcal{S}$
4 return v^*, π^* ;

We end this section by reporting the pseudo-code for the value iteration (VI) algorithm in finite-horizon ERM-MDP (Algorithm 8). This algorithm is an adaptation of the standard VI algorithm to the finite-horizon ($T < \infty$) setting. It uses the results of Theorems 5.3.3 and 5.3.4, and first computes the optimal value function v_t^* with dynamic program ($t = 1, 2, \dots, T$) according to (5.9), and then obtains the optimal policy as a policy greedy to v^* by solving the optimization (5.10).

5.3.2 Infinite Horizon ERM-MDP

We now turn to derive an algorithm that can solve the ERM-MDP objective (5.3) when the horizon T is large or infinite. Solving ERM-MDP in the *infinite-horizon* setting is considerably more challenging than in finite-horizon because the risk level and optimal policy are both time-dependent. The simplest way to address this issue is to truncate the horizon at some $T' < \infty$ and resort to an arbitrary policy for any steps $\tau > T'$. The main limitation of this approach is that T' may need to be very large to achieve a reasonably small approximation error.

In Algorithm 9, we propose an approximation that is superior to the simple truncation of the planning horizon, described above. Algorithm 9 first computes the optimal risk-neutral value function $\bar{v}_\infty^*$ and (stationary) policy $\bar{\pi}_\infty^*$ using value or policy iteration algorithms [103]. It uses policy $\hat{\pi}_0^* = \bar{\pi}_\infty^*$ to act for all time-steps $\tau > T'$ and value function $\hat{v}_0^* = \bar{v}_\infty^*$ to approximate $\max_{\pi \in \Pi_{\text{MR}}} \text{ERM}_{\beta\gamma^{T'}}[\tilde{\mathfrak{R}}_\infty^\pi]$. This approach takes advantage of the fact that the risk level $\beta \cdot \gamma^\tau$

in (5.6) and (5.7) approaches 0 as τ gets larger, which means that the ERM value function becomes close to the risk-neutral $\hat{v}_0^*$.

Algorithm 9: VI for infinite-horizon ERM-MDP

Input: Risk level $\beta > 0$, bounded horizon $T' < \infty$ based on Theorem 5.3.5

Output: Value function $\hat{v}^* = (\hat{v}_t^*)_{t=0}^{T'}$ and policy $\hat{\pi}^* = (\hat{\pi}_t^*)_{t=0}^{T'}$

- 1 Compute $\hat{v}_0^*$ and $\hat{\pi}_0^*$ as the optimal solutions to the risk-neutral infinite-horizon discounted problem ;
 - 2 $(\hat{v}_t^*)_{t=0}^{T'}, (\hat{\pi}_t^*)_{t=1}^{T'} \leftarrow \text{Algorithm 8}(\beta, T', \hat{v}_0^*)$
 - 3 **return** $\hat{v}^*, \hat{\pi}^*$
-

To quantify the quality of a policy $\hat{\pi}^*$ returned by Algorithm 9, we now derive a bound on its performance loss. In particular, we focus on how quickly the error decreases as a function of the planning horizon T' . This bound can be used both to determine the planning horizon and to quantify the improvement of Algorithm 9 over simple truncation.

Theorem 5.3.5. *The performance loss of the policy $\hat{\pi}^*$ returned by Algorithm 9 decreases with T' as*

$$\text{ERM}_\beta \left[\tilde{\mathfrak{R}}_\infty^{\pi^*} \right] - \text{ERM}_\beta \left[\tilde{\mathfrak{R}}_\infty^{\hat{\pi}^*} \right] \leq \frac{\beta \cdot \gamma^{2T'} \cdot \Delta_{\mathfrak{R}}^2}{8}, \quad (5.11)$$

where π^* is optimal in (5.3) and $\Delta_{\mathfrak{R}}$ is the range of the return random variable $\tilde{\mathfrak{R}}_\infty$. Therefore, Algorithm 9 runs in $O(S^2 A \log(1/\delta))$ time to compute a δ -optimal policy as

$$\text{ERM}_{\beta}^{\pi^*, s} \left[\tilde{\mathfrak{R}}_\infty^{\pi^*} \right] - \text{ERM}_{\beta}^{\hat{\pi}^*, s} \left[\tilde{\mathfrak{R}}_\infty^{\hat{\pi}^*} \right] \leq \delta.$$

The proof of Theorem 5.3.5 reported in Appendix C.2.4 uses Hoeffding's lemma to bound the error between ERM and expectation, and then propagates it backward using standard DP techniques. Analysis analogous to Theorem 5.3.5 shows that when we simply truncate the planning horizon at T' and follow an arbitrary policy thereafter, the performance loss decreases proportionally to $\gamma^{T'}$ as opposed to $\gamma^{2T'}$ in (5.11). As a result, simple truncation requires a planning horizon T' that is at least twice longer than the one used by Algorithm 9 to achieve the same performance.

Remark 5.3.6 (Quadratic dependence on $\Delta_{\mathfrak{R}}$). An attentive reader may be puzzled by the fact that the bound in Theorem 5.3.5 scales quadratically with the range of the returns $\Delta_{\mathfrak{R}}$. Given the quadratic dependence, one can make the relative error arbitrarily small just by shrinking the rewards appropriately. This is indeed true but is less useful than it may seem at the first blush. Since ERM is not positively homogeneous, scaling the rewards can change the optimal policy, unlike in risk-neutral MDPs. To avoid changing the set of optimal policies when scaling the rewards, one also needs to scale the risk parameter β appropriately as dictated by Theorem 5.3.2. When both r and β are scaled simultaneously, the relative error in Theorem 5.3.5 does not change.

In practice, one can compute bounds that are tighter than the one in Theorem 5.3.5 by computing both an upper-bound on the optimal value function and a lower-bound on the value of the policy. It is easy to see that v_{∞}^* is an upper-bound on v^* , which can be used to compute an upper-bound on v_0^* , and therefore, an upper-bound on the performance loss. According to Theorem 5.3.5, given an arbitrary desired tolerance δ , one can select $T' \geq \frac{1}{2\log(\delta)} \log(\frac{8\delta}{\beta\Delta_{\mathfrak{R}}^2})$ to compute a δ -optimal policy. We give more details on this in Appendix C.2.

5.4 Discounted MDPs with EVaR Objective

In this section, we analyze the EVaR-MDP objective and propose a new DP algorithm to solve it. As mentioned in Section 5.2, EVaR is preferable to ERM because it is coherent and approximates both VaR and CVaR well. We report the proofs of this section in Appendix C.3.

The objective in this section is to compute a policy that maximizes the EVaR of the return random variable $\tilde{\mathfrak{R}}_T^{\pi}$ at some given risk level $\alpha \in [0, 1]$. In other words, we are interested in solving the optimization problem in (2.19) with the risk measure $\psi[\cdot]$ set to $\text{EVaR}_{\alpha}[\cdot]$:

$$\max_{\pi \in \Pi_{\text{HR}}} \text{EVaR}_{\beta}^{\pi, s_0} \left[\sum_{k=0}^{T-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] = \max_{\pi \in \Pi_{\text{HR}}} \text{EVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_T^{\pi}(s_0) \right]. \quad (5.12)$$

It is important to note that the objective in (5.12) differs from prior work on EVaR in MDPs, which has focused on the nested EVaR objective [1, 2, 48], and thus, does not approximate the

static formulation in (5.12) well [70].

The main challenge in solving (5.12) is that EVaR does not satisfy the tower property (or equivalently, it is *not* dynamically consistent) and cannot be directly optimized using a DP. Our main contribution in this section is to derive an algorithm that solves EVaR-MDP in time that is polynomial in the problem size and the desired accuracy. The main idea of our algorithm is to reduce EVaR-MDP to a specific sequence of ERM-MDP problems.

Using the definition of EVaR in (5.2), we may reformulate the EVaR-MDP objective (5.12) as

$$\max_{\pi \in \Pi_{\text{MR}}} \text{EVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_T^{\pi} \right] = \sup_{\beta > 0} h(\beta) \quad (5.13)$$

where the objective function $h: \mathbb{R} \rightarrow \mathbb{R}$ is defined as

$$h(\beta) = \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_{\beta} \left[\tilde{\mathfrak{R}}_T^{\pi} \right] + \beta^{-1} \cdot \log(\alpha) \right).$$

We switch the notation to Markov policies, Π_{MR} , because we will show in Corollary 5.4.2 that an optimal policy for EVaR-MDP belongs to this class. The equality in (5.13) follows by swapping the order of max and sup operators. The connection that the reformulation in (5.13) establishes between the objectives of EVaR-MDP and ERM-MDP allows us to directly carry over the following properties from the ERM-MDP setting to EVaR-MDP.

Theorem 5.4.1. *Let π^* be an optimal solution to the EVaR-MDP in (5.12) and suppose that the supremum is attained. Then, there exists a risk level $\beta^* \in (0, \infty]$ such that π^* is optimal for ERM-MDP in (5.3) with $\beta = \beta^*$. Proof reported in Appendix C.3.1.*

A similar argument holds also when the supremum is not attained, but requires additional technical developments, which we leave it for a future extended version of this work.

Corollary 5.4.2. *There exists an optimal Markov deterministic policy for EVaR-MDP. The result follows directly from Theorem 5.4.1.*

We are now ready to describe our algorithm for solving the EVaR-MDP objective. Our algorithm,

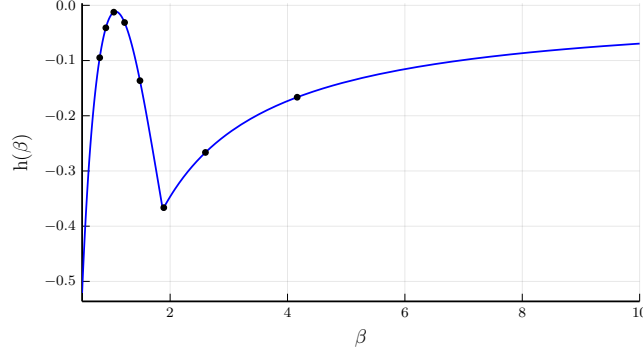


Figure 5.1: Function h for the EVaR-MDP described in Proposition C.3.1. The circles indicate the discretized $h(\beta_k)$ according to (5.14) with $\alpha = 0.5$, $\delta = 0.1$, and $\Delta_{\mathfrak{R}} = 1$.

whose pseudo-code is shown in Algorithm 10, optimizes the single-dimensional objective function h in (5.13). Because the function h is not concave in general (see Proposition C.3.1 in Appendix C.3), we cannot use standard one-dimensional algorithms like Brent’s method or BFGS. However, we leverage the fact that h is the sum of a non-increasing function and a concave function, with a discrete grid $\{\beta_k\}_{k=1}^K$ to search over the risk level β that can approximate the optimal policy in polynomial time. We define the grid values for each $k \in 1:K-1$ as

$$\beta_1 = \frac{8\delta}{\Delta_{\mathfrak{R}}^2}, \quad \beta_{k+1} = \beta_k \cdot \frac{\log(\alpha)}{\beta_k \delta + \log(\alpha)}, \quad (5.14)$$

where $\delta > 0$ is the desired approximation error and $K \in \mathbb{N}$ is sufficiently large to ensure that

$$\beta_K \geq \frac{-\log(\alpha)}{\delta}. \quad (5.15)$$

We also assume that β_K is trimmed so that (5.15) holds with equality. The grid values in (5.14) are constructed to minimize the suboptimality bound δ in Theorem 5.4.3 below. Figure 5.1 depicts an example of a function h for an MDP described in the proof of Proposition C.3.1. The black circles are the values of h at the grid points $\{\beta_k\}_{k=1}^K$ constructed according to (5.14) with $\delta = 0.1$ and $\Delta_{\mathfrak{R}} = 1$.

The following theorem shows that Algorithm 10 runs in time that is polynomial in $1/\delta$ and computes a policy $\hat{\pi}^*$ whose return has an EVaR that is δ -close to optimal.

Algorithm 10: EVaR-MDP

Input: Risk level $\alpha > 0$, horizon T , tolerable error $\delta > 0$

Output: EVaR-MDP optimized policy $\hat{\pi}^*$

```
1 Initialize the discretization set  $B$  with Eq. (5.14) given  $\delta$  and  $\alpha$ 
2 for  $k = 1, \dots, K$  do
3   | Select appropriate ERM-MDP: Algorithm 8 for  $T < \infty$ , and Algorithm 9 for  $T = \infty$ .
4   |  $(\hat{v}^k, \hat{\pi}^k) \leftarrow \text{ERM-MDP}(\beta_k, T)$ 
5   |  $\hat{v}^k \leftarrow \hat{v}^k + \frac{\log(\alpha)}{\beta_k}$  ;
6  $\star \leftarrow \arg \max_{k \in 1:K(\delta)} \{\hat{v}^k\}$ 
7 return  $\hat{v}^\star, \hat{\pi}^\star$  ;
```

Theorem 5.4.3. *For any $\delta > 0$, Algorithm 10 runs in $O(S^2 A(\frac{\log(1/\delta)}{\delta})^2)$ time and returns a policy $\hat{\pi}^*$ such that*

$$\text{EVaR}_\alpha \left[\tilde{\mathfrak{R}}_\infty^{\pi^*} \right] - \text{EVaR}_\alpha \left[\tilde{\mathfrak{R}}_\infty^{\hat{\pi}^*} \right] \leq \delta,$$

where π^* is optimal for (5.12). Proof reported in Appendix C.3.2.

Theorem 5.4.3 establishes the time complexity that Algorithm 10 needs to compute a δ -optimal EVaR policy. Note that the bound in Theorem 5.4.3 takes into account both the errors due to the discretization in (5.14) and the truncated horizon in Theorem 5.3.5 when solving the ERM-MDPs.

We report Algorithm 10 for solving EVaR-MDP because it is conceptually simple and relatively easy to analyze. However, significant computational improvements are possible in this setting. One approach to accelerate Algorithm 10 is by realizing that Algorithm 9 computes value functions for multiple risk levels $\beta, \gamma\beta, \gamma^2\beta, \dots$. For example, running Algorithm 9 with $\beta = 0.5$ computes v_0 with risk level $\beta = 0.5$, v_1 with risk level $\beta = 0.5\gamma$, v_2 with risk level $\beta = 0.5\gamma^2$ and so on. This observation can significantly reduce the computational effort while introducing an additional small error due to the effective approximate horizon T' being different for different grids over the risk level β .

5.5 Numerical Evaluation

In this section, we evaluate our EVaR-MDP algorithm numerically on several tabular MDPs. We focus on the EVaR-MDP objective for two reasons. First, as discussed in Section 5.2, EVaR is a

more practical risk measure than ERM because it is coherent and approximates the popular VaR and CVaR well. Second, the EVaR-MDP algorithm (Algorithm 10) also evaluates the ERM-MDP algorithm (Algorithm 9) since it uses it as a subroutine.

We assume that the objective is to solve an EVaR-MDP for a risk level $\alpha = 0.1$. That is, we seek to find a policy π that maximizes $\text{EVaR}_{0.1} \left[\tilde{\mathfrak{R}}_T^\pi \right]$. The risk level $\alpha = 0.1$ is a common choice in the risk-averse literature and the results are qualitatively insensitive to its choice. The numerical evaluation assumes a finite horizon $T = 100$, which makes it possible to evaluate the risk of $\tilde{\mathfrak{R}}_T^\pi$ by simulation. We sample 100,000 episodes of $\tilde{\mathfrak{R}}_T^\pi$ for this evaluation.

To understand how the components of Algorithm 10 contribute to the quality of its solution, we perform a small ablation study that compares it with two simplified algorithms: **1)** *naive grid* that uses a uniform grid of values β_k , $k = 1:K$, such that $\beta_1 = 0$ and $\beta_K = 10$, instead of what we propose in (5.14), and K is set to the same value as in (5.14), and **2)** *naive level* that uses the optimized grid but does not adjust the risk level with the time-step when it solves ERM-MDPs. Algorithm 10 uses the optimized grid in (5.14) with δ and $\Delta_{\mathfrak{R}}$ values given in Appendix C.4.

In addition to the ablation study, we also compare Algorithm 10 with several risk-averse algorithms that optimize objectives related to EVaR-MDPs. Specifically, we compare it with *risk-neutral* MDP, *nested CVaR* [12], and *nested EVaR* (related to Ahmadi et al. 2), both with $\alpha = 0.1$, and finally *ERM* (Algorithm 8) and *nested ERM*, both with $\beta = 0.5$. The parameter α was chosen to match the EVaR objective, but the parameter $\beta = 0.5$ is chosen arbitrarily since no general method exists to find a β that matches a given α . All the above methods compute Markov policies. We also compare Algorithm 10 with *augmented CVaR* [33] that computes a history-dependent policy for CVaR-MDPs. We implemented the augmented CVaR method using the faster quantile-based approach described in section 4 of Li et al. [79]. All algorithms were implemented in Julia with the exception of augmented MDP which was implemented both in R and Julia. The R implementation using quantiles was significantly faster than the implementation of the original algorithm [33] in Julia 1.8.

As described in the introduction the augmented CVaR-MDP [33] may not compute an optimal

Table 5.1: $\text{EVaR}_{0.1} \left[\tilde{\mathfrak{R}}_T^\pi \right]$ for π returned by each method.

Method	MR	GR	INV1	INV2	RS
Algorithm 10	-6.73	5.34	67.4	189	303
Naive grid	-6.87	5.37	43.2	189	303
Naive level	-10.00	4.17	64.6	188	217
Risk neutral	-6.53	2.29	40.6	186	300
Nested CVaR	-10.00	-0.02	-0.0	132	217
Nested EVaR	-10.00	4.61	-0.0	164	217
ERM	-6.72	5.19	50.7	178	217
Nested ERM	-10.00	4.76	24.9	150	217
Augmented CVaR	-7.06	3.64	49.0	82	93

policy [62]. While an augmented CVaR-MDP is guaranteed to evaluate policies correctly, the dynamic program overestimates the true optimal value function and computes suboptimal policies. This is one possible reason why EVaR-MDP achieves a better CVaR objective than the augmented CVaR-MDP. We do not compare with the augmented EVaR-MDP [93, 94] for two main reasons. First, this algorithm is even slower than the augmented CVaR-MDP because one needs to solve a conic optimization instead of a linear optimization in each time step. Second, this augmented EVaR-MDP is not guaranteed to compute a correct (or even approximately correct) value function even when the policy is fixed [62].

To obtain a holistic picture of the relative performance of the algorithms, we selected a diverse set of domains with varying numbers of actions, discount factors, and levels of uncertainty. These domains have all been used in risk-averse and robust RL literature and are as follows: *machine replacement* (MR) [43], *gamblers ruin* (GR) [13, 79], two classic *inventory* management problems (INV1) and (INV2) [66], and *river-swim* (RS) [121].

Table 5.1 summarizes $\text{EVaR}_{0.1} \left[\tilde{\mathfrak{R}}_T^\pi \right]$ for policies π computed by the algorithms described above. Bold font indicates results within a 95% confidence interval of the best policy. The variation in these results is due to the simulation used to estimate the risk. We can make the following observations from the results. First, the particular design of Algorithm 10 is important because it outperforms its ablated versions significantly in some domains. Second, the results confirm that none of the nested risk measures can optimize the static EVaR-MDP well. Even the risk-neutral

Table 5.2: $\text{CVaR}_{0.1} \left[\tilde{\mathfrak{R}}_T^\pi \right]$ for π returned by each method.

Method	MR	GR	INV1	INV2	RS
Algorithm 10	-4.62	7.87	76.6	195	382
Naive grid	-4.63	7.91	47.8	195	381
Naive level	-10.00	7.41	73.1	194	217
Risk neutral	-4.56	5.47	52.3	193	379
Nested CVaR	-10.00	0.00	0.0	135	217
Nested EVaR	-10.00	7.12	0.0	169	217
ERM	-4.58	7.64	56.0	182	217
Nested ERM	-10.00	7.27	28.3	153	217
Augmented CVaR	-4.83	8.27	55.1	82	101

policy often outperforms the nested risk measures. Finally in Table 5.2, we show that the results are similar when compared in terms of the $\text{CVaR}_{0.1} \left[\tilde{\mathfrak{R}}_T^\pi \right]$ objective. This is not surprising since EVaR is often a good proxy for CVaR [4]. Note that *augmented CVaR* is guaranteed to be optimal for CVaR when the discretization is sufficiently fine, which significantly increases the computation time.

It is also important to discuss the run-time of the algorithms summarized in Table 5.3. We implemented all of them in Julia and ran each one in less than 30 seconds on a laptop computer with the exception of *augmented CVaR* which ran for up to 10 minutes. The difference in run-time between solving the nested risk measures and computing the ERM-MDP optimal value function (described in Theorem 5.3.3) is negligible since they all evaluate nearly identical dynamic programs. However, Algorithm 10 in our experiments typically needs to solve between 20 and 50 ERM-MDP problems, one for each β_k , $k = 1:K$. This additional computation is significant, but we believe it can be addressed. As described in Section 5.4, there are ways to significantly speed up Algorithm 10, but we decided to focus on algorithms that are conceptually simple and can be analyzed in this work and leave computational concerns for future work.

Table 5.3: Run-time for the algorithms in seconds.

Method	MR	GR	INV1	INV2	RS
Algorithm 10	2.70	6.35	1.14	0.96	3.87
Naive grid	2.64	6.30	1.05	0.88	3.81
Naive level	2.79	6.38	1.19	0.92	3.95
Risk neutral	0.00	0.00	0.18	0.20	0.00
Nested CVaR	0.01	0.01	0.26	0.16	0.01
Nested EVaR	0.01	0.03	0.66	0.06	0.01
ERM	0.00	0.00	0.24	0.16	0.00
Nested ERM	0.01	0.01	0.10	0.02	0.01
Augmented CVaR	14.8	29.01	780	120	22.9

5.6 Chapter Summary

We analyzed discounted MDPs with two risk measures: ERM and EVaR that had not been studied in discounted multi-stage decision-making literature. This lack of interest is surprising because their properties make them especially suitable for dynamic decision-making. In this chapter, we derived the first exact DP formulation for ERM in discounted MDPs. We also showed that the optimal value function and an optimal deterministic Markov policy exist for ERM-MDP, and can be computed using value iteration. We showed that EVaR-MDP also has deterministic optimal policies, proposed a new polynomial-time algorithm for computing them, and demonstrated the algorithms numerically. Our numerical results showed that our EVaR algorithm performs consistently well across several domains and risk measures.

CHAPTER 6

Conclusion

Risk measure (ψ)	Finite horizon	Infinite horizon
$\mathbb{E}, \min, \max$	Π_{MD}	Π_{SD}
ERM, EVaR	Π_{MD}	Π_{MD}
VaR, CVaR	Π_{HD}	Π_{HD}

Table 6.1: Policy set that contains the optimal policy of risk-sensitive MDPs.

This dissertation proposed several new approaches toward constructing an accurate and computation-efficient algorithm for solving static risk-sensitive MDP. More specifically, we identify gaps in recent works, proposed practical algorithms to solve VaR (Algorithms 3 and 5), CVaR (Algorithm 11), EVaR (Algorithm 10) and ERM (Algorithms 8 and 9) MDP accurately. Table 6.1 demonstrates the known smallest policy set that contains the optimal policy for finite and infinite horizon discounted risk-averse MDP.

Optimizing static risk-sensitive objectives in (2.19) in MDP is difficult because they do not conform to standard dynamic programming equations commonly used in RL algorithms. Recent approaches introduce dynamic programming decompositions that augment the state space with risk levels, which have gained popularity in the RL community [29, 30, 33, 46, 47, 93, 105, 118]. In Chapter 3, we show that these widely used decompositions for Conditional-Value-at-Risk (CVaR) and Entropic-Value-at-Risk (EVaR) are inherently suboptimal regardless of the discretization level. Specifically, we prove that a saddle point property assumed in prior literature may be violated. In contrast, a valid decomposition exists for Value-at-Risk (VaR) and our proof highlights how VaR fundamentally differs from CVaR and EVaR. These findings are critical since risk-averse

algorithms are deployed in high-stakes environments where correctness is paramount.

The VaR MDP framework presented in Chapter 3 is a powerful approach. However, the augmented state space in this VaR DP decomposition is continuous, making direct computation impractical. In Chapter 4, we address this by approximating the risk level with carefully defined discrete functions that provide both lower and upper bounds for the quantile value function. We guarantee that the performance of our proposed policy (π), will be bounded by these lower and upper value functions. Moreover, as the number of uniform discretizations increases, we show that these bounds converge, indicating that the discretized VaR MDP’s performance approaches the VaR MDP’s optimal policy. Furthermore, Section 4.4 introduces a novel Q-learning algorithm for quantile optimization in MDPs, providing strong convergence and performance guarantees. This algorithm leverages a new, simplified dynamic program (DP) decomposition for quantile MDPs explained in Section 4.3.2. Compared with Chapter 3, this new DP decomposition requires neither known transition probabilities nor solving complex saddle point equations and serves as a suitable foundation for other model-free RL algorithms. Our numerical results in tabular domains show that our Q-learning algorithm converges to its DP variant and outperforms earlier algorithms. Additionally, we apply the same discretization intuition to optimize CVaR MDPs [13], which also suffer from a continuous augmented state space, as detailed in Appendix B.4.2.5.

The algorithms presented in Chapters 3 and 4 may be computationally intensive and return policies that depend on history. In Chapter 5, we derive a new DP formulation for discounted risk-averse MDPs with Entropic Risk Measure (ERM) and Entropic Value at Risk (EVaR) objectives. Our formulation for ERM, which is possible because of our novel definition of value function with time-dependent risk levels, can approximate optimal policies in a time that is polynomial in the approximation error.

Using an optimized discretization scheme, we further leverage the ERM algorithm to optimize the EVaR objective δ -optimally in polynomial time. Unlike VaR and CVaR MDP, whose optimal policies are history-dependent, we proved that ERM and EVaR MDP commit to Markovian optimal policy, which is easier to optimize. Moreover, since EVaR tightest Chernoff approximates of VaR

and CVaR [4], our EVaR MDP performs well even when evaluated in terms of VaR and CVaR. Our numerical results show the viability of our formulations and algorithms in discounted MDPs. This research empowers autonomous systems with customized objectives aligning with a decision-maker’s interests. Exploring risk-sensitive objectives in reinforcement learning advances the development of autonomous systems and deepens our understanding of human decision-making. Additionally, it provides decision scientists with valuable insights derived from theoretically proven optimal decision and learning rules.

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CHAPTER A

Supplementary Materials for Chapter 3

A.1 Proof of Proposition 3.3.1

Suppose that $\alpha > 0$; the decomposition for $\alpha = 0$ holds readily because $\text{CVaR}_0[\tilde{x}] = \text{ess inf}[\tilde{x}]$.

To streamline the notation, Define a random variable $\tilde{x} = r(\tilde{s}, \tilde{a}, \tilde{s}')$ over a random space $\Omega = \mathcal{S} \times \mathcal{A} \times \mathcal{S}$ with a probability distribution $\mathbf{q} \in \Delta_m$ such that $q_{s,a,s'} = \hat{p}_s \cdot \pi(s, a) \cdot p(s, a, s')$. The value $\mathbf{x}$ is the vector representation of the random variable $\tilde{x}$ and $\boldsymbol{\xi}_s = \boldsymbol{\xi}_{s,\cdot,\cdot} \in \mathbb{R}^{S \cdot A}$ for $\boldsymbol{\xi} \in \mathbb{R}^m$ is a vector that corresponds to the subset of the elements of Ω in which the first element is some $s \in \mathcal{S}$. The vectors $\mathbf{x}_s = \mathbf{x}_{s,\cdot,\cdot} \in \mathbb{R}^{S \cdot A}$ and $\mathbf{q}_s = \mathbf{q}_{s,\cdot,\cdot} \in \mathbb{R}^{S \cdot A}$ are defined analogously to $\boldsymbol{\xi}_s$.

Starting with the CVaR definition in (2.3) and introducing an auxiliary variable $\boldsymbol{\zeta}$ we get that

$$\begin{aligned} \text{CVaR}_\alpha[\tilde{x}] &= \min_{\boldsymbol{\xi} \in \Delta_m} \{ \mathbf{x}^\top \boldsymbol{\xi} \mid \alpha \boldsymbol{\xi} \leq \mathbf{q} \} = \min_{\boldsymbol{\xi} \in \Delta_m, \boldsymbol{\zeta} \in \mathbb{R}^S} \{ \mathbf{x}^\top \boldsymbol{\xi} \mid \alpha \boldsymbol{\xi} \leq \mathbf{q}, \zeta_s = \mathbf{1}^\top \boldsymbol{\xi}_s, \forall s \in \mathcal{S} \} \\ &= \min_{\boldsymbol{\xi} \in \Delta_m, \boldsymbol{\zeta} \in \Delta_S} \{ \mathbf{x}^\top \boldsymbol{\xi} \mid \alpha \boldsymbol{\xi} \leq \mathbf{q}, \zeta_s = \mathbf{1}^\top \boldsymbol{\xi}_s, \alpha \zeta_s \leq \hat{p}_s, \forall s \in \mathcal{S} \} \\ &= \min_{\boldsymbol{\xi} \in \mathbb{R}_+^\Omega, \boldsymbol{\zeta} \in \mathcal{Z}_C} \{ \mathbf{x}^\top \boldsymbol{\xi} \mid \alpha \boldsymbol{\xi} \leq \mathbf{q}, \zeta_s = \mathbf{1}^\top \boldsymbol{\xi}_s, \forall s \in \mathcal{S} \}. \end{aligned}$$

In the derivation above, we replaced the infimum by a minimum because Ω is finite, introduced a new variable $\boldsymbol{\zeta}$, derived implied constraints on $\boldsymbol{\zeta}$, and then dropped superfluous constraints on $\boldsymbol{\xi}$. Continuing with the derivation above and noticing that the constraints on each $\boldsymbol{\xi}_s$ are independent

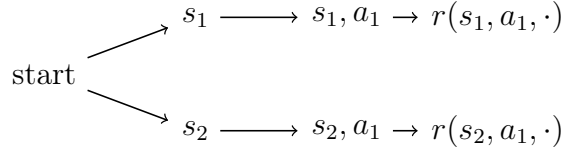


Figure A.1: Rewards of the MDP M_E used in the proof of Theorem 3.4.1. The dot indicates that the rewards are independent of the next state.

given ζ , we get that

$$\begin{aligned}
\text{CVaR}_\alpha[\tilde{x}] &= \min_{\xi \in \mathbb{R}_+^\Omega, \zeta \in \mathcal{Z}_C} \left\{ \sum_{s \in \mathcal{S}} \mathbf{x}_s^\top \xi_s \mid \alpha \xi \leq \mathbf{q}, \zeta_s = \mathbf{1}^\top \xi_s, \forall s \in \mathcal{S} \right\} \\
&\stackrel{(a)}{=} \min_{\zeta \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \inf_{\xi_s \in \mathbb{R}_+^{\Omega_s}} \{ \mathbf{x}_s^\top \xi_s \mid \alpha \xi_s \leq \mathbf{q}_s, \zeta_s = \mathbf{1}^\top \xi_s \} \\
&\stackrel{(b)}{=} \min_{\zeta \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \zeta_s \cdot \min_{\chi \in \Delta_{\mathcal{S} \cdot \mathcal{A}}} \{ \mathbf{x}_s^\top \chi \mid \alpha \hat{p}_s^{-1} \zeta_s \chi_{a,s'} \leq \hat{p}_s^{-1} q_{s,a,s'}, \forall a \in \mathcal{A}, s' \in \mathcal{S} \} \\
&\stackrel{(c)}{=} \min_{\zeta \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \zeta_s \cdot \text{CVaR}_{\alpha \zeta_s \hat{p}_s^{-1}}[\tilde{x} \mid \tilde{s} = s].
\end{aligned}$$

The step (a) follows from the interchangeability principle [113, theorem 7.92], and the step (b) follows by substituting $\xi_{s,a,s'} = \zeta_s \chi_{a,s'}$ taking care when $\zeta_s = 0$ and multiplying both sides of the inequality by $\hat{p}_s^{-1} > 0$. Finally, in step (c), the random variable $\tilde{x} = r(\tilde{s}, \tilde{a}, \tilde{s}')$ conditional on $\tilde{s} = s$ is distributed according to $q_{s,a,s'} \hat{p}_s^{-1}$ and the equality follows from the definition of CVaR in (2.3). $\square$

A.2 Proof of Theorem 3.4.1

Consider an MDP M_E depicted in Fig. A.1 with $\mathcal{S} = \{s_1, s_2\}$ and $\mathcal{A} = \{a_1\}$ and a reward function $r(s_1, a_1, \cdot) = 1$ and $r(s_2, a_1, \cdot) = 0$. We abbreviate the rewards to $r(s_1)$ and $r(s_2)$ because they only depend on the originating state. The initial distribution is $\hat{p}_{s_1} = \hat{p}_{s_2} = 0.5$. We finally let $\alpha = 0.75$.

Because $\mathcal{Z}_E \subseteq \mathcal{Z}_C$, the right-hand side of (3.10) can be lower-bounded by CVaR as

$$\begin{aligned} \min_{\xi \in \mathcal{Z}_E} \sum_{s \in \mathcal{S}} \xi_s \text{EVaR}_{\alpha \xi_s \hat{p}_s^{-1}} [r(s, a_1, \tilde{s}')] &= \min_{\xi \in \mathcal{Z}_E} \sum_{s \in \mathcal{S}} \xi_s r(s) \\ &\geq \min_{\xi \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \xi_s r(s) = \text{CVaR}_\alpha [r(\tilde{s}, a_1, \tilde{s}')] . \end{aligned} \quad (\text{A.1})$$

The first equality holds from the positive homogeneity and cash invariance properties of EVaR, and the last equality follows from the dual representation of CVaR [57].

Because $\text{EVaR}_\alpha [\tilde{x}] \leq \text{CVaR}_\alpha [\tilde{x}]$ for each $\alpha \in [0, 1]$ and $\tilde{x} \in \mathbb{X}$ (see [4, proposition 3.2]), we can further lower-bound (A.1) as

$$\text{EVaR}_\alpha [r(\tilde{s}, a_1, \tilde{s}')] \leq \text{CVaR}_\alpha [r(\tilde{s}, a_1, \tilde{s}')] \leq \min_{\xi \in \mathcal{Z}_E} \sum_{s \in \mathcal{S}} \xi_s \text{EVaR}_{\alpha \xi_s \hat{p}_s^{-1}} [r(s, a_1, \tilde{s}')] . \quad (\text{A.2})$$

Therefore, (3.10) holds with an inequality.

To prove by contradiction that the inequality in (3.10) is strict, suppose that

$$\text{EVaR}_\alpha [r(\tilde{s}, a_1, \tilde{s}')] = \min_{\xi \in \mathcal{Z}_E} \sum_{s \in \mathcal{S}} \xi_s \text{EVaR}_{\alpha \xi_s \hat{p}_s^{-1}} [r(s, a_1, \tilde{s}')] . \quad (\text{A.3})$$

Equalities (A.3) and (A.2) imply that $\text{EVaR}_\alpha [r(\tilde{s}, a_1, \tilde{s}')] = \text{CVaR}_\alpha [r(\tilde{s}, a_1, \tilde{s}')] which is false in general [4].$

We now show that EVaR does not equal CVaR even for the categorical distribution of $\tilde{s}$. The CVaR of the return in M_E reduces from (2.3) to

$$\text{CVaR}_\alpha [r(\tilde{s}, a_1, \tilde{s}')] = \min_{\xi \in \mathcal{Z}_C} \sum_{s \in \mathcal{S}} \xi_s r(s) = \max \left\{ 0, \frac{\hat{p}_{s_1} + \alpha - 1}{\alpha} \right\} . \quad (\text{A.4})$$

Since $1 - \alpha = 0.25 < 0.5 = \hat{p}_{s_1}$, then the optimal ξ^* in (A.4) is

$$\xi^* = \begin{pmatrix} \frac{\hat{p}_{s_1} + \alpha - 1}{\alpha} \\ \frac{1 - \hat{p}_{s_1}}{\alpha} \end{pmatrix} .$$

Since $\text{KL}(\boldsymbol{\xi}^* \parallel \hat{\mathbf{p}}) < -\log \alpha$, we have that $\boldsymbol{\xi}^*$ is in the relative interior of the EVaR feasible region in (2.5), and, therefore, there exists an $\epsilon > 0$ such that

$$\text{EVaR}_\alpha [r(\tilde{s}, a_1, \tilde{s}')] = \text{CVaR}_\alpha [r(\tilde{s}, a_1, \tilde{s}')] - \epsilon < \text{CVaR}_\alpha [r(\tilde{s}, a_1, \tilde{s}')] ,$$

which proves the desired inequality. $\square$

A.3 Proof of Corollary 3.4.2

We start by proposing a new decomposition for EVaR.

Proposition A.3.1. *Given a random variable $\tilde{x} \in \mathbb{X}$ and a discrete variable $\tilde{y}: \Omega \rightarrow \mathcal{N} = \{1, \dots, N\}$, with probabilities denoted as $\{\hat{p}_i\}_{i=1}^N$, for any $\alpha \in (0, 1]$ we have that*

$$\text{EVaR}_\alpha [\tilde{x}] = \inf_{\boldsymbol{\zeta} \in (0,1]^N} \min_{\boldsymbol{\xi} \in \mathcal{Z}'_{\text{E}}(\boldsymbol{\zeta})} \sum_i \xi_i \text{EVaR}_{\zeta_i} [\tilde{x} \mid \tilde{y} = i] ,$$

where

$$\mathcal{Z}'_{\text{E}}(\boldsymbol{\zeta}) = \left\{ \boldsymbol{\xi} \in \Delta_N \mid \boldsymbol{\xi} \ll \hat{\mathbf{p}}, \sum_{i=1}^N \xi_i (\log(\xi_i / \hat{p}_i) - \log(\zeta_i)) \leq -\log \alpha \right\} .$$

Proof. Let $\mathbf{q}$ denote the joint probability distribution of $\tilde{x}$ and $\tilde{y}$. The proof exploits the chain rule of relative entropy (e.g., Cover and Thomas [37, theorem 2.5.3]), which states that for any probability distributions $\boldsymbol{\eta}, \mathbf{q} \in \Delta_\Omega$ with $\boldsymbol{\eta} \ll \mathbf{q}$ that

$$\text{KL}(\boldsymbol{\eta} \parallel \mathbf{q}) = \text{KL}(\boldsymbol{\eta}(\tilde{y}) \parallel \mathbf{q}(\tilde{y})) + \text{KL}(\boldsymbol{\eta}(\tilde{x}|\tilde{y}) \parallel \mathbf{q}(\tilde{x}|\tilde{y})), \quad (\text{A.5})$$

where the conditional relative entropy is defined as

$$\text{KL}(\boldsymbol{\eta}(\tilde{x}|\tilde{y}) \parallel \mathbf{q}(\tilde{x}|\tilde{y})) = \mathbb{E}^\boldsymbol{\eta} \left[\log \frac{\boldsymbol{\eta}(\tilde{x}|\tilde{y})}{\mathbf{q}(\tilde{x}|\tilde{y})} \right] .$$

with $\mathbb{E}^\boldsymbol{\eta}[f(\tilde{x}, \tilde{y})]$ as a shorthand notation to indicate that $(\tilde{x}, \tilde{y}) \sim \boldsymbol{\eta}$. We can now decompose

EVaR from its definition in (2.5) as

$$\begin{aligned}
\text{EVaR}_\alpha [\tilde{x}] &= \inf_{\boldsymbol{\eta} \in \Delta_{m \cdot N} : \boldsymbol{\eta} \ll \mathbf{q}} \left\{ \mathbb{E}^\boldsymbol{\eta}[\tilde{x}] \mid \text{KL}(\boldsymbol{\eta} \mid \mathbf{q}) \leq -\log \alpha \right\} \\
&\stackrel{(a)}{=} \inf_{\boldsymbol{\eta} \in \Delta_{m \cdot N} : \boldsymbol{\eta} \ll \mathbf{q}} \left\{ \mathbb{E}^\boldsymbol{\eta}[\tilde{x}] \mid \text{KL}(\boldsymbol{\eta}(\tilde{y}) \parallel \mathbf{q}(\tilde{y})) + \text{KL}(\boldsymbol{\eta}(\tilde{x}|\tilde{y}) \parallel \mathbf{q}(\tilde{x}|\tilde{y})) \leq -\log \alpha \right\} \\
&= \inf_{\boldsymbol{\eta} \in \Delta_{m \cdot N} : \boldsymbol{\eta} \ll \mathbf{q}} \left\{ \mathbb{E}^\boldsymbol{\eta}[\tilde{x}] \mid \text{KL}(\boldsymbol{\eta}(\tilde{y}) \parallel \mathbf{q}(\tilde{y})) + \mathbb{E}^\boldsymbol{\eta} \left[\mathbb{E}^\boldsymbol{\eta} \left[\log \frac{\boldsymbol{\eta}(\tilde{x}|\tilde{y})}{\mathbf{q}(\tilde{x}|\tilde{y})} \right] \mid \tilde{y} \right] \leq -\log \alpha \right\} \\
&\stackrel{(b)}{=} \inf_{\boldsymbol{\eta} \in \Delta_{m \cdot N}, \zeta \in (0,1]^N : \boldsymbol{\eta} \ll \mathbf{q}} \left\{ \mathbb{E}^\boldsymbol{\eta}[\mathbb{E}^\boldsymbol{\eta}[\tilde{x} \mid \tilde{y}]] \mid \begin{array}{l} \text{KL}(\boldsymbol{\eta}(\tilde{y}) \parallel \mathbf{q}(\tilde{y})) + \mathbb{E}^\boldsymbol{\eta}[-\log(\zeta_{\tilde{y}})] \leq -\log \alpha \\ \mathbb{P}_\boldsymbol{\eta}[\mathbb{E}^\boldsymbol{\eta}[\log(\boldsymbol{\eta}(\tilde{x}|\tilde{y})/\mathbf{q}(\tilde{x}|\tilde{y})) \mid \tilde{y}] \leq -\log(\zeta_{\tilde{y}})] = 1 \end{array} \right\} \\
&\stackrel{(c)}{=} \inf_{\boldsymbol{\xi} \in \Delta_N, \zeta \in (0,1]^N : \boldsymbol{\xi} \ll \hat{\mathbf{p}}} \left\{ \mathbb{E}^\boldsymbol{\xi}[\text{EVaR}_{\zeta_{\tilde{y}}}[\tilde{x}|\tilde{y}]] \mid \text{KL}(\boldsymbol{\xi} \parallel \hat{\mathbf{p}}) + \mathbb{E}^\boldsymbol{\xi}[-\log(\zeta_{\tilde{y}})] \leq -\log \alpha \right\} \\
&= \inf_{\boldsymbol{\xi} \in \Delta_N, \zeta \in (0,1]^N : \boldsymbol{\xi} \ll \hat{\mathbf{p}}} \left\{ \sum_i \xi_i \text{EVaR}_{\zeta_i}[\tilde{x} \mid \tilde{y} = i] \mid \sum_{i=1}^N \xi_i \log(\xi_i/\hat{p}_i) - \sum_{i=1}^N \xi_i \log(\zeta_i) \leq -\log \alpha \right\}.
\end{aligned}$$

Here, we decompose the relative entropy of $\boldsymbol{\eta}$ and $\mathbf{q}$ using (A.5) in step (a) and then use the tower property of the expectation operator in the next step. In step (b), we introduce a variable ζ_i for each realization of $\tilde{y} = i$ with $i \in \mathcal{N}$ to decouple the influence of $\boldsymbol{\eta}(\tilde{x}|\tilde{y})$, under each $\tilde{y}$, in the inequality constraint. Finally, we replace the conditional EVaR definition by solving for $\boldsymbol{\eta}(\tilde{x}|\tilde{y})$ for a given $\boldsymbol{\zeta}$ in step (c), and representing $\boldsymbol{\eta}(\tilde{y})$ using $\boldsymbol{\xi}$. $\square$

The first part of our theorem follows directly from Proposition A.3.1. Suppose that $\alpha > 0$; the result follows for $\alpha = 0$ because $\text{EVaR}_0[\cdot]$ reduces to ess inf . Then, the second part of the corollary holds as

$$\begin{aligned}
\text{EVaR}_\alpha [r(\tilde{s}, \tilde{a}, \tilde{s}')] &= \\
&= \inf_{\zeta \in (0,1]^N, \boldsymbol{\xi} \in \Delta_N} \left\{ \sum_{s \in \mathcal{S}} \xi_s \text{EVaR}_{\zeta_s} [r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \mid \sum_{s \in \mathcal{S}} \xi_s \log \frac{\xi_s}{\zeta_s \hat{p}_s} \leq -\log \alpha \right\} \\
&\leq \inf_{\zeta \in (0,1]^N, \boldsymbol{\xi} \in \Delta_N} \left\{ \sum_{s \in \mathcal{S}} \xi_s \text{EVaR}_{\zeta_s} [r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \mid \sum_{s \in \mathcal{S}} \xi_s \log \frac{\xi_s}{\zeta_s \hat{p}_s} \leq -\log \alpha, \boldsymbol{\xi} \leq \alpha^{-1} \hat{\mathbf{p}} \right\} \\
&\leq \inf_{\boldsymbol{\xi} \in \Delta_N} \left\{ \sum_{s \in \mathcal{S}} \xi_s \text{EVaR}_{\alpha \xi_s \hat{p}_s^{-1}} [r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \mid \boldsymbol{\xi} \leq \alpha^{-1} \hat{\mathbf{p}} \right\}.
\end{aligned}$$

The first inequality follows from adding a constraint on the pairs on the $\boldsymbol{\xi}$ considered by the infimum. The second inequality follows by fixing $\zeta_s = \hat{\zeta}_s$ with $\hat{\zeta}_s = \alpha \xi_s \hat{p}_s^{-1}$ for each $s \in \mathcal{S}$. This is an upper bound because $\hat{\zeta}_s$ is feasible in the infimum:

$$\sum_{s \in \mathcal{S}} \xi_s \log \frac{\xi_s}{\hat{\zeta}_s \hat{p}_s} = -\log \alpha \leq -\log \alpha.$$

The value $\hat{\zeta}_s$ is well-defined since $\hat{p}_s > 0$ and the constraint $\boldsymbol{\xi} \leq \alpha^{-1} \hat{\mathbf{p}}$ ensures that $\hat{\zeta}_s \leq 1$. Also, we can relax the constraint $\zeta_s > 0 \Rightarrow \xi_s > 0$ to $\xi_s \geq 0$ because $\text{EVaR}_0[\tilde{x}] = \lim_{\alpha \rightarrow 0} \text{EVaR}_\alpha[\tilde{x}]$, and, therefore, the infimum is not affected. Finally, the inequality in the corollary follows immediately by further upper bounding the decomposition above by adding a constraint. $\square$

A.4 Proof of Theorem 3.5.3

The equality develops from Theorem 3.5.2 as

$$\begin{aligned} \max_{\pi \in \Pi} \text{VaR}_\alpha^{\tilde{a} \sim \pi(\tilde{s})}[r(\tilde{s}, \tilde{a}, \tilde{s}')] &= \max_{\pi \in \Pi} \sup_{\zeta \in \Delta_{\mathcal{S}}: \alpha \cdot \zeta \leq \hat{\mathbf{p}}} \min_{s \in \mathcal{S}} \left(\text{VaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim \pi(s)}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \right) \\ &= \sup_{\zeta \in \Delta_{\mathcal{S}}: \alpha \cdot \zeta \leq \hat{\mathbf{p}}} \max_{\pi \in \Pi} \min_{s \in \mathcal{S}} \left(\text{VaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim \pi(s)}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \right) \\ &= \sup_{\zeta \in \Delta_{\mathcal{S}}: \alpha \cdot \zeta \leq \hat{\mathbf{p}}} \min_{s \in \mathcal{S}} \left(\max_{\mathbf{d} \in \Delta_A} \text{VaR}_{\alpha \zeta_s \hat{p}_s^{-1}}^{\tilde{a} \sim \mathbf{d}}[r(s, \tilde{a}, \tilde{s}') \mid \tilde{s} = s] \right), \end{aligned}$$

where we first change the order of maximum and supremum, followed by changing the order of $\max_\pi \min_s$ with $\min_s \max_\pi$. The latter is a direct consequence of the interchangeability property of the maximum operation [114, proposition 2.2]. $\square$

A.5 Proof of Proposition 3.5.4

The proof mainly relies on correcting the decomposition of the lower quantile proposed in [79].

Proposition A.5.1. *Given an $\tilde{x} \in \mathbb{X}$, suppose that a random variable $\tilde{y}: \Omega \rightarrow \mathcal{N} = \{1, \dots, N\}$*

is distributed as $\hat{\mathbf{p}} = (\hat{p}_i)_{i=1}^N$ with $\hat{p}_i > 0$. Then:

$$Q_\alpha(\tilde{x}) = \sup_{\zeta \in [0,1]^N} \left\{ \min_{i \in \mathcal{N}: \zeta_i < 1} Q_{\zeta_i}(\tilde{x} \mid \tilde{y} = i) \mid \sum_{i=1}^N \zeta_i \hat{p}_i < \alpha \right\}, \quad (\text{A.6})$$

where we interpret the supremum to be minus infinity if its feasible set is empty, which only occurs if $\alpha = 0$.

Proof. First, we decompose the lower quantile using its definition as

$$\begin{aligned} Q_\alpha(\tilde{x}) &= \sup \{z \mid \mathbb{P}[\tilde{x} < z] < \alpha\} \stackrel{(a)}{=} \sup_{z \in \mathbb{R}} \left\{ z \mid \sum_{i=1}^N \mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] \hat{p}_i < \alpha \right\} \\ &\stackrel{(b)}{=} \sup_{z \in \mathbb{R}, \zeta \in [0,1]^N} \left\{ z \mid \sum_{i=1}^N \zeta_i \hat{p}_i < \alpha, \mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] < \zeta_i, \forall i \in \mathcal{N} : \zeta_i < 1 \right\} \\ &\stackrel{(c)}{=} \sup_{z \in \mathbb{R}, \zeta \in [0,1]^N} \left\{ z \mid z < Q_{\zeta_i}(\tilde{x} \mid \tilde{y} = i), \forall i \in \mathcal{N} : \zeta_i < 1, \sum_{i=1}^N \zeta_i \hat{p}_i < \alpha \right\} \\ &\stackrel{(d)}{=} \sup_{\zeta \in [0,1]^N} \left\{ \sup_{z \in \mathbb{R}} \{z \mid z < Q_{\zeta_i}(\tilde{x} \mid \tilde{y} = i), \forall i \in \mathcal{N} : \zeta_i < 1\} \mid \sum_{i=1}^N \zeta_i \hat{p}_i < \alpha \right\} \\ &\stackrel{(e)}{=} \sup_{\zeta \in [0,1]^N} \left\{ \min_{i \in \mathcal{N}: \zeta_i < 1} Q_{\zeta_i}(\tilde{x} \mid \tilde{y} = i) \mid \sum_{i=1}^N \zeta_i \hat{p}_i < \alpha \right\}. \end{aligned}$$

We decompose the probability $\mathbb{P}[\tilde{x} < z]$ in terms of the conditional probabilities $\mathbb{P}[\tilde{x} < z \mid \tilde{y} = i]$ in step (a) and then lower-bound them by an auxiliary variable ζ_i in step (b). In step (c), we exploit the following equivalence:

$$\mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] < \zeta_i \quad \Leftrightarrow \quad z < Q_{\zeta_i}(\tilde{x} \mid \tilde{y} = i)$$

The direction $\Leftarrow$ in the equivalence follows from the definition of $Q_{\zeta_i}(\tilde{x} \mid \tilde{y} = i)$:

$$z < Q_{\zeta_i}(\tilde{x} \mid \tilde{y} = i) = \inf \{z \mid \mathbb{P}[\tilde{x} \leq z \mid \tilde{y} = i] \geq \zeta_i\} \Rightarrow \mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] < \zeta_i.$$

The direction $\Rightarrow$ follows from the definition of VaR (see equation (2.2)), which implies that VaR

upper-bounds any z that satisfies the left-hand condition:

$$\mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] < \zeta_i \Rightarrow Q_{\zeta_i}(\tilde{x} \mid \tilde{y} = i) = \sup \{z \in \mathbb{R} \mid \mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] < \zeta_i\} \geq z,$$

yet $Q_{\zeta_i}(\tilde{x} \mid \tilde{y} = i) \neq z$ otherwise since $\mathbb{P}[\tilde{x} < z \mid \tilde{y} = i]$ is right continuous, there must exist some $\epsilon > 0$ for which $\mathbb{P}[\tilde{x} < z + \epsilon \mid \tilde{y} = i] < \zeta_i$ hence:

$$z = \sup \{z \in \mathbb{R} \mid \mathbb{P}[\tilde{x} < z \mid \tilde{y} = i] < \zeta_i\} \geq z + \epsilon > z,$$

which leads to a contradiction. In step (e), we solve for z . Finally, we obtain the form in (A.6). $\square$

CHAPTER B

Supplementary Materials for Chapter 4

B.1 Proofs of Section 4.2

B.1.1 Proof of Lemma 4.2.1

Proof. We aim to show that for $\tilde{x} \in \mathbb{X}, \alpha \in [0, 1]$,

$$\arg \min_{y \in \mathbb{R}} \mathbb{E}[\max(\alpha(\tilde{x} - y), -(1 - \alpha)(\tilde{x} - y))] = [\mathbf{q}_\alpha^-(\tilde{x}), \mathbf{q}_\alpha^+(\tilde{x})] \cap \mathbb{R},$$

where $[\mathbf{q}_0^-(\tilde{x}), \mathbf{q}_0^+(\tilde{x})] \cap \mathbb{R} = (-\infty, \mathbf{q}_0^+(\tilde{x})]$ and $[\mathbf{q}_1^-(\tilde{x}), \mathbf{q}_1^+(\tilde{x})] = [\mathbf{q}_0^-(\tilde{x}), \infty)$. For the case $\alpha \in (0, 1)$, we refer the reader to [60, Thm. 9]. We study the case $\alpha = 0$ as a similar set of arguments holds when $\alpha = 1$. By definition of $\mathbf{q}_0^+(\tilde{x})$, for all $\bar{y} \in \mathbb{R}$ such that $\bar{y} \leq \mathbf{q}_0^+(\tilde{x})$, $\mathbb{P}[\tilde{x} < \bar{y}] \leq \mathbb{P}[\tilde{x} < \mathbf{q}_0^+(\tilde{x})] \leq 0$ (i.e., equals zero) and therefore, by the law of total probability:

$$0 \leq \mathbb{E}[\ell_0(\tilde{x} - \bar{y})] = \mathbb{E}[0|\tilde{x} \geq \bar{y}]\mathbb{P}[\tilde{x} \geq \bar{y}] + \mathbb{E}[-(\tilde{x} - \bar{y})|\tilde{x} < \bar{y}]\mathbb{P}[\tilde{x} < \bar{y}] = 0.$$

Hence, $\arg \min_{y \in \mathbb{R}} \mathbb{E}[\ell_0(\tilde{x} - y)] \supseteq (-\infty, \mathbf{q}_0^+(\tilde{x})]$.

By definition of $\mathbf{q}_0^+(\tilde{x})$, for all $\bar{y} \in \mathbb{R}$ such that $\bar{y} > \mathbf{q}_0^+(\tilde{x})$, $\mathbb{P}[\tilde{x} < \bar{y}] > 0$ and:

$$\mathbb{E}[\ell_0(\tilde{x} - \bar{y})] = \mathbb{E}[0|\tilde{x} \geq \bar{y}]\mathbb{P}[\tilde{x} \geq \bar{y}] + \mathbb{E}[-(\tilde{x} - \bar{y})|\tilde{x} < \bar{y}]\mathbb{P}[\tilde{x} < \bar{y}] = \mathbb{E}[\bar{y} - \tilde{x}|\tilde{x} < \bar{y}]\mathbb{P}[\tilde{x} < \bar{y}] > 0.$$

The last inequality exploits two facts. First, by left continuity of $h(z) := \mathbb{P}[\tilde{x} < \bar{y} + z]$, $\mathbb{P}[\tilde{x} < \bar{y}] >$

0 implies that there must be some $\epsilon > 0$ for which $\mathbb{P}[\tilde{x} \leq \bar{y} - \epsilon] > 0$. Second, by the Markov inequality $\mathbb{E}[\bar{y} - \tilde{x} | \bar{y} - \tilde{x} > 0] \geq \epsilon \mathbb{P}[\bar{y} - \tilde{x} \geq \epsilon | \bar{y} - \tilde{x} > 0] = \epsilon \mathbb{P}[\bar{y} - \tilde{x} \geq \epsilon] / \mathbb{P}[\bar{y} - \tilde{x} > 0] > 0$. Hence, $\arg \min_{y \in \mathbb{R}} \mathbb{E}[\ell_0(\tilde{x} - y)] \subseteq (-\infty, \mathbf{q}_0^+(\tilde{x})]$. We have shown $\arg \min_{y \in \mathbb{R}} \mathbb{E}[\ell_0(\tilde{x} - y)] = (-\infty, \mathbf{q}_0^+(\tilde{x})]$, which ends the proof for $\alpha = 0$ since $\mathbf{q}_0^-(\tilde{x}) = -\infty$. $\square$

B.1.2 From VaR state-action to VaR state value function

Proposition B.1.1. *The optimal value functions satisfy for each $s \in \mathcal{S}$, $\alpha \in [0, 1]$, and $t \in [T]$ that*

$$v_t^*(s, \alpha) = \max_{a \in \mathcal{A}} q_t^*(s, \alpha, a).$$

Proof. The result follows straightforwardly from the definition of v_t^* and q_t^* . Namely,

$$\begin{aligned} v_t^*(s, \alpha) &= \max_{\pi \in \Pi_{\text{HD}}^t} \text{VaR}_{\alpha}^{\pi, s} \left[\sum_{k=0}^{t-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] \\ &= \max_{a \in \mathcal{A}, \pi \in \Pi_{\text{HD}}^t : \pi_0(s) = a} \text{VaR}_{\alpha}^{\pi, s} \left[\sum_{k=0}^{t-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] = \max_{a \in \mathcal{A}} q_t^*(s, \alpha, a). \end{aligned}$$

$\square$

B.2 Proofs of Section 4.3

B.2.1 Proof of Theorem 4.3.1

Proof. This proof extends the results obtained in [62, Appx. C] to the case where there exist (s, a, s') tuples for which $p(s, a, s') = 0$, and reparameterizes the representation. The result on the definition and optimality of π^* comes directly from [62]. Specifically, by [62, Appx. C], it holds that $\ddot{q}_t = q_t^*, \forall t \in [T]$, for all sequences $\ddot{q} := (\ddot{q}_t)_{t=0}^T$ such that $\ddot{q}_0(s, \alpha, a) = \text{VaR}_{\alpha}[0]$ and

$$\ddot{q}_{t+1}(s, \alpha, a) = r(s, a) + \gamma \max_{\zeta \in \Xi_{sa}(\alpha)} \min_{s' \in \mathcal{S} : p(s, a, s') > 0} \max_{a' \in \mathcal{A}} \ddot{q}_t \left(s', \frac{\alpha \zeta_{s'}}{p(s, a, s')}, a' \right), \quad \forall t \in [T-1],$$

where

$$\Xi_{sa}(\alpha) := \{\zeta \in [0, 1]^S \mid \sum_{s' \in \mathcal{S}} \zeta_{s'} = 1, \alpha \zeta_{s'} \leq p(s, a, s'), \quad \forall s' \in \mathcal{S}\}.$$

By construction of $q = (q_t)_{t=0}^T$ from the theorem statement, we have $q_0 = \ddot{q}_0$, so it remains to establish $B_{\max} q_t = \ddot{q}_{t+1}$ for all $t \in [T-1]$. By mathematical induction, assuming that $q_t = \ddot{q}_t$ for some $t \in [T-1]$, we show that $q_{t+1} := B_{\max} q_t = \ddot{q}_{t+1}$ in two steps : (1) $B_{\max} q_t \leq \ddot{q}_{t+1}$ and (2) $B_{\max} q_t \geq \ddot{q}_{t+1}$.

Step 1: Establishing $B_{\max} q_t \leq \ddot{q}_{t+1}$.

Let $o^* \in \mathcal{O}_{sa}(\alpha)$ (which is non-empty given that $0 \in \mathcal{O}_{sa}(\alpha)$) be an optimal point for operator $B_{\max}$ in Eq. (4.5) and consider the function:

$$\begin{aligned} g(o) &:= \min_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} q_t(s', o_{s'}, a') \\ &= \min_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} q_t^*(s', o_{s'}, a') \\ &= \min_{s' \in \mathcal{S}} \max_{\pi \in \Pi_{\text{HD}}^t} \text{VaR}_{o_{s'}}^{\pi, s'} \left[\sum_{k=0}^{t-1} \gamma^k \cdot r(\tilde{s}_k, \tilde{a}_k) \right]. \end{aligned}$$

By properties of VaR and min operators, g is non-decreasing in o . Therefore, $\max_{o \in \mathcal{O}_{sa}(\alpha)} g(o)$ is achieved inside $\{o \in [0, 1]^S \mid \sum_{s' \in \mathcal{S}} o_{s'} p(s, a, s') = \alpha\}$ and necessarily, $\sum_{s' \in \mathcal{S}} o_{s'}^* p(s, a, s') = \alpha$.

Now let $\zeta \in \mathbb{R}^S$ be defined as:

$$\zeta_{s'} := \begin{cases} \frac{o_{s'}^* p(s, a, s')}{\alpha} & \text{if } \alpha > 0 \\ p(s, a, s') & \text{otherwise.} \end{cases}$$

for all $s' \in \mathcal{S}$. We aim to show that $\zeta \in \Xi_{sa}(\alpha)$. When $\alpha = 0$, the claim follows by definition of a transition kernel p . When $\alpha > 0$, we get

- $\sum_{s' \in \mathcal{S}} \zeta_{s'} = \frac{\sum_{s' \in \mathcal{S}} o_{s'}^* p(s, a, s')}{\alpha} = 1$
- $p(s, a, s') \geq 0, o_{s'}^* \geq 0 \implies \zeta_{s'} = \frac{o_{s'}^* p(s, a, s')}{\alpha} \geq 0, \forall s' \in \mathcal{S}$

- $(\zeta_{s'} \geq 0 \ \forall s' \in \mathcal{S} , \ \sum_{s' \in \mathcal{S}} \zeta_{s'} = 1) \implies (\zeta_{s'} \leq 1 \ \forall s' \in \mathcal{S})$
- $o_{s'}^* \leq 1 \implies \alpha \zeta_{s'} = o_{s'}^* p(s, a, s') \leq p(s, a, s') , \forall s' \in \mathcal{S}$

In both cases, $\alpha \zeta_{s'} = o_{s'}^* p(s, a, s'), \forall s' \in \mathcal{S}$. Indeed, for $\alpha = 0$, the condition $\sum_{s' \in \mathcal{S}} o_{s'}^* p(s, a, s') = \alpha$ implies $o_{s'}^* = 0$ whenever $p(s, a, s') > 0$, so $o_{s'}^* p(s, a, s') = 0 = \alpha \zeta_{s'}$ for all $s' \in \mathcal{S}$. We can thus deduce:

$$\begin{aligned}
B_{\max} q_t(s, \alpha, a) &= r(s, a) + \gamma \max_{o \in \mathcal{O}_{sa}(\alpha)} \min_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} q_t(s', o_{s'}, a') \\
&= r(s, a) + \gamma \min_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} q_t(s', o_{s'}^*, a') \\
&\leq r(s, a) + \gamma \min_{s' \in \mathcal{S}: p(s, a, s') > 0} \max_{a' \in \mathcal{A}} q_t(s', o_{s'}^*, a') \\
&= r(s, a) + \gamma \min_{s' \in \mathcal{S}: p(s, a, s') > 0} \max_{a' \in \mathcal{A}} q_t \left(s', \frac{\alpha \zeta_{s'}}{p(s, a, s')}, a' \right) \\
&\leq r(s, a) + \gamma \max_{\zeta \in \Xi_{sa}(\alpha)} \min_{s' \in \mathcal{S}: p(s, a, s') > 0} \max_{a' \in \mathcal{A}} q_t \left(s', \frac{\alpha \zeta_{s'}}{p(s, a, s')}, a' \right) = \ddot{q}_{t+1}(s, \alpha, a).
\end{aligned}$$

Step 2: Establishing $B_{\max} q_t \geq \ddot{q}_{t+1}$. We proceed similarly. Let an optimal $\zeta^* \in \Xi_{sa}(\alpha)$ (it exists since $\zeta := p(s, a, \cdot)$ is always feasible) satisfying:

$$\ddot{q}_{t+1}(s, \alpha, a) = r(s, a) + \gamma \min_{s' \in \mathcal{S}: p(s, a, s') > 0} \max_{a' \in \mathcal{A}} q_t \left(s', \frac{\alpha \zeta_{s'}^*}{p(s, a, s')}, a' \right),$$

and define $o \in \mathbb{R}^{\mathcal{S}}$ as

$$o_{s'} := \begin{cases} \frac{\alpha \zeta_{s'}^*}{p(s, a, s')} & \text{if } p(s, a, s') > 0, \\ 1 & \text{otherwise.} \end{cases}$$

To check if $o \in \mathcal{O}_{sa}(\alpha)$, we remark that for any $s' \in \mathcal{S}$ with $p(s, a, s') > 0$,

$$\begin{aligned}
\frac{\alpha \zeta_{s'}^*}{p(s, a, s')} &\geq 0 & [\alpha \geq 0, \zeta^* \geq 0, p(s, a, s') > 0] \\
\frac{\alpha \zeta_{s'}^*}{p(s, a, s')} &\leq 1, & [\alpha \zeta_{s'}^* \leq p(s, a, s')]
\end{aligned}$$

so $o_{s'} \in [0, 1]$ when $p(s, a, s') > 0$. Otherwise, $o_{s'} = 1 \in [0, 1]$. Additionally,

$$\sum_{s' \in \mathcal{S}} o_{s'} p(s, a, s') = \sum_{s' \in \mathcal{S}: p(s, a, s') > 0} \frac{\alpha \zeta_{s'}^*}{p(s, a, s')} \cdot p(s, a, s') = \sum_{s' \in \mathcal{S}: p(s, a, s') > 0} \alpha \zeta_{s'}^* = \alpha,$$

so $o \in \mathcal{O}_{sa}(\alpha)$. We can thus establish:

$$\begin{aligned} B_{\max} q_t(s, \alpha, a) &= r(s, a) + \gamma \max_{o \in \mathcal{O}_{sa}(\alpha)} \min_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} q_t(s', o_{s'}, a') \\ &\geq r(s, a) + \gamma \min_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} q_t(s', o_{s'}, a') \\ &\stackrel{(a)}{=} r(s, a) + \gamma \min_{s' \in \mathcal{S}: p(s, a, s') > 0} \max_{a' \in \mathcal{A}} q_t(s', o_{s'}, a') \\ &= r(s, a) + \gamma \min_{s' \in \mathcal{S}: p(s, a, s') > 0} \max_{a' \in \mathcal{A}} q_t\left(s', \frac{\alpha \zeta_{s'}^*}{p(s, a, s')}, a'\right) \\ &= r(s, a) + \gamma \max_{\zeta \in \Xi_{sa}(\alpha)} \min_{s' \in \mathcal{S}: p(s, a, s') > 0} \max_{a' \in \mathcal{A}} q_t\left(s', \frac{\alpha \zeta_{s'}}{p(s, a, s')}, a'\right) \\ &= \ddot{q}_{t+1}(s, \alpha, a). \end{aligned}$$

Equality (a) stems from the implications: $p(s, a, s') = 0 \implies o_{s'} = 1 \implies q_t(s', o_{s'}, a') = \infty$ and for any $c \in \bar{\mathbb{R}}$, $\min(\infty, c) = c$. We now conclude the proof by induction, as $q_{t+1} := B_{\max} q_t = \ddot{q}_{t+1} \forall t \in [T - 1]$. $\square$

B.2.2 Proof of Theorem 4.3.2

Before diving into the proof of the theorem, we show the following general lemmas which will eventually be used for establishing Theorem 4.3.2.

Lemma B.2.1. *For all $t \in [T]$ and $s \in \mathcal{S}$, $\alpha \mapsto v_t^*(s, \alpha)$ is non-decreasing, right-continuous on $[0, 1]$.*

Proof. Proposition B.1.1 indicates that for any $t \in [T]$,

$$v_t^*(s, \alpha) = \text{VaR}_{\alpha}^{\pi^*, s} \left[\sum_{k=0}^{t-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right].$$

By [57, Lem. A.19], $\alpha \mapsto \text{VaR}_\alpha[\tilde{x}]$ is non-decreasing and right-continuous for any $\tilde{x} \in \mathbb{X}$, so the result follows. $\square$

Lemma B.2.2. *Let $f : [0, 1] \rightarrow \bar{\mathbb{R}}$ be non-decreasing and $\tilde{u}$ be a uniform random variable over $[0, 1]$. Then, we have that $\text{VaR}_\alpha[f(\tilde{u})] = f(\alpha)$ for all $\alpha \in [0, 1)$ where $f(\alpha)$ is right-continuous. Moreover, if $f(1) = \infty$, the equality also holds at $\alpha = 1$.*

Proof. By definition of VaR (see Eq. (4.1)), we have: $\text{VaR}_\alpha[f(\tilde{u})] = \max\{z \in \bar{\mathbb{R}} \mid \mathbb{P}[f(\tilde{u}) < z] \leq \alpha\}$. When $\alpha = 1$, $\mathbb{P}[f(\tilde{u}) < z] \leq \alpha$ for all $z \in \bar{\mathbb{R}}$ so that $\text{VaR}_\alpha[f(\tilde{u})] = \infty$ and the second part of the statement holds. Let thus $\alpha \in [0, 1)$ be such that $f(\alpha)$ is right-continuous at α . By assumption on f being non-decreasing, $\tilde{u} \geq \alpha$ implies $f(\tilde{u}) \geq f(\alpha)$ and we can establish:

$$\begin{aligned} \mathbb{P}[\tilde{u} \geq \alpha] &\leq \mathbb{P}[f(\tilde{u}) \geq f(\alpha)] \\ \iff 1 - \mathbb{P}[\tilde{u} < \alpha] &\leq 1 - \mathbb{P}[f(\tilde{u}) < f(\alpha)] \\ \iff 1 - \alpha \leq 1 - \mathbb{P}[f(\tilde{u}) < f(\alpha)] &\quad [\mathbb{P}[\tilde{u} < \alpha] = \mathbb{P}[\tilde{u} \leq \alpha] = \alpha] \\ \iff \alpha \geq \mathbb{P}[f(\tilde{u}) < f(\alpha)]. \end{aligned}$$

As a result, $f(\alpha) \leq \text{VaR}_\alpha[f(\tilde{u})]$.

On the other hand, the right-continuity of f at α ensures that for all $\epsilon > 0$ there exists a $\delta > 0$ such that $f(\alpha + \delta) < f(\alpha) + \epsilon$. Thus

$$\begin{aligned} \mathbb{P}[f(\tilde{u}) < f(\alpha) + \epsilon] &\geq \mathbb{P}[f(\tilde{u}) \leq f(\alpha + \delta)] && [\text{By construction: } f(\alpha + \delta) < f(\alpha) + \epsilon] \\ &\geq \mathbb{P}[\tilde{u} \leq \alpha + \delta] && [\tilde{u} \leq \alpha + \delta \implies f(\tilde{u}) \leq f(\alpha + \delta)] \\ &= \alpha + \delta \\ &> \alpha. \end{aligned}$$

Hence, $\text{VaR}_\alpha[f(\tilde{u})] \leq f(\alpha) + \epsilon$ for all $\epsilon > 0$. Setting $\epsilon \rightarrow 0$, $\text{VaR}_\alpha[f(\tilde{u})] \leq f(\alpha)$.

We conclude that $\text{VaR}_\alpha[f(\tilde{u})] = f(\alpha)$ for all $\alpha \in [0, 1)$ at which $f(\alpha)$ is right-continuous. $\square$

The result below directly follows from Lemmas B.2.1 and B.2.2.

Corollary B.2.3. *For all $t \in [T]$ and $s \in \mathcal{S}$, we have $\text{VaR}_\alpha[v_t^*(s, \tilde{u})] = v_t^*(s, \alpha)$, where $\tilde{u}$ is a uniform random variable on $[0, 1]$.*

Lemma B.2.4. *Let $S \in \mathbb{N}$ non-decreasing functions $f_i: [0, 1] \rightarrow \bar{\mathbb{R}}, i \in 1:S$ with each $f_i(\alpha) \in \mathbb{R}$ for all $\alpha \in (0, 1)$. Let also $\tilde{y}$ be a discrete random variable on $1:S$ with probability mass function $\hat{p}_i := \mathbb{P}[\tilde{y} = i], \forall i \in 1:S$, and $\tilde{u}$ an independent random variable with uniform distribution on $[0, 1]$. Then, we have*

$$\text{VaR}_\alpha[f_{\tilde{y}}(\tilde{u})] = \max_{\mathbf{o} \in [0, 1]^S} \left\{ \min_{i \in 1:S} \text{VaR}_{o_i}[f_i(\tilde{u})] \mid \sum_{j=1}^S o_j \hat{p}_j \leq \alpha \right\}. \quad (\text{B.1})$$

Proof. This proof closely follows that of [62, Thm. 5.1] but relaxes the assumption $\hat{p}_i > 0$ and simplifies the notation. Let $\mathfrak{J} := \{i \in 1:S \mid \hat{p}_i > 0\}$. We decompose VaR based on its definition in Eq. (4.1):

$$\begin{aligned} \text{VaR}_\alpha[f_{\tilde{y}}(\tilde{u})] &= \max \left\{ \tau \in \bar{\mathbb{R}} \mid \mathbb{P}[f_{\tilde{y}}(\tilde{u}) < \tau] \leq \alpha \right\} \\ &\stackrel{(b)}{=} \max \left\{ \tau \in \bar{\mathbb{R}} \mid \sum_{i \in \mathfrak{J}} \mathbb{P}[f_{\tilde{y}}(\tilde{u}) < \tau \mid \tilde{y} = i] \cdot \hat{p}_i \leq \alpha \right\} \\ &= \max \left\{ \tau \in \bar{\mathbb{R}} \mid \sum_{i \in \mathfrak{J}} \mathbb{P}[f_i(\tilde{u}) < \tau] \cdot \hat{p}_i \leq \alpha \right\} \\ &\stackrel{(c)}{=} \max \left\{ \tau \in \bar{\mathbb{R}} \mid \exists \mathbf{o} \in [0, 1]^S, \mathbb{P}[f_i(\tilde{u}) < \tau] \leq o_i, \forall i \in \mathfrak{J}, \sum_{j \in \mathfrak{J}} o_j \hat{p}_j \leq \alpha \right\} \\ &= \max \left\{ \tau \in \bar{\mathbb{R}} \mid \exists \mathbf{o} \in [0, 1]^S, \mathbb{P}[f_i(\tilde{u}) < \tau] \leq o_i, \forall i \in \mathfrak{J}, \sum_{j=1}^S o_j \hat{p}_j \leq \alpha \right\} \\ &\stackrel{(d)}{=} \max_{\mathbf{o} \in [0, 1]^S} \left\{ \max \left\{ \tau \in \bar{\mathbb{R}} \mid \mathbb{P}[f_i(\tilde{u}) < \tau] \leq o_i, \forall i \in \mathfrak{J} \right\} \mid \sum_{j=1}^S o_j \hat{p}_j \leq \alpha \right\} \\ &= \max_{\mathbf{o} \in [0, 1]^S} \left\{ \max_{i \in \mathfrak{J}} \bigcap \left\{ \tau \in \bar{\mathbb{R}} \mid \mathbb{P}[f_i(\tilde{u}) < \tau] \leq o_i \right\} \mid \sum_{j=1}^S o_j \hat{p}_j \leq \alpha \right\} \\ &\stackrel{(e)}{=} \max_{\mathbf{o} \in [0, 1]^S} \left\{ \min_{i \in \mathfrak{J}} \max \left\{ \tau \in \bar{\mathbb{R}} \mid \mathbb{P}[f_i(\tilde{u}) < \tau] \leq o_i \right\} \mid \sum_{j=1}^S o_j \hat{p}_j \leq \alpha \right\} \end{aligned}$$

$$\stackrel{(f)}{=} \max_{\mathbf{o} \in [0,1]^S} \left\{ \min_{i \in \mathfrak{I}} \text{VaR}_{o_i} [f_i(\tilde{u})] \mid \sum_{j=1}^S o_j \hat{p}_j \leq \alpha \right\}. \quad (\text{B.2})$$

In the derivation above, step (b) follows from the law of total probability and omitting zero probability events. Then we lower-bound them by an auxiliary variable o_i in step (c). In step (d) we replace the joint maximum over τ and $\mathbf{o}$ by sequential max, and then we replace the max of an intersection by the minimum of the maxima of sets in (e). The equality in (e) holds because $\tau \mapsto \mathbb{P}[f_i(\tilde{u}) < \tau]$ is monotone and, therefore, the sets $\{\tau \in \bar{\mathbb{R}} \mid \mathbb{P}[f_i(\tilde{u}) < \tau \mid \tilde{y} = i] \leq o_i\}$ are nested. Step (f) holds by definition of VaR.

It remains to show that (B.2) equals (B.1). Suppose that $o^* \in [0,1]^S$ is optimal in (B.2) and construct $\bar{o} \in [0,1]^S$ as

$$\bar{o}_i = \begin{cases} o_i^* & \text{if } i \in \mathfrak{I}, \\ 1 & \text{otherwise,} \end{cases} \quad \forall i \in 1:S.$$

Since $\bar{o}$ is feasible in (B.1) with the same objective, (B.2) $\leq$ (B.1). To show that (B.2) $\geq$ (B.1) suppose that $o^* \in [0,1]^S$ is optimal in (B.1). The inequality then holds because o^* is feasible in (B.2) and because

$$\min_{i \in \mathfrak{I}} \text{VaR}_{o_i^*} [f_i(\tilde{u})] \geq \min_{i \in 1:S} \text{VaR}_{o_i^*} [f_i(\tilde{u})].$$

□

Lemma B.2.5. *Assume non-decreasing functions $f_i: [0,1] \rightarrow \bar{\mathbb{R}}, i \in 1:S$ with each $f_i(\alpha) \in \mathbb{R}$ for all $\alpha \in (0,1)$. Then, it holds that $\max_{i \in 1:S} \text{VaR}_{\tilde{u}_1} [f_i(\tilde{u}_2)] = \max_{i \in 1:S} f_i(\tilde{u}_1)$ almost surely, where $\tilde{u}_1$ and $\tilde{u}_2$ are two independent uniform random variables on $[0,1]$.*

Proof. By Lemma B.2.2, for any $j \in 1:S$, $\text{VaR}_{\bar{\alpha}} [f_j(\tilde{u})] = f_j(\bar{\alpha})$ at any value of $\bar{\alpha} \in [0,1)$ where $f_j(\cdot)$ is right-continuous. This implies that for all $\alpha \in [0,1)$ where all f_i 's are right-continuous, $\max_{i \in 1:S} \text{VaR}_{\alpha} [f_i(\tilde{u})] = \max_{i \in 1:S} f_i(\alpha)$ since the maximum is necessarily right-continuous then. For $i \in 1:S$, f_i is monotone on the interval $(0,1)$, so by Froda's theorem [109, Thm. 4.30], the number of discontinuities of f_i must be at most countable on $(0,1)$ so therefore also on $[0,1]$.

This implies that the number of points α at which some f_i from $i \in 1:S$ is discontinuous is at most countable. We thus conclude that $\max_{i \in 1:S} f_i(\tilde{u}_1) = \max_{i \in 1:S} \text{VaR}_{\tilde{u}_1}[f_i(\tilde{u}_2)]$ with probability one. $\square$

Lemma B.2.6. *Let S non-decreasing functions $f_i: [0, 1] \rightarrow \bar{\mathbb{R}}, i \in 1:S$ with each $f_i(\alpha) \in \mathbb{R}$ for all $\alpha \in (0, 1)$. Then $\text{VaR}_\alpha[\max_{i \in 1:S} f_i(\tilde{u})] = \max_{i \in 1:S} \text{VaR}_\alpha[f_i(\tilde{u})]$, where $\tilde{u}$ is a uniform random variable on $[0, 1]$*

Proof. First, the claim trivially applies to $\alpha = 1$ since $\text{VaR}_1[\tilde{x}] = \infty$ for all random variables $\tilde{x} \in \mathbb{X}$. Thus, we focus on the case $\alpha \in [0, 1)$. Since $\alpha \mapsto \text{VaR}_\alpha[\cdot]$ is non-decreasing (see [57, Lem. A.19]), $\text{VaR}_\alpha[\max_{i \in 1:S} f_i(\tilde{u})] \geq \text{VaR}_\alpha[f_j(\tilde{u})]$ for all $j \in 1:S$, hence $\text{VaR}_\alpha[\max_{i \in 1:S} f_i(\tilde{u})] \geq \max_{i \in 1:S} \text{VaR}_\alpha[f_i(\tilde{u})]$. We are therefore left with showing that

$$\text{VaR}_\alpha[\max_{i \in 1:S} f_i(\tilde{u})] \leq \max_{i \in 1:S} \text{VaR}_\alpha[f_i(\tilde{u})]. \quad (\text{B.3})$$

We do so by contradiction. Assume that

$$\text{VaR}_\alpha[\max_{i \in 1:S} f_i(\tilde{u})] > \max_{i \in 1:S} \text{VaR}_\alpha[f_i(\tilde{u})] =: \nu^\star.$$

Applying Lemma B.2.5, $\max_{i \in 1:S} f_i(\tilde{u}) = \max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)]$ almost surely, so we must have $\nu^\star < \text{VaR}_\alpha[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)]]$ where $\tilde{u}_2$ is uniformly distributed on $[0, 1]$. By definition of $\text{VaR}_\alpha[\cdot]$ (Eq. (4.1)), this implies that there exists $\epsilon > 0$ such that:

$$\mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^\star + \epsilon \right] \leq \alpha.$$

Since $\nu^\star = \max_{i \in 1:S} \text{VaR}_\alpha[f_i(\tilde{u})]$ and $\alpha \mapsto \text{VaR}_\alpha[f_i(\tilde{u})]$ is non-decreasing, we must have

$$\max_{i \in 1:S} \text{VaR}_{\alpha'}[f_i(\tilde{u})] \leq \nu^\star < \nu^\star + \epsilon, \quad (\text{B.4})$$

for all $\alpha' \leq \alpha$. In addition, by the law of total probability:

$$\begin{aligned}
\alpha &\geq \mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^* + \epsilon \right] \\
&= \mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^* + \epsilon \mid \tilde{u} > \alpha \right] \mathbb{P}[\tilde{u} > \alpha] \\
&\quad + \mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^* + \epsilon \mid \tilde{u} \leq \alpha \right] \mathbb{P}[\tilde{u} \leq \alpha] \\
&= (1 - \alpha) \mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^* + \epsilon \mid \tilde{u} > \alpha \right] + \alpha \mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^* + \epsilon \mid \tilde{u} \leq \alpha \right] \\
&= (1 - \alpha) \mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^* + \epsilon \mid \tilde{u} > \alpha \right] + \alpha,
\end{aligned}$$

so necessarily:

$$\mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^* + \epsilon \mid \tilde{u} > \alpha \right] = 0.$$

Yet, since $\max_{i \in 1:S} \text{VaR}_{\alpha'}[f_i(\tilde{u}_2)]$ is right-continuous, non-decreasing in α' and evaluates at ν^* for $\alpha' = \alpha$, there must be a $\delta > 0$ such that $\max_{i \in 1:S} \text{VaR}_{\alpha'}[f_i(\tilde{u}_2)] \leq \nu^* + \epsilon$ for all $\alpha' \leq \alpha + \delta$. This leads to a contradiction since:

$$\begin{aligned}
0 < \delta &= \mathbb{P}[\tilde{u} \in (\alpha, \alpha + \delta)] \\
&= \mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^* + \epsilon \mid \tilde{u} \in (\alpha, \alpha + \delta) \right] \mathbb{P}[\tilde{u} \in (\alpha, \alpha + \delta)] \\
&= \mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^* + \epsilon, \tilde{u} \in (\alpha, \alpha + \delta) \right] \\
&\leq \mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^* + \epsilon, \tilde{u} > \alpha \right] \\
&= \mathbb{P} \left[\max_{i \in 1:S} \text{VaR}_{\tilde{u}}[f_i(\tilde{u}_2)] < \nu^* + \epsilon \mid \tilde{u} > \alpha \right] (1 - \alpha) = 0.
\end{aligned}$$

□

We are now ready to prove Theorem 4.3.2, whose statement is recalled below.

Theorem. *Let a sequence $q^u = (q_t^u)_{t=0}^T$ be such that $q_0^u(s, \alpha, a) = \text{VaR}_\alpha[0]$ and $q_{t+1}^u(s, \alpha, a) := B_u q_t^u(s, \alpha, a)$ for $t \in [T - 1]$. Then, $q_t^u = q_t^*$ for all $t \in [T]$, where q_t^* is defined in Eq. (4.4).*

Moreover, if a policy $\pi^u = (\pi_k^u)_{k=0}^{T-1}$ is greedy for q^u as in Eq. (4.9), then it maximizes the value-at-risk objective (4.3).

Proof. We start with demonstrating that $q_t^* = q_t^u$, $\forall t \in [T]$, recursively with mathematical induction. Assuming that $q_t^u = q_t^*$ we want to show that $q_{t+1}^u := B_u q_t^u = q_{t+1}^*$ as written in Eq. (4.7). We then prove the optimality of π^* constructed using $\hat{\alpha}^u$.

Step 1: For all $s \in \mathcal{S}$, $\alpha \in [0, 1]$ and $a \in \mathcal{A}$, $q_0^*(s, \alpha, a) = q_0^u(s, \alpha, a) = \text{VaR}_\alpha[0]$ so the base case holds. Assume that $q_t^u = q_t^*$ for some $t \in [T - 1]$. Then, one can derive:

$$\begin{aligned}
q_{t+1}^*(s, \alpha, a) &= r(s, a) + \gamma \cdot \max_{o \in \mathcal{O}_{sa}(\alpha)} \min_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} q_t^*(s', o_{s'}, a') \quad [\text{By Theorem 4.3.1}] \\
&= r(s, a) + \gamma \cdot \max_{o \in \mathcal{O}_{sa}(\alpha)} \min_{s' \in \mathcal{S}} v_t^*(s', o_{s'}) \quad [\text{By Proposition B.1.1}] \\
&= r(s, a) + \gamma \cdot \max_{o \in \mathcal{O}_{sa}(\alpha)} \min_{s' \in \mathcal{S}} \text{VaR}_{o_{s'}}[v_t^*(s', \tilde{u})] \quad [\text{By Corollary B.2.3}] \\
&= r(s, a) + \gamma \cdot \text{VaR}_\alpha^{a,s}[v_t^*(\tilde{s}_1, \tilde{u})] \quad [\text{By Lemma B.2.4}] \\
&= \text{VaR}_\alpha^{a,s}[r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q_t^*(\tilde{s}_1, \tilde{u}, a')] \quad [\text{By Proposition B.1.1}] \\
&= \text{VaR}_\alpha^{a,s}[r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q_t^u(\tilde{s}_1, \tilde{u}, a')] \quad [\text{By inductive assumption}] \\
&= B_u q_t^u(s, \alpha, a) = q_{t+1}^u(s, \alpha, a) \quad [\text{By Eq. (4.7)}].
\end{aligned}$$

This confirms that q^* satisfies $q_{t+1}^* = B_u q_t^* = B_u q_t^u = q_{t+1}^u$ for all $t \in [T - 1]$, so that $q^* = q^u$.

Step 2: We now show that $\hat{\alpha}_k^u(\cdot)$ constructed according to:

$$\hat{\alpha}_{k+1}^u(h_{k+1}) := \min \left\{ o \in [0, 1] \mid \max_{a \in \mathcal{A}} q_{T-k-1}^u(s_{k+1}, o, a) \geq \frac{q_{T-k}^u(s_k, \hat{\alpha}_k^u(h_k), a_k) - r(s_k, a_k)}{\gamma} \right\}$$

defines an optimal policy. Namely,

$$\hat{\alpha}_{k+1}^u(h_{k+1}) := \min \left\{ o \in [0, 1] \mid \max_{a \in \mathcal{A}} q_{T-k-1}^u(s_{k+1}, o, a) \geq \frac{q_{T-k}^u(s_k, \hat{\alpha}_k^u(h_k), a_k) - r(s_k, a_k)}{\gamma} \right\}.$$

Based on what has been shown in Step 1, we can interchangeably write q_{T-k-1}^u or q_{T-k-1}^* in the

construction of $\hat{\alpha}_k^u, k \in [T-1]$, so that

$$\hat{\alpha}_{k+1}^u(h_{k+1}) := \min \left\{ o \in [0, 1] \mid \max_{a \in \mathcal{A}} q_{T-k-1}^*(s_{k+1}, o, a) \geq \frac{q_{T-k}^*(s_k, \hat{\alpha}_k^u(h_k), a_k) - r(s_k, a_k)}{\gamma} \right\}.$$

By Lemma B.2.1, $\alpha \mapsto v_t^*(s, \alpha)$ is right-continuous and non-decreasing so the minimum above is well-defined. We are left to check that $\hat{\alpha}_k^u(\cdot)$ leads to an associated $\hat{\delta}^{ka}(h_k) \in \mathcal{O}_{sa}(\hat{\alpha}_k^u(h_k))$ satisfying:

$$q_{T-k}(s, \alpha_k(h_k), a) = r(s, a) + \gamma \cdot \min_{s' \in \mathcal{S}} \max_{a' \in \mathcal{A}} q_{T-k-1}(s', \alpha_{k+1}(\langle h_k, a, s' \rangle), a') \quad (\text{B.5})$$

This can be done in two steps.

We can first show that for all $h_k \in \mathcal{H}_k, a \in \mathcal{A}$, the vector $\hat{\delta}^k \in \mathbb{R}^{\mathcal{S}}$ composed of $\hat{\delta}_{s'}^k := \hat{\alpha}_{k+1}^u(\langle h_k, a, s' \rangle)$ for all $s' \in \mathcal{S}$, is in $\mathcal{O}_{sa}(\hat{\alpha}_k^u(h_k))$. To do so, we make use of the fact that for all t the maximum over $\max_{o \in \mathcal{O}_{sa}}$ in (4.5) is achieved thus implying that it is achieved at $T-k$ by some $\hat{\delta}^{k*}$ when $s = s_k$ and $\alpha = \hat{\alpha}_k^u(h_k)$. Namely, $\hat{\delta}^{k*}$ satisfies:

$$q_{T-k}^*(s_k, \hat{\alpha}_k^u(h_k), a) = \min_{s' \in \mathcal{S}} r(s_k, a) + \gamma v_{T-k-1}^*(s', \hat{\delta}_{s'}^{k*}) \leq r(s_k, a) + \gamma v_{T-k-1}^*(s', \hat{\delta}_{s'}^{k*}), \forall s' \in \mathcal{S}$$

where we replaced $\max_{a' \in \mathcal{A}} q_{T-k-1}^*(s', \hat{\delta}_{s'}^{k*}, a')$ with $v_{T-k-1}^*(s', \hat{\delta}_{s'}^{k*})$. We can therefore easily conclude that:

$$\sum_{s' \in \mathcal{S}} \hat{\delta}_{s'}^k p(s_k, a, s') = \sum_{s' \in \mathcal{S}} \hat{\alpha}_{k+1}^u(\langle h_k, a, s' \rangle) p(s_k, a, s') \sum_{s' \in \mathcal{S}} \hat{\delta}_{s'}^{k*} p(s_k, a, s') \leq \hat{\alpha}_k^u(h_k),$$

which establishes that $\hat{\delta}^k \in \mathcal{O}_{sa}(\hat{\alpha}_k^u(h_k))$.

We can then show that equation (B.5), with q_{T-k} and q_{T-k-1} respectively replaced by their equivalent q_{T-k}^* and q_{T-k-1}^* (based on Theorem 4.3.1), is satisfied based on:

$$q_{T-k}^*(s, \hat{\alpha}_k^u(h_k), a) = \max_{o \in \mathcal{O}_{sa}(\hat{\alpha}_k^u(h_k))} \min_{s' \in \mathcal{S}} \left(r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q_{T-k-1}^*(s', o_{s'}, a') \right) \quad [\text{By Eq. (4.5)}]$$

$$\begin{aligned}
&\geq \min_{s' \in \mathcal{S}} \left(r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q_{T-k-1}^*(s', \hat{\alpha}_{k+1}^u(\langle h_k, a, s' \rangle), a') \right) \quad [\hat{o}^k \in \mathcal{O}_{sa}(\hat{\alpha}_k^u(h_k))] \\
&= \min_{s' \in \mathcal{S}} \left(r(s, a) + \gamma \cdot v_{T-k-1}^*(s', \hat{\alpha}_{k+1}^u(\langle h_k, a, s' \rangle)) \right) \quad [\text{By Proposition B.1.1}] \\
&\geq \min_{s' \in \mathcal{S}} \left(r(s, a) + \gamma \cdot \frac{q_{T-k}^*(s, \hat{\alpha}_k^u(h_k), a) - r(s, a)}{\gamma} \right) \quad [\text{Definition of } \hat{\alpha}_k^u] \\
&= \min_{s' \in \mathcal{S}} q_{T-k}^*(s, \hat{\alpha}_k^u(h_k), a) = q_{T-k}^*(s, \hat{\alpha}_k^u(h_k), a),
\end{aligned}$$

The above inequalities must therefore be equalities, so equation (B.5) holds for $\hat{\alpha}_k^u(h_k)$. As a result, the policy π^* constructed using $\hat{\alpha}^u$ is optimal. $\square$

B.3 Proofs of Section 4.4

Lemma B.3.1. *For any monetary risk measure $\varrho: \mathbb{X} \rightarrow \mathbb{R}$ where $\mathbb{X}$ is defined in a finite outcome space $\omega \in \Omega$, it holds that:*

$$|\varrho(\tilde{x}) - \varrho(\tilde{y})| \leq \max_{\omega \in \Omega} |\tilde{x}(\omega) - \tilde{y}(\omega)|.$$

Proof. Define $\epsilon := \max_{\omega \in \Omega} |\tilde{x}(\omega) - \tilde{y}(\omega)| \geq 0$. We prove that $\varrho(\tilde{x}) - \varrho(\tilde{y}) \leq \epsilon$. The second inequality $\varrho(\tilde{y}) - \varrho(\tilde{x}) \leq \epsilon$ follows analogously. Let $\tilde{z} := \max\{\tilde{x}, \tilde{y}\}$. Then, for all $\omega \in \Omega$:

$$\begin{aligned}
\tilde{x}(\omega) &\leq \max\{\tilde{x}(\omega), \tilde{y}(\omega)\} \\
&= \tilde{z}(\omega) \\
&= \tilde{y}(\omega) + \max\{\tilde{x}(\omega) - \tilde{y}(\omega), 0\} \\
&\leq \tilde{y}(\omega) + |\tilde{x}(\omega) - \tilde{y}(\omega)| \quad [\tilde{x}(\omega) - \tilde{y}(\omega) \leq |\tilde{x}(\omega) - \tilde{y}(\omega)|, 0 \leq |\tilde{x}(\omega) - \tilde{y}(\omega)|] \\
&\leq \tilde{y}(\omega) + \max_{\omega' \in \Omega} |\tilde{x}(\omega') - \tilde{y}(\omega')| \\
&= \tilde{y}(\omega) + \epsilon.
\end{aligned}$$

As a result, $\tilde{x} \leq \tilde{z} \leq \tilde{y} + \epsilon$. Since ϱ is a monetary risk measure, it is monotonous and translation invariant, so that

$$\varrho(\tilde{x}) \leq \varrho(\tilde{z}) \leq \varrho(\tilde{y} + \epsilon) = \varrho(\tilde{y}) + \epsilon,$$

and $\varrho(\tilde{x}) - \varrho(\tilde{y}) \leq \epsilon$. □

B.3.1 Proof of Lemma 4.4.2

Proof. The inner inequalities follow from

$$\begin{aligned}
\mathbf{q}_{\underline{f}(\alpha)}^+(\tilde{x}) &\leq \mathbf{q}_{\alpha}^+(\tilde{x}) && \text{[By monotonicity]} \\
&= \text{VaR}_{\alpha}[\tilde{x}] \\
&= \max \left\{ \tau \in \bar{\mathbb{R}} \mid \mathbb{P}[\tilde{x} < \tau] \leq \alpha \right\} \\
&\leq \sup \left\{ \tau \in \bar{\mathbb{R}} \mid \mathbb{P}[\tilde{x} < \tau] < \bar{f}(\alpha) \right\} \\
&= \mathbf{q}_{\bar{f}(\alpha)}^-(\tilde{x}). && \text{[[57, Appx. A.3, Def. A.24]]}
\end{aligned}$$

The outer inequalities follow from Lemma 4.2.1. □

B.3.2 Proof of Theorem 4.4.3

Proof. The proof is broken down into two parts. First, we address the bounding on q^* , then we demonstrate the stated properties of our constructed policy.

Step 1: Upper and lower bound on q^* . We prove by induction on $t \in [T]$ that for all $(s, \alpha, a) \in \mathcal{S} \times (0, 1) \times \mathcal{A}$:

$$t\bar{R} \leq \underline{q}_t^u(s, \alpha, a) \leq q_t^*(s, \alpha, a) \leq \bar{q}_t^u(s, \alpha, a) \leq t\bar{R},$$

or more succinctly, that $t\bar{R} \leq \underline{q}_t^u \leq q_t^* \leq \bar{q}_t^u \leq t\bar{R}$. At $t = 0$, the bounds are obtained by definition given that $\bar{q}_0^u = \underline{q}_0^u = q_0^* = 0$. Assume that the statement holds for some $t \in [T - 1]$ and let's check if the proposition is preserved at $t + 1$.

Case 1: $\bar{f}(\alpha) < 1$. For all $\alpha \in (0, 1)$ such that $\bar{f}(\alpha) < 1$, we have:

$$q_{t+1}^*(s, \alpha, a) = \text{VaR}_{\alpha}^{a,s}[r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q_t^*(\tilde{s}_1, \tilde{u}, a')] \quad \text{[By Theorem 4.3.2]}$$

$$\begin{aligned}
&= \mathbf{q}_\alpha^{+,a,s}[r(s,a) + \gamma \cdot \max_{a' \in \mathcal{A}} q_t^*(\tilde{s}_1, \tilde{u}, a')] && [\text{By definition of } \text{VaR}_\alpha] \\
&\leq \mathbf{q}_\alpha^{+,a,s}[r(s,a) + \gamma \cdot \max_{a' \in \mathcal{A}} \bar{q}_t^u(\tilde{s}_1, \tilde{u}, a')] && [\text{By inductive assumption}] \\
&\leq \mathbf{q}_{\bar{f}(\alpha)}^{-,a,s}[r(s,a) + \gamma \cdot \max_{a' \in \mathcal{A}} \bar{q}_t^u(\tilde{s}_1, \tilde{u}, a')] && [\text{By Lemma 4.4.2}]
\end{aligned}$$

Finally, we exploit the elicibility of $\mathbf{q}_{\bar{f}(\alpha)}$ (see Lemma 4.2.1), due to the random variable $r(s,a) + \gamma \cdot \max_{a' \in \mathcal{A}} \bar{q}_t^u(s', \tilde{u}, a')$ being supported on the interval $[(t+1)\underline{R}, (t+1)\bar{R}]$, to obtain:

$$\begin{aligned}
q_{t+1}^*(s, \alpha, a) &\leq \mathbf{q}_{\bar{f}(\alpha)}^{-,a,s}[r(s,a) + \gamma \cdot \max_{a' \in \mathcal{A}} \bar{q}_t^u(\tilde{s}_1, \tilde{u}, a')] \\
&= \min \arg \min_q \mathbb{E}^{a,s} \left[\ell_{\bar{f}(\alpha)} \left(r(s,a) + \gamma \cdot \max_{a' \in \mathcal{A}} \bar{q}_t^u(\tilde{s}_1, \tilde{u}, a') - q \right) \right] \\
&= \min \left\{ (\mathcal{B}_{\mathbf{u}}^{\bar{f}} \bar{q}_t^u)(s, \alpha, a) \right\} \\
&\leq \bar{q}_{t+1}^u(s, \alpha, a),
\end{aligned}$$

where the last inequality is due to the fact that $\bar{q}_{t+1}^u \in (\mathcal{B}_{\mathbf{u}}^{\bar{f}} \bar{q}_t^u)$ by construction:

$$\bar{q}_{t+1}^u \in (\mathcal{B}_{\mathbf{u}}^{\bar{f}} \bar{q}_t^u), \quad \underline{q}_{t+1}^u \in (\mathcal{B}_{\mathbf{u}}^f \underline{q}_t^u), \quad \forall t \in [T-1]. \quad (\text{B.6})$$

Moreover, we have that:

$$\arg \min_q \mathbb{E}^{a,s} \left[\ell_{\bar{f}(\alpha)} \left(r(s,a) + \gamma \cdot \max_{a' \in \mathcal{A}} \bar{q}_t^u(\tilde{s}_1, \tilde{u}, a') - q \right) \right] \subset (-\infty, (t+1)\bar{R}]$$

since it is a quantile of $r(s,a) + \gamma \cdot \max_{a' \in \mathcal{A}} \bar{q}_t^u(s', \tilde{u}, a') \leq (t+1)\bar{R}$. This confirms the statement for $\bar{f}(\alpha) < 1$. *Case 2:* $\bar{f}(\alpha) = 1$. We instead rely on the following inequalities:

$$\begin{aligned}
q_{t+1}^*(s, \alpha, a) &= \max_{\pi \in \Pi_{\text{HR}}} \text{VaR}_\alpha^{(a, \pi_{1:t}), s} \left[\sum_{k=0}^t \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] \\
&\leq \sum_{k=0}^t \gamma^k \bar{R} \leq (t+1)\bar{R} = \bar{R} + \max_{s \in \mathcal{S}, a' \in \mathcal{A}} \bar{q}_t^u(s, 1, a') = \bar{q}_{t+1}^u(s, \alpha, a).
\end{aligned}$$

A similar set of arguments applies to the lower bound. Namely,

$$\begin{aligned}
q_t^*(s, \alpha, a) &= \mathbf{q}_\alpha^{+,a,s}[r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q_{t-1}^*(\tilde{s}_1, \tilde{u}, a')] && [\text{By Theorem 4.3.2}] \\
&\geq \mathbf{q}_\alpha^{+,a,s}[r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} \underline{q}_{t-1}^u(\tilde{s}_1, \tilde{u}, a')] && [\text{By inductive assumption}] \\
&\geq \mathbf{q}_{\underline{f}(\alpha)}^{+,a,s}[r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} \underline{q}_{t-1}^u(\tilde{s}_1, \tilde{u}, a')] && [\text{By Lemma 4.4.2}],
\end{aligned}$$

Similarly, we exploit the elicibility and the fact that $\underline{q}_{t+1}^u \in (\mathcal{B}_{\underline{u}}^f \underline{q}_t^u)$ to write

$$\begin{aligned}
q_{t+1}^*(s, \alpha, a) &\geq \mathbf{q}_{\underline{f}(\alpha)}^{+,a,s}[r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} \underline{q}_t^u(\tilde{s}_1, \tilde{u}, a')] \\
&= \max \arg \min_q \mathbb{E}^{a,s} \left[\ell_{\underline{f}(\alpha)} \left(r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} \underline{q}_t^u(\tilde{s}_1, \tilde{u}, a') - q \right) \right] \\
&= \max \left\{ (\mathcal{B}_{\underline{u}}^f \underline{q}_t^u)(s, \alpha, a) \right\} \\
&\geq \underline{q}_{t+1}^u(s, \alpha, a),
\end{aligned}$$

where the difference lies in the second inequality, i.e. one needs to exploit the monotonicity of $\alpha \mapsto \mathbf{q}_\alpha^+(\tilde{x})$. The case where $\underline{f}(\alpha) = 0$ also follows naturally:

$$q_{t+1}^*(s, \alpha, a) = \max_{\pi \in \Pi_{\text{HR}}} \text{Var}_\alpha^{(a, \pi_{1:t}), s} \left[\sum_{k=0}^t \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] \geq \sum_{k=0}^t \gamma^k \underline{R} \geq (t+1) \underline{R} = \underline{q}_{t+1}^u(s, \alpha, a).$$

This completes our inductive proposition: $t\underline{R} \leq \underline{q}_t^u \leq q_t^* \leq \bar{q}_t^u \leq t\bar{R}$.

Step 2: Bounds on the performance of $\underline{\pi}_k(h_k)$. We start by proving that

$$\max_{a \in \mathcal{A}} \underline{q}_T^u(s_0, \alpha_0, a) \leq \text{Var}_{\alpha_0}^{\underline{\pi}, s_0} \left[\sum_{k=0}^{T-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right].$$

The rest follows based on the bounding properties of $\bar{q}_T^u$ and $\underline{q}_T^u$.

First, we can confirm that $\underline{q}^k$ composed using $\underline{q}_{s'}^k := \underline{\alpha}_{k+1}^u(\langle h_k, a, s' \rangle)$ is in $\mathcal{O}_{sa}(\underline{\alpha}_k^u(h_k))$. To do so,

we exploit the fact that:

$$\begin{aligned}
q_{T-k}^u(s_k, \underline{\alpha}_k^u(h_k), a) &\stackrel{\text{B.6}}{\in} \arg \min_{q \in \mathbb{R}} \mathbb{E}^{a, s_k} \left[\ell_{f(\underline{\alpha}_k^u(h_k))} \left(r(s_k, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q_{T-k-1}^u(\tilde{s}_1, \tilde{u}, a') - q \right) \right] \\
&\stackrel{4.4.2}{\leq} \text{VaR}_{\underline{f}(\underline{\alpha}_k^u(h_k))}^{a, s_k} [r(s_k, a) + \gamma \max_{a' \in \mathcal{A}} q_{T-k-1}^u(\tilde{s}_1, \tilde{u}_1, a')] \\
&\stackrel{4.4.2}{\leq} \text{VaR}_{\underline{\alpha}_k^u(h_k)}^{a, s_k} [r(s_k, a) + \gamma \max_{a' \in \mathcal{A}} q_{T-k-1}^u(\tilde{s}_1, \tilde{u}_1, a')] \\
&\stackrel{\text{B.2.4}}{=} \max_{o \in \mathcal{O}_{s_k a}(\underline{\alpha}_k^u(h_k))} \min_{s' \in \mathcal{S}} r(s_k, a) + \gamma \text{VaR}_{o_{s'}} [\max_{a' \in \mathcal{A}} q_{T-k-1}^u(s', \tilde{u}_1, a')] \\
&= \min_{s' \in \mathcal{S}} r(s_k, a) + \gamma \text{VaR}_{o_{s'}^*} [\max_{a' \in \mathcal{A}} q_{T-k-1}^u(s', \tilde{u}_1, a')] \\
&\stackrel{\text{B.2.6}}{=} \min_{s' \in \mathcal{S}} r(s_k, a) + \gamma \max_{a' \in \mathcal{A}} \text{VaR}_{o_{s'}^*} [q_{T-k-1}^u(s', \tilde{u}_1, a')] \\
&\stackrel{\text{B.2.2}}{=} \min_{s' \in \mathcal{S}} r(s_k, a) + \gamma \max_{a' \in \mathcal{A}} q_{T-k-1}^u(s', o_{s'}^*, a') \\
&\leq r(s_k, a) + \gamma \max_{a' \in \mathcal{A}} q_{T-k-1}^u(s', o_{s'}^*, a') \quad , \forall s' \in \mathcal{S}
\end{aligned}$$

for optimal $o^* \in \mathcal{O}_{s_k a}(\underline{\alpha}_k^u(h_k))$. This implies that

$$\max_{a' \in \mathcal{A}} q_{T-k-1}^u(s', o_{s'}^*, a') \geq \frac{q_{T-k}^u(s_k, \underline{\alpha}_k^u(h_k), a) - r(s_k, a)}{\gamma}$$

By construction of $\underline{o}^k$, we have that $\underline{o}_{s'}^k \leq o_{s'}^*$, hence

$$\sum_{s' \in \mathcal{S}} \underline{o}_{s'}^k p(s_k, a, s') \leq \sum_{s' \in \mathcal{S}} o_{s'}^* p(s_k, a, s') \leq \underline{\alpha}_k^u(h_k).$$

Let us now construct a series of policies $\underline{\pi}_k^t$ defined as:

$$\underline{\pi}_k^t(h_k) \in \arg \max_{a \in \mathcal{A}} q_{t-k}^u(s_k, \underline{\alpha}_k^{u,t}(h_k), a),$$

where $\underline{\alpha}_0^{u,t-1}(s') := \underline{\alpha}_1^{u,t}(\langle s, a, s' \rangle)$ and $\underline{\pi}_{k-1}^{t-1}(h_{1:k}) := \underline{\pi}_k^t(h_k)$ are defined without loss of generality for all k, α, t, h_k . The superscript t represents the horizon of the sub-problem, we may omit the superscript when it represents the final objective decision horizon T as $\underline{\pi} = \underline{\pi}^T$ and $\underline{\alpha}^u = \underline{\alpha}^{u,T}$.

Letting

$$v_t^\pi(s, \alpha) := \text{VaR}_\alpha^{\pi, s} \left[\sum_{k=0}^{t-1} r(\tilde{s}_k, \tilde{a}_k) \right] \quad (\text{B.7})$$

and $\underline{v}_t^u(s, \alpha) := \max_{a \in \mathcal{A}} \underline{q}_t^u(s, \alpha, a)$, we now prove by induction on t that $v_t^\pi(s, \alpha) \geq \underline{v}_t^u(s, \alpha) \forall t \in [T]$. This is obviously the case at $t = 0$ since $v_0^\pi(s, \alpha) = 0 = \underline{q}_0^u(s, \alpha, a)$ for all $s, a, \alpha \in (0, 1)$, and π . Now assuming that for some $t \in 1 : T$, $v_{t-1}^{\pi^{t-1}}(s, \alpha) \geq \underline{v}_{t-1}^u(s, \alpha)$, letting $\underline{\alpha}_0^{u, T}(s) = \alpha$ and $a = \underline{\pi}_0^t(s)$ for $v_t^\pi(s, \alpha)$ we obtain:

$$\begin{aligned} v_t^{\pi^t}(s, \alpha) &\stackrel{(\text{B.7})}{=} \text{VaR}_\alpha^{\pi^t, s} \left(\sum_{k=0}^{t-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right) \\ &\stackrel{\text{B.2.4}}{=} \max_{o \in \mathcal{O}_{sa}(\alpha)} \min_{s' \in \mathcal{S}} r(s, a) + \gamma \text{VaR}_{o_{s'}}^s \left[\sum_{k=1}^{t-1} \gamma^{k-1} r(\tilde{s}_k, \pi_k^t(\tilde{h}_k) | \tilde{a}_0 = a, \tilde{s}_1 = s') \right] \end{aligned}$$

Let's rewrite the second term as follows

$$\begin{aligned} &\text{VaR}_{o_{s'}}^s \left[\sum_{k=1}^{t-1} \gamma^{k-1} r(\tilde{s}_k, \underline{\pi}_k^t(\tilde{h}_k)) | \tilde{a}_0 = a, \tilde{s}_1 = s' \right] \\ &= \text{VaR}_{o_{s'}}^s \left[\sum_{k=1}^{t-1} \gamma^{k-1} r(\tilde{s}_k, \underline{\pi}_{k-1}^{t-1}(h_{1:k} | \underline{\alpha}_0^{u, t-1}(\tilde{s}_1) = \underline{\alpha}_1^{u, t}(\langle s, \tilde{a}_0, \tilde{s}_1 \rangle))) | \tilde{a}_0 = a, \tilde{s}_1 = s' \right] \\ &= \text{VaR}_{o_{s'}}^{s'} \left[\sum_{k'=0}^{t-2} \gamma^{k'} r(\tilde{s}_{k'}, \underline{\pi}_{k'}^{t-1}(h_{k'})) \right] \\ &= \text{VaR}_{o_{s'}}^{\pi^{t-1}, s'} \left[\sum_{k'=0}^{t-2} \gamma^{k'} r(\tilde{s}_{k'}, \tilde{a}_{k'}) \right] \end{aligned}$$

Now we have

$$\begin{aligned} v_t^{\pi^t}(s, \alpha) &= \max_{o \in \mathcal{O}_{sa}(\alpha)} \min_{s' \in \mathcal{S}} r(s, a) + \text{VaR}_{o_{s'}}^{\pi^{t-1}, s'} \left[\sum_{k'=0}^{t-2} \gamma^{k'} r(\tilde{s}_{k'}, \tilde{a}_{k'}) \right] && [\text{From derivation above}] \\ &\geq \min_{s' \in \mathcal{S}} r(s, a) + \text{VaR}_{\underline{\alpha}_1^{u, t}(\langle s, a, s' \rangle)}^{\pi^{t-1}, s'} \left[\sum_{k'=0}^{t-2} \gamma^{k'} r(\tilde{s}_{k'}, \tilde{a}_{k'}) \right] && [\text{Property of max}] \\ &= \min_{s' \in \mathcal{S}} r(s, a) + \gamma \underline{v}_{t-1}^{\pi^{t-1}}(s', \underline{\alpha}_1^{u, t}(\langle s, a, s' \rangle)) && [\text{By Eq. (B.7)}] \\ &\geq \min_{s' \in \mathcal{S}} r(s, a) + \gamma \underline{v}_{t-1}^u(s', \underline{\alpha}_1^{u, t}(\langle s, a, s' \rangle)) && [\text{By Inductive assumption}] \end{aligned}$$

$$\geq \min_{s' \in \mathcal{S}} \underline{v}_t^u(s, \alpha) = \underline{v}_t^u(s, \alpha) \quad [\text{By Eq. (B.6)}].$$

We could conclude with induction that:

$$\text{VaR}_{\alpha_0}^{\pi, s_0} \left[\sum_{k=0}^{T-1} \gamma^k r(\tilde{s}_k, \tilde{a}_k) \right] = v_T^{\pi^T}(s_0, \alpha_0) \geq \underline{v}_T^u(s_0, \alpha_0) = \max_{a \in \mathcal{A}} \underline{q}_T^u(s_0, \alpha_0, a).$$

□

B.3.3 Proof of Proposition 4.4.5

Proof. This result follows by induction from showing that the constructed $\underline{q}$ satisfies the conditions identified in Theorem 4.4.3 under $\underline{f}$ as defined in Example 4.4.4. Naturally, the initial condition is satisfied:

$$\underline{q}_0(s, \alpha, a) = \underline{q}_0^d(s, J \cdot \underline{f}(\alpha), a) = 0.$$

To prove the inductive step for any $t \in 1:T-1$, we have that if $\underline{f}(\alpha) = 0$ then :

$$\underline{q}_{t+1}(s, \alpha, a) = \underline{q}_{t+1}^d(s, 0, a) = \underline{R} + \min_{s \in \mathcal{S}, a \in \mathcal{A}} \underline{q}_t^d(s, 0, a) = \underline{R} + \min_{s \in \mathcal{S}, a \in \mathcal{A}} \underline{q}_t(s, 0, a)$$

since $\underline{f}(0) = 0$. If $\underline{f}(\alpha) = j/J$ with $j \geq 1$, then

$$\begin{aligned} \underline{q}_{t+1}(s, \alpha, a) &= \underline{q}_{t+1}^d(s, J \cdot \underline{f}(\alpha), a) \\ &\in \arg \min_{x \in \mathbb{R}} \frac{1}{J} \sum_{j'=0}^{J-1} \mathbb{E}^{a,s} \left[\ell_{j/J} \left(r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} \underline{q}_t^d(\tilde{s}_1, j', a') - x \right) \right] \\ &= \arg \min_{x \in \mathbb{R}} \frac{1}{J} \sum_{j'=0}^{J-1} \mathbb{E}^{a,s} \left[\ell_{j/J} \left(r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} \underline{q}_t^u(\tilde{s}_1, j'/J, a') - x \right) \right] \\ &= \arg \min_{x \in \mathbb{R}} \mathbb{E}^{a,s} \left[\ell_{j/J} \left(r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} \underline{q}_t^u(\tilde{s}_1, \underline{f}(\tilde{u}), a') - x \right) \right] \\ &= \mathcal{B}_{\underline{u}}^f \underline{q}_t. \end{aligned}$$

Moreover, since $\underline{f}$ is right-continuous and non-decreasing $\underline{q}$ is right-continuous and non-decreasing by construction. $\square$

B.3.4 Why not Huber's Loss

A common differentiable function used in quantile regression is the *Huber's* quantile regression loss function and is commonly defined as (and in general differs from the Moreau envelope of the quantile loss function):

$$e_{\alpha}^{\kappa}(\delta) := \begin{cases} -(1 - \alpha)(\delta + \kappa) + \frac{1}{2}(1 - \alpha)\kappa & \text{if } \delta < -\kappa \\ (1 - \alpha) \left(\frac{\delta^2}{2\kappa} \right) & \text{if } \delta \in [-\kappa, 0] \\ \alpha \left(\frac{\delta^2}{2\kappa} \right) & \text{if } \delta \in (0, \kappa) \\ \alpha(\delta - \kappa) + \frac{1}{2}\alpha\kappa & \text{if } \delta > \kappa \end{cases} \quad (\text{B.8})$$

Although some definitions of Huber's loss are scaled by a positive constant compared with (B.8), such scaling does not affect the minimizer.

The next proposition demonstrates that it is not sufficient to consider the popular Huber's loss function in place of the quantile loss function. Suppose that one defines a risk measure as

$$\xi(\tilde{x}) := \arg \min_{m \in \mathbb{R}} \mathbb{E}[e_{\alpha}^{\kappa}(\tilde{x} - m)] \quad (\text{B.9})$$

where Huber's loss e_{α}^{κ} is defined in (B.8).

Proposition B.3.2. *The minimization problem in (B.9) may not have a unique solution.*

Proof. Consider a risk level of $\alpha = 0.5$, $\kappa = 0.5$, and random variable $\tilde{x}$ such that $\mathbb{P}[\tilde{x} = -1] = \mathbb{P}[\tilde{x} = 1] = 1/2$. Then the risk measure defined in (B.9) becomes

$$\xi(\tilde{x}) := \arg \min_{m \in \mathbb{R}} \mathbb{E}[e_{\alpha}^{\kappa}(\tilde{x} - m)] = \arg \min_{m \in \mathbb{R}} \mathbb{E} \left[e_{1/2}^{1/2}(\tilde{x} - m) \right].$$

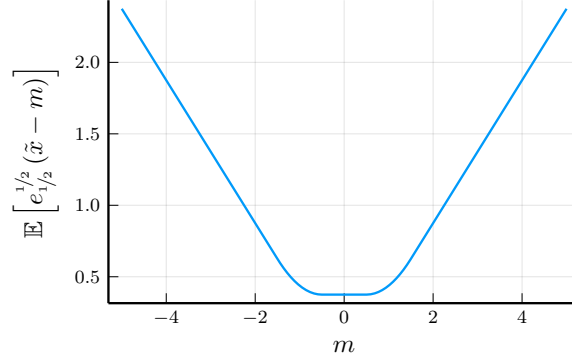


Figure B.1: A example used to prove the non-uniqueness of an optimal solution in Proposition B.3.2.

Simple algebraic manipulation shows that the minimization above does not have a unique optimal solution and instead $m \in [-1/2, 1/2]$ is optimal, as Fig. B.1 illustrates. $\square$

B.3.5 Proof of Lemma 4.4.6

Before proving Lemma 4.4.6, first define the notions it invokes.

Definition B.3.3. Given $L, \mu \in \mathbb{R}_{++}$, a differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$ is μ -strongly convex if

$$f(x) \geq f(y) + f'(y)(x - y) + \frac{1}{2}\mu(x - y)^2, \quad \forall x, y \in \mathbb{R}.$$

It has an L -Lipschitz continuous derivative if

$$|f'(x) - f'(y)| \leq L|x - y|, \quad \forall x, y \in \mathbb{R}.$$

We always have $L \geq \mu$ [91, Thm. 2.1.10].

The next lemma focuses on the properties of the loss function, after which we can prove Lemma 4.4.6.

Lemma B.3.4. Suppose that $\kappa \in (0, 1]$. Then the function ℓ_α^κ is continuously differentiable and satisfies for all $\delta_1, \delta_2 \in \mathbb{R}$ that

$$\min\{\alpha, 1 - \alpha\} \kappa \leq \frac{|\partial \ell_\alpha^\kappa(\delta_2) - \partial \ell_\alpha^\kappa(\delta_1)|}{|\delta_2 - \delta_1|} \leq \frac{\max\{\alpha, 1 - \alpha\}}{\kappa}. \quad (\text{B.10})$$

Moreover, the function ℓ_α^κ is strongly convex with constant $\min\{\alpha, 1 - \alpha\} \kappa$ and has a Lipschitz-continuous derivative with constant $\max\{\alpha, 1 - \alpha\} \kappa^{-1}$.

Proof. The inequality in (B.10) follows from the fact that $\partial\ell_\alpha^\kappa$, derived in (4.12), is a piecewise linear function and $\kappa \leq 1$. Then, the function ℓ_α^κ is strongly convex from the strong monotonicity of its derivative in (4.12) by [106, Exercise 12.59]. The function ℓ_α^κ has a Lipschitz continuous derivative immediately by the upper bound in (B.10). $\square$

With Definition B.3.3 and Lemma B.3.4, we now ready prove Lemma 4.4.6.

Proof. By Lemma B.3.4 and [91, Lem. 2.1.6], we have that the objective in (B.12) is also strongly convex with constant μ given in the lemma. By Lemma 4.4.6 and the linearity of the expectation operator, the derivative of (B.12) is also Lipschitz continuous with constant L given in the lemma. $\square$

B.3.6 Proof of Theorem 4.4.8

B.3.6.1 Properties of Soft-Quantile Measure

Definition B.3.5. A risk measure $\hat{q}_\alpha^\kappa$ is a *shortfall risk measure* (see [56, 129]) defined as:

$$\hat{q}_\alpha^\kappa(\tilde{x}) := \sup\{m \in \mathbb{R} : \mathbb{E}[\partial\ell_\alpha^\kappa(\tilde{x} - m)] \geq 0\}, \quad (\text{B.11})$$

where $\partial\ell_\alpha^\kappa(\cdot)$ is defined in (4.12).

Lemma B.3.6. *The measure $\hat{q}_\alpha^\kappa$ satisfies monotonicity, and translation invariance, and is elicitable as:*

$$\hat{q}_\alpha^\kappa(\tilde{x}) := \arg \min_{m \in \mathbb{R}} \mathbb{E}[\ell_\alpha^\kappa(\tilde{x} - m)] \quad (\text{B.12})$$

with ℓ_α^κ as defined in (4.11).

Proof. Monotonicity and translation invariance follow naturally from the properties of shortfall risk measures as defined in [56, 129], after confirming that $\partial\ell_\alpha^\kappa(\delta)$ is increasing and non-constant.

Elicitability follows from Theorem 4.3 in [15] after confirming that $\partial \ell_\alpha^\kappa(\delta)$ is strictly increasing and left continuous. $\square$

B.3.6.2 Standard Operator Convergence Results

Our convergence analysis follows the framework presented in Section 4 of [19], which we summarize in this section. The Q-learning algorithm considers the following iteration for some random sequence $\tilde{z}_i: \Omega \rightarrow \mathbb{R}^\mathcal{N}$ where $\mathcal{N} = \{1, \dots, n\}$ defined as

$$\begin{aligned}\tilde{z}_{i+1}(b) &= (1 - \tilde{\theta}_i(b)) \cdot \tilde{z}_i(b) + \tilde{\theta}_i(b) \cdot ((H\tilde{z}_i)(b) + \tilde{\phi}_i(b)), & i = 0, 1, \dots, \\ &= \tilde{z}_i(b) + \tilde{\theta}_i(b) \cdot ((H\tilde{z}_i)(b) + \tilde{\phi}_i(b) - \tilde{z}_i(b)), & i = 0, 1, \dots,\end{aligned}\tag{B.13}$$

for all $b \in \mathcal{N}$, where $H: \mathbb{R}^\mathcal{N} \rightarrow \mathbb{R}^\mathcal{N}$ is some possibly non-linear operator, $\tilde{\theta}_i: \Omega \rightarrow \mathbb{R}_+$ is a step size, and $\tilde{\phi}_i: \Omega \rightarrow \mathbb{R}^\mathcal{N}$ is some random noise sequence. The *random* history $\mathcal{F}_i$ at iteration $i = 1, \dots$ is denoted by

$$\mathcal{F}_i = \left(\tilde{z}_0, \dots, \tilde{z}_i, \tilde{\phi}_0, \dots, \tilde{\phi}_{i-1}, \tilde{\theta}_0, \dots, \tilde{\theta}_i \right).$$

To study the convergence of the algorithm, we may also need to define a weighted maximum norm for $x \in \mathbb{R}^\mathcal{N}$ and weights $w \in \mathbb{R}_{++}^\mathcal{N}$, i.e. $w \in \mathbb{R}^\mathcal{N}$ with $w(b) > 0$ for all $b \in \mathcal{N}$, as

$$\|x\|_w := \max_{b \in \mathcal{N}} \frac{|x(b)|}{w(b)}.$$

The weighted maximum norm is useful when analyzing the convergence of non-discounted MDPs. Its importance is in the fact that a non-negative matrix with a sub-unit spectral radius is a contraction in the weighted norm, but may not be a contraction in the plain maximum norm.

Definition B.3.7 (Star-contraction). An operator $H: \mathbb{R}^\mathcal{N} \rightarrow \mathbb{R}^\mathcal{N}$ is a *weighted maximum norm star-contraction* if there exist $z^* \in \mathbb{R}^\mathcal{N}$, $w \in \mathbb{R}_{++}^\mathcal{N}$, $\chi \in [0, 1)$ such that

$$\|Hz - z^*\|_w \leq \chi \cdot \|z - z^*\|_w.$$

Note that the original name for star-contraction is pseudo-contraction. We use the term star-contraction because of its close resemblance to star-convexity.

The following assumption on the noise of the stochastic process will be needed to ensure the convergence of our algorithm.

Assumption B.3.8 (Assm. 4.3 [19]). The noise terms in (B.13) satisfy for each $i = 1, 2, \dots$ that:

- (1) Random errors are conditionally unbiased, a.s.:

$$\mathbb{E}[\tilde{\phi}_i(b) \mid \mathcal{F}_i] = 0, \quad \forall b \in 1:n,$$

- (2) There exists a norm $\|\cdot\|$ on $\mathbb{R}^{\mathcal{N}}$ and $c, g \in \mathbb{R}$ such that

$$\mathbb{E}[\tilde{\phi}_i(b)^2 \mid \mathcal{F}_i] \leq c + g \cdot \|\tilde{z}_i\|^2, \quad \forall b \in 1:n, \quad \text{a.s.}$$

Proposition B.3.9. [Proposition 4.4 [19]] Let $\tilde{z}_i, i = 1, \dots$ be the sequence generated by the iteration in (B.13). Assume that

1. The step-sizes $\tilde{\theta}_i, \forall i = 1, \dots$ satisfy almost surely that $\tilde{\theta}_i \geq 0$ and

$$\sum_{t=0}^{\infty} \tilde{\theta}_i(b) = \infty, \quad \sum_{i=0}^{\infty} \tilde{\theta}_i^2(b) < \infty, \quad \forall b \in \mathcal{N}.$$

2. The noise terms $\tilde{\phi}_i, i = 1, \dots$ satisfy Assumption B.3.8.

3. The operator H in (B.13) is a weighted maximum norm star-contraction as in Definition B.3.7.

Then, $\tilde{z}_i$ converges to z^* , a fixed point of H , with probability 1:

$$\mathbb{P} \left[\lim_{i \rightarrow \infty} \tilde{z}_i = z^* \right] = 1.$$

B.3.6.3 Operator Definitions

We will need the following operators for any $\xi > 0$ and $\tilde{j}' \sim U([J-1])$. Let $b = (t, s, j, a)$ for $t \in [T], s \in \mathcal{S}, j \in [J-1], a \in \mathcal{A}$:

$$\begin{aligned}
(Gq)(b) &:= \begin{cases} q(b) - \underline{R} \cdot t & \text{if } j = 0 \vee t = 0, \\ \left. \frac{d}{dx} \mathbb{E}^{a,s} \left[\ell_{j/J}^\kappa \left(r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q(t-1, \tilde{s}_1, \tilde{j}', a') - x \right) \right] \right|_{x=q(t,s,j,a)} & \text{otherwise,} \end{cases} \\
&= \begin{cases} q(b) - \underline{R} \cdot t & \text{if } j = 0 \vee t = 0, \\ -\mathbb{E}^{a,s} \left[\partial \ell_{\frac{j}{J}}^\kappa \left(r(s, a) + \gamma \max_{a' \in \mathcal{A}} q(t-1, \tilde{s}_1, \tilde{j}', a') - q(t, s, j, a) \right) \right] & \text{otherwise,} \end{cases} \\
(G_{s'}q)(b) &:= \begin{cases} q(b) - \underline{R} \cdot t & \text{if } j = 0 \vee t = 0, \\ -\mathbb{E} \left[\partial \ell_{\frac{j}{J}}^\kappa \left(r(s, a) + \gamma \max_{a' \in \mathcal{A}} q(t-1, s', \tilde{j}', a') - q(t, s, j, a) \right) \right] & \text{otherwise,} \end{cases} \\
Hq &:= q - \xi \cdot Gq, \\
H_{s'}q &:= q - \xi \cdot G_{s'}q.
\end{aligned} \tag{B.14}$$

Consider a random sequence of inputs $((\tilde{t}_i, \tilde{s}_i, \tilde{j}_i, \tilde{a}_i, \tilde{s}'_i), \tilde{\beta}_i, \tilde{q}_i)_{i=0}^\infty$ in Algorithm 5. We can define a real-valued random variable $\tilde{\phi}$ for $t \in [T], s \in \mathcal{S}, j \in [J-1], a \in \mathcal{A}$ as

$$\begin{aligned}
\tilde{\phi}_i(t, s, j, a) &:= \begin{cases} (H_{\tilde{s}'_i} \tilde{q}_i^d)(t, s, j, a) - (H \tilde{q}_i^d)(t, s, j, a) & \text{if } (\tilde{t}_i, \tilde{s}_i, \tilde{j}_i, \tilde{a}_i) = (t, s, j, a), \\ 0 & \text{otherwise.} \end{cases} \\
\tilde{\theta}_i(t, s, j, a) &:= \begin{cases} \frac{\tilde{\beta}_i}{\xi} & \text{if } (\tilde{t}_i, \tilde{s}_i, \tilde{j}_i, \tilde{a}_i) = (t, s, j, a), \\ 0 & \text{otherwise.} \end{cases}
\end{aligned} \tag{B.15}$$

Lemma B.3.10. *The random sequence of iterations followed by Algorithm 5 satisfies*

$$\begin{aligned}
\tilde{q}_0^d(t, s, j, a) &= t \underline{R}, \quad a.s., \\
\tilde{q}_{i+1}^d(t, s, j, a) &= \tilde{q}_i^d(t, s, j, a) + \tilde{\theta}_i(t, s, j, a) \cdot (H \tilde{q}_i^d + \tilde{\phi}_i - \tilde{q}_i^d)(t, s, j, a), \quad \forall i \in \mathbb{N}, \quad a.s.,
\end{aligned}$$

where the terms are defined in (B.14) and (B.15).

Proof. We prove the claim by induction on i . The base case holds immediately from the definition.

To prove the inductive case, suppose that $i \in \mathbb{N}$ and we prove the result in the following cases.

Case 1a: Suppose that $\tilde{b} = (\tilde{t}_i, \tilde{s}_i, \tilde{j}_i, \tilde{a}_i) = (t, s, j, a) = b$ and $t_i > 0, j_i > 0$, then by algebraic manipulation:

$$\begin{aligned}
\tilde{q}_{i+1}^d(b) &= \tilde{q}_i^d(b) + \theta_i(b)((H\tilde{q}_i^d)(b) + \tilde{\phi}_i(b) - \tilde{q}_i^d(b)) \\
&= \tilde{q}_i^d(b) + \theta_i(b)((H\tilde{q}_i^d)(b) + (H_{\tilde{s}'_i}\tilde{q}_i^d)(b) - (H\tilde{q}_i^d)(b) - \tilde{q}_i^d(b)) \\
&= \tilde{q}_i^d(b) + \theta_i(b)((H_{\tilde{s}'_i}\tilde{q}_i^d)(b) - \tilde{q}_i^d(b)) \\
&= \tilde{q}_i^d(b) + \theta_i(b)((\tilde{q}_i^d - \xi G_{\tilde{s}'_i}\tilde{q}_i^d)(b) - \tilde{q}_i^d(b)) \\
&= \tilde{q}_i^d(b) + \frac{\beta_i}{J} \sum_{j' \in [J-1]} \partial \ell_{\frac{j}{J}}^{\kappa} \left(r(s, a) + \gamma \max_{a' \in \mathcal{A}} \tilde{q}_i^d(t-1, \tilde{s}'_i, j', a') - \tilde{q}_i^d(b) \right).
\end{aligned}$$

Case 1b: Suppose that $\tilde{b} = (\tilde{t}_i, \tilde{s}_i, \tilde{j}_i, \tilde{a}_i) = (t, s, j, a) = b$ and $t_i = 0 \vee j_i = 0$, then by algebraic manipulation:

$$\begin{aligned}
\tilde{q}_{i+1}^d(b) &= \tilde{q}_i^d(b) + \theta_i(b)((H\tilde{q}_i^d)(b) + \tilde{\phi}_i(b) - \tilde{q}_i^d(b)) \\
&= \tilde{q}_i^d(b) + \theta_i(b)((H\tilde{q}_i^d)(b) + (H_{\tilde{s}'_i}\tilde{q}_i^d)(b) - (H\tilde{q}_i^d)(b) - \tilde{q}_i^d(b)) \\
&= \tilde{q}_i^d(b) + \theta_i(b)((H_{\tilde{s}'_i}\tilde{q}_i^d)(b) - \tilde{q}_i^d(b)) \\
&= \tilde{q}_i^d(b) + \theta_i(b)((\tilde{q}_i^d - \xi G_{\tilde{s}'_i}\tilde{q}_i^d)(b) - \tilde{q}_i^d(b)) \\
&= \tilde{q}_i^d(b) + \theta_i(b)(\tilde{q}_i^d(b) - \xi \tilde{q}_i^d(b) + \xi \underline{R}t - \tilde{q}_i^d(b)) \\
&= \tilde{q}_i^d(b) - \theta_i(b)\xi(\tilde{q}_i^d(b) - \underline{R}t) \\
&= \tilde{q}_i^d(b) - \beta_i(\tilde{q}_i^d(b) - \underline{R}t).
\end{aligned}$$

Case 2: Suppose that $\tilde{b} = (\tilde{t}_i, \tilde{s}_i, \tilde{j}_i, \tilde{a}_i) \neq (t, s, j, a) = b$, then by algebraic manipulation, the

algorithm does not change the q-function:

$$\begin{aligned}
\tilde{q}_{i+1}^d(b) &= \tilde{q}_i^d(b) + \theta_i(b) \cdot ((H\tilde{q}_i^d)(b) + \tilde{\phi}_i(b) - \tilde{q}_i^d(b)) \\
&= \tilde{q}_i^d(b) + 0 \cdot ((H\tilde{q}_i^d)(b) + \tilde{\phi}_i(b) - \tilde{q}_i^d(b)) \\
&= \tilde{q}_i^d(b).
\end{aligned}$$

□

B.3.6.4 Operator H is a contraction

We use the following weights $w \in \mathbb{R}^{[T] \times \mathcal{S} \times [J-1] \times \mathcal{A}}$ with a weighted max norm to prove the contraction properties in this section: where the weights are defined as

$$w(t, s, j, a) := 2^t, \quad \forall t \in [T], s \in \mathcal{S}, j \in [J-1], a \in \mathcal{A}. \quad (\text{B.16})$$

Lemma B.3.11. *The operator $\underline{B}_\kappa^d$ is a weighted max norm contraction for w defined in (B.16):*

$$\|\underline{B}_\kappa^d x - \underline{B}_\kappa^d y\|_w \leq \frac{1}{2} \|x - y\|_w, \quad \forall x, y \in \mathbb{R}^{\mathcal{S}}.$$

Proof. The operator $\underline{B}_\kappa^d$ is equivalently defined using a shortfall risk measure $\hat{\mathbf{q}}_\alpha^\kappa$ from Lemma B.3.6 for each $b = (t, s, j, a)$ as

$$(\underline{B}_\kappa^d q)(b) := \begin{cases} \underline{R} \cdot t & \text{if } j = 0 \vee t = 0, \\ (\hat{\mathbf{q}}_{\frac{j}{J}}^\kappa)^{a,s} [r(s, a) + \gamma \max_{a' \in \mathcal{A}} q(t-1, \tilde{s}_1, \tilde{j}', a')] & \text{otherwise,} \end{cases}$$

We analyze the following two cases.

Case 1: For each $t \in 1:T, s \in \mathcal{S}, j \in 1:(J-1), a \in \mathcal{A}$:

$$0 \leq \frac{1}{w(t, s, j, a)} \left| (\underline{B}_\kappa^d x)(t, s, j, a) - (\underline{B}_\kappa^d y)(t, s, j, a) \right|$$

$$\begin{aligned}
& \stackrel{(a)}{=} \frac{1}{w(t, s, j, a)} \left| (\hat{\mathbf{q}}_{j/J}^\kappa)^{a,s} \left[\max_{a' \in \mathcal{A}} \gamma x(t-1, \tilde{s}_1, \tilde{j}', a') \right] - (\hat{\mathbf{q}}_{j/J}^\kappa)^{a,s} \left[\max_{a' \in \mathcal{A}} \gamma y(t-1, \tilde{s}_1, \tilde{j}', a') \right] \right| \\
& \stackrel{(b)}{\leq} \frac{1}{w(t, s, j, a)} \max_{a' \in \mathcal{A}^{\mathcal{S} \times [J-1]}} \left| (\hat{\mathbf{q}}_{j/J}^\kappa)^{a,s} [\gamma x(t-1, \tilde{s}_1, \tilde{j}', a'(\tilde{s}_1, \tilde{j}'))] - (\hat{\mathbf{q}}_{j/J}^\kappa)^{a,s} [\gamma y(t-1, \tilde{s}_1, \tilde{j}', a'(\tilde{s}_1, \tilde{j}'))] \right| \\
& \stackrel{(c)}{\leq} \frac{1}{w(t, s, j, a)} \cdot \max_{s' \in \mathcal{S}, j' \in [J-1], a' \in \mathcal{A}} |\gamma x(t-1, s', j', a') - \gamma y(t-1, s', j', a')| \\
& = \frac{1}{w(t, s, j, a)} \cdot \gamma \cdot \max_{s' \in \mathcal{S}, j' \in [J-1], a' \in \mathcal{A}} |x(t-1, s', j', a') - y(t-1, s', j', a')| \\
& \stackrel{(d)}{\leq} \max_{s' \in \mathcal{S}, j' \in [J-1], a' \in \mathcal{A}} \frac{|x(t-1, s', j', a') - y(t-1, s', j', a')|}{2w(t-1, s', j', a')}.
\end{aligned}$$

In step (a), we use the translation invariance of the risk measure to cancel out $r(s, a)$. Step (b) follows by upper bounding the difference using the monotonicity of $\hat{\mathbf{q}}_{j/J}^\kappa$, step (c) follows from Lemmas B.3.1 and B.3.6, step (d) follows from the definition of w , its dependence on t only, and $\gamma \leq 1$.

Case 2: For $t = 0 \vee j = 0$:

$$\frac{1}{w(t, s, j, a)} |(\underline{B}_\kappa^d x)(t, s, j, a) - (\underline{B}_\kappa^d y)(t, s, j, a)| = \frac{1}{2^t} |\underline{R}t - \underline{R}t| = 0.$$

Then, using the equalities above and $\mathcal{K} = [T] \times \mathcal{S} \times [J-1] \times \mathcal{A}$ we get that

$$\begin{aligned}
\|\underline{B}_\kappa^d x - \underline{B}_\kappa^d y\|_w &= \max_{b \in \mathcal{K}} \frac{1}{w(b)} |(\underline{B}_\kappa^d x)(b) - (\underline{B}_\kappa^d y)(b)| \\
&\leq \max_{t \in 1:T, s' \in \mathcal{S}, j' \in 1:J-1, a' \in \mathcal{A}} \frac{|x(t-1, s', j', a') - y(t-1, s', j', a')|}{2 \cdot w(t-1, s', j', a')} \\
&\leq \max_{t \in [T], s' \in \mathcal{S}, j' \in [J-1], a' \in \mathcal{A}} \frac{|x(t, s', j', a') - y(t, s', j', a')|}{2 \cdot w(t, s', j', a')} \\
&= \frac{1}{2} \|x - y\|_w.
\end{aligned}$$

□

The following lemma establishes that the gradient update is a convex combination of the starting value and an optimal solution for an appropriate step size.

Lemma B.3.12. *Suppose that $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable μ -strongly convex function with an L -*

Lipschitz continuous gradient. Consider $x_i \in \mathbb{R}$ and a gradient update for any step size $\xi \in (0, 1/L]$:

$$x_{i+1} := x_i - \xi \cdot f'(x_i).$$

Then $\exists l \in [1/L, 1/\mu]$ such that $\xi/l \in (0, 1]$ and

$$x_{i+1} = (1 - \xi/l) \cdot x_i + \xi/l \cdot x^*,$$

where $x^* = \arg \min_{x \in \mathbb{R}} f(x)$ (unique from strong convexity).

Proof. Assume that $f'(x_i) \neq 0$ hence $x_i \neq x^*$; otherwise the result holds trivially. Then construct $l \neq 0$ as

$$l := \frac{x_i - x^*}{f'(x_i)}.$$

Substituting the definition of l into the gradient update, we get that

$$x_{i+1} := x_i - \xi \cdot f'(x_i) = (1 - \xi/l) x_i + \xi/l \cdot x^*,$$

as desired.

It remains to show that $l \in [1/L, 1/\mu]$ and $\xi/l \in (0, 1]$. Using that $f'(x^*) = 0$ and strong convexity [91, Thm. 2.1.10] and Lipschitz continuity of the derivative:

$$\frac{1}{L} |f'(x_i)| = \frac{1}{L} |f'(x_i) - f'(x^*)| \leq |x_i - x^*| \leq \frac{1}{\mu} |f'(x_i) - f'(x^*)| = \frac{1}{\mu} |f'(x_i)|. \quad (\text{B.17})$$

Next, we analyze two cases.

Case 1: Suppose that $x^* > x_i$. Then $f'(x_i) > f'(x^*) = 0$ because f' is increasing for a strongly

convex f , and therefore, (B.17) becomes

$$\begin{aligned} -\frac{1}{L}f'(x_i) &\leq x^* - x_i \leq -\frac{1}{\mu}f'(x_i), \\ -\frac{1}{L} &\geq \frac{x^* - x_i}{f'(x_i)} \geq -\frac{1}{\mu}, \\ -\frac{1}{L} &\geq -l \geq -\frac{1}{\mu}, \\ \frac{1}{L} &\leq l \leq \frac{1}{\mu}. \end{aligned}$$

In addition, $\xi \in (0, 1/L] \implies \xi/l \in (0, 1]$.

Case 2: Suppose that $x^* < x_i$. Then $f'(x_i) > f'(x^*) = 0$ because f' is increasing for a strongly convex f , and therefore, (B.17) becomes

$$\begin{aligned} \frac{1}{L}f'(x_i) &\leq x_i - x^* \leq \frac{1}{\mu}f'(x_i), \\ \frac{1}{L} &\leq \frac{x_i - x^*}{f'(x_i)} \leq \frac{1}{\mu}, \\ \frac{1}{L} &\leq l \leq \frac{1}{\mu}. \end{aligned}$$

In addition, $\xi \in (0, 1/L] \implies \xi/l \in (0, 1]$. □

Theorem B.3.13. *A fixed point $x^* = \underline{B}_\kappa^d x^*$ exists and satisfies $x^* = Hx^*$. Let $\bar{\mu} := J^{-1}\kappa$, $\bar{L} := \kappa^{-1}$, and $\xi \in (0, \min(1, 1/\bar{L}))$ in the definition of H . Then, H is a weighted max norm star contraction:*

$$\|Hx - Hx^*\|_w \leq \left(1 - \frac{\bar{\mu}\xi}{2}\right) \cdot \|x - x^*\|_w,$$

for w defined in (B.16).

Proof. The fixed point x^* exists from Lemma B.3.11 and the Banach fixed point theorem. The operator H takes a gradient step towards $\underline{B}_\kappa^d$.

Case 1: Fix some $b = (t, s, j, a)$ with $t \in [T]$, $s \in \mathcal{S}$, $j \in [J-1]$, $a \in \mathcal{A}$ and suppose that $t > 0$ and $j > 0$. Fix some q and define

$$f(y) = \mathbb{E}^{s,a} \left[\ell_{j/J}^\kappa \left(r(s, a) + \gamma \cdot \max_{a' \in \mathcal{A}} q(t-1, \tilde{s}_1, \tilde{j}', a') - y \right) \right], \quad (\text{B.18})$$

The function f is strongly convex with Lipschitz gradient with parameters $\bar{\mu}$ and $\bar{L}$ based on Lemma 4.4.6 and since

$$\bar{\mu} \leq \min \left\{ \frac{j}{J}, 1 - \frac{j}{J} \right\} \kappa \leq \max \left\{ \frac{j}{J}, 1 - \frac{j}{J} \right\} \kappa^{-1} \leq \bar{L}, \forall j \in 1:J-1.$$

Let $y^* \in \arg \min_{y \in \mathbb{R}} f(y)$. Then, $\exists l \in [\frac{1}{\bar{L}}, \frac{1}{\bar{\mu}}]$ such that

$$\begin{aligned} (Hq)(b) &= (q - \xi Gq)(b) = q(b) - \xi \cdot f'(q(b)) = (1 - \xi/l) \cdot q(b) + \xi/l \cdot y^* \\ &= (1 - \xi/l) \cdot q(b) + \xi/l \cdot (\underline{B}_\kappa^d q)(b), \end{aligned} \tag{B.19}$$

from algebraic manipulation and application of Lemma B.3.12 to the function f in (B.18) which satisfies the requisite strong convexity and Lipschitz continuity properties.

The fixed point of x^* of $\underline{B}_\kappa^d$ is a fixed point of H from (B.19)

$$(Hx^*)(b) = \left(1 - \frac{\xi}{l}\right) x^*(b) + \frac{\xi}{l} (\underline{B}_\kappa^d x^*)(b) = \left(1 - \frac{\xi}{l}\right) x^*(b) + \frac{\xi}{l} (x^*)(b) = x^*(b).$$

Finally, we get using (B.19) that

$$\begin{aligned} |(Hx)(b) - (Hx^*)(b)| &= |(1 - \xi/l)x(b) + \xi/l(\underline{B}_\kappa^d x)(b) - ((1 - \xi/l)x^*(b) + \xi/l(\underline{B}_\kappa^d x^*)(b))| \\ &= |(1 - \xi/l)(x - x^*)(b) + \xi/l(\underline{B}_\kappa^d x - \underline{B}_\kappa^d x^*)(b)| \\ &\leq (1 - \xi/l)|(x - x^*)(b)| + \xi/l|(\underline{B}_\kappa^d x - \underline{B}_\kappa^d x^*)(b)| \\ &\leq (1 - \xi/l)|(x - x^*)(b)| + \frac{1}{2}\xi/l|(x - x^*)(b)| \\ &= (1 - \xi/2l)|(x - x^*)(b)|. \end{aligned}$$

Here, we used the triangle inequality for absolute values and the fact that x^* is a fixed point of $\underline{B}_\kappa^d$ and Lemma B.3.11. Hence,

$$\frac{1}{w(b)} |(Hx - Hx^*)(b)| \leq \left(1 - \frac{\bar{\mu}\xi}{2}\right) \frac{1}{w(b)} |(x - x^*)(b)|. \tag{B.20}$$

Case 2: Fix some $b = (t, s, j, a)$ with $t \in [T], s \in \mathcal{S}, j \in [J - 1], a \in \mathcal{A}$ and suppose that $t = 0 \vee j = 0$. Then, from $x^\star(b) = \underline{R} \cdot t$, we have that if $\xi < 1$:

$$\begin{aligned} |(Hx)(b) - (Hx^\star)(b)| &= |x(b) - \xi(x(b) - \underline{R} \cdot t) - \underline{R} \cdot t| \\ &= (1 - \xi)|x(b) - \underline{R} \cdot t| = (1 - \xi)|x(b) - x^\star(b)|. \end{aligned}$$

Hence,

$$\frac{1}{w(b)}|(Hx - Hx^\star)(b)| \leq (1 - \xi) \frac{1}{w(b)}|(x - x^\star)(b)| \leq (1 - \bar{\mu}\xi/2) \frac{1}{w(b)}|(x - x^\star)(b)| \quad (\text{B.21})$$

since $\bar{\mu}\xi/2 \leq \kappa J^{-1}\xi/2 \leq \kappa\xi \leq \xi$ because $\kappa \leq 1$.

Conclusion: Putting (B.20) and (B.21) together with the definition of the weighted norm, we get the desired star contraction rate. $\square$

B.3.6.5 Noise Properties

The history $\mathcal{F}_i$ at an iteration $i \in \mathbb{N}$ is defined as

$$\mathcal{F}_i := \left(\tilde{q}_0^d, \dots, \tilde{q}_i^d, \tilde{\phi}_0, \dots, \tilde{\phi}_{i-1}, \tilde{\theta}_0, \dots, \tilde{\theta}_i \right). \quad (\text{B.22})$$

Recall from Assumption 4.4.7 that

$$\mathcal{G}_i := (\tilde{\beta}_l, (\tilde{t}_l, \tilde{s}_l, \tilde{j}_l, \tilde{a}_l, \tilde{s}'_l))_{l=0}^i,$$

and

$$\mathbb{P} \left[\tilde{s}'_i = s' \mid \mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i \right] = p(\tilde{s}_i, \tilde{a}_i, s'), \quad \forall s' \in \mathcal{S},$$

almost surely, where $\mathcal{G}_{i-1} := (\tilde{\beta}_l, (\tilde{t}_l, \tilde{s}_l, \tilde{j}_l, \tilde{a}_l, \tilde{s}'_l))_{l=0}^{i-1}$.

Lemma B.3.14. *Let Ω be an appropriate sample space. Then for each $\omega_1, \omega_2 \in \Omega$ and $i = 1, \dots$:*

$$(\mathcal{G}_{i-1}(\omega_1) = \mathcal{G}_{i-1}(\omega_2)) \wedge (\tilde{b}_i(\omega_1) = \tilde{b}_i(\omega_2)) \wedge (\tilde{\beta}_i(\omega_1) = \tilde{\beta}_i(\omega_2))$$

$$\implies$$

$$\mathcal{F}_i(\omega_1) = \mathcal{F}_i(\omega_2) = \bar{\mathcal{F}}_i(\mathcal{G}_{i-1}(\omega_1), \tilde{b}_i(\omega_1), \tilde{\beta}_i(\omega_1)), \text{ a.s.}$$

for some $\bar{\mathcal{F}}_i$ operator that maps a tuple $((\beta_l, (t_l, s_l, j_l, a_l, s'_l))_{l=0}^{i-1}, (t_i, s_i, j_i, a_i), \beta_i)$ to some

$$\left(\underline{q}_0^d, \dots, \underline{q}_i^d, \phi_0, \dots, \phi_{i-1}, \theta_0, \dots, \theta_i \right) .$$

Proof. We proceed by induction on i . To prove the base step for $i = 0$:

$$\mathcal{F}_0(\omega_1) = (\tilde{q}_0^d(\omega_1), \tilde{\theta}_0(\omega_1)) = (t\underline{R}, \tilde{\theta}_0(\omega_1)) = (t\underline{R}, \tilde{\theta}_0(\omega_2)) = \mathcal{F}_0(\omega_2).$$

Here, $\tilde{\theta}_0(\omega_1) = \tilde{\theta}_0(\omega_2)$ because $\tilde{\beta}_0(\omega_1) = \tilde{\beta}_0(\omega_2)$, and $\tilde{b}_0(\omega_1) = \tilde{b}_0(\omega_2)$.

To prove the inductive step, assume that the property holds for i and prove it for $i + 1$. That is, suppose that $l = i + 1$

$$(\mathcal{G}_{l-1}(\omega_1) = \mathcal{G}_{l-1}(\omega_2)) \wedge (\tilde{b}_l(\omega_1) = \tilde{b}_l(\omega_2)) \wedge (\tilde{\beta}_l(\omega_1) = \tilde{\beta}_l(\omega_2)).$$

Then from the inductive assumption:

$$\mathcal{F}_l = \mathcal{F}_l, \quad \forall l = 1, \dots, i,$$

and for $b = \tilde{b}_{i+1}(\omega_1) = \tilde{b}_{i+1}(\omega_2)$:

$$\begin{aligned} (\tilde{\phi}_i(b))(\omega_1) &= (H_{\tilde{s}'_i(\omega_1)} \tilde{q}_i^d(\omega_1))(b) - (H \tilde{q}_i^d(\omega_1))(b) \\ &= (H_{\tilde{s}'_i(\omega_2)} \tilde{q}_i^d(\omega_2))(b) - (H \tilde{q}_i^d(\omega_2))(b) \\ &= (\tilde{\phi}_i(b))(\omega_2). \end{aligned}$$

$$(\tilde{\theta}_{i+1}(b))(\omega_1) = \frac{\tilde{\beta}_{i+1}(\omega_1)}{\xi} = \frac{\tilde{\beta}_{i+1}(\omega_2)}{\xi} = (\tilde{\theta}_{i+1}(b))(\omega_2),$$

and for $b \neq \tilde{b}_{i+1}(\omega_1) = \tilde{b}_{i+1}(\omega_2)$:

$$(\tilde{\phi}_i(b))(\omega_1) = 0 = (\tilde{\phi}_i(b))(\omega_2).$$

$$(\tilde{\theta}_{i+1}(b))(\omega_1) = 0 = (\tilde{\theta}_{i+1}(b))(\omega_2).$$

In addition,

$$\begin{aligned} (\tilde{q}_{i+1}^d(b))(\omega_1) &= (\tilde{q}_i^d(b))(\omega_1) + (\tilde{\theta}_i(b))(\omega_1) \cdot (H\tilde{q}_i^d(\omega_1) + \tilde{\phi}_i(\omega_1) - \tilde{q}_i^d(\omega_1))(t, s, j, a) \\ &= (\tilde{q}_i^d(b))(\omega_2) + (\tilde{\theta}_i(b))(\omega_2) \cdot (H\tilde{q}_i^d(\omega_2) + \tilde{\phi}_i(\omega_2) - \tilde{q}_i^d(\omega_2))(b), \\ &= (\tilde{q}_{i+1}^d(b))(\omega_2). \end{aligned}$$

Putting the inequalities above together we get the desired equality:

$$\begin{aligned} \mathcal{F}_{i+1}(\omega_1) &= \left(\tilde{q}_0^d(\omega_1), \dots, \tilde{q}_{i+1}^d(\omega_1), \tilde{\phi}_0(\omega_1), \dots, \tilde{\phi}_i(\omega_1), \tilde{\theta}_0(\omega_1), \dots, \tilde{\theta}_{i+1}(\omega_1) \right) \\ &= \left(\tilde{q}_0^d(\omega_2), \dots, \tilde{q}_{i+1}^d(\omega_2), \tilde{\phi}_0(\omega_2), \dots, \tilde{\phi}_i(\omega_2), \tilde{\theta}_0(\omega_1), \dots, \tilde{\theta}_{i+1}(\omega_1) \right) \\ &= \mathcal{F}_{i+1}(\omega_2). \end{aligned}$$

□

Lemma B.3.15. *Under Assumption 4.4.7:*

$$\mathbb{P} \left[\tilde{s}'_i = s' \mid \mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i, \mathcal{F}_i \right] = p(\tilde{s}_i, \tilde{a}_i, s'), \quad a.s.,$$

for each $s' \in \mathcal{S}$ and $i \in \mathbb{N}$.

Proof. Using Lemma B.3.14, we have that

$$\mathbb{P} \left[\mathcal{F}_i = \bar{\mathcal{F}}(\mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i) \mid \mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i \right] = 1.$$

Hence, from Assumption 4.4.7 and the law of total probability, for each $s' \in \mathcal{S}, i \in \mathbb{N}$:

$$\begin{aligned}
p(\tilde{s}_i, \tilde{a}_i, s') &= \mathbb{P} \left[\tilde{s}'_i = s' \mid \mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i \right] \\
&= \mathbb{P} \left[\tilde{s}'_i = s' \mid \mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i, \mathcal{F}_i = \bar{\mathcal{F}}(\mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i) \right] \mathbb{P} \left[\mathcal{F}_i = \bar{\mathcal{F}}(\mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i) \mid \mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i \right] \\
&= \mathbb{P} \left[\tilde{s}'_i = s' \mid \mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i, \mathcal{F}_i = \bar{\mathcal{F}}(\mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i) \right] \\
&= \mathbb{P} \left[\tilde{s}'_i = s' \mid \mathcal{G}_{i-1}, \tilde{b}_i, \tilde{\beta}_i, \mathcal{F}_i \right] \text{ a.s.}
\end{aligned}$$

□

Lemma B.3.16. *The noise $\tilde{\phi}_i$ in (B.15) satisfies almost surely*

$$\mathbb{E}[\tilde{\phi}_i(t, s, j, a) \mid \mathcal{F}_i] = 0, \quad \forall t \in [T], s \in \mathcal{S}, j \in [J-1], a \in \mathcal{A}, i \in \mathbb{N},$$

where $\mathcal{F}_i$ is the history defined in (B.22).

Proof. Let $b := (t, s, j, a)$ and $i \in \mathbb{N}$ be arbitrary. We decompose the expectation using the law of total expectation to get that

$$\mathbb{E}[\tilde{\phi}_i(b) \mid \mathcal{F}_i] = \mathbb{E}[\tilde{\phi}_i(b) \mid \mathcal{F}_i, \tilde{b}_i \neq b] \cdot \mathbb{P}[\tilde{b}_i \neq b \mid \mathcal{F}_i] + \mathbb{E}[\tilde{\phi}_i(b) \mid \mathcal{F}_i, \tilde{b}_i = b] \cdot \mathbb{P}[\tilde{b}_i = b \mid \mathcal{F}_i] \text{ a.s.,} \quad (\text{B.23})$$

where $\tilde{b}_i := (\tilde{s}_i, \tilde{a}_i, \tilde{t}_i, \tilde{j}_i)$.

The first r.h.s. term in (B.23) is, from the definition of $\tilde{\phi}_i(b)$,

$$\mathbb{E}[\tilde{\phi}_i(b) \mid \mathcal{F}_i, \tilde{b}_i \neq b] = \mathbb{E}[0 \mid \mathcal{F}_i, \tilde{b}_i \neq b] = 0, \quad \text{a.s.} \quad (\text{B.24})$$

We now analyze two cases to evaluate the second r.h.s. term in (B.23).

Case 1: $j > 0 \wedge t > 0$. Then almost surely:

$$\begin{aligned}
\mathbb{E}[\tilde{\phi}_i(b) \mid \mathcal{F}_i, \tilde{b}_i = b] &= \mathbb{E}[(H_{\tilde{s}'_i \tilde{q}_i^d})(b) - (H_{\tilde{q}_i^d})(b) \mid \mathcal{F}_i, \tilde{b}_i = b] \\
&= \xi \cdot \mathbb{E}[-(G_{\tilde{s}'_i \tilde{q}_i^d})(b) + (G_{\tilde{q}_i^d})(b) \mid \mathcal{F}_i, \tilde{b}_i = b] \\
&= \xi \cdot \mathbb{E}[-(G_{\tilde{s}'_i \tilde{q}_i^d})(b) + (G_{\tilde{q}_i^d})(b) \mid \mathcal{F}_i, \tilde{b}_i = b] \\
&= \xi \cdot \mathbb{E}[\mathbb{E}[-(G_{\tilde{s}'_i \tilde{q}_i^d})(b) \mid \mathcal{F}_i, \tilde{b}_i = b, \tilde{\beta}_i, \mathcal{G}_{i-1}] + (G_{\tilde{q}_i^d})(b) \mid \mathcal{F}_i, \tilde{b}_i = b] \quad (\text{B.25}) \\
&\stackrel{(a)}{=} \xi \cdot \mathbb{E}[\mathbb{E}^{a,s}[-(G_{\tilde{s}_1 \tilde{q}_i^d})(b)] + (G_{\tilde{q}_i^d})(b) \mid \mathcal{F}_i, \tilde{b}_i = b] \\
&= \xi \cdot \mathbb{E}[-(G_{\tilde{q}_i^d})(b) + (G_{\tilde{q}_i^d})(b) \mid \mathcal{F}_i, \tilde{b}_i = b] \\
&= 0.
\end{aligned}$$

To clarify, when $\tilde{s}_1$ is used in an expectation with a superscript, such as $\mathbb{E}^{a,s}$, then it does not represent a sample $\tilde{s}_i$ with $i = 1$, but instead it represents the transition from $\tilde{s}_0 = s$ to $\tilde{s}_1$ distributed as $p(s, a, \cdot)$.

Step (a) above follows from Lemma B.3.15 given that the randomness of $(G_{\tilde{s}'_i \tilde{q}_i^d})(b)$ only comes from $\tilde{s}'_i$ when conditioning on $\mathcal{F}_i, \tilde{b}_i = b, \tilde{\beta}_i$, and $\mathcal{G}_{i-1}$.

Case 2: $j = 0 \vee t = 0$. Directly from the definition of the operators in (B.14):

$$\begin{aligned}
\mathbb{E}[\tilde{\phi}_i(b) \mid \mathcal{F}_i, \tilde{b}_i = b] &= \mathbb{E}[(H_{\tilde{s}'_i \tilde{q}_i^d})(b) - (H_{\tilde{q}_i^d})(b) \mid \mathcal{F}_i, \tilde{b}_i = b] \\
&= \xi \cdot \mathbb{E}[-(G_{\tilde{s}'_i \tilde{q}_i^d})(b) + (G_{\tilde{q}_i^d})(b) \mid \mathcal{F}_i, \tilde{b}_i = b] \quad (\text{B.26}) \\
&= 0.
\end{aligned}$$

Substituting (B.24), and the appropriate case, (B.25) or (B.26), into (B.23) proves the desired equality. $\square$

Lemma B.3.17. *The noise $\tilde{\phi}_i$ in (B.15) satisfies*

$$\mathbb{E}[(\tilde{\phi}_i(t, s, j, a))^2 \mid \mathcal{F}_i] \leq c + g \|\tilde{q}_i^d\|_\infty^2, \quad \forall t \in [T], s \in \mathcal{S}, j \in [J-1], a \in \mathcal{A}, i \in \mathbb{N},$$

almost surely, for some $c, g \in \mathbb{R}_+$ where $\mathcal{F}_i$ is the history defined in (B.22).

Proof. Let $b := (t, s, j, a)$ and $i \in \mathbb{N}$ be arbitrary. We decompose the expectation using the law of total expectation to get almost surely

$$\mathbb{E}[\tilde{\phi}_i(b)^2 \mid \mathcal{F}_i] = \mathbb{E}[\tilde{\phi}_i(b)^2 \mid \mathcal{F}_i, \tilde{b}_i \neq b] \cdot \mathbb{P}[\tilde{b}_i \neq b \mid \mathcal{F}_i] + \mathbb{E}[\tilde{\phi}_i(b)^2 \mid \mathcal{F}_i, \tilde{b}_i = b] \cdot \mathbb{P}[\tilde{b}_i = b \mid \mathcal{F}_i], \quad (\text{B.27})$$

where $\tilde{b}_i := (\tilde{t}_i, \tilde{s}_i, \tilde{a}_i, \tilde{j}_i, \tilde{a}_i)$.

The first r.h.s. term in (B.27) is, from the definition of $\tilde{\phi}_i(b)$,

$$\mathbb{E}[\tilde{\phi}_i(b)^2 \mid \mathcal{F}_i, \tilde{b}_i \neq b] = \mathbb{E}[0 \mid \mathcal{F}_i, \tilde{b}_i \neq b] = 0, \quad \text{a.s.} \quad (\text{B.28})$$

We now analyze two cases to evaluate the second r.h.s. term in (B.27).

Case 1: Assume that $j > 0, t > 0$. Then from the definitions of the operators in (B.14):

$$\begin{aligned} \mathbb{E}[(\tilde{\phi}_i(b))^2 \mid \mathcal{F}_i, \tilde{b}_i = b] &= \mathbb{E} \left[\left((H_{\tilde{s}_i'} \tilde{q}_i^{\text{d}})(b) - (H \tilde{q}_i^{\text{d}})(b) \right)^2 \mid \mathcal{F}_i, \tilde{b}_i = b \right] \\ &= \xi^2 \mathbb{E} \left[\left(-(G_{\tilde{s}_i'} \tilde{q}_i^{\text{d}})(b) + (G \tilde{q}_i^{\text{d}})(b) \right)^2 \mid \mathcal{F}_i, \tilde{b}_i = b \right] \\ &= \xi^2 \mathbb{E} \left[\mathbb{E} \left[\left(-(G_{\tilde{s}_i'} \tilde{q}_i^{\text{d}})(b) + (G \tilde{q}_i^{\text{d}})(b) \right)^2 \mid \mathcal{F}_i, \tilde{b}_i = b, \tilde{\beta}_i, \mathcal{G}_{i-1} \right] \mid \mathcal{F}_i, \tilde{b}_i = b \right] \\ &\stackrel{(a)}{=} \xi^2 \mathbb{E} \left[\mathbb{E}^{a,s} \left[\left((G_{\tilde{s}_1} \tilde{q}_i^{\text{d}})(b) - (G \tilde{q}_i^{\text{d}})(b) \right)^2 \mid \mathcal{F}_i, \tilde{b}_i = b \right] \right]. \end{aligned}$$

To clarify, when $\tilde{s}_1$ is used in an expectation with a superscript, such as $\mathbb{E}^{a,s}$, then it does not represent a sample $\tilde{s}_i$ with $i = 1$, but instead it represents the transition from $\tilde{s}_0 = s$ to $\tilde{s}_1$ distributed as $p(s, a, \cdot)$.

Step (a) above follows from Lemma B.3.15 given that the randomness of $-(G_{\tilde{s}_i'} \tilde{q}_i^{\text{d}})(b) + (G \tilde{q}_i^{\text{d}})(b)$ only comes from $\tilde{s}_i'$ when conditioning on $\mathcal{F}_i, \tilde{b}_i = b, \tilde{\beta}_i$, and $\mathcal{G}_{i-1}$. Then, continuing the derivation:

$$\begin{aligned} &\mathbb{E}[(\tilde{\phi}_i(b))^2 \mid \mathcal{F}_i, \tilde{b}_i = b] \\ &= \xi^2 \mathbb{E} \left[\mathbb{E}^{a,s} \left[\left((G_{\tilde{s}_1} \tilde{q}_i^{\text{d}})(b) - (G \tilde{q}_i^{\text{d}})(b) \right)^2 \mid \mathcal{F}_i, \tilde{b}_i = b \right] \right] \end{aligned}$$

$$\begin{aligned}
&\stackrel{(a)}{=} \xi^2 \mathbb{E} \left[\mathbb{E}^{a,s} \left[\left(\mathbb{E} \left[\partial \ell_{\frac{j}{J}}^\kappa \left(\tilde{\delta}_i(\tilde{s}_1, \tilde{j}') \right) \mid \tilde{s}_1 \right] - \mathbb{E}^{a,s} \left[\partial \ell_{\frac{j}{J}}^\kappa \left(\tilde{\delta}_i(\tilde{s}_1, \tilde{j}') \right) \right] \right)^2 \mid \mathcal{F}_i, \tilde{b}_i = b \right] \right] \\
&\stackrel{(b)}{=} \xi^2 \mathbb{E} \left[\left(\mathbb{E}^{a,s} \left[\left(\mathbb{E} \left[\partial \ell_{\frac{j}{J}}^\kappa \left(\tilde{\delta}_i(\tilde{s}_1, \tilde{j}') \right) \mid \tilde{s}_1 \right] \right)^2 \right] - \left(\mathbb{E}^{a,s} \left[\partial \ell_{\frac{j}{J}}^\kappa \left(\tilde{\delta}_i(\tilde{s}_1, \tilde{j}') \right) \right] \right)^2 \right) \mid \mathcal{F}_i, \tilde{b}_i = b \right] \\
&\leq \xi^2 \mathbb{E} \left[\mathbb{E}^{a,s} \left[\left(\mathbb{E} \left[\partial \ell_{\frac{j}{J}}^\kappa \left(\tilde{\delta}_i(\tilde{s}_1, \tilde{j}') \right) \mid \tilde{s}_1 \right] \right)^2 \mid \mathcal{F}_i, \tilde{b}_i = b \right] \right] \\
&\stackrel{(c)}{\leq} \xi^2 \mathbb{E} \left[\max_{j' \in [J-1], s' \in \mathcal{S}} \partial \ell_{\frac{j}{J}}^\kappa (\tilde{\delta}_i(s', j'))^2 \mid \mathcal{F}_i, \tilde{b}_i = b \right] \\
&\stackrel{(d)}{\leq} \xi^2 \mathbb{E} \left[\max_{j' \in [J-1], s' \in \mathcal{S}} \left(\left| \partial \ell_{\frac{j}{J}}^\kappa (\tilde{\delta}_i(s', j')) - \partial \ell_{\frac{j}{J}}^\kappa (0) \right| \right)^2 \mid \mathcal{F}_i, \tilde{b}_i = b \right] \\
&\stackrel{(e)}{\leq} \xi^2 \mathbb{E} \left[\max_{j' \in [J-1], s' \in \mathcal{S}} \left(\max \left\{ \frac{j}{J}, 1 - \frac{j}{J} \right\} \kappa^{-1} |\tilde{\delta}_i(s', j')| \right)^2 \mid \mathcal{F}_i, \tilde{b}_i = b \right] \\
&\leq \xi^2 \kappa^{-2} \cdot \mathbb{E} \left[\max_{j' \in [J-1], s' \in \mathcal{S}} \tilde{\delta}_i(s', j')^2 \mid \mathcal{F}_i, \tilde{b}_i = b \right] \\
&\stackrel{(f)}{\leq} \xi^2 \kappa^{-2} \cdot (2\|r\|_\infty^2 + 8 \cdot \|\underline{\tilde{q}}_i^d\|_\infty^2).
\end{aligned}$$

Step (a) above follows by substituting $(G_{\tilde{s}_1, \underline{\tilde{q}}_i^d})(b) - (G\underline{\tilde{q}}_i^d)(b)$ and replacing

$$\tilde{\delta}_i(s', j') := r(s, a) + \gamma \max_{a' \in \mathcal{A}} \tilde{q}_i^d(t-1, s', j', a') - \underline{\tilde{q}}_i^d(t, s, j, a). \quad (\text{B.29})$$

The equality in step (b) holds because for a random variable $\tilde{x} := \mathbb{E}[\partial \ell_{\frac{j}{J}}^\kappa (\tilde{\delta}_i(\tilde{s}_1, \tilde{j}')) \mid \tilde{s}_1]$, the variance satisfies $\mathbb{E}[(\tilde{x} - \mathbb{E}[\tilde{x}])^2] = \mathbb{E}[\tilde{x}^2] - (\mathbb{E}[\tilde{x}])^2$. Step (c) upper bounds the expectation by a supremum, and step (d) uses $\partial \ell_{\frac{j}{J}}^\kappa (0) = 0$ from the definition in (4.12). Step (e) uses Lemma B.3.4 to bound the derivative difference as a function of the step size. Finally, step (f) derives the final upper bound since

$$\|r\|_\infty = \max_{s \in \mathcal{S}, a \in \mathcal{A}} |r(s, a)|, \quad \|\underline{\tilde{q}}_i^d\|_\infty = \max_{b \in \mathcal{K}} |\underline{\tilde{q}}_i^d(b)|,$$

where $\mathcal{K} = [T] \times \mathcal{S} \times [J-1] \times \mathcal{A}$ and

$$\begin{aligned}
\max_{j' \in [J-1], s' \in \mathcal{S}} \tilde{\delta}_i(s', j')^2 &\leq (\|r\|_\infty + 2\|\underline{\tilde{q}}_i^d\|_\infty)^2 \leq (\|r\|_\infty + 2\|\underline{\tilde{q}}_i^d\|_\infty)^2 + (\|r\|_\infty - 2\|\underline{\tilde{q}}_i^d\|_\infty)^2 \\
&= 2\|r\|_\infty^2 + 8\|\underline{\tilde{q}}_i^d\|_\infty^2,
\end{aligned}$$

and because $\underline{\tilde{q}}_i^d$ is measurable on $\mathcal{F}_i$.

Case 2: Assume that $t = 0 \vee j = 0$. Directly from the definition of the operators in (B.14):

$$\begin{aligned}\mathbb{E}[(\tilde{\phi}_i(b))^2 \mid \mathcal{F}_i, \tilde{b}_i = b] &= \mathbb{E}[(H_{\tilde{s}_i \tilde{q}_i^d})(b) - (H_{\underline{q}_i^d})(b))^2 \mid \mathcal{F}_i, \tilde{b}_i = b] \\ &= \xi \cdot \mathbb{E}[(-(G_{\tilde{s}_i \tilde{q}_i^d})(b) + (G_{\underline{q}_i^d})(b))^2 \mid \mathcal{F}_i, \tilde{b}_i = b] \\ &= 0.\end{aligned}$$

Finally, we can confirm that

$$\begin{aligned}\mathbb{E}[\tilde{\phi}_i(b)^2 \mid \mathcal{F}_i] &= \mathbb{E}[\tilde{\phi}_i(b)^2 \mid \mathcal{F}_i, \tilde{b}_i \neq b] \cdot \mathbb{P}[\tilde{b}_i \neq b \mid \mathcal{F}_i] + \mathbb{E}[\tilde{\phi}_i(b)^2 \mid \mathcal{F}_i, \tilde{b}_i = b] \cdot \mathbb{P}[\tilde{b}_i = b \mid \mathcal{F}_i] \\ &\leq 0 \cdot \mathbb{P}[\tilde{b}_i \neq b \mid \mathcal{F}_i] + \xi^2 \kappa^{-2} \cdot (2\|r\|_\infty^2 + 8 \cdot \|\tilde{q}_i^d\|_\infty^2) \cdot \mathbb{P}[\tilde{b}_i = b \mid \mathcal{F}_i] \\ &\leq \xi^2 \kappa^{-2} \cdot (2\|r\|_\infty^2 + 8 \cdot \|\tilde{q}_i^d\|_\infty^2).\end{aligned}$$

□

B.3.6.6 Main Proof

Proof of Theorem 4.4.8. We verify that the sequence of our q-learning iterates satisfies the properties in Proposition B.3.9.

- The step size condition in Theorem 4.4.8 guarantees that we satisfy property (a) in Proposition B.3.9
- We satisfy property (b) in Proposition B.3.9 because
 - Lemma B.3.16 shows that we satisfy property (1) in Assumption B.3.8
 - Lemma B.3.17 shows that we satisfy property (2) in Assumption B.3.8
- Theorem B.3.13 shows that we satisfy property (c) in Proposition B.3.9

□

B.4 Empirical Results

The code used to generate all plots can be found in <https://github.com/MonkieDein/DRA-Q-LA>.

B.4.1 Domain Details

For each domain, we provide CSV files and Julia JLD files in the supplementary material with the exact specifications of the domains we use. Domain detail for six out of seven of our domains include (Machine Replacement (MR), Gambler’s Ruin (GR), Inventory1 (INV1), Inventory2 (INV2), Riverswim (RS), and Population Management (POP)) can be found in Appendix C.4. The Cliffwalk (CW) domain is similar to the one described in [122, Ex. 6.6], with a minor modification. In this version, the agent transitions to each adjacent direction with a 10%-probability instead of always following the selected direction (see Fig. 2.9). The initial state s_0 specification can be found in Table B.1. We initialize all environments with a discount factor of $\gamma = 0.9$ and a horizon $T = 100$.

	MR	GR	INV1	INV2	RS	POP	CW
s_0	1	5	10	20	9	44	37

Table B.1: Initial state for each domain

B.4.2 Algorithmic Details

B.4.2.1 Algorithm 4

Line 6 is implemented with $\epsilon = 10^{-14}$ to account for the non-associative property of floating point arithmetic with $\tau = \frac{q_t^d(s, j, a^*) - r}{\gamma}$ as:

$$j \leftarrow \arg \min \left\{ j' \in [J - 1] \mid \max_{a' \in \mathcal{A}} q_{t-1}^d(s', j', a') \geq \tau - \epsilon |\tau| \right\} .$$

B.4.2.2 Algorithm 5

Without loss of generality, the VaR-Q-value function is trained with a standardized scaled reward function $\hat{r}(s, a) \leftarrow \frac{r(s, a) - \underline{R}}{\bar{R} - \underline{R}}$. We also remove the time indices to reduce computational overhead and initialize the Q-value function with $\hat{q}^d \leftarrow (1 - \gamma)^{-1}$. The VaR-Q-value function is then unscaled via $\tilde{q}^d \leftarrow \hat{q}^d \cdot (\bar{R} - \underline{R}) + \frac{\underline{R}}{1 - \gamma}$ before being compared with the DP variant $\underline{q}^d$. The learning rate is defined as $\beta_i \leftarrow 100 \cdot (0.1^{i \cdot 0.0003})$ for i -th occurrence of sample (s, a) across all domains. For a fair comparison between domains, we sample a transition for every (s, a) -pair at each iteration.

B.4.2.3 Nested VaR (nVaR)

also known as dynamic VaR, nVaR is solved via the following DP for each $s \in \mathcal{S}$ and $t \in [T - 1]$ as

$$v_{t+1}(s) = \max_{a \in \mathcal{A}} \text{VaR}_{\alpha_0} [r(s, a) + \gamma \cdot v_t(\tilde{s}')],$$

where $\tilde{s}' \sim p(s, a, \cdot)$. Then, we evaluate a greedy policy $\pi_t: \mathcal{S} \rightarrow \mathcal{A}, k \in [T - 1]$ constructed to satisfy

$$\pi_k(s) \in \arg \max_{a \in \mathcal{A}} \text{VaR}_{\alpha_0} [r(s, a) + \gamma \cdot v_{T-k}(\tilde{s}')].$$

B.4.2.4 Distributional VaR (VaR-IQN)

It uses the Markov action-selection strategy proposed by [39, 72] with $J = 4096$ uniform quantile discretization [40, 108]:

$$q_{t+1}(s, \alpha_j, a) = \text{VaR}_{\alpha_j} \left[r(s, a) + \gamma \max_{a' \in \mathcal{A}} \text{VaR}_{\alpha_0} [q_t(\tilde{s}', \tilde{u}, a')] \right] \quad \forall \alpha_j = \frac{2j+1}{J} \text{ where } j \in [J - 1],$$

where $\tilde{u}$ refer to the discretized uniform distribution satisfy $\mathbb{P}[\tilde{u} = \frac{2j+1}{J}] = \frac{1}{J} \forall j \in [J - 1]$ follows greedy policy

$$\pi_k(s) \in \arg \max_{a \in \mathcal{A}} \text{VaR}_{\alpha_0} [q_{T-k}(s, \tilde{u}, a)].$$

In contrast to our algorithm, an optimal action a is selected w.r.t the initial Markov risk level of interest α_0 instead of the quantile-dependent risk level α_j [81].

B.4.2.5 Conditional Value at Risk (CVaR)

Follows the dynamic program described in [13],

$$\begin{aligned} \max_{\pi} \text{CVaR}_{\alpha} [\tilde{\mathfrak{R}}_T^{\pi}(s)] &:= \max_{\pi} \sup_{z \in \mathbb{R}} \left(z - \alpha^{-1} \mathbb{E} \left[z - \tilde{\mathfrak{R}}_T^{\pi}(s) \right]_+ \right) \\ &= \sup_{z \in \mathbb{R}} \left(z - \alpha^{-1} \min_{\pi} \mathbb{E} \left[z - \tilde{\mathfrak{R}}_T^{\pi}(s) \right]_+ \right), \end{aligned}$$

instead of solving the objective directly, they suggested a bi-level optimization. On the first level (inner optimization), given fixed z it requires optimizing for $v_T^*(s, z) := \min_{\pi} \mathbb{E} \left[z - \tilde{\mathfrak{R}}_T^{\pi}(s) \right]_+$. The second level (outer optimization) is optimized through $z \in \mathbb{R}$ or over a discretization set $z \in \mathcal{Z}$ and solves for the objective $\max_{\pi} \text{CVaR}_{\alpha} [\tilde{\mathfrak{R}}_T^{\pi}(s)] = \sup_{z \in \mathbb{R}} (z - \alpha^{-1} v_T^*(s, z))$.

For inner optimization, we initialize $v_0^{\pi}(s, z) = \mathbb{E} \left[z - \tilde{\mathfrak{R}}_0^{\pi}(s) \right]_+ = \mathbb{E} [z - 0]_+ = \max\{0, z\}$ for all π . We can compute the optimal value function $v_t^*(s, z) = \min_{\pi} \mathbb{E} \left[z - \tilde{\mathfrak{R}}_t^{\pi}(s) \right]_+$ using dynamic program, for each $t \in \{T-1, \dots, 0\}$, $s \in \mathcal{S}$ and $z \in \mathbb{R}$ as

$$\hat{v}_{t+1}^*(s, z) = \gamma \cdot \min_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \hat{v}_t^*(s', \frac{z - r(s, a)}{\gamma}) \cdot p(s, a, s').$$

We proof the dynamic program via mathematical induction: Let $v_t^*(s, z) = \min_{\pi} \mathbb{E} \left[z - \tilde{\mathfrak{R}}_t^{\pi}(s) \right]_+$, we show that following our dynamic program above we get

$$\begin{aligned} \hat{v}_{t+1}^*(s, z) &= \gamma \cdot \min_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} v_t^*(s', \frac{z - r(s, a)}{\gamma}) \cdot p(s, a, s') \\ &= \gamma \cdot \min_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \min_{\pi} \mathbb{E} \left[\frac{z - r(s, a)}{\gamma} - \tilde{\mathfrak{R}}_t^{\pi}(s') \right]_+ \cdot p(s, a, s') \\ &= \gamma \cdot \min_{a \in \mathcal{A}} \min_{\pi} \mathbb{E} \left[\frac{z - r(s, a)}{\gamma} - \tilde{\mathfrak{R}}_t^{\pi}(\tilde{s}') \right]_+ \end{aligned}$$

$$\begin{aligned}
&= \min_{\pi} \mathbb{E} \left[z - r(s, a) - \gamma \cdot \tilde{\mathfrak{R}}_t^{\pi}(\tilde{s}') \right]_+ \\
&= \min_{\pi} \mathbb{E} \left[z - \tilde{\mathfrak{R}}_{t+1}^{\pi}(\tilde{s}') \right]_+ = v_{t+1}^*(s, z) .
\end{aligned}$$

However, solving this dynamic program is computationally infeasible for arbitrary MDPs because $z \in \mathbb{R}$ has infinitely many possible values.

Since it is not computationally feasible to solve for all $z \in \mathbb{R}$ has infinitely many possible values, with the same intuition presented in Section 4.4.1. We propose an approximation with two steps to ensure finite discretization: (1) rounding z to a finite decimal precision d , and (2) ensuring a closed set for the input parameter z .

(1) The simplest method to tackle infinite elements is via rounding, we replace the optimization of $z \in \mathbb{R}$ to an approximation with rounding to decimal d as an approximation, i.e. $\lceil x \rceil_d$ is denoted as the ceiling function (rounding x up) to d digits of decimal. Let $\underline{v}_T^*(s, z) \leq v_T^*(s, z) \leq \bar{v}_T^*(s, z)$ for all $s \in \mathcal{S}, z \in \mathbb{R}$, where $\underline{v}_T^*(s, z)$ represents an underestimation and $\bar{v}_T^*(s, z)$ represents an overestimation of the value function we have that

$$\sup_{z \in \mathbb{R}} (z - \alpha^{-1} \bar{v}_T^*(s, z)) \leq \max_{\pi} \text{CVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_T^{\pi}(s) \right] = \sup_{z \in \mathbb{R}} (z - \alpha^{-1} v_T^*(s, z)) \leq \sup_{z \in \mathbb{R}} (z - \alpha^{-1} \underline{v}_T^*(s, z)) .$$

To provide guarantee, the approximation will have to underestimate the true CVaR, therefore, needed to solve for an overestimation of the true value function.

(2) In the discounted setting, it suffices to compute the value function for a closed set $\mathcal{Z}_t^d := \{L, L+10^{-d}, \dots, U\} \subset [L, U]$ for an MDP with horizon t where $L = \lceil \frac{\underline{R}(1-\gamma^t)}{1-\gamma} \rceil_d$ and $U = \lceil \frac{\bar{R}(1-\gamma^t)}{1-\gamma} \rceil_d$, because the value function and behavior of z outside this range, i.e. $z \in (-\infty, L) \cup (U, \infty)$ are well defined. Specifically,

- For $z \leq L$, the value function for all states s is given by $\min_{\pi} \mathbb{E} \left[z - \tilde{\mathfrak{R}}^{\pi}(s) \right]_+ = 0$.
- For $z \geq U$, the value function for all states s satisfies $\min_{\pi} \mathbb{E} \left[z - \tilde{\mathfrak{R}}^{\pi}(s) \right]_+ = (z - U) + \min_{\pi} \mathbb{E} \left[U - \tilde{\mathfrak{R}}^{\pi}(s) \right]_+ = (z - U) + v_t^*(s, U)$.

This approach can also be extended to the non-discounted finite-horizon case, where the bounds simplify to $L = t\underline{R}$ and $U = t\bar{R}$.

The discretized dynamic program is defined as for each $t \in \{T - 1, \dots, 0\}$, $s \in \mathcal{S}$ and $d \in \mathbb{N}$

$$\bar{v}_{t+1}^*(s, z) = \gamma \cdot \min_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \bar{v}_t^*(s', \mathcal{Z}_t^d(\lceil \frac{z - r(s, a)}{\gamma} \rceil_d)) \cdot p(s, a, s'), \quad (\text{B.30})$$

where $\mathcal{Z}_t(x)$ denotes the clamp function limits a value to a range between the range of set $\mathcal{Z}_t^d$.

Algorithm 11: CVaR MDP

Require: Risk level $\alpha \in (0, 1]$, discount factor $\gamma \in [0, 1]$

- 1: $U_0 \leftarrow 0$
 - 2: $\bar{v}_0^*(s, 0) \leftarrow 0$, $\forall s \in \mathcal{S}$
 - 3: **for all** $t \in 0 : T - 1$ **do**
 - 4: $U_{t+1} \leftarrow \bar{R} + \gamma \cdot U_t$
 - 5: $\bar{v}_{t+1}^*(s, z) \leftarrow \gamma \cdot \min_{a \in \mathcal{A}} \mathbb{E}[(z - U_t)_+ + \bar{v}_t^*(s', \mathcal{Z}_t^d(\lceil \frac{z - r(s, a)}{\gamma} \rceil_d))] , \forall s \in \mathcal{S}, z \in \mathcal{Z}_t^d$
 - 6: $\pi_{t+1}^*(s, z) \leftarrow \arg \min_{a \in \mathcal{A}} \gamma \cdot \mathbb{E}[(z - U_t)_+ + \bar{v}_t^*(s', \mathcal{Z}_t^d(\lceil \frac{z - r(s, a)}{\gamma} \rceil_d))] , \forall s \in \mathcal{S}, z \in \mathcal{Z}_t^d$
 - 7: **end for**
 - 8: $z^* \leftarrow \max_{z \in \mathcal{Z}_T^d} (z - \frac{1}{\alpha} (\sum_{s_0 \in \mathcal{S}} \bar{v}_T^*(s_0, z) \cdot p(s_0)))$
 - 9: **return** $\bar{v}^*, \pi^*, z^*$
-

B.4.2.6 Quantile Based CVaR (Chow)

We implemented the algorithm described in [33, 62] with $J = 4096$ uniform discretization as

$$(B_{\max} q)(s, \alpha_j, a) := r(s, a) + \gamma \cdot \min_{\zeta \in \mathcal{Z}_{sa}(\alpha_j)} \sum_{s' \in \mathcal{S}} \zeta_{s'} \cdot \max_{a' \in \mathcal{A}} q(s', \frac{\alpha_j \zeta_{s'}}{p(s, a, s')}, a') \quad \forall \alpha_j = \frac{j}{J} \text{ where } j \in [J],$$

$$\mathcal{Z}_{sa}(\alpha) := \{\xi \in \Delta_{\mathcal{S}} \mid \alpha \zeta_{s'} \leq p(s, a, s')\}.$$

which follow a greedy history-dependent policy described in [33]. We use the same discretization level as our algorithm. It is important to note that this algorithm could over-approximate the true static CVaR value function due to the duality gap, so it may perform badly, as described in Section 3.3.

B.4.2.7 EVaR

Algorithm described in Chapter 5 and known to perform well when evaluated with static CVaR and EVaR. We implemented the algorithm with the time-dependent policy described there, and with fixed ERM discretization $\beta_j = 100 \cdot (0.99^j) \ \forall j \in [3000]$, instead of a domain-dependent discretization.

B.4.3 More empirical results

Fig. B.2 extends Fig. 4.2 demonstrates that the performance of $\underline{\pi}$ across all the domains to understand how the selection of $\underline{q}^d$ in Algorithm 4 contribute to the quality of the solution. More specifically, $\underline{\pi}$ lies within $[\underline{q}^d, \bar{q}^d]$, whereas $\bar{\pi}$ may performs worse than $\underline{q}^d$. Furthermore, as the discretization level increases, the bounding gap $\bar{q}^d - \underline{q}^d$ shrinks, suggesting that $\underline{\pi}$ converges to π^* .

Fig. B.3 extends Fig. 4.1 compares our algorithm with other related algorithms (detailed in Appendix B.4.2) for all the domains on quantile levels $\alpha_0 \in \{0.05, 0.15, \dots, 0.85, 0.95\}$. As we can see, our algorithm consistently outperforms all other algorithms across all tested domains and quantile levels illustrating the robustness of our algorithm.

Fig. B.4 shows that for all the domains, both the value function and the performance of the policy for the κ -soft quantile Q-learning (Algorithm 5) with $\kappa \in \{10^{-4}, 10^{-8}, 10^{-12}, 0\}$ and uniform discretization of $J = 256$ converges to DP variant Eq. (4.10) after 20,000 iterations. Not only do the value functions for the Q-learning converge closely to the DP's, but the performance of the policy implied by the Q-learning value function also matches the policy from the DP variant.

Fig. B.5 extends Fig. 4.3 demonstrates for all the domains that the value function achieve from κ -soft quantile Q-learning (Algorithm 5) for $\kappa \in \{10^{-4}, 10^{-8}, 10^{-12}, 0\}$ and uniform discretization of $J = 256$ converges to the DP variant Eq. (4.10). We use the Wasserstein-1 distance of quantile function defined as

$$W_1(\underline{q}^d, \tilde{q}^d) := \frac{1}{J} \sum_{j \in [J-1]} |\max_{a \in \mathcal{A}} \{ \underline{q}^d(s_0, j, a) \} - \max_{a \in \mathcal{A}} \{ \tilde{q}^d(s_0, j, a) \}|$$

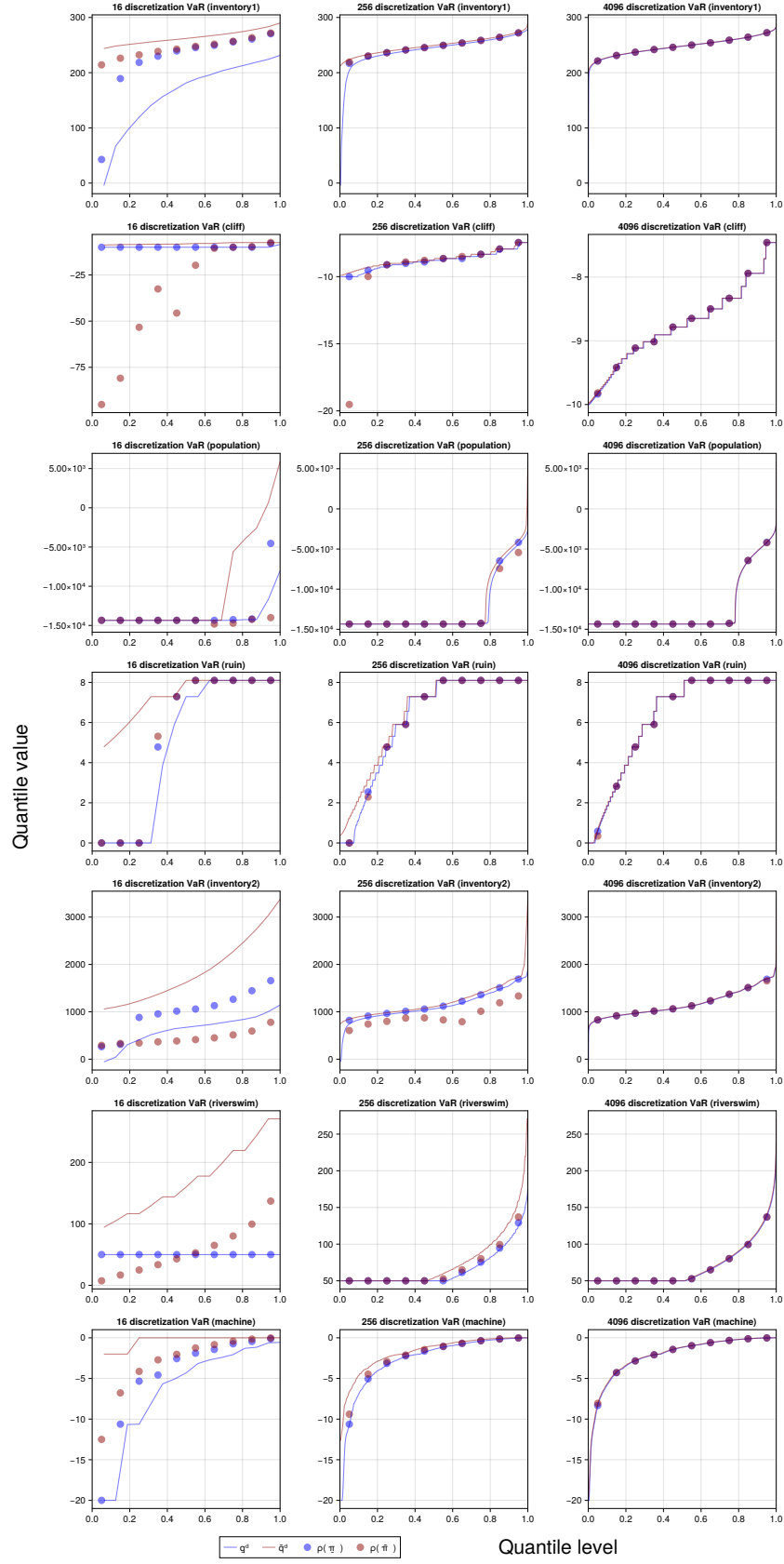


Figure B.2: Approximation bound q^d and its respective π policy performance

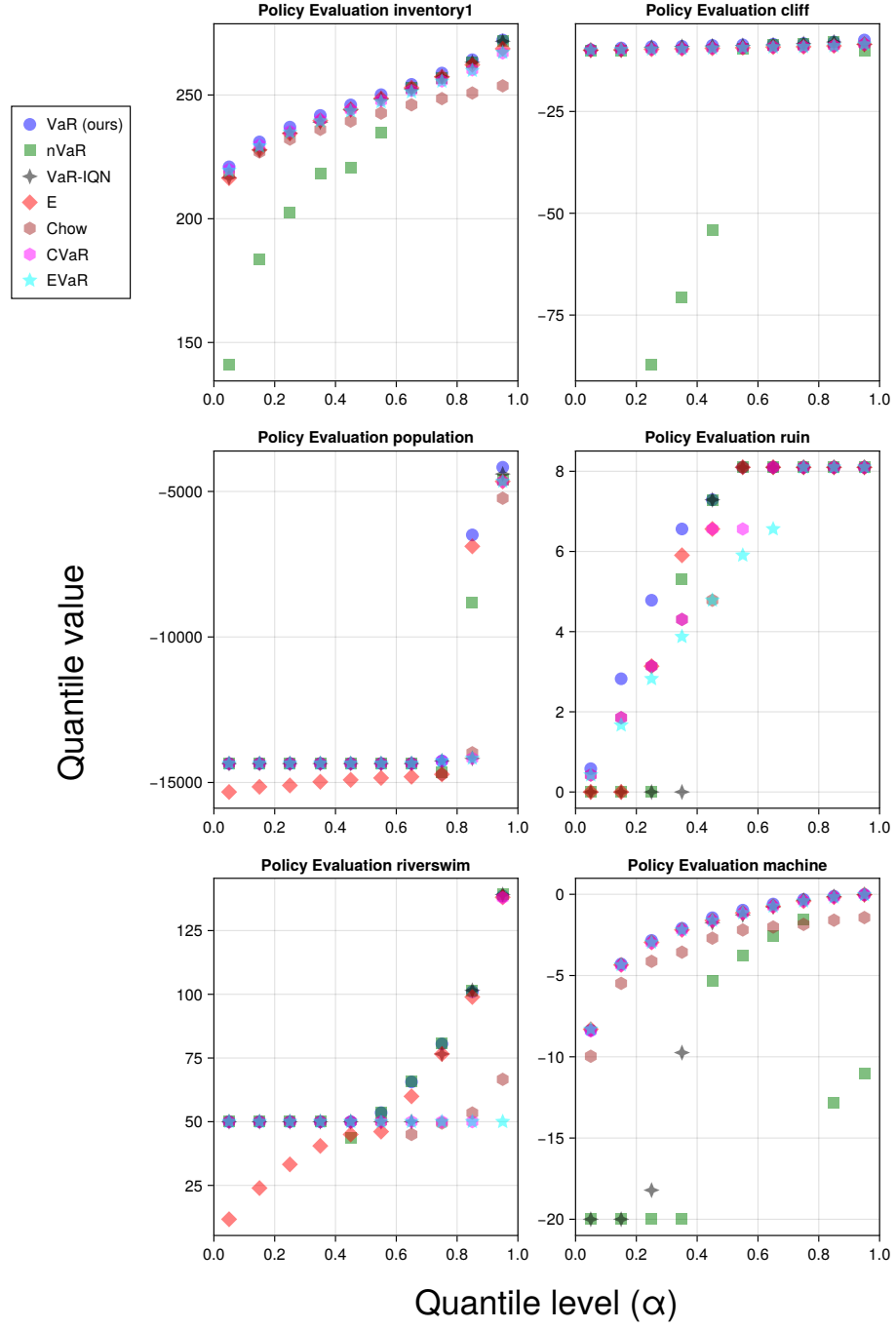


Figure B.3: Policy performance evaluation $\text{VaR}_\alpha(\tilde{\mathfrak{R}}_T^\pi)$

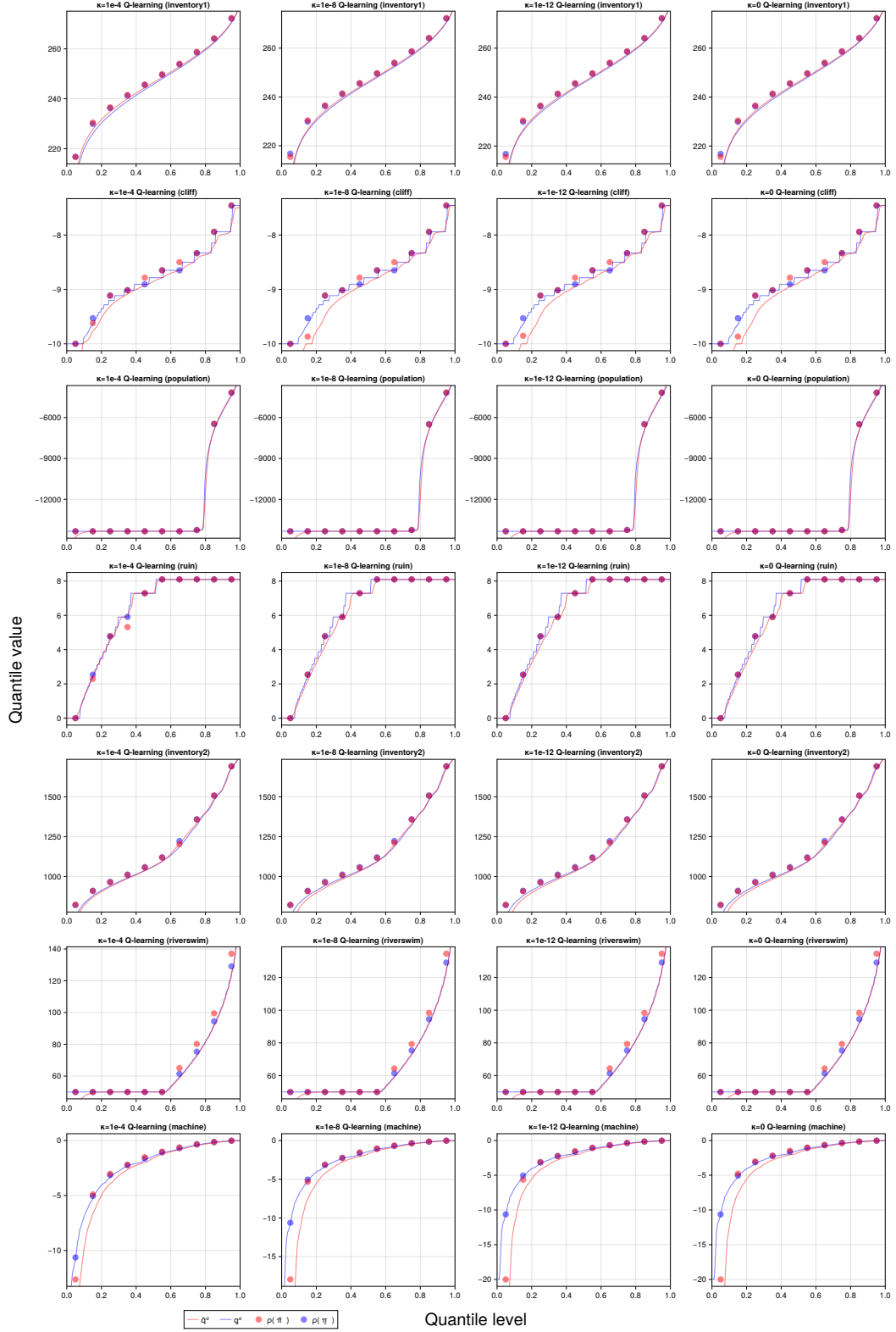


Figure B.4: Q-learning $\tilde{q}$ vs DP q value function and policy performance after 20,000 iterations ($J = 256$)

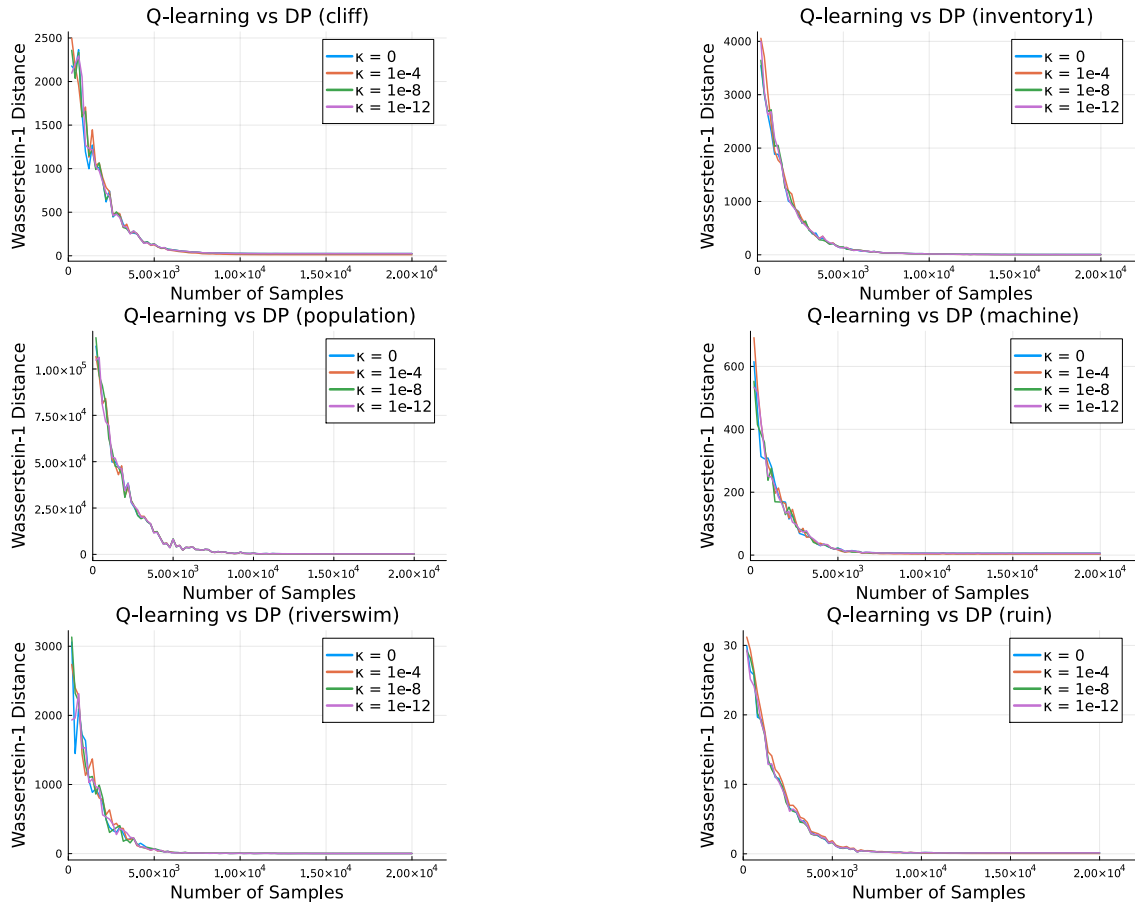


Figure B.5: Q-learning vs DP value function Wasserstein distance.

to evaluate the differences between two value functions. From Fig. B.5 we can see that for all domains, the Wasserstein-1 distances of the value functions were far apart at the beginning and quickly converges to zero as the number of samples for each (s, a) pair increase.

CHAPTER C

Supplementary Materials for Chapter 5

C.1 Proofs of Section 5.2

The following proposition states a simple, but important property of the expectation operator which plays a crucial role in formulating the dynamic programs. The property is known under several different names, including *the tower property*, *the law of total expectation*, and *the law of iterated expectations*.

Proposition C.1.1 (Tower Property for Expectation (e.g., Proposition 3.4 in [107])). *Any two random variables $\tilde{x}_1, \tilde{x}_2 \in \mathbb{X}$ satisfy that*

$$\mathbb{E}[\tilde{x}_1] = \mathbb{E}[\mathbb{E}[\tilde{x}_1 \mid \tilde{x}_2]] .$$

A convenient way to represent ERM is to use its *certainty equivalent* form. This form relates the risk measure to the popular expected utility framework for decision-making [17]. In the expected utility framework, one prefers a lottery (or a random reward) $\tilde{x}_1 \in \mathbb{X}$ over $\tilde{x}_2 \in \mathbb{X}$ if and only if

$$\mathbb{E}[u(\tilde{x}_1)] \geq \mathbb{E}[u(\tilde{x}_2)] ,$$

for some increasing *utility function* $u: \mathbb{R} \rightarrow \mathbb{R}$.

The expected utility $\mathbb{E}[u(\tilde{x})]$ is difficult to interpret because its units are incompatible with $\tilde{x}$. A more interpretable characterization of the expected utility is to use the *certainty equivalent* $z \in \mathbb{R}$,

which is defined as the certain quantity that achieves the same expected utility as $\tilde{x}$:

$$\mathbb{E}[u(z)] = \mathbb{E}[u(\tilde{x})], \quad \text{and therefore,} \quad z = u^{-1}(\mathbb{E}[u(\tilde{x})]) . \quad (\text{C.1})$$

Algebraic manipulation from (C.1) then shows that ERM for any $\tilde{x} \in \mathbb{X}$ can be represented as the certainty equivalent

$$\text{ERM}_\beta [\tilde{x}] = u^{-1}(\mathbb{E}[u(\tilde{x})]) , \quad (\text{C.2})$$

for the utility function $u: \mathbb{R} \rightarrow \mathbb{R}$ (see definition 2.1 in [17]) defined as

$$u(x) = \beta^{-1} - \beta^{-1} \cdot \exp(-\beta \cdot x) .$$

Because the function u is strictly increasing, its inverse $u^{-1}: \mathbb{R} \rightarrow \mathbb{R}$ exists and equals to

$$u^{-1}(z) = -\beta^{-1} \cdot \log(1 - \beta \cdot z) .$$

C.1.1 Proof of Theorem 5.2.2

Proof. The property is trivially true when $\beta = 0$ from Proposition C.1.1 since $\text{ERM}_0 [=] \mathbb{E}$. The property then follows by algebraic manipulation for $\beta > 0$ using the certainty equivalent representation in (C.2) as

$$\begin{aligned} \text{ERM}_\beta [\text{ERM}_\beta [\tilde{x}_1 \mid \tilde{x}_2]] &= \text{ERM}_\beta [u^{-1}(\mathbb{E}[u(\tilde{x}_1) \mid \tilde{x}_2])] \\ &= u^{-1}(\mathbb{E}[u(u^{-1}(\mathbb{E}[u(\tilde{x}_1) \mid \tilde{x}_2]))]) \\ &= u^{-1}(\mathbb{E}[\mathbb{E}[u(\tilde{x}_1) \mid \tilde{x}_2]]) \\ &= u^{-1}(\mathbb{E}[u(\tilde{x}_1)]) && \text{Proposition C.1.1} \\ &= \text{ERM}_\beta [\tilde{x}_1] . \end{aligned}$$

□

C.1.2 Proof of Lemma 5.2.5

Proof. Recall that the Hoeffding's lemma shows that for any $\forall \lambda \in \mathbb{R}$, we may write [26, 87]

$$0 < \mathbb{E}[e^{\lambda \tilde{x}}] \leq \exp \left(\lambda \cdot \mathbb{E}[\tilde{x}] + \frac{\lambda^2 (x_{\max} - x_{\min})^2}{8} \right) .$$

Applying log to both sides of the inequality above gives

$$\log (\mathbb{E}[e^{\lambda \tilde{x}}]) \leq \lambda \mathbb{E}[\tilde{x}] + \frac{\lambda^2 (x_{\max} - x_{\min})^2}{8} .$$

Then, variable substitution $\lambda = -\beta$ and algebraic manipulation shows that

$$\begin{aligned} \log (\mathbb{E}[e^{-\beta \tilde{x}}]) &\leq -\beta \cdot \mathbb{E}[\tilde{x}] + \frac{\beta^2 (x_{\max} - x_{\min})^2}{8} \\ -\frac{1}{\beta} \log (\mathbb{E}[e^{-\beta \tilde{x}}]) &\geq \mathbb{E}[\tilde{x}] - \frac{\beta (x_{\max} - x_{\min})^2}{8} \end{aligned}$$

Substituting the definition of ERM into the inequality above yields then the first desired inequality:

$$\mathbb{E}[\tilde{x}] - \frac{\beta (x_{\max} - x_{\min})^2}{8} \leq \text{ERM}_{\beta} [\tilde{x}] .$$

The second inequality in the lemma's statement, $\text{ERM}_{\beta} [\tilde{x}] \leq \mathbb{E}[\tilde{x}]$, follows immediately from the Donsker-Varadhan's Variational Formula. $\square$

C.1.3 Proof of Lemma 5.2.6

Proof. We first prove that $\max_{a \in \mathcal{A}} g(a) \leq \max_{d \in \Delta^{\mathcal{A}}} \text{ERM}_{\beta} [g(\tilde{x}) \mid \tilde{x} \sim d]$. Let $a^* \in \arg \max_{a \in \mathcal{A}} g(a)$ be an optimal action. We now construct a policy $\bar{d} \in \Delta^{\mathcal{A}}$ as $\bar{d}(a^*) = 1$ and $\bar{d}(a) = 0, \forall a \in \mathcal{A} \setminus \{a^*\}$.

Substituting $\bar{d}$ in the definition of ERM yields that

$$\begin{aligned}
\text{ERM}_\beta [g(\tilde{x}) \mid \tilde{x} \sim \bar{d}] &= -\beta^{-1} \cdot \log \left(\mathbb{E} [\exp(-\beta \cdot g(\tilde{x})) \mid \tilde{x} \sim \bar{d}] \right) \\
&= -\beta^{-1} \cdot \log \left(\exp(-\beta \cdot g(a^*)) \right) \\
&= g(a^*) .
\end{aligned} \tag{C.3}$$

Using (C.3) and the fact that $\bar{d}$ is a valid probability distribution in Δ^A , we obtain the desired inequality as

$$\max_{a \in \mathcal{A}} g(a) = g(a^*) = \text{ERM}_\beta [g(\tilde{x}) \mid \tilde{x} \sim \bar{d}] \leq \max_{d \in \Delta^A} \text{ERM}_\beta [g(\tilde{x}) \mid \tilde{x} \sim d] .$$

To prove the converse inequality $\max_{a \in \mathcal{A}} g(a) \geq \max_{d \in \Delta^A} \text{ERM}_\beta [g(\tilde{x}) \mid \tilde{x} \sim d]$, we define d^* as an optimal distribution $d^* \in \arg \max_{d \in \Delta^A} \text{ERM}_\beta [g(\tilde{x}) \mid \tilde{x} \sim d]$. It will be convenient to use the dual representation of $\text{ERM}_\beta [g(\tilde{x}) \mid \tilde{x} \sim d]$, which for any $d \in \Delta^A$ is defined as (see e.g., [4])

$$\text{ERM}_\beta [g(\tilde{x}) \mid \tilde{x} \sim d] = \inf_{\bar{d} \in \Delta^A, \bar{d} \ll d} \left\{ \mathbb{E}[g(\tilde{x}) \mid \tilde{x} \sim \bar{d}] + \frac{1}{\beta} \text{KL}(\bar{d} \parallel d) \right\} ,$$

where KL is the KL-divergence and $\ll$ denotes the absolute continuity of probability measures.

Using this dual representation, we get the following upper-bound on $\text{ERM}_\beta [g(\tilde{x}) \mid \tilde{x} \sim d^*]$:

$$\begin{aligned}
\text{ERM}_\beta [g(\tilde{x}) \mid \tilde{x} \sim d^*] &= \inf_{\bar{d} \in \Delta^A} \left\{ \mathbb{E}[g(\tilde{x}) \mid \tilde{x} \sim \bar{d}] + \frac{1}{\beta} \text{KL}(\bar{d} \parallel d^*) \mid \bar{d} \ll d^* \right\} \\
&\leq \mathbb{E}[g(\tilde{x}) \mid \tilde{x} \sim d^*] + \frac{1}{\beta} \text{KL}(d^* \parallel d^*) \\
&\stackrel{\text{(a)}}{=} \mathbb{E}[g(\tilde{x}) \mid \tilde{x} \sim d^*] \\
&\stackrel{\text{(b)}}{\leq} \max_{a \in \mathcal{A}} g(a) ,
\end{aligned}$$

where **(a)** holds because $\text{KL}(d \parallel d) = 0$, and **(b)** follows because A is finite, and thus, for each $d \in \Delta^A$, we have

$$\max_{a \in \mathcal{A}} g(a) \geq \mathbb{E}[g(\tilde{x}) \mid \tilde{x} \sim d] .$$

This proves the second desired inequality since $d^* \in \Delta^A$ and concludes the proof. $\square$

The result in Lemma 5.2.6 can be further generalized to a broader class of risk measures [44].

C.1.4 Proof of Lemma 5.2.9

Proof. Using positive quasi-homogeneity (Theorem 5.3.2) of ERM and algebraic manipulation, we can reformulate (5.2) for $c \cdot \tilde{x}$ as

$$\begin{aligned}
\text{EVaR}_\alpha [c \cdot \tilde{x}] &= \sup_{\beta > 0} \left(\text{ERM}_\beta [c \cdot \tilde{x}] + \frac{\log(1 - \alpha)}{\beta} \right) \\
&= \sup_{\beta > 0} \left(c \cdot \text{ERM}_{c \cdot \beta} [\tilde{x}] + \frac{\log(1 - \alpha)}{\beta} \right) \\
&= c \cdot \sup_{\beta > 0} \left(\text{ERM}_{c \cdot \beta} [\tilde{x}] + \frac{\log(1 - \alpha)}{c \cdot \beta} \right) \\
&= c \cdot \sup_{\beta > 0} \left(\text{ERM}_{c \cdot \beta} [\tilde{x}] + \frac{\log(1 - \alpha)}{c \cdot \beta} \right) \\
&= c \cdot \sup_{\tau > 0} \left(\text{ERM}_\tau [\tilde{x}] + \frac{\log(1 - \alpha)}{\tau} \right) \\
&= c \cdot \text{EVaR}_\alpha [\tilde{x}]
\end{aligned}$$

We used the variable substitution $\tau = c \cdot \beta$. Therefore, if the supremum is attained at τ^* for $c \cdot \tilde{x}$, it is attained at $\beta^* = c^{-1} \cdot \tau^*$ for $c \cdot \tilde{x}$.

Note that the derivation above also confirms that EVaR is positively homogeneous. $\square$

C.2 Proofs of Section 5.3

C.2.1 Proof of Theorem 5.3.2

Proof. The property is trivially true for $c = 0$ or $\beta = 0$ because $\text{ERM}_\beta[0] = 0$ and $\text{ERM}_0[\cdot] = \mathbb{E}[\cdot]$. For $c > 0$ and $\beta > 0$, the property follows by rearranging the terms as

$$\begin{aligned} \text{ERM}_{\beta \cdot c}[\tilde{x}] &= -\frac{1}{\beta c} \log(\mathbb{E}[e^{-\beta \cdot c \cdot \tilde{x}}]) \implies c \cdot \text{ERM}_{\beta \cdot c}[\tilde{x}] = -\frac{1}{\beta} \log(\mathbb{E}[e^{-\beta \cdot c \cdot \tilde{x}}]) \\ &= \text{ERM}_\beta[c \cdot \tilde{x}]. \end{aligned}$$

□

C.2.2 Proof of Theorem 5.3.3

Proof. The proof is divided into two parts: the proof for v^π (Eq. 5.8) and a proof for $v^\star$ (Eq. 5.9).

Proof for v^π : For any fixed $\pi \in \Pi_{\text{MR}}$, we prove the claim for v^π by backward induction on t from $t = 0$ to $t = T$. The base case of the induction with $t = 0$ is trivial because by definition $v_0(s) = 0, \forall s \in \mathcal{S}$. To prove the inductive step, we first assume that any function $v_{t'}^\pi, t' = 0 : (t-1)$ defined by (5.6) satisfies (5.8), and then show that the same is true for v_t^π . By the induction hypothesis we can substitute the definition of $v_{t-1}^\pi(\tilde{s}')$ from (5.6) into (5.8) and with change in variable $\tau = T - t$ we derive

$$\begin{aligned} v_t^\pi(s) &= \text{ERM}_{\beta \cdot \gamma^{T-t}}[r(s, \tilde{a}_\tau) + \gamma \cdot v_{t-1}^\pi(\tilde{s}')] \\ &= \text{ERM}_{\beta \cdot \gamma^{T-t}}[r(s, \tilde{a}_\tau) + \gamma \cdot \text{ERM}_{\beta \cdot \gamma^{T-(t-1)}}[\tilde{\mathfrak{R}}_{(T-(t-1)):T}^\pi(\tilde{s}')]] \\ &\stackrel{\tau=T-t}{=} \text{ERM}_{\beta \cdot \gamma^\tau}[r(s, \tilde{a}_\tau) + \gamma \cdot \text{ERM}_{\beta \cdot \gamma^{\tau+1}}[\tilde{\mathfrak{R}}_{\tau+1:T}^\pi(\tilde{s}')]] \\ &= \text{ERM}_{\beta \cdot \gamma^\tau} \left[r(s, \tilde{a}_\tau) + \gamma \cdot \text{ERM}_{\beta \cdot \gamma^{\tau+1}} \left[\sum_{\tau'=\tau+1}^{T-1} \gamma^{\tau'-\tau-1} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_{\tau+1} = \tilde{s}' \right] \right] \\ &\stackrel{(a)}{=} \text{ERM}_{\beta \cdot \gamma^\tau} \left[r(s, \tilde{a}_\tau) + \text{ERM}_{\beta \cdot \gamma^\tau} \left[\sum_{\tau'=\tau+1}^{T-1} \gamma^{\tau'-\tau} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_{\tau+1} = \tilde{s}' \right] \right] \end{aligned}$$

$$\begin{aligned}
&= \text{ERM}_{\beta, \gamma^\tau} \left[r(\tilde{s}_\tau, \tilde{a}_\tau) + \text{ERM}_{\beta, \gamma^\tau} \left[\sum_{\tau'=\tau+1}^{T-1} \gamma^{\tau'-\tau} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_{\tau+1} = \tilde{s}', \tilde{a}_\tau \right] \mid \tilde{s}_\tau = s \right] \\
&\stackrel{(b)}{=} \text{ERM}_{\beta, \gamma^\tau} \left[\text{ERM}_{\beta, \gamma^\tau} \left[r(\tilde{s}_\tau, \tilde{a}_\tau) + \sum_{\tau'=\tau+1}^{T-1} \gamma^{\tau'-\tau} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_{\tau+1} = \tilde{s}', \tilde{a}_\tau \right] \mid \tilde{s}_\tau = s \right] \\
&= \text{ERM}_{\beta, \gamma^\tau} \left[\text{ERM}_{\beta, \gamma^\tau} \left[\sum_{\tau'=\tau}^{T-1} \gamma^{\tau'-\tau} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_{\tau+1} = \tilde{s}', \tilde{a}_\tau \right] \mid \tilde{s}_\tau = s \right] \\
&\stackrel{(c)}{=} \text{ERM}_{\beta, \gamma^\tau} \left[\sum_{\tau'=\tau}^{T-1} \gamma^{\tau'-\tau} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_\tau = s \right] \\
&= \text{ERM}_{\beta, \gamma^\tau} \left[\tilde{\mathfrak{R}}_{\tau:T}^\pi(s) \right] \stackrel{\tau=T-t}{=} \text{ERM}_{\beta, \gamma^{T-t}} \left[\tilde{\mathfrak{R}}_{(T-t):T}^\pi(s) \right]
\end{aligned}$$

where (a), (b), and (c) come from the positive quasi-homogeneity (Theorem 5.3.2), translation invariance (A-2), and tower (Theorem 5.2.2) properties of ERM. This derivation proves the inductive step and shows that any function v^π that satisfies the Bellman equation in (5.8) satisfies the definition of value function in (5.6), and thus, is unique.

Proof for v^ :* The proof of the Bellman equation for the optimal value function v^* proceeds by backward induction analogously to the proof of (5.8) with the difference that it incorporates the optimization over actions. As before, the base case with $t = 0$ is trivial because $v_0^*(s) = 0$, $\forall s \in \mathcal{S}$ by definition. To prove the inductive step, we first assume that any function $v_{t'}^*$, $t' = 0 : (t-1)$ defined by (5.7) satisfies (5.9), and then show that the same is true for v_t^* .

In the proof of the inductive step, we use Lemma 5.2.6, which shows how the maximization over actions can be replaced by a maximization over randomized policies that are distributions over actions as

$$\begin{aligned}
v_t^*(s) &= \max_{a \in \mathcal{A}} \text{ERM}_{\beta, \gamma^{T-t}} \left[r(s, \tilde{a}_t) + \gamma \cdot v_{t-1}^*(\tilde{s}') \mid \tilde{a}_t = a \right] \\
&= \max_{d \in \Delta^{\mathcal{A}}} \text{ERM}_{\beta, \gamma^{T-t}} \left[r(s, \tilde{a}_t) + \gamma \cdot v_{t-1}^*(\tilde{s}') \mid \tilde{a}_t \sim d \right] .
\end{aligned} \tag{C.4}$$

By the induction hypothesis we can substitute the definition of $v_{t-1}^*(\tilde{s}')$ from (5.7) into (C.4) and

with change in variable $\tau = T - t$ we derive

$$\begin{aligned}
v_t^*(s) &= \max_{d \in \Delta^A} \text{ERM}_{\beta, \gamma^{T-t}} \left[r(s, \tilde{a}_\tau) + \gamma \max_{\pi \in \Pi_{\text{MR}}^{(T-t+2):T}} \text{ERM}_{\beta, \gamma^{T-(t-1)}} \left[\tilde{\mathfrak{R}}_{(T-(t-1)):T}^\pi(s) \right] \right] \\
&\stackrel{\tau=T-t}{=} \max_{d \in \Delta^A} \text{ERM}_{\beta, \gamma^\tau} \left[r(s, \tilde{a}_\tau) + \gamma \max_{\pi \in \Pi_{\text{MR}}^{\tau+2:T}} \text{ERM}_{\beta, \gamma^{\tau+1}} \left[\tilde{\mathfrak{R}}_{(\tau+1):T}^\pi(s) \right] \right] \\
&= \max_{d \in \Delta^A} \text{ERM}_{\beta, \gamma^\tau} \left[r(s, \tilde{a}_\tau) + \gamma \max_{\pi \in \Pi_{\text{MR}}^{\tau+2:T}} \text{ERM}_{\beta, \gamma^{\tau+1}} \left[\sum_{\tau'=\tau+1}^{T-1} \gamma^{\tau'-\tau-1} r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_{\tau+1} = \tilde{s}' \right] \right] \\
&\stackrel{(a)}{=} \max_{d \in \Delta^A} \text{ERM}_{\beta, \gamma^\tau} \left[r(s, \tilde{a}_\tau) + \max_{\pi \in \Pi_{\text{MR}}^{\tau+2:T}} \text{ERM}_{\beta, \gamma^\tau} \left[\sum_{\tau'=\tau+1}^{T-1} \gamma^{\tau'-\tau} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_{\tau+1} = \tilde{s}' \right] \right] \\
&= \max_{d \in \Delta^A} \text{ERM}_{\beta, \gamma^\tau} \left[r(\tilde{s}_\tau, \tilde{a}_\tau) + \max_{\pi \in \Pi_{\text{MR}}^{\tau+2:T}} \text{ERM}_{\beta, \gamma^\tau} \left[\sum_{\tau'=\tau+1}^{T-1} \gamma^{\tau'-\tau} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_{\tau+1} = \tilde{s}', \tilde{a}_\tau \right] \mid \tilde{s}_\tau = s \right] \\
&\stackrel{(b)}{=} \max_{d \in \Delta^A} \text{ERM}_{\beta, \gamma^\tau} \left[\max_{\pi \in \Pi_{\text{MR}}^{\tau+2:T}} \text{ERM}_{\beta, \gamma^\tau} \left[r(\tilde{s}_\tau, \tilde{a}_\tau) + \sum_{\tau'=\tau+1}^{T-1} \gamma^{\tau'-\tau} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_{\tau+1} = \tilde{s}', \tilde{a}_\tau \right] \mid \tilde{s}_\tau = s \right] \\
&= \max_{d \in \Delta^A} \text{ERM}_{\beta, \gamma^\tau} \left[\max_{\pi \in \Pi_{\text{MR}}^{\tau+2:T}} \text{ERM}_{\beta, \gamma^\tau} \left[\sum_{\tau'=\tau}^{T-1} \gamma^{\tau'-\tau} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_{\tau+1} = \tilde{s}', \tilde{a}_\tau \right] \mid \tilde{s}_\tau = s \right] \\
&\stackrel{(c)}{=} \max_{\pi \in \Pi_{\text{MR}}^{\tau+1:T}} \text{ERM}_{\beta, \gamma^\tau} \left[\sum_{\tau'=\tau}^{T-1} \gamma^{\tau'-\tau} \cdot r(\tilde{s}_{\tau'}, \tilde{a}_{\tau'}) \mid \tilde{s}_\tau = s \right] \\
&= \max_{\pi \in \Pi_{\text{MR}}^{\tau+1:T}} \text{ERM}_{\beta, \gamma^\tau} \left[\tilde{\mathfrak{R}}_{\tau:T}^\pi(s) \right] \stackrel{\tau=T-t}{=} \max_{\pi \in \Pi_{\text{MR}}^{(T-t+1):T}} \text{ERM}_{\beta, \gamma^{T-t}} \left[\tilde{\mathfrak{R}}_{(T-t):T}^\pi(s) \right],
\end{aligned}$$

where (a) comes from the positive quasi-homogeneity (Theorem 5.3.2) property of ERM, (b) comes from the translation invariance (Property A-2) of ERM, and (c) comes from the monotonicity (Property A-1) and tower (Theorem 5.2.2) properties of ERM. This derivation proves the inductive step and shows that any function v^* that satisfies the Bellman equation in (5.9) satisfies the definition of value function in (5.7), and thus, is unique. $\square$

C.2.3 Proof of Theorem 5.3.4

Proof. Following the notation of Section 2.4.1, let $\mathcal{H}_\tau$ be the set of all histories up to time τ inclusively. Let the optimal history-dependent value function be $u_t^*: \mathcal{H}_{T-t} \rightarrow \mathbb{R}, t \in 0:T$. The

value function $u^* = (u_t^*)_{t=0}^T$ is achieved by the optimal history-dependent policy because the state and actions are finite, and thus, the space of randomized history-dependent policies is compact.

The proof proceeds in three steps.

(i) First let $\tau := T - t$, we show that v^* attains the return of the optimal history-dependent value function:

$$u_t^*(h_\tau) = v_t^*(s_\tau) \quad \forall h_\tau \in \mathcal{H}_\tau, \quad t = 0:T,$$

where s_τ is the τ -th and final state in the history h_τ . This result is a consequence of the dynamic programming formulation in Theorem 5.3.3. An argument analogous to the proof of Theorem 4.4.2(a) in [103] shows that $u_t^*(h_\tau)$ depends only on s_τ , which is the final state in the history h_τ .

Using the standard backward-induction argument on t , we assume that $u_{t-1}^*(h_{\tau+1}) = v_{t-1}^*(s_{\tau+1})$ holds and then prove that $u_t^*(h_\tau) = v_t^*(s_\tau)$. Let $d^* \in \Delta^A$ be part of a decision rule achieving u^* that decides about the actions that should be taken at the remaining horizon t . Applying Theorem 5.3.3 to the optimal history-dependent value function u^* , we may write

$$\begin{aligned} u_t^*(h_\tau) &\stackrel{(a)}{=} \text{ERM}_{\beta, \gamma^{T-t}} [r(s_\tau, \tilde{a}_\tau) + \gamma \cdot u_{t-1}^*(\langle h_\tau, \tilde{a}_\tau, \tilde{s}' \rangle) \mid \tilde{a}_\tau \sim d^*] \\ &\stackrel{(b)}{=} \text{ERM}_{\beta, \gamma^{T-t}} [r(s_\tau, \tilde{a}_\tau) + \gamma \cdot v_{t-1}^*(\tilde{s}') \mid \tilde{a}_\tau \sim d^*] \\ &= \max_{d \in \Delta^A} \text{ERM}_{\beta, \gamma^{T-t}} [r(s_\tau, \tilde{a}_\tau) + \gamma \cdot v_{t-1}^*(\tilde{s}') \mid \tilde{a}_\tau \sim d] \\ &\stackrel{(c)}{=} \max_{a \in \mathcal{A}} \text{ERM}_{\beta, \gamma^{T-t}} [r(s_\tau, a) + \gamma \cdot v_{t-1}^*(\tilde{s}')] \stackrel{(d)}{=} v_t^*(s_\tau), \end{aligned}$$

where (a) follows from the fact that (i) the reward function depends only on the current state and not the full history and (ii) the history at time-step $\tau + 1$, $h_{\tau+1}$ is constructed by appending action $\tilde{a}_\tau$ and state $\tilde{s}'$ to the history at time-step τ : $h_{\tau+1} = \langle h_\tau, \tilde{a}_\tau, \tilde{s}' \rangle$, (b) comes from the inductive hypothesis, (c) is the result of Lemma 5.2.6, and (d) comes from (5.9) in Theorem 5.3.3. This result shows that the optimal history-dependent randomized u^* and the optimal Markov deterministic v^* value functions are equal in ERM-MDP.

(ii) The second part of the proof is to show that the value function of any (optimal) policy

$\pi^* \in \Pi_{\text{MD}}$ that is a solution to the optimization problem (5.3) is equal to the optimal value function v^* :

$$v_t^{\pi^*}(s) = v_t^*(s), \quad \forall s \in \mathcal{S}, t = 0:T.$$

This result follows using the standard backward induction argument and algebraic manipulation from Theorem 5.3.3. The derivation relies on the fact that $\mathcal{A}$ is finite and the maximum in (5.9) exists and is achievable.

(iii) Here we show that any greedy policy to the optimal value function v^* is an optimal policy, that is, a solution to the optimization problem (5.3). This is trivial from parts (i) and (ii) because the value function of the greedy policy to v^* is v^* , and we know from part (ii) that any policy that solves (5.3) also has value v^* . Thus, the greedy policy is in fact optimal. $\square$

C.2.4 Proof of Theorem 5.3.5

Proof. It is important to reiterate the following definitions:

1. $\bar{\pi}_\infty^*$ and $\bar{v}_\infty^* = v_\infty^{\bar{\pi}_\infty^*}$ are the optimal policy and value function of the MDP in the risk-neutral infinite-horizon discounted setting.
2. $\tilde{\pi}^* = \{\tilde{\pi}_t^*\}_{t=0}^{T'}$ and $\tilde{v}^* = v^{\tilde{\pi}^*} = \{\tilde{v}_t^*\}_{t=0}^{T'}$ are the optimal policy and value function of the finite-horizon discounted ERM-MDP with risk level β , discount factor γ , horizon T' , and the value function at time-step T' set to $\tilde{v}_0^* = v_\infty^*$. In other words, $\tilde{\pi}^*$ and $\tilde{v}^*$ are the outputs of Algorithm 8 with inputs $T = T'$ and initialization $\tilde{v}_0^* = v_\infty^*$.
3. $\hat{\pi}^* = \{\hat{\pi}_t^*\}_{t=0}^{T'}$ and its value $\hat{v}^* = \{\hat{v}_t^*\}_{t=0}^{T'}$ are the policy and value function returned by Algorithm 9. It uses policy $\hat{\pi}_0^*$, for all time step $\tau \geq T'$.

In this proof, we introduce a new set of value function w_τ and policy $\underline{\pi}_\tau$, where the subscript τ represents time steps instead of the remaining horizon t . This notation facilitates the analysis of infinite-horizon MDPs under ERM by explicitly capturing changes in risk levels at each step. We define $\tau = T - t$, so in the finite-horizon case, we have $v_t = w_{T-t} = w_\tau$. However, in

the infinite-horizon setting where $T = \infty$, expressing w_τ in terms of v becomes problematic, as $v_{T-\tau} = v_{\infty-\tau} = v_\infty$, which can be misleading. This distinction is crucial because the ERM-optimal policy may not be stationary in the infinite-horizon case.

The value function for this notation analogous to Eq. (5.6) and Eq. (5.7) are

$$w_\tau^\pi(s) = \text{ERM}_{\beta \cdot \gamma^\tau} [\mathfrak{R}_{\tau:T}^\pi(s)], \quad (\text{C.5})$$

$$w_\tau^*(s) = \max_{\pi \in \Pi_{\text{MR}}^{\tau:T}} \text{ERM}_{\beta \cdot \gamma^\tau} [\mathfrak{R}_{\tau:T}^\pi(s)]. \quad (\text{C.6})$$

The value iteration for this notation analogous to Eq. (5.9) and the optimal policy Eq. (5.10) for finite horizon ERM dynamic program with $\check{w}_T^*(s_T) = 0$, $\forall s_T \in \mathcal{S}$ for all $\tau \in (T-1) : 0$

$$w_\tau^*(s) = \max_{a \in \mathcal{A}} \text{ERM}_{\beta \cdot \gamma^\tau} [r(s, a) + \gamma \cdot w_{\tau+1}^*(\tilde{s}')], \quad (\text{C.7})$$

$$\check{\pi}_\tau^*(s) \in \arg \max_{a \in \mathcal{A}} \text{ERM}_{\beta \cdot \gamma^\tau} [r(s, a) + \gamma w_{\tau+1}^*(\tilde{s}')]. \quad (\text{C.8})$$

Now we define each value function in the new notation as:

1. $\underline{\pi}^* = \{\pi_\tau^*\}_{\tau=0}^\infty$ and $w^* = w^{\pi^*} = \{w_\tau^*\}_{\tau=0}^\infty$ are the optimal policy and optimal value function of the ERM-MDP in the infinite-horizon discounted setting with $T = \infty$ and $\gamma \in (0, 1)$. In other words, π^* is a solution to the ERM-MDP optimization (5.3).
2. $\underline{\pi}_\infty^*$ and $w_\infty^* = w_\infty^{\pi_\infty^*}$ are the optimal policy and value function of the MDP in the risk-neutral infinite-horizon discounted setting.
3. $\check{\underline{\pi}}^* = \{\check{\pi}_\tau^*\}_{\tau=0}^{T'}$ and $\check{w}^* = w^{\check{\pi}^*} = \{\check{w}_\tau^*\}_{\tau=0}^{T'}$ are the optimal policy and value function of the finite-horizon discounted ERM-MDP with risk level β , discount factor γ , horizon T' , and the value function at time-step T' set to $\check{w}_{T'}^* = w_\infty^*$.
4. $\hat{\underline{\pi}}^* = \{\hat{\pi}_\tau^*\}_{\tau=0}^\infty$ and its value $\hat{w}^* = \{\hat{w}_\tau^*\}_{\tau=0}^\infty$ are the policy and value function that are constructed as follows:

$$\hat{\pi}_\tau^* = \begin{cases} \tilde{\pi}_\tau^* & \text{if } 0 \leq \tau < T', \\ \pi_\infty^* & \text{if } \tau \geq T'. \end{cases} \quad \hat{w}_\tau^* = w_{\hat{\pi}_\tau^*}^* = \begin{cases} \tilde{w}_\tau^* = w_{\tilde{\pi}_\tau^*}^* & \text{if } 0 \leq \tau < T', \\ w_\infty^* = w_{\pi_\infty^*}^* & \text{if } \tau \geq T'. \end{cases}$$

This policy and value function is analogous to Algorithm 9, with opposite ordering as $\hat{\pi}_{(T'-\tau)_+}^* = \hat{\pi}_\tau^*$ and $\hat{v}_{(T'-\tau)_+}^* = \hat{w}_\tau^*$.

Our goal is to prove an upper-bound on the difference between the $\text{ERM}_\beta[\cdot]$ of the returns of policies π^* and $\hat{\pi}^*$:

$$\text{ERM}_\beta \left[\tilde{\mathfrak{R}}_\infty^{\pi^*}(s_0) \right] - \text{ERM}_\beta \left[\tilde{\mathfrak{R}}_\infty^{\hat{\pi}^*}(s_0) \right] \quad \text{or equivalently} \quad w_0^{\pi^*}(s_0) - w_0^{\hat{\pi}^*}(s_0),$$

where s_0 denotes the initial states.

As the first step of the proof, we bound the difference between the infinite-horizon γ -discounted risk-neutral and ERM value functions of any policy π , at any time-step $\tau = 0:\infty$, and any state $s \in \mathcal{S}$ as

$$\begin{aligned} 0 &\stackrel{(a)}{\leq} w_\infty^\pi(s) - w_\tau^\pi(s) = \mathbb{E}[\tilde{\mathfrak{R}}_{0:\infty}^\pi(s)] - \text{ERM}_{\beta \cdot \gamma^\tau} \left[\tilde{\mathfrak{R}}_{\tau:\infty}^\pi(s) \right] \\ &= \mathbb{E}[\tilde{\mathfrak{R}}_{0:\infty}^\pi(s)] - \text{ERM}_{\beta \cdot \gamma^\tau} \left[\tilde{\mathfrak{R}}_{0:\infty}^\pi(s) \right] \stackrel{(b)}{\leq} \frac{\beta \cdot \gamma^\tau \cdot \Delta_{\mathfrak{R}}^2}{8}. \end{aligned} \tag{C.9}$$

(a) holds because $\mathbb{E}$ is an upper-bound on the ERM, and **(b)** follows from the result of Lemma 5.2.5.

We now bound the difference between the optimal value function of the infinite-horizon discounted ERM-MDP, $w_{T'}^*$, and the value function returned by Algorithm 9, $w_{T'}^{\hat{\pi}^*}$, at the planning horizon T' and for any state $s \in \mathcal{S}$ as follows:

$$\begin{aligned} w_{T'}^*(s) - w_{T'}^{\hat{\pi}^*}(s) &\stackrel{(a)}{\leq} w_{T'}^*(s) - w_{\hat{\pi}_{T'}^*}^\infty(s) + \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8} \\ &\stackrel{(b)}{\leq} w_{\pi^*}^\infty(s) - w_{\hat{\pi}_{T'}^*}^\infty(s) + \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8} \\ &\stackrel{(c)}{\leq} \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8}. \end{aligned}$$

(a) follows from the RHS of (C.9) and the fact that $\hat{\pi}_t^* = \pi_\infty^*$, $\forall t \geq T'$, **(b)** comes from the fact

that using the LHS of (C.9), we have $0 \leq w_\infty^* - w_{T'}^{\pi^*}$, and (c) is true because $\hat{\pi}_{T'}^*(s) = \pi_\infty^*(s) \in \arg \max_{\pi \in \Pi_{\text{MR}}} w_\infty^\pi(s), \forall s \in \mathcal{S}$, and thus, $w_\infty^*(s) - w_{T'}^{\hat{\pi}^*}(s) \leq 0$ for all s .

As the second step of the proof, we construct the value function $u_t \in \mathbb{R}^{\mathcal{S}}$ from $\hat{\pi}^*$ for all $\tau \in 0:T'$ and $s \in \mathcal{S}$ as

$$\begin{aligned} u_{T'}(s) &= w_{T'}^{\hat{\pi}^*}(s) - \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8} = w_\infty^{\pi^*}(s) - \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8} = w_\infty^*(s) - \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8}, \\ u_\tau(s) &= \max_{a \in \mathcal{A}} \text{ERM}_{\beta \cdot \gamma^\tau} [r(s, a) + \gamma \cdot u_{\tau+1}(\tilde{s}'_{\tau+1, a})] \\ &= \text{ERM}_{\beta \cdot \gamma^\tau} [r(s, \hat{\pi}_\tau^*(s)) + \gamma \cdot u_{\tau+1}(\tilde{s}'_{\tau+1, \hat{\pi}_\tau^*(s)})], \end{aligned} \quad (\text{C.10})$$

where $\tilde{s}'_{\tau+1, a}$ denotes the random variable representing the state at time $\tau + 1$ that follows by taking action $a \in \mathcal{A}$ in state s at time τ .

Note that u_t has been constructed such that (i) it is a lower-bound on $\hat{w}_\tau^* = w_\tau^{\hat{\pi}^*}$ and (ii) $\hat{\pi}^*$ is its greedy policy.

(i) is true because $w_\tau^{\hat{\pi}^*} = w_\tau^{\check{\pi}^*} = \check{w}_\tau^*$, which is the optimal finite-horizon ERM value function when we set $\check{w}_{T'}^* = w_\infty^*$, and u_t has been constructed as the optimal finite-horizon ERM value function when we set its value at the planning horizon T' by a lower-bound of w_∞^* from (C.9): $u_{T'} = w_\infty^* - \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8}$. We now provide a formal proof for this. From (C.9), we have $w_{T'}^{\hat{\pi}^*}(s) \geq u_{T'}(s)$ for all $s \in \mathcal{S}$. Then, assuming $w_{\tau+1}^{\hat{\pi}^*}(s) \geq u_{\tau+1}(s)$ for all $s \in \mathcal{S}$ (inductive hypothesis), we can use backward induction on τ to show that for all $s \in \mathcal{S}$, we have

$$\begin{aligned} &w_\tau^{\hat{\pi}^*}(s) - u_\tau(s) \\ &= \text{ERM}_{\beta \cdot \gamma^\tau} [r(s, \hat{\pi}_\tau^*(s)) + \gamma \cdot w_{\tau+1}^{\hat{\pi}^*}(\tilde{s}'_{\tau+1, \hat{\pi}_\tau^*(s)})] - \text{ERM}_{\beta \cdot \gamma^\tau} [r(s, \hat{\pi}_\tau^*(s)) + \gamma \cdot u_{\tau+1}(\tilde{s}'_{\tau+1, \hat{\pi}_\tau^*(s)})] \\ &\stackrel{(a)}{=} \text{ERM}_{\beta \cdot \gamma^\tau} [\gamma \cdot w_{\tau+1}^{\hat{\pi}^*}(\tilde{s}'_{\tau+1, \hat{\pi}_\tau^*(s)})] - \text{ERM}_{\beta \cdot \gamma^\tau} [\gamma \cdot u_{\tau+1}(\tilde{s}'_{\tau+1, \hat{\pi}_\tau^*(s)})] \\ &\stackrel{(b)}{=} \gamma \cdot \left(\text{ERM}_{\beta \cdot \gamma^{\tau+1}} [w_{\tau+1}^{\hat{\pi}^*}(\tilde{s}'_{\tau+1, \hat{\pi}_\tau^*(s)})] - \text{ERM}_{\beta \cdot \gamma^{\tau+1}} [u_{\tau+1}(\tilde{s}'_{\tau+1, \hat{\pi}_\tau^*(s)})] \right) \\ &\stackrel{(c)}{\geq} 0. \end{aligned} \quad (\text{C.11})$$

(a) is by subtracting the constant reward from both terms. This can be done because ERM is trans-

lation invariant. **(b)** follows from the positive quasi-homogeneity of ERM (see Theorem 5.3.2). **(c)** follows from the monotonicity of ERM and the inductive hypothesis.

(ii) is true because $\hat{\pi}_\tau^*$ is a greedy policy to $\hat{w}_\tau^*$, and since subtracting a constant from all states does not change the greedy policy, it is also a greedy policy to u_τ . The last equality in (C.10) is the result of this fact.

As the third step of the proof, we show that for each $s \in \mathcal{S}$ and $\tau = 0:T'$, we have

$$w_\tau^*(s) - u_\tau(s) \leq \gamma^{T'-\tau} \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8}. \quad (\text{C.12})$$

To prove (C.12) by induction, we first show that the inequality (C.12) holds for $\tau = T'$, that is, $w_{T'}^*(s) - u_{T'}(s) \leq \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8}$, $\forall s \in \mathcal{S}$, as follows:

$$\begin{aligned} w_{T'}^*(s) - u_{T'}(s) &= w_{T'}^*(s) - w_\infty^*(s) + \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8} \\ &= w_{T'}^{\pi^*}(s) - w_\infty^*(s) + \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8} \\ &\stackrel{(a)}{\leq} w_\infty^{\pi^*}(s) - w_\infty^*(s) + \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8} \stackrel{(b)}{\leq} \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{R}}^2}{8}. \end{aligned} \quad (\text{C.13})$$

(a) follows from the LHS of (C.9) that $0 \leq w_\infty^{\pi^*}(s) - w_{T'}^{\pi^*}(s)$, and **(b)** comes from the fact that w_∞^* is the optimal value function of the infinite-horizon discounted risk-neutral MDP, and thus, $w_\infty^*(s) \leq w_\infty^{\pi^*}(s)$.

Now assuming that (C.12) holds for $\tau + 1$ for each $s \in \mathcal{S}$ (inductive hypothesis), we use backward induction on τ and show that for all $s \in \mathcal{S}$, we have

$$\begin{aligned} &w_\tau^*(s) - u_\tau(s) \\ &\stackrel{(a)}{=} \text{ERM}_{\beta \cdot \gamma^\tau} \left[r(s, \underline{\pi}_\tau^*(s)) + \gamma \cdot w_{\tau+1}^*(\tilde{s}'_{\tau+1, \underline{\pi}_\tau^*(s)}) \right] - \text{ERM}_{\beta \cdot \gamma^\tau} \left[r(s, \hat{\pi}_\tau^*(s)) + \gamma \cdot u_{\tau+1}(\tilde{s}'_{\tau+1, \hat{\pi}_\tau^*(s)}) \right] \\ &\stackrel{(b)}{\leq} \text{ERM}_{\beta \cdot \gamma^\tau} \left[r(s, \underline{\pi}_\tau^*(s)) + \gamma \cdot w_{\tau+1}^*(\tilde{s}'_{\tau+1, \underline{\pi}_\tau^*(s)}) \right] - \text{ERM}_{\beta \cdot \gamma^\tau} \left[r(s, \underline{\pi}_\tau^*(s)) + \gamma \cdot u_{\tau+1}(\tilde{s}'_{\tau+1, \underline{\pi}_\tau^*(s)}) \right] \\ &\stackrel{(c)}{=} \text{ERM}_{\beta \cdot \gamma^\tau} \left[\gamma \cdot w_{\tau+1}^{\pi^*}(\tilde{s}'_{\tau+1, \underline{\pi}_\tau^*(s)}) \right] - \text{ERM}_{\beta \cdot \gamma^\tau} \left[\gamma \cdot u_{\tau+1}(\tilde{s}'_{\tau+1, \underline{\pi}_\tau^*(s)}) \right] \\ &\stackrel{(d)}{=} \gamma \cdot \left(\text{ERM}_{\beta \cdot \gamma^{\tau+1}} \left[w_{\tau+1}^{\pi^*}(\tilde{s}'_{\tau+1, \underline{\pi}_\tau^*(s)}) \right] - \text{ERM}_{\beta \cdot \gamma^{\tau+1}} \left[u_{\tau+1}(\tilde{s}'_{\tau+1, \underline{\pi}_\tau^*(s)}) \right] \right). \end{aligned} \quad (\text{C.14})$$

(a) holds by the definition of w_τ^* and u_τ , (b) follows from $\hat{\pi}^*$ being greedy to u , (c) is by subtracting the constant reward from both terms which can be done because ERM is translation invariant, and finally (d) follows from the positive quasi-homogeneity of ERM (see Theorem 5.3.2).

Now we can write the following sequence of inequalities:

$$\begin{aligned}
w_{\tau+1}^{\pi^*}(s) - u_{\tau+1}(s) &\stackrel{(a)}{\leq} \gamma^{T'-\tau-1} \cdot \frac{\beta \cdot \gamma^{T'} \cdot V^2}{8}, \quad \forall s \in \mathcal{S} \\
\text{ERM}_{\beta \cdot \gamma^{\tau+1}} \left[w_{\tau+1}^{\pi^*}(\tilde{s}) \right] - \text{ERM}_{\beta \cdot \gamma^{\tau+1}} \left[u_{\tau+1}(\tilde{s}) \right] &\stackrel{(b)}{\leq} \gamma^{T'-\tau-1} \cdot \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{H}}^2}{8} \\
\gamma \cdot (\text{ERM}_{\beta \cdot \gamma^{\tau+1}} \left[w_{\tau+1}^{\pi^*}(\tilde{s}) \right] - \text{ERM}_{\beta \cdot \gamma^{\tau+1}} \left[u_{\tau+1}(\tilde{s}) \right]) &\leq \gamma^{T'-\tau} \cdot \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{H}}^2}{8}. \tag{C.15}
\end{aligned}$$

(a) follows from the inductive hypothesis and (b) comes from the monotonicity and translation invariance of ERM.

We can conclude the induction by combining (C.14) and (C.15).

Now that we proved (C.12), we can set $\tau = 0$ in it and use the fact that for all $\tau \in 0:T'$, we have $u_\tau(s) \leq w_{\frac{\hat{\pi}^*}{\tau}}(s)$, to write

$$\begin{aligned}
w_0^*(s_0) - u_0(s_0) &\leq \gamma^{T'} \cdot \frac{\beta \cdot \gamma^{T'} \cdot \Delta_{\mathfrak{H}}^2}{8} \\
\Rightarrow w_0^{\pi^*}(s_0) - w_0^{\hat{\pi}^*}(s_0) &\leq \frac{\beta \cdot \gamma^{2T'} \cdot \Delta_{\mathfrak{H}}^2}{8}.
\end{aligned}$$

Finally, we change notation from w back to v ,

$$v_{T'}^{\pi^*}(s_0) - v_T^{\hat{\pi}^*}(s_0) \leq \frac{\beta \cdot \gamma^{2T'} \cdot \Delta_{\mathfrak{H}}^2}{8} \tag{C.16}$$

and

$$\text{ERM}_{\beta} \left[\tilde{\mathfrak{R}}_{\infty}^{\pi^*}(s_0) \right] - \text{ERM}_{\beta} \left[\tilde{\mathfrak{R}}_{\infty}^{\hat{\pi}^*}(s_0) \right] \leq \frac{\beta \cdot \gamma^{2T'} \cdot \Delta_{\mathfrak{H}}^2}{8}$$

which concludes the proof. $\square$

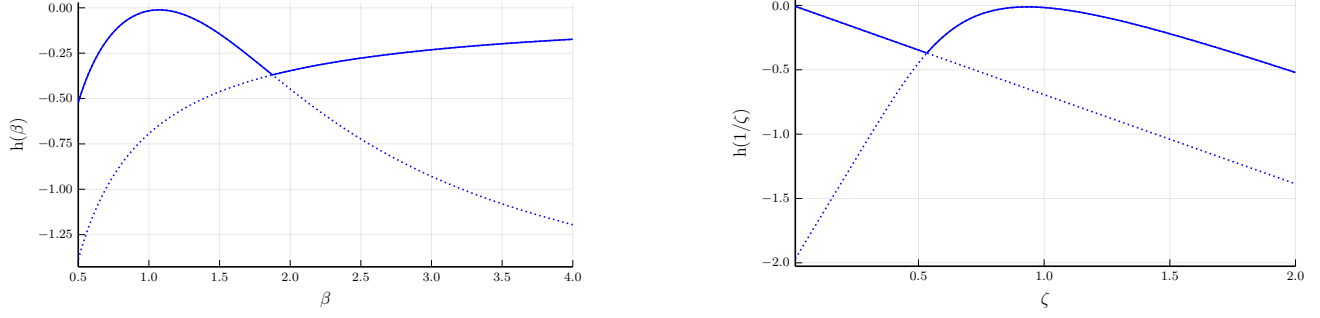


Figure C.1: Plots of $h(\beta)$ (left) and $\zeta \mapsto h(\zeta^{-1})$ (right) for $\alpha = 0.5$, which are used in the proof of Proposition C.3.1.

C.3 Proofs of Section 5.4

Before reporting the proofs for the theorems in Section 5.4, we state some results that highlight certain properties of EVaR. Recall that the function $h: \mathbb{R} \rightarrow \mathbb{R}$ is defined in (5.13) as

$$h(\beta) = \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_{\beta} \left[\tilde{\mathfrak{R}}_T^{\pi} \right] + \beta^{-1} \cdot \log(\alpha) \right) .$$

Because solving EVaR-MDP reduces to computing $\max_{\beta \geq 0} h(\beta)$, it would be ideal if it were concave or at least quasi-concave. Without the maximum in the definition of h (e.g., when $|\Pi_{\text{MR}}| = 1$), it is easy to see that the function $\zeta \mapsto h(\zeta^{-1})$ is concave [4], and thus, h is quasi-concave. Unfortunately, the following proposition shows that h is not necessarily quasi-concave when $|\Pi_{\text{MR}}| > 1$ (with the maximum), which precludes the use of more efficient optimization techniques for solving $\max_{\beta \geq 0} h(\beta)$.

Proposition C.3.1. *There exists an MDP and $\alpha \in [0, 1)$ such that the function $h: \mathbb{R} \rightarrow \mathbb{R}$ defined in (5.13) is neither concave nor convex either in β or β^{-1} .*

Proof. We show the property by constructing a counter-example for which the function h is not concave. Consider an EVaR-MDP with states $\mathcal{S} = \{s_0, s_1, s_2, s_3\}$, actions $\mathcal{A} = \{a_1, a_2\}$, a finite-horizon objective with $T = 2$ and $\gamma = 1$, and the following parameters:

$$p(\cdot \mid s_0, a_1) = [0, 0, 1, 0], \quad p(\cdot \mid s_0, a_2) = [0, 0.02, 0, 0.98], \quad r(\cdot, a_i) = [0, -2, 0, 1], \quad \forall i \in \{1, 2\} .$$

The transition probabilities from s_1, s_2, s_3 are irrelevant because the horizon is $T = 2$. Since the reward is independent of the action and only depends on the state, the returns of the policies depend only on the action they take at state s_0 . Thus, we only have two returns for all the policies in this MDP. Setting the confidence parameter to $\alpha = 0.5$, the plots in Appendix C.3 show that neither $h(\beta)$ (*left*) nor $\zeta \mapsto h(\zeta^{-1})$ (*right*) is concave. In each plot, the functions for the two individual returns are indicated by dotted lines and are concave, but their maximum, shown by the solid line, is not. $\square$

The following technical lemma will be useful when analyzing the optimal EVaR solution.

Lemma C.3.2. *Assume a fixed $\alpha \in (0, 1)$ and a random variable $\tilde{x} \in \mathbb{X}$. Then, either the supremum in (5.2) is attained at some $\beta^* > 0$ or*

$$\text{EVaR}_\alpha[\tilde{x}] = \lim_{\beta \rightarrow \infty} (\text{ERM}_\beta[\tilde{x}] + \beta^{-1} \cdot \log(\alpha)).$$

Proof. The lemma follows directly from Proposition 2.11 in [6]. $\square$

C.3.1 Proof of Theorem 5.4.1

Proof. Recall that we assume that the supremum in (5.13) is attained for some β^* . Using the existence of an optimal β^* we get by algebraic manipulation that

$$\begin{aligned} \pi^* &\in \arg \max_{\pi \in \Pi_{\text{MR}}} \text{EVaR}_\alpha \left[\tilde{\mathfrak{R}}_T^\pi \right] \\ &= \arg \max_{\pi \in \Pi_{\text{MR}}} \max_{\beta > 0} \left(\text{ERM}_\beta \left[\tilde{\mathfrak{R}}_T^\pi \right] + \beta^{-1} \cdot \log(\alpha) \right) \\ &\supseteq \arg \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_{\beta^*} \left[\tilde{\mathfrak{R}}_T^\pi \right] + (\beta^*)^{-1} \cdot \log(\alpha) \right) \\ &= \arg \max_{\pi \in \Pi_{\text{MR}}} \text{ERM}_{\beta^*} \left[\tilde{\mathfrak{R}}_T^\pi \right]. \end{aligned} \tag{C.17}$$

$\square$

C.3.2 Proof of Theorem 5.4.3

Before stating the proof of Theorem 5.4.3, we report some results that we use there. The following lemma shows how to decompose the approximation error of Algorithm 10.

Lemma C.3.3. *Let π^* be the optimal solution to (5.12) and $\hat{\pi}^*$ be the policy returned by Algorithm 10 when it is executed with a grid $\beta_1 < \dots < \beta_K$ and calls to Algorithm 9 with horizon T' . Then, for any $\alpha \in (0, 1)$, the approximation error of Algorithm 10 can be bounded as*

$$\begin{aligned} \text{EVaR}_\alpha \left[\tilde{\mathfrak{R}}_\infty^{\pi^*} \right] - \text{EVaR}_\alpha \left[\tilde{\mathfrak{R}}_\infty^{\hat{\pi}^*} \right] &\leq \frac{\beta_K \cdot \gamma^{2T'} \cdot \Delta_{\mathfrak{R}}^2}{8} + \\ &\max \left\{ \sup_{\beta \in (0, \beta_1)} h(\beta) - h(\beta_1), \max_{k \in 1:K-1} \left(\sup_{\beta \in [\beta_k, \beta_{k+1})} h(\beta) - h(\beta_k) \right), \sup_{\beta \in [\beta_K, \infty)} h(\beta) - h(\beta_K) \right\}, \end{aligned} \quad (\text{C.18})$$

where the function h is defined by (5.13). Moreover, the bound for the finite-horizon objective is the same except the last term that depends on T' is zero.

Proof. First, recall that Algorithm 10 calls Algorithm 9 for each value $\beta_1, \dots, \beta_K$ in the grid and Algorithm 9 returns an approximately optimal policy $\hat{\pi}^*$ for the ERM-MDP with the corresponding risk level β_k and its corresponding value function $\hat{v}^k$. In the following derivation, we use $\hat{\pi}_t^k$ and $\hat{v}_t^k$ for $t = 0:T', k = 1:K$ for the policy and value function, respectively, computed for β_k , by Algorithm 9. That is, the value function $\hat{v}_t^k$ uses ERM in the time steps $\tau = 0:T'-1$ and the standard risk-neutral value function $\hat{v}_T^k = v_\infty^*$ thereafter. In contrast, the value $v^{\hat{\pi}^k}$ refers to the true ERM value function of the policy $\hat{\pi}^k$.

Using arguments analogous to (C.11), one can show for each $k = 1:K$ that

$$v_{T'}^{\hat{\pi}^k}(s) \leq \hat{v}_{T'}^k(s), \quad (\text{C.19})$$

for each $s \in \mathcal{S}$. Similarly, using arguments analogous to (C.13), one can show that

$$\hat{v}_{T'}^k(s) \leq v_{T'}^{\hat{\pi}^k}(s) + \frac{\beta_k \cdot \gamma^{2T'} \cdot \Delta_{\mathfrak{R}}^2}{8}, \quad (\text{C.20})$$

for each $s \in \mathcal{S}$.

Given the definition of $\hat{v}$, we can also define the EVaR objective function $h: \mathbb{R} \rightarrow \mathbb{R}$ and its approximation in the discrete points $(\beta_k)_{k=1}^K$ as

$$\begin{aligned} h(\beta) &= \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_{\beta} \left[\tilde{\mathfrak{R}}_{\infty}^{\pi} \right] + \beta^{-1} \cdot \log(\alpha) \right) \\ &= \max_{\pi \in \Pi_{\text{MR}}} \left(v_{T'}^{\pi}(s_0) + \beta^{-1} \cdot \log(\alpha) \right) \\ \tilde{h}_k &= \hat{v}_{T'}^k(s_0) + \beta_k^{-1} \cdot \log(\alpha) . \end{aligned}$$

The bound in (C.19) then implies for $k = 1:K$ that

$$h(\beta_k) \leq \tilde{h}_k. \quad (\text{C.21})$$

Note that $\hat{\pi}^*$ refers to the EVaR-MDP policy computed by Algorithm 10 and, therefore, $\hat{\pi}^* = \hat{\pi}^{k^*}$ for the optimal k^* . Then, assuming that $k^* \in \arg \max_{k=1:K} \tilde{h}_k$ in Algorithm 10, we get that $\text{EVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_{\infty}^{\pi^*} \right] - \text{EVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_{\infty}^{\hat{\pi}^*} \right] = \delta$ for

$$\begin{aligned} \delta &= \text{EVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_{\infty}^{\pi^*} \right] - \sup_{\beta > 0} \left(\text{ERM}_{\beta} \left[\tilde{\mathfrak{R}}_{\infty}^{\hat{\pi}^*} \right] + \beta^{-1} \cdot \log(\alpha) \right) \\ &\leq \text{EVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_{\infty}^{\pi^*} \right] - v_{T'}^{\hat{\pi}^*}(s_0) - \beta_{k^*}^{-1} \cdot \log(\alpha) && \text{Substitute feasible } \beta_{k^*} \\ &= \text{EVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_{\infty}^{\pi^*} \right] - v_{T'}^{\pi^{k^*}}(s_0) - \beta_{k^*}^{-1} \cdot \log(\alpha) && \text{Choice of } \hat{\pi}^* \\ &\leq \text{EVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_{\infty}^{\pi^*} \right] - \hat{v}_{T'}^{k^*}(s_0) - \beta_{k^*}^{-1} \cdot \log(\alpha) + \frac{\beta_{k^*} \cdot \gamma^{2T'} \cdot \Delta_{\mathfrak{R}}^2}{8} && \text{From (C.20)} \\ &\leq \text{EVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_{\infty}^{\pi^*} \right] - \hat{v}_{T'}^{k^*}(s_0) - \beta_{k^*}^{-1} \cdot \log(\alpha) + \frac{\beta_K \cdot \gamma^{2T'} \cdot \Delta_{\mathfrak{R}}^2}{8} && \text{Because } \beta_K \geq \beta_{k^*} \\ &= \text{EVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_{\infty}^{\pi^*} \right] - \max_{k=1:K} \tilde{h}_k + \frac{\beta_K \cdot \gamma^{2T'} \cdot \Delta_{\mathfrak{R}}^2}{8} && \text{From the optimality of } k^* \\ &\leq \text{EVaR}_{\alpha} \left[\tilde{\mathfrak{R}}_{\infty}^{\pi^*} \right] - \max_{k=1:K} h(\beta_k) + \frac{\beta_K \cdot \gamma^{2T'} \cdot \Delta_{\mathfrak{R}}^2}{8} && \text{From (C.21)} \\ &\leq \sup_{\beta > 0} h(\beta) - \max_{k=1:K} h(\beta_k) + \frac{\beta_K \cdot \gamma^{2T'} \cdot \Delta_{\mathfrak{R}}^2}{8}. && \text{From the definition of } \pi^* \end{aligned}$$

The lemma then follows by decomposing the supremum above as

$$\sup_{\beta > 0} h(\beta) = \max \left\{ \sup_{\beta \in (0, \beta_1)} h(\beta), \sup_{\beta \in [\beta_k, \beta_{k+1})} h(\beta), \sup_{\beta \in [\beta_k, \infty)} h(\beta) \right\}.$$

□

The following three lemmas now bound each one of the terms in the maximum in (C.18).

Lemma C.3.4. *The function $h: \mathbb{R} \rightarrow \mathbb{R}$ defined in (5.13) satisfies that*

$$\sup_{\beta \in (0, \beta_1)} h(\beta) - h(\beta_1) \leq \frac{\beta_1 \cdot \Delta_{\mathfrak{R}}^2}{8}.$$

Therefore, for any $\delta > 0$, $\sup_{\beta \in (0, \beta_1)} h(\beta) - h(\beta_1) \leq \delta$ when

$$\beta_1 \leq \frac{8\delta}{\Delta_{\mathfrak{R}}^2}.$$

Proof. Because the function $\beta \mapsto \text{ERM}_{\beta}[\tilde{x}]$ is non-increasing as shown in Lemma 5.2.4 and $\beta^{-1} \cdot \log(\alpha)$ is increasing for $\beta > 0$, we derive the bound as

$$\begin{aligned} \sup_{\beta \in (0, \beta_1)} h(\beta) - h(\beta_1) &= \sup_{\beta \in (0, \beta_1)} \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_{\beta}[\tilde{\mathfrak{R}}_{\infty}^{\pi}] + \beta^{-1} \cdot \log(\alpha) \right) - \\ &\quad \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_{\beta_1}[\tilde{\mathfrak{R}}_{\infty}^{\pi}] + \beta_1^{-1} \cdot \log(\alpha) \right) \\ &\leq \sup_{\beta \in (0, \beta_1)} \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_0[\tilde{\mathfrak{R}}_{\infty}^{\pi}] + \beta^{-1} \cdot \log(\alpha) \right) - \\ &\quad \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_{\beta_1}[\tilde{\mathfrak{R}}_{\infty}^{\pi}] + \beta_1^{-1} \cdot \log(\alpha) \right) \\ &\leq \max_{\pi \in \Pi_{\text{MR}}} \text{ERM}_0[\tilde{\mathfrak{R}}_{\infty}^{\pi}] - \max_{\pi \in \Pi_{\text{MR}}} \text{ERM}_{\beta_1}[\tilde{\mathfrak{R}}_{\infty}^{\pi}] \\ &\leq \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_0[\tilde{\mathfrak{R}}_{\infty}^{\pi}] - \text{ERM}_{\beta_1}[\tilde{\mathfrak{R}}_{\infty}^{\pi}] \right). \end{aligned}$$

The lemma then follows readily by algebraic manipulation from Lemma 5.2.5 because $\text{ERM}_0[\tilde{\mathfrak{R}}_{\infty}^{\pi}] = \mathbb{E}[\tilde{\mathfrak{R}}_{\infty}^{\pi}]$. □

Lemma C.3.5. *The function $h: \mathbb{R} \rightarrow \mathbb{R}$ defined in (5.13) satisfies that*

$$\sup_{\beta \in [\beta_k, \beta_{k+1})} h(\beta) - h(\beta_k) \leq (\beta_{k+1}^{-1} - \beta_k^{-1}) \cdot \log(\alpha)$$

for each $k \in 1:K$. Therefore, for any $\delta > 0$, $\sup_{\beta \in [\beta_k, \beta_{k+1})} h(\beta) - h(\beta_k) \leq \delta$ when

$$\beta_{k+1} \leq \frac{\beta_k \cdot \log(\alpha)}{\beta_k \delta + \log(\alpha)} \quad (\text{C.22})$$

and $\beta_k \delta + \log(\alpha) < 0$. If $\beta_k \delta + \log(\alpha) \geq 0$, then β_{k+1} is not bounded from above.

Proof. From the definition of h and because the function $\beta \mapsto \text{ERM}_\beta[\tilde{x}]$ is non-increasing as shown in Lemma 5.2.4 and $\beta^{-1} \cdot \log(\alpha)$ is increasing for $\beta > 0$, we have

$$\begin{aligned} \sup_{\beta \in [\beta_k, \beta_{k+1})} h(\beta) - h(\beta_k) &= \sup_{\beta \in [\beta_k, \beta_{k+1})} \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_\beta \left[\tilde{\mathfrak{R}}_\infty^\pi \right] + \beta^{-1} \cdot \log(\alpha) \right) - \\ &\quad \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_{\beta_k} \left[\tilde{\mathfrak{R}}_\infty^\pi \right] + \beta_k^{-1} \cdot \log(\alpha) \right) \\ &\leq \sup_{\beta \in [\beta_k, \beta_{k+1})} \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_\beta \left[\tilde{\mathfrak{R}}_\infty^\pi \right] + \beta_{k+1}^{-1} \cdot \log(\alpha) \right) - \\ &\quad \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_{\beta_k} \left[\tilde{\mathfrak{R}}_\infty^\pi \right] + \beta_k^{-1} \cdot \log(\alpha) \right) \\ &\leq \sup_{\beta \in [\beta_k, \beta_{k+1})} \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_\beta \left[\tilde{\mathfrak{R}}_\infty^\pi \right] - \text{ERM}_{\beta_k} \left[\tilde{\mathfrak{R}}_\infty^\pi \right] \right) + \\ &\quad (\beta_{k+1}^{-1} \cdot \log(\alpha) - \beta_k^{-1} \cdot \log(\alpha)) \\ &\leq \beta_{k+1}^{-1} \cdot \log(\alpha) - \beta_k^{-1} \cdot \log(\alpha). \end{aligned}$$

The lemma then follows readily by algebraic manipulation. □

It is important to note that the multiplicative steps in Lemma C.3.5 increase with an increasing k . In particular, when $\delta\beta_k = -\log(\alpha)$, the constraint on β_{k+1} becomes vacuous with $\beta_{k+1} \leq \infty$. At this point, we know that β_k is the last grid point that needs to be evaluated in order to guarantee an error of δ .

Lemma C.3.6. *The function $h: \mathbb{R} \rightarrow \mathbb{R}$ defined in (5.13) satisfies that*

$$\sup_{\beta \in [\beta_K, \infty)} h(\beta) - h(\beta_K) \leq \frac{-\log(\alpha)}{\beta_K}.$$

Therefore, for any $\delta > 0$, $\sup_{\beta \in [\beta_K, \infty)} h(\beta) - h(\beta_K) \leq \delta$ when

$$\beta_K \geq \frac{-\log(\alpha)}{\delta}.$$

Proof. From the definition of h and because $\beta \mapsto \text{ERM}_\beta[\tilde{x}]$ is non-increasing and $\beta^{-1} \cdot \log(\alpha)$ is increasing for $\beta > 0$, we have

$$\begin{aligned} \sup_{\beta \in [\beta_K, \infty)} h(\beta) - h(\beta_K) &\leq \sup_{\beta \in [\beta_K, \infty)} \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_\beta \left[\tilde{\mathfrak{R}}_\infty^\pi \right] + \beta^{-1} \cdot \log(\alpha) \right) - \\ &\quad \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_{\beta_K} \left[\tilde{\mathfrak{R}}_\infty^\pi \right] + \beta_K^{-1} \cdot \log(\alpha) \right) \\ &\stackrel{(a)}{\leq} \sup_{\beta \in [\beta_K, \infty)} \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_\beta \left[\tilde{\mathfrak{R}}_\infty^\pi \right] \right) - \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_{\beta_K} \left[\tilde{\mathfrak{R}}_\infty^\pi \right] + \beta_K^{-1} \cdot \log(\alpha) \right) \\ &\leq \sup_{\beta \in [\beta_K, \infty)} \max_{\pi \in \Pi_{\text{MR}}} \left(\text{ERM}_\beta \left[\tilde{\mathfrak{R}}_\infty^\pi \right] - \text{ERM}_{\beta_K} \left[\tilde{\mathfrak{R}}_\infty^\pi \right] \right) - \beta_K^{-1} \cdot \log(\alpha) \\ &\leq -\beta_K^{-1} \cdot \log(\alpha). \end{aligned}$$

(a) follows because $\beta^{-1} \cdot \log(\alpha)$ is negative for all $\beta \in [\beta_K, \infty)$. The lemma then follows readily by algebraic manipulation. $\square$

Equipped with the above lemmas, we are now ready to prove Theorem 5.4.3.

Proof. Suppose that Algorithm 10 is executed with the grid defined by (5.14) and (5.15), and with T' set as

$$T' = \frac{\log(8\delta) - \log(\beta_K \Delta_{\mathfrak{R}}^2)}{2 \log \gamma}. \tag{C.23}$$

Then, Lemmas C.3.3 to C.3.6 show that

$$\text{EVaR}_\alpha \left[\tilde{\mathfrak{R}}_\infty^{\pi^*} \right] - \text{EVaR}_\alpha \left[\tilde{\mathfrak{R}}_\infty^{\hat{\pi}^*} \right] \leq 2\delta.$$

It remains to show that the Algorithm 10 runs in time that is polynomial in $1/\delta$.

First, note that Algorithm 9 runs in time that is $O(S^2AT')$, assuming that v^∞ is computed using value iteration for some fixed $\gamma < 1$. Then using the choice of T' in (C.23), we have that evaluating a single β_k takes $O(S^2A \log(1/\delta))$ time.

Second, we need to upper-bound the value K since Algorithm 10 examines each one of these values. To emphasize that K is a function of δ , we denote it as K_δ in the remainder of the proof. To upper-bound K_δ , we first construct a lower-bound on each β_{k+1} , $\forall k \in 1:K_\delta-1$ using definition (5.14) and the fact that $\beta_1 \leq \beta_k$ as

$$\begin{aligned}\beta_{k+1} &= \beta_k \cdot \frac{\log(\alpha)}{\beta_k \cdot \delta + \log(\alpha)} \\ &\geq \beta_k \cdot \frac{\log(\alpha)}{\beta_1 \cdot \delta + \log(\alpha)} \\ &\geq \beta_1 \cdot \left(\frac{\log(\alpha)}{\beta_1 \cdot \delta + \log(\alpha)} \right)^k.\end{aligned}\tag{C.24}$$

Here, we assume that β_1 is sufficiently small such that $\beta_k \cdot \delta + \log(\alpha) < 0$. Otherwise, we can use $K_\delta = 1$ to achieve the desired approximation error δ .

Recall that K_δ is chosen such that (5.15) is satisfied:

$$\beta_{K_\delta} \geq \frac{-\log(\alpha)}{\delta}.$$

Substituting the lower-bound on β_{K_δ} from (C.24) we get that the sufficient condition for K_δ is that

$$\beta_1 \cdot \left(\frac{\log(\alpha)}{\beta_1 \cdot \delta + \log(\alpha)} \right)^{K_\delta-1} \geq \frac{-\log(\alpha)}{\delta}.\tag{C.25}$$

Next, define a variable z as follows and substitute the value for β_1 from (5.14) to get

$$z = \frac{\beta_1 \cdot \delta}{-\log(\alpha)} = \frac{8\delta^2}{-\Delta_{\mathfrak{H}}^2 \cdot \log(\alpha)}.\tag{C.26}$$

From the assumption that $\beta_k \cdot \delta + \log(\alpha) < 0$ and the fact that $\alpha \in (0, 1)$, we get that $-\log(\alpha) \in$

$(0, \infty)$ and, thus, $z \in (0, 1)$. Substituting the variable z into (C.25) yields that the sufficient condition for K_δ is that

$$\left(\frac{1}{1-z}\right)^{K_\delta-1} \geq \frac{1}{z}.$$

Taking the log of both sides and algebraic manipulation realizing that $(1-z)^{-1} \in (0, \infty)$ gives that it is sufficient to choose K_δ such that

$$K_\delta = \frac{\log \frac{1}{z}}{\log \frac{1}{1-z}} + 1 = \frac{\log \frac{1}{z}}{\sum_{n=1}^{\infty} \frac{z^n}{n}} + 1 \leq \frac{1}{z} \cdot \log \frac{1}{z} + 1.$$

The derivation above follows by a MacLaurin expansion of the denominator which is valid because $z \in (0, 1)$. The last inequality follows because $z > 0$. Then, substituting the expression for z from (C.26), we get that

$$K_\delta \in O\left(\frac{\log(1/\delta)}{\delta^2}\right).$$

The complexity statement in the theorem then follows from the fact that running Algorithm 9 for each K_δ takes $O(S^2 A \log(1/\delta))$ time. □

C.4 Numerical Results: Details

C.4.1 Domain Details

For each domain, we also provide CSV files in supplementary material with the exact specifications of the domains we use. The states in our tabular domains are identified with integer values $0, \dots, S - 1$, and actions for each state s are identified also with integer values $0, \dots, A_s - 1$. Note that the action counts may be state-dependent. In our experiments, we assume that the reward $r: \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ depends both on the originating and the destination state. Each CSV file has the following columns: “idstatefrom”, “idaction”, “idstateto”, “probability”, and “reward”. Each row entry specifies a transition from “idstatefrom” after taking an action “idaction” to state “idstateto” with the associated probability and reward. It is important to note that each combination (“idstatefrom”, “idaction”, “idstateto”) is not necessarily unique; repeated combinations indicate different transitions to the same state. These transitions need to be properly accounted for when computing the risk of $r(\tilde{s}, \tilde{a}, \tilde{s}')$ since the associated rewards may be different.

C.4.1.1 Machine Replacement

This is the domain with the nominal transition probabilities described in [43]. We use the same discount factor $\gamma = 0.9$ and the time horizon $T = 100$. The initial state s_0 is that the machine is in the repair state R_1 indexed as “idstate” = 1. The exact definition of the problem is given in `machine.csv`.

C.4.1.2 Gambler’s Ruin

This domain is based on a problem given in [13]. In this problem, a gambler starts with an initial capital of c_0 and can invest some of it in each time period. This investment doubles with a probability p and is lost with a probability $1 - p$. The reward is zero until a target wealth level c_f is achieved. The reward in the absorbing state c_f is 1.0. The state in this problem is the current,

and the action is the investment. We use the initial capital $c_0 = 7$, the target capital $c_f = 10$, the probability of winning $p = 0.7$, and the discount factor $\gamma = 0.95$. For this domain, we use a longer horizon $T = 200$. The precise definition of the problem is given in `ruin.csv`.

C.4.1.3 Inventory Management

This is a classical single-product stochastic inventory control problem [103]. The states $\mathcal{S} = \{0, \dots, S_{\max}\}$ represent the current stock of the product. The actions $\mathcal{A}_s = \{0, \dots, \min\{A_{\max}, (S_{\max} - s)\}\}$ for each state $s \in \mathcal{S}$ represent the amount of product ordered. The integer-valued random variable D represents the random demand. At any time step t , the next state S_{t+1} is a random variable computed as

$$S_{t+1} = [s_t + a_t - D]_+.$$

The amount of product sold computed in a time step is

$$l_t = s_t - S_{t+1} + a_t.$$

The revenue u and expenses e are computed as

$$\begin{aligned} u_t &= l_t \cdot p \\ x_t &= \begin{cases} c^h \cdot s_t & \text{if } a_t = 0 \\ c^h \cdot s_t + c^f + c^v \cdot a_t & \text{otherwise} \end{cases}. \end{aligned}$$

Here, p is the purchase price and c^h , c^f , and c^v are holding, fixed, and variable costs, respectively. The reward is then $r_t = u_t - x_t$.

The specific parameters that we use for the two inventory problems are summarized in Table C.1. The time horizon for both problems is $T = 100$. The exact specifications of the two inventory domains are given in `inventory1.csv` and `inventory2.csv`.

Table C.1: Parameters of the inventory management problems.

Parameter	INV1	INV2
γ	0.9	0.9
$S_{\max}$	100	40
$A_{\max}$	50	10
Distribution D	Categorical	Poisson, $\lambda = 30$
p	16	4.99
c^h	0.3	0.05
c^f	2	0.49
c^v	5	2.49

Table C.2: Parameters of Algorithm 10 for each benchmark problem.

Domain	Tolerance δ	Scale $(1 - \gamma) \cdot \Delta_{\mathfrak{R}}$
MR	2	20
GR	0.5	1
INV1	5	5
INV2	1	1
RS	1	1

C.4.1.4 Riverswim

This is an adapted version of the riverswim problem described in [121]. The discount factor in this problem is $\gamma = 0.9$ and the horizon is $T = 100$.

C.4.2 Algorithms

C.4.2.1 Algorithm 10

We implemented the algorithm in Julia, closely following the pseudo-code in Algorithm 10. The grid of values β_k are selected according to (5.14) with the parameters δ and $\Delta_{\mathfrak{R}}$ given in Table C.2. We chose the value $\Delta_{\mathfrak{R}}$ based on the reward function structure of the problem and chose the tolerance value δ accordingly to be about 10% of the value function span. For small problems, like MR, we reduced δ even further. Anecdotally, δ has a smaller impact on the solution’s quality than $\Delta_{\mathfrak{R}}$, but can significantly increase the computation time. The ERM-MDP sub-problem is solved exactly, which is possible because the horizon is finite.

C.4.2.2 Naive Grid

Follows the same approach as Algorithm 10, but uses $\beta_k, k = 1:K$ computed as

$$\beta_1 = 10^5, \quad \beta_k = \frac{10 - \beta_1}{K - 1}.$$

The value K is chosen to be the same for each domain as the optimized K in (5.14).

C.4.2.3 Naive Level

Follows the same approach as Algorithm 10, but computes the value v^k for β^k by solving the following dynamic program for each $s \in \mathcal{S}$ and $t = 1:T$ as

$$v_t^k(s) = \max_{a \in \mathcal{A}} \text{ERM}_{\beta_k} [r_{sa} + \gamma \cdot v_{t-1}^k(\tilde{s}'_{sa})],$$

where $\tilde{s}'_{sa}$ is the random variable that represents the state that follows after taking an action a in state s .

C.4.2.4 Nested EVaR, CVaR, ERM

For any risk measure $\psi: \mathbb{X} \rightarrow \mathbb{R}$, like CVaR and EVaR, solve computes the value v^* by solving the following dynamic program for each $s \in \mathcal{S}$ and $t = 1:T$ as

$$v_t^*(s) = \max_{a \in \mathcal{A}} \psi [r_{sa} + \gamma \cdot v_{t+1}^*(\tilde{s}'_{sa})],$$

where $\tilde{s}'_{sa}$ is the random variable that represents the state that follows after taking an action a in state s . Then, we evaluate a greedy policy $\pi_t^*: \mathcal{S} \rightarrow \mathcal{A}, t = 0:T-1$ constructed to satisfy

$$\pi_t^*(s) \in \max_{a \in \mathcal{A}} \psi [r_{sa} + \gamma \cdot v_{t-1}^*(\tilde{s}'_{sa})].$$

For EVaR and CVaR, we use $\alpha = 0.1$ and for ERM, we use $\beta = 0.5$.

Table C.3: $\text{VaR}_{0.1} \left[\tilde{\mathfrak{R}}_T^\pi \right]$ for π returned by each method.

Method	MR	GR	INV1	INV2	RS
Algorithm 10	-2.82	10.80	87.80	202	500
Naive grid	-2.90	10.80	52.60	202	501
Naive level	-10.00	11.40	83.30	201	217
Risk neutral	-2.90	12.60	67.50	202	499
Nested CVaR	-10.00	0.00	0.00	138	217
Nested EVaR	-10.00	10.30	0.00	173	217
ERM	-3.00	9.75	62.40	187	217
Nested ERM	-10.00	10.30	32.20	157	217
Augmented CVaR	-3.18	12.56	55.80	82	110

C.4.2.5 ERM

We solve the optimal ERM value function and policy as described in Section 5.3 using $\beta = 0.5$

C.4.2.6 Augmented CVaR

We implemented the tabular version of the algorithm described in [33]. We chose the discretization as recommended in [33] with the maximum number of points so that the computation finished in at most 10 minutes (about 20 times longer than the computation of other methods). One of the sources of complexity in this algorithm is that one needs to solve a linear program for every evaluation of the Bellman operator.

C.4.3 Results

Tables C.3 and C.4 show additional results for the algorithms and domains that we have compared. The results are broadly consistent with the results for other risk measures, and we include them for the sake of completeness.

Table C.4: $\mathbb{E} \left[\tilde{\mathfrak{R}}_T^\pi \right]$ for π returned by each method.

Method	MR	GR	INV1	INV2	RS
Algorithm 10	-1.01	14.30	114.00	218	873
Naive grid	-1.01	14.30	63.20	219	873
Naive level	-10.00	15.80	107.00	217	217
Risk neutral	-0.98	17.10	128.00	219	871
Nested CVaR	-10.00	0.00	0.00	142	217
Nested EVaR	-10.00	14.60	0.00	182	217
ERM	-0.99	14.20	76.40	197	217
Nested ERM	-10.00	14.60	39.70	163	217
Augmented CVaR	-2.36	14.55	69.68	135	101

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